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from matter to materials

Luminosity Determination with Restgas Background from the Target for the $\overline{\text{P}}$ANDA Experiment

Dissertation

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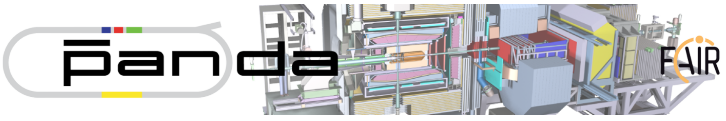

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Abstract

Understanding the dynamics of the strong interaction and the nature of exotic hadronic matter represents a fundamental frontier in modern physics. The $\overline{\text{P}}$ANDA experiment at the Facility for Antiproton and Ion Research (FAIR) is designed to explore this regime with unprecedented precision. However, a strict prerequisite for achieving its primary physics goals, particularly the absolute cross section measurements and high-precision resonance energy scans, is the highly accurate determination of the time-integrated luminosity. While this primary measurement relies on antiproton-proton elastic scattering, the specific conditions of the $\overline{\text{P}}$ANDA fixed-target setup introduce a critical challenge. The interaction of the antiproton beam with residual gas (Restgas) creates a continuously distributed background along the beam pipe. This complex, non-localized interaction region fundamentally breaks standard point-like approximations and severely limits the achievable experimental precision.

This thesis establishes a comprehensive, data-driven methodology that transforms this Restgas background distribution from a systematic liability into a precisely controlled observable. First, the software framework, *LuminosityFit*, was extended to handle the Restgas distribution during the luminosity determination process. However, the accuracy of the determination is strictly bound to the prior knowledge of the longitudinal Restgas density profile. To dynamically constrain this dominant systematic uncertainty, this work evaluates the upgrade of the $\overline{\text{P}}$ANDA Micro Vertex Detector (MVD) with High-Voltage Monolithic Active Pixel Sensors (HV-MAPS). Simulations demonstrate that the significantly reduced material budget of HV-MAPS (thinned to $50\,\mu$m) minimizes multiple scattering, achieving an exceptional longitudinal vertex resolution of approximately $20\,\mu$m. To fully exploit this hardware potential, a specialized off-IP reconstruction chain was developed to efficiently reconstruct highly displaced vertices originating from beam-Restgas interactions.

Ultimately, by integrating the experimentally measured Restgas profile back into the luminosity determination chain, the overall systematic uncertainty including the Restgas background is better than 0.5%. This research proves that the synergy of advanced fitting frameworks and high-precision HV-MAPS tracking provides a robust solution, ensuring that normalization uncertainties will not limit the $\overline{\text{P}}$ANDA experiment's capability to achieve its physics goals.

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Motivation

Understanding the fundamental building blocks of visible matter remains one of the most profound challenges in modern physics. At the microscopic level, the strong interaction, described by Quantum Chromodynamics (QCD), governs the dynamics of quarks and gluons which form the conventional hadrons, such as baryons (e.g. protons and neutrons) or mesons (e.g. pions). These hadrons have been extensively studied. However, the mechanism by which the strong interaction dynamically generates the bulk of their masses remains a subject of intense theoretical and experimental investigation. Beyond conventional hadronic states, QCD explicitly permits the existence of exotic matter configurations, such as glueballs (composed entirely of interacting gluons), hybrid hadrons (containing an actively excited gluon field), and multi-quark states like tetraquarks and pentaquarks.

The $\overline{\text{P}}$ANDA experiment at the Facility for Antiproton and Ion Research (FAIR) represents the future precision frontier of hadron physics designed to explore the uncharted territory of so-called exotic hadrons. By utilizing the brilliant high-intensity antiproton beams, $\overline{\text{P}}$ANDA addresses fundamental questions regarding the strong interaction, including charmonium spectroscopy, the search for these elusive gluonic excitations and exotic states, and the investigation of complex nucleon structure [1].

To achieve these ambitious physics goals and reliably identify extremely rare exotic signals against overwhelming conventional backgrounds, the experiment requires not only high-statistics data samples but also unprecedented precision in normalization. The absolute cross section, which serves as the critical link between theoretical QCD predictions and experimental observables, is linearly dependent on the absolute time-integrated luminosity. Consequently, the precision of any cross section measurement in the $\overline{\text{P}}$ANDA experiment is inherently limited by the accuracy of the luminosity determination [2]. Furthermore, a crucial component of the $\overline{\text{P}}$ANDA physics program involves high-precision resonance energy scans to determine the precise mass and width of newly discovered hadronic states. The accurate determination of resonance line shapes in these energy scans depends heavily on the normalization by the relative time-integrated luminosity measured across different beam momenta [3].

For the $\overline{\text{P}}$ANDA experiment, luminosity is determined by measuring the rate of antiproton-proton ($\bar{p}p$) elastic scattering, a reference process with a well-understood physical cross section [4]. While the theoretical framework for this reference channel is firmly established, its experimental realization faces severe challenges. Unlike standard collider experiments with highly focused interaction points (IP), $\overline{\text{P}}$ANDA operates with a fixed-target setup using a Cluster-Jet target. The intense hydrogen

jet causes the evaporation of hydrogen into the surrounding beam pipe which is called the residual gas or Restgas distribution. Interactions between the antiproton beam and the Restgas create continuously distributed interaction vertices along the beam axis [5]. This phenomenon results in a complex, extended target profile that deviates significantly from an ideal point-like target. Previous reconstruction algorithms, which naively approximate the IP as an extended point at the coordinate origin [6, 7], are fundamentally insufficient for this complex geometric situation. Neglecting the extensive spread of the target profiles leads to severe systematic biases in the luminosity determination, ultimately limiting the achievable precision of cross section measurements and resonance scans.

This thesis addresses these critical challenges by developing a comprehensive, data-driven methodology that transforms the Restgas distribution from an unknown, limiting background contribution into a precisely monitored and continuously measurable observable. Central to this approach is the refinement of the software framework to explicitly account for the complex, smeared geometry of the beam-Restgas overlap through the introduction of an "Effective Acceptance Map". To constrain the dominant systematic uncertainties associated with the Restgas density profile, this work evaluates a hardware-driven solution: upgrading the $\overline{\text{P}}$ANDA Micro Vertex Detector (MVD) with state-of-the-art High-Voltage Monolithic Active Pixel Sensors (HV-MAPS). Leveraging the significantly reduced material budget of HV-MAPS alongside a specialized off-IP reconstruction chain, this thesis establishes the feasibility of extracting the longitudinal Restgas distribution directly from tracking data. By integrating this measured profile back into the luminosity determination, the systematic uncertainty is successfully suppressed to better than 0.5%. Ultimately, this research provides the robust luminosity determination essential for fulfilling the high-precision physics goals of the $\overline{\text{P}}$ANDA experiment, ensuring that the deviation in the luminosity determination will not limit the experiment's ability to map the exotic hadron spectrum and explore the fundamental nature of the strong interaction.

Chapter 1

Introduction

1.1 The Standard Model

Elementary Particles

The elementary particles of the Standard Model (SM), which constitute the matter content and mediate the interactions, are summarized in Figure 1.1.

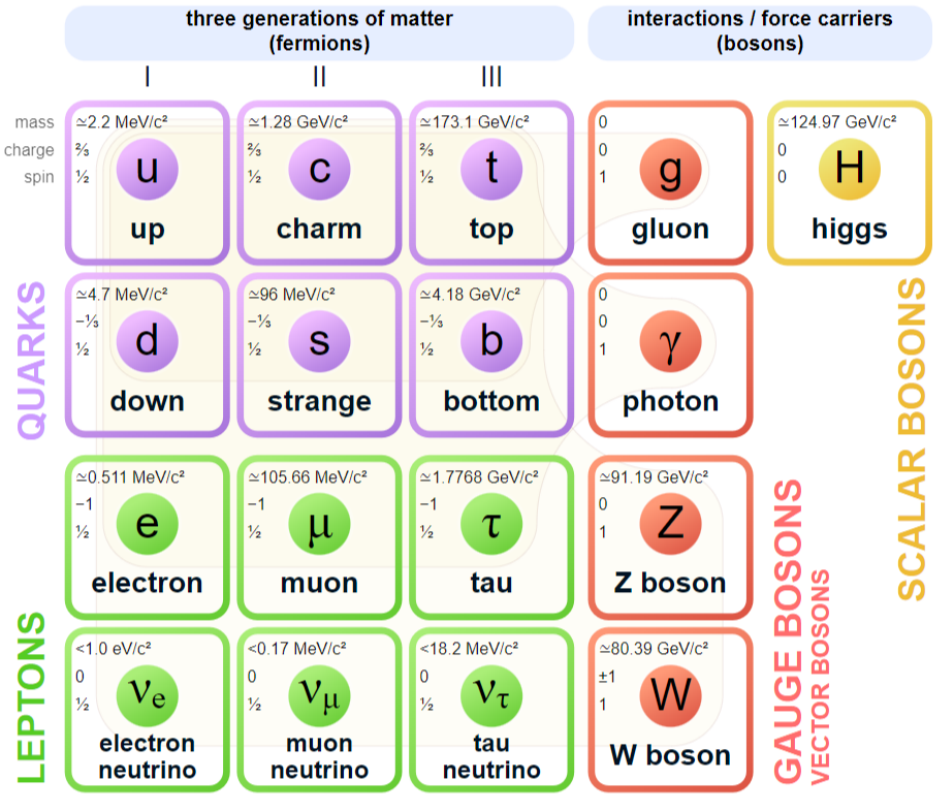


Figure 1.1: Standard model of particle physics. Figure adapted from [8].

The fundamental matter of the universe consists of **fermions**, which are half-integer spin (spin 1/2) particles that obey Fermi-Dirac statistics and the Pauli exclusion principle. These fermions are organized into three distinct generations. Particles across different generations share identical gauge quantum numbers and interaction properties, differing fundamentally only in their masses. The fermionic sector is further subdivided into two main categories: leptons and quarks. Leptons are fermions that do not participate in the strong interaction. This group comprises three electrically charged leptons: the electron (e), muon (μ), and tau (τ), which interact via both the electromagnetic and weak forces, alongside their three corresponding neutral neutrinos (ν_e, ν_μ, ν_τ), which interact exclusively through the weak force. In contrast, quarks come in six distinct flavors: up (u), down (d), charm (c), strange (s), top (t), and bottom (b). While quarks also participate in the electromagnetic and weak interactions, they uniquely carry *color charge*, subjecting them to the strong interaction. A defining characteristic of quarks is *color confinement*, they cannot exist as isolated free particles and are perpetually bound into color-neutral composite states known as hadrons, such as baryons (e.g. protons and neutrons) and mesons.

The fundamental interactions between these matter particles are mediated by integer-spin (spin 1) **gauge bosons**, which emerge naturally from the requirement of local gauge invariance in the SM Lagrangian. The electromagnetic interaction is mediated by the massless photon (γ), while the weak interaction is propagated by the massive $W^\pm$ and Z^0 bosons. The substantial masses of these weak vector bosons ($m_W \approx 80.4\,\text{GeV}$, $m_Z \approx 91.2\,\text{GeV}$) dictate the extremely short range of the weak force. Meanwhile, the strong interaction is mediated by eight massless gluons (g). Unlike photons, gluons carry color charge themselves, which permits gluon-gluon self-interactions and gives rise to the unique phenomenological property of *asymptotic freedom*.

Finally, the **Higgs boson (H)** is a massive scalar particle (spin 0) that is absolutely essential for the mathematical and physical consistency of the Standard Model. It represents the physical excitation of the Higgs field, a complex scalar doublet that permeates the vacuum. Through the mechanism of spontaneous symmetry breaking, the non-zero vacuum expectation value of this field provides the rest masses for the $W^\pm$ and Z^0 bosons. Furthermore, the fermions acquire their constituent masses via Yukawa couplings to the Higgs field, where the coupling strength directly dictates the specific mass of each fundamental particle.

Symmetries and Quantum Numbers

The Standard Model of particle physics is a relativistic quantum field theory that describes three of the four known fundamental forces: the electromagnetic, weak, and strong interactions. Mathematically, the SM is constructed as a gauge theory based on the local gauge invariance of the Lagrangian under the symmetry group:

$$SU(3)_C \times SU(2)_L \times U(1)_Y \tag{1.1}$$

Here, $SU(3)_C$ corresponds to the strong interaction acting on the color charge C, while the electroweak sector is described by the spontaneously broken symmetry

$SU(2)_L \times U(1)_Y$, acting on the weak isospin I and hypercharge Y [9].

The connection between these abstract mathematical symmetries and physical observables is established by Noether's Theorem [10]. It states that every continuous symmetry of the Lagrangian corresponds to a conserved quantum number. Consequently, the conservation of color charge, weak isospin, and hypercharge is a direct manifestation of the fundamental gauge symmetries defined in Equation 1.1. Beyond these continuous symmetries, discrete symmetries also play a crucial role in classifying particle states and constraining decay channels.

Fundamental Quantum Numbers Certain intrinsic properties define the identity of every elementary particle and dictate its participation in fundamental interactions. Among these, **spin (J)** represents the intrinsic angular momentum of a particle, quantized in units of $\hbar$. It crucially determines the statistical behavior of the particle: fermions, possessing half-integer spin, obey the Pauli exclusion principle [11], whereas bosons, with integer spin, follow Bose-Einstein statistics [12]. Another defining property is the **electric charge (Q)**, which determines the coupling strength to the electromagnetic field (γ). As an exactly conserved quantity derived from the $U(1)_{EM}$ gauge symmetry, it is related to the weak isospin (I_3) and hypercharge (Y) via the Gell-Mann-Nishijima formula, $Q = I_3 + Y/2$. Furthermore, **baryon number (B)** and **lepton number (L)** serve as additive quantum numbers associated with global $U(1)$ symmetries. Baryons, such as protons and neutrons, carry $B = 1$, while antibaryons have $B = -1$ (with constituent quarks carrying $B = 1/3$). Similarly, leptons and antileptons carry $L = 1$ and $L = -1$, respectively. Within the framework of the Standard Model, both B and L are strictly conserved in all perturbative processes.

Flavor Symmetries The strong interaction also exhibits approximate symmetries that arise from the relatively small mass differences between the lighter quark flavors. Historically introduced by Heisenberg to describe the proton and neutron as two states of a single nucleon doublet, **isospin (I)** [13] emerges from the approximate mass degeneracy of the up (u) and down (d) quarks. The third component of isospin, I_3, distinguishes these states, assigning $I_3 = +1/2$ to the u quark and $I_3 = -1/2$ to the d quark. While isospin is a conserved quantity in strong interactions, it is explicitly violated by both electromagnetic and weak interactions. Similarly, the concept of **strangeness (S)** [14] was introduced to explain the unusually long lifetimes of certain heavy hadrons, such as Kaons and the Λ baryon. It is defined by the net number of strange quarks, $S = -(n_s - n_{\bar{s}})$. Strangeness is conserved in strong and electromagnetic processes but is violated by the weak interaction, which enables the decay of strange particles into non-strange final states.

Discrete Symmetries Unlike continuous gauge symmetries, discrete symmetries describe non-continuous physical transformations of a system and play a vital role in constraining particle interactions. **Parity (P)** corresponds to a spatial inversion of the coordinate system ($\vec{x} \to -\vec{x}$). As a multiplicative quantum number, it is strictly conserved in strong and electromagnetic interactions. However, the weak

interaction maximally violates parity, a phenomenon famously demonstrated in β-decay experiments, by coupling exclusively to left-handed fermions and right-handed antifermions. Closely related is **charge conjugation (C)** [15], an operator that transforms a particle into its corresponding antiparticle. This operation reverses all internal additive quantum numbers (such as Q and B) while keeping spacetime properties like mass and spin invariant. Only strictly neutral bosons, such as the π^0, γ, and η, are true eigenstates of the C operator. Similar to parity, C-parity is conserved in strong and electromagnetic processes but violated in weak interactions. Finally, because charged particles (like $\pi^\pm$) are not C-eigenstates, **G-parity (G)** [16] was introduced as a valid generalization for isospin multiplets. Defined as the product of charge conjugation and a 180° rotation in isospin space ($G = Ce^{i\pi I_y}$), G-parity is highly useful for constraining hadronic transitions since it is conserved by the strong interaction. For instance, the selection rule $G(\pi) = -1$ strictly forbids decays such as $\rho \to 3\pi$, given that the ρ meson carries $G = +1$.

Electromagnetic and Weak Interactions

The electromagnetic interaction is the force responsible for governing the behavior of electrically charged particles and shaping the structure of atoms and molecules. It is described by Quantum Electrodynamics (QED), an Abelian gauge theory based on the $U(1)_{EM}$ symmetry. The force is mediated by the massless photon (γ), which couples to the electric charge. Due to the vanishing mass of the carrier particle, the electromagnetic interaction has an infinite range, decreasing in strength with the inverse square of the distance.

In contrast, the weak interaction is responsible for radioactive decays (such as β-decay) and nuclear fusion processes. It is unique among the fundamental forces as it is the only interaction capable of changing the flavor of quarks and violating parity (P) symmetry. The weak force is short-range ($\sim 10^{-18}$ m), a consequence of being mediated by the massive gauge bosons $W^\pm$ and Z^0.

Despite these striking differences at low energies, the electromagnetic and weak interactions are unified within the Glashow-Weinberg-Salam (GWS) theory [17, 18, 19]. In this framework, the physical gauge bosons are constructed as linear combinations of the fundamental gauge fields: the isotriplet $W_\mu^{1,2,3}$ associated with the weak isospin of the $SU(2)_L$ group (with coupling constant g) and the singlet B_μ associated with the weak hypercharge of the $U(1)_Y$ group (with coupling constant g').

The charged current interactions are mediated by the $W^\pm$ bosons, which correspond to the complex linear combinations of the first two components of the isotriplet:

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \tag{1.2}$$

For the neutral sector, the spontaneous symmetry breaking mechanism mixes the neutral fields W_μ^3 and B_μ to form the massless photon (γ) and the massive Z^0 boson [9]:

$$\begin{pmatrix} \gamma \\ Z^0 \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}, \tag{1.3}$$

where θ_W is the Weinberg weak mixing angle. As geometrically illustrated in Figure 1.2, this mixing can be intuitively visualized using a right-angled triangle, where the fundamental gauge couplings g ($SU(2)_L$ coupling constant) and g' ($U(1)_Y$ coupling constant) relate to the adjacent and opposite sides of θ_W, respectively. This visual representation highlights how the electromagnetic coupling constant e is not an independent parameter, but rather emerges directly from the mixing of the two gauge groups, satisfying the trigonometric relationship:

$$e = g \sin\theta_W = g' \cos\theta_W \tag{1.4}$$

The specific value of the weak mixing angle θ_W is a free parameter in the Standard Model and cannot be predicted theoretically. It has been determined through precision experimental measurements, such as neutrino scattering [20] and the precise determination of the Z-boson mass at electron-positron colliders (e.g., LEP, SLC) [21, 22]. These experiments yield an effective value of $\sin^2\theta_W \approx 0.23$ at the electroweak scale, which determines the relative strength of the neutral current interactions. A direct consequence of this mixing is the mass difference between the heavy vector bosons. While the photon remains massless, the masses of the Z^0 and $W^\pm$ bosons are connected by the relation:

$$m_Z = \frac{m_W}{\cos\theta_W} \tag{1.5}$$

This relationship is crucial for the consistency of the Standard Model and connects the symmetry breaking sector to the observable gauge boson masses.

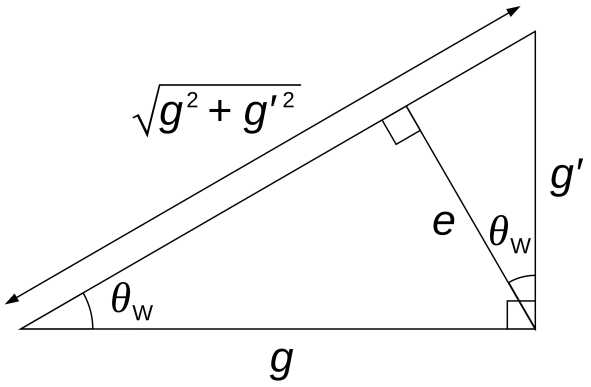


Figure 1.2: Schematic representation of the Weinberg weak mixing angle θ_W and its relation to the electroweak gauge couplings. The $SU(2)_L$ coupling constant is denoted as g, while the $U(1)_Y$ coupling constant is g'. Figure adapted from [23].

Strong interaction

The strong interaction, described theoretically by Quantum Chromodynamics (QCD), is the fundamental force responsible for binding quarks into hadrons and holding the atomic nucleus together. It is mediated by massless spin-1 gauge bosons known as

gluons. Unlike the electrically neutral photon in QED, gluons themselves carry color charge, a feature that arises from the non-Abelian structure of the underlying $SU(3)_C$ gauge symmetry. In this theory, quarks transform under the fundamental representation ($\mathbf{3}$), while antiquarks transform under the complex conjugate representation ($\bar{\mathbf{3}}$). The gluons, carrying a color and an anticolor index, span the tensor product space:

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1} \tag{1.6}$$

This mathematical decomposition implies nine possible color-anticolor combinations. As indicated in Figure 1.3, the six outer points represent gluons with distinct color-anticolor indices, while the center represents the two linear combinations of diagonal states (black circles) and a colorless singlet state (open circle). However, experimentally, only eight gluons are observed in nature. These correspond to the color octet ($\mathbf{8}$), which consists of the traceless 3×3 Hermitian matrices. In the standard Gell-Mann basis, these independent gluon states are represented as [24]:

$$\begin{aligned}
|g1\rangle &= (r\bar{b} + b\bar{r})/\sqrt{2}, & |g2\rangle &= -i(r\bar{b} - b\bar{r})/\sqrt{2}, \\
|g3\rangle &= (r\bar{g} + g\bar{r})/\sqrt{2}, & |g4\rangle &= -i(r\bar{g} - g\bar{r})/\sqrt{2}, \\
|g5\rangle &= (b\bar{g} + g\bar{b})/\sqrt{2}, & |g6\rangle &= -i(b\bar{g} - g\bar{b})/\sqrt{2}, \\
|g7\rangle &= (r\bar{r} - b\bar{b})/\sqrt{2}, & |g8\rangle &= (r\bar{r} + b\bar{b} - 2g\bar{g})/\sqrt{6}.
\end{aligned} \tag{1.7}$$

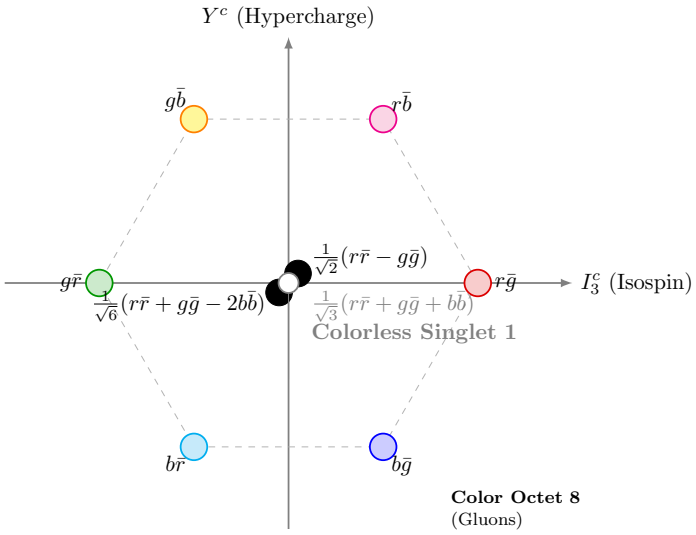


Figure 1.3: The weight diagram of the gluon octet and singlet in $SU(3)_C$ color space.

The gluon octet is characterized by these eight states being linearly independent. Crucially, because these gluons carry net color charge, they can interact with each other. This self-interaction is the defining feature of the non-Abelian nature of QCD, giving rise to unique phenomena such as *asymptotic freedom* at high energies and *color confinement* at low energies.

The ninth state arising from the decomposition is the color singlet (**1**), which is completely symmetric and colorless:

$$|g_{\text{singlet}}\rangle = (r\bar{r} + g\bar{g} + b\bar{b})/\sqrt{3} \tag{1.8}$$

Unlike the octet gluons, this singlet state carries no net color charge and is invariant under $SU(3)_C$ rotations. Consequently, it would not couple to the other gluons (no self-interaction) but would couple to all baryons with equal strength. If such a massless vector boson existed, it would mediate an infinite-range force between hadrons, similar to the electromagnetic force. Since the strong interaction is experimentally confirmed to be short-range (~ 1 fm), the existence of a physical color singlet gluon is excluded from the Standard Model [25].

The Brout-Englert-Higgs Mechanism

In the gauge sector described above, explicit mass terms for the gauge bosons are strictly forbidden by local gauge invariance, which would otherwise render the theory non-renormalizable. The generation of mass is instead achieved through the Brout-Englert-Higgs (BEH) mechanism, which introduces a complex scalar doublet field ϕ with a self-interacting potential $V(\phi)$ [26, 27, 28]:

$$V(\phi) = \mu^2(\phi^\dagger\phi) + \lambda(\phi^\dagger\phi)^2, \quad \text{with} \quad \phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}. \tag{1.9}$$

In this expression, μ represents the quadratic mass parameter and $\lambda > 0$ denotes the quartic self-coupling strength, which ensures the stability of the vacuum at large field values. By choosing a negative mass squared parameter, $\mu^2 < 0$, the potential does not have a minimum at the origin, instead, it develops a degenerate set of minima forming a "Mexican hat" shape as shown in Figure 1.4. The field acquires a non-zero vacuum expectation value (VEV), defined as:

$$v = \sqrt{-\frac{\mu^2}{\lambda}} \approx 246.22 \text{ GeV}. \tag{1.10}$$

The acquisition of this value v induces the Spontaneous Symmetry Breaking (SSB) of the electroweak symmetry:

$$SU(2)_L \times U(1)_Y \to U(1)_{EM}, \tag{1.11}$$

where $U(1)_{EM}$ represents the residual unbroken symmetry group of electromagnetism. This unbroken continuous symmetry ensures that the generator of electromagnetism (the electric charge Q) remains conserved, ultimately giving rise to the massless photon.

Through this process, three of the four degrees of freedom of the Higgs doublet are absorbed by the gauge fields, becoming the longitudinal polarization components of the $W^\pm$ and Z^0 bosons, thereby generating their observed masses ($m_W \approx 80.4$ GeV, $m_Z \approx 91.2$ GeV). The remaining physical degree of freedom manifests as a massive scalar particle, the Higgs boson (H), whose mass is given by $m_H = \sqrt{2\lambda}v = \sqrt{-2\mu^2}$.

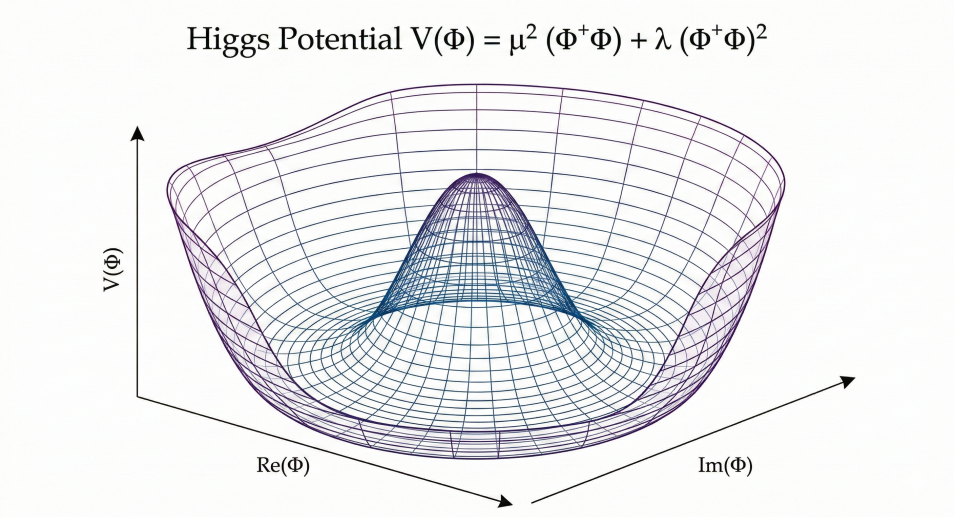


Figure 1.4: The Higgs potential, $V(\phi)$, as a function of the scalar field ϕ. When $\mu^2 < 0$, the potential takes on a "Mexican hat" shape, with a degenerate set of minima at a non-zero vacuum expectation value.

Furthermore, fermions acquire mass via Yukawa couplings to the Higgs field. However, it is important to distinguish this mechanism from the origin of most visible mass in the universe. While the Higgs mechanism generates the current masses of quarks (e.g., $m_u, m_d \sim$ MeV), these account for less than 2% of the total nucleon mass. The overwhelming majority of the proton's mass arises from the energy density of the gluon fields and the dynamics of chiral symmetry breaking within the strong interaction.

The existence of the Higgs boson was experimentally confirmed in 2012 by the ATLAS and CMS collaborations at the Large Hadron Collider (LHC) [29, 30]. This discovery identified the last missing piece of the Standard Model and provided the first experimental evidence of a fundamental scalar field in nature.

1.2 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the non-Abelian gauge theory describing the strong interaction, corresponding to the $SU(3)_C$ sector of the Standard Model. It governs the dynamics of quarks and gluons, the fundamental constituents of hadrons. Unlike QED, where the force carriers (photons) are electrically neutral, the force carriers of QCD (gluons) carry color charge, leading to a rich and complex dynamical structure [31].

The Lagrangian and Self-Interaction

The dynamics of the strong interaction are encoded in the Quantum Chromodynamics (QCD) Lagrangian density, which is constructed to be invariant under local

$SU(3)_C$ gauge transformations. The classical Lagrangian is given by:

$$\mathcal{L}_{\text{QCD}} = \sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f)\psi_f - \frac{1}{4} G^a_{\mu\nu} G^{\mu\nu}_a. \tag{1.12}$$

Here, ψ_f represents the quark field spinor of flavor f (a triplet in color space). The parameter m_f denotes the current mass of the quark, and γ^μ represents the Dirac matrices which define the relativistic spin structure of the theory. The gauge covariant derivative,

$$D_\mu = \partial_\mu - i g_s \frac{\lambda^a}{2} A^a_\mu, \tag{1.13}$$

introduces the coupling to the eight gluon gauge fields A^a_μ. The coupling strength is determined by the strong coupling constant g_s, which is conventionally expressed as the strong fine-structure constant $\alpha_s = g_s^2/4\pi$. The matrices λ^a ($a = 1, \ldots, 8$) are the Gell-Mann matrices, which serve as the generators of the $SU(3)_C$ group in the fundamental representation, satisfying the normalization $\text{Tr}(\lambda^a \lambda^b) = 2\delta^{ab}$.

A defining feature that distinguishes QCD from QED lies in the non-Abelian structure of the gluon field strength tensor:

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{1.14}$$

Here, f^{abc} are the structure constants of the $SU(3)$ group, defined by the commutation relations of the generators: $[\frac{\lambda^a}{2}, \frac{\lambda^b}{2}] = i f^{abc} \frac{\lambda^c}{2}$. These constants are fully antisymmetric indices that quantify the non-commutative nature of the color charges.

To elucidate the interactions, the QCD Lagrangian can be expanded into kinetic and interaction terms:

$$\begin{aligned} \mathcal{L}_{\text{QCD}} = & \sum_f \bar{\psi}_f (i\gamma^\mu \partial_\mu - m_f)\psi_f - \frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2 \\ & + g_s \sum_f \bar{\psi}_f \gamma^\mu \frac{\lambda^a}{2} \psi_f A^a_\mu \\ & - \frac{g_s}{2} f^{abc} (\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) A^\mu_b A^\nu_c - \frac{g_s^2}{4} f^{abc} f_{ade} A^\mu_b A^\nu_c A^d_\mu A^e_\nu. \end{aligned} \tag{1.15}$$

The first row represents the kinetic energy terms for the free quarks and gluons, which lead to their respective propagators. The second row describes the fundamental quark-gluon interaction ($\bar{q}qg$ vertex). The last row, which are proportional to the structure constants f^{abc}, arise directly from the non-Abelian nature of the theory. These terms generate three-gluon (ggg) and four-gluon ($gggg$) vertices, implying that gluons, unlike photons, carry color charge and can interact with themselves. These self-interactions are the root cause of the unique scaling behavior of the strong force, such as asymptotic freedom and confinement. The Feynman diagrams corresponding to these fundamental QCD interactions are illustrated in Figure 1.5.

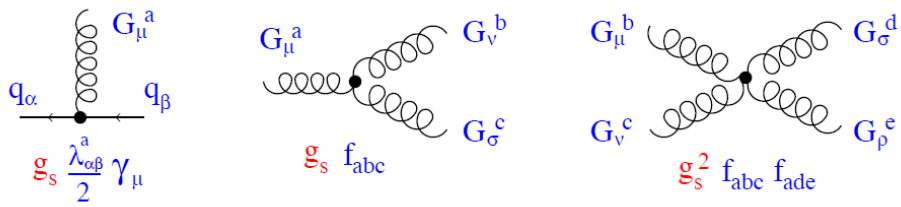


Figure 1.5: Interaction vertices of QCD: vertex of quark-gluon, 3 gluons and 4 gluons. Figure adapted from [32].

The Running Coupling Constant and Asymptotic Freedom

The self-interaction of gluons leads to the anti-screening of the color charge. Consequently, the effective strong coupling constant $\alpha_s = g_s^2/4\pi$ depends strongly on the momentum transfer scale Q^2, a phenomenon known as the running coupling constant. To the first-order approximation in perturbative QCD, this evolution is given by [33]:

$$\alpha_s(Q^2) = \frac{12\pi}{(33 - 2n_f)\ln(Q^2/\Lambda^2_{\text{QCD}})}, \tag{1.16}$$

where n_f denotes the number of active quark flavors and $\Lambda_{\text{QCD}} \approx 200 - 300$ MeV is the fundamental QCD scale parameter.

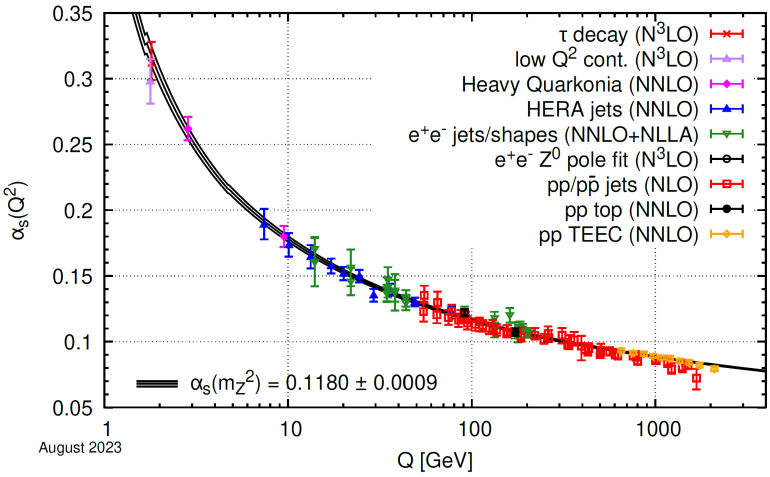


Figure 1.6: Summary of measurements of α_s as a function of the energy scale Q. The respective degree of QCD perturbation theory used in the extraction of α_s is indicated in brackets (NLO, NNLO, etc.). The curve represents the QCD prediction for the combined world average value of $\alpha_s(M_Z^2)$. Figure adapted from the Particle Data Group [33].

As illustrated in Figure 1.6, this equation governs two contrasting phenomenological regimes depending on the momentum transfer scale, Q^2.

In the high-energy, or short-distance, limit where $Q^2 \gg \Lambda^2_{\text{QCD}}$, the theory exhibits **asymptotic freedom**. In this regime, the strong coupling constant α_s asymptoti-

cally approaches zero. Consequently, quarks and gluons interact weakly and behave essentially as quasi-free particles, enabling highly precise theoretical predictions via perturbative QCD [34, 35]. Conversely, in the low-energy, or large-distance, regime where $Q^2 \sim \Lambda^2_{\text{QCD}}$, the physical behavior is drastically different. As the energy scale approaches the mass of the nucleon, α_s grows to order unity ($\mathcal{O}(1)$), leading to **color confinement**. In this strong-coupling domain, standard perturbative expansions completely break down. The effective potential between color charges grows linearly with their separation distance, making it energetically impossible to isolate free quarks or gluons. Instead, these fundamental partons remain permanently confined within color-neutral bound states, such as hadrons.

Chiral Symmetry Breaking and Hadron Mass

In the framework of QCD, quarks and gluons cannot exist as isolated free particles due to color confinement. Instead, they must hadronize to form color-neutral (color-singlet) bound states, collectively known as hadrons. The two primary categories of these bound states are baryons, which consist of three valence quarks (qqq, such as protons and neutrons), and mesons, which are composed of a valence quark-antiquark pair ($q\bar{q}$). When investigating these composite systems, it is important to distinguish the sources of hadron mass. While the Higgs mechanism generates the current quark masses (e.g., $m_u \approx 2.2$ MeV), these account for less than 2% of the nucleon mass ($m_p \approx 938$ MeV). The bulk of the visible mass in the universe arises dynamically from the complex internal structure of the hadron governed by QCD [36].

In the modern view of QCD, a hadron is far more complex than a static bound state. This is conceptually illustrated in Figure 1.7 [37]. While the three valence quarks—depicted as the larger spheres—determine the defining quantum numbers of the proton, they account for merely a fraction of its physical mass. Rather, the proton is a highly dynamic, relativistic many-body system teeming with a fluctuating background sea of virtual gluons (the force carriers) and transient quark-antiquark pairs (sea quarks). Dictated by mass-energy equivalence ($E = mc^2$), the bulk of the nucleon mass does not arise from the intrinsic rest masses of these fundamental constituents. Instead, it is dominated by the complex dynamics of the strong interaction: it originates primarily from the vast kinetic energy of the confined quarks—both valence and sea—coupled with the immense energy density stored within the mediating massless gluon field. Remarkably, these nearly massless constituents interact so intensely that their dynamics generate approximately 98% of the proton's total mass [38].

The fundamental mechanism governing this dynamic mass generation is Spontaneous Chiral Symmetry Breaking (χSB). In the theoretical limit where the N_f lightest quark flavors (typically the up and down quarks, $N_f = 2$) are completely massless, the QCD Lagrangian exhibits a global chiral symmetry denoted by the group $SU(N_f)_L \times SU(N_f)_R$. Physically, this mathematical structure implies that the left-handed (L) and right-handed (R) helicity states of the quarks can be rotated independently in flavor space without altering the physics of the system. However, this symmetry is spontaneously broken down to the diagonal vector subgroup, $SU(N_f)_V$, where left- and right-handed quarks must transform simultaneously. This breaking

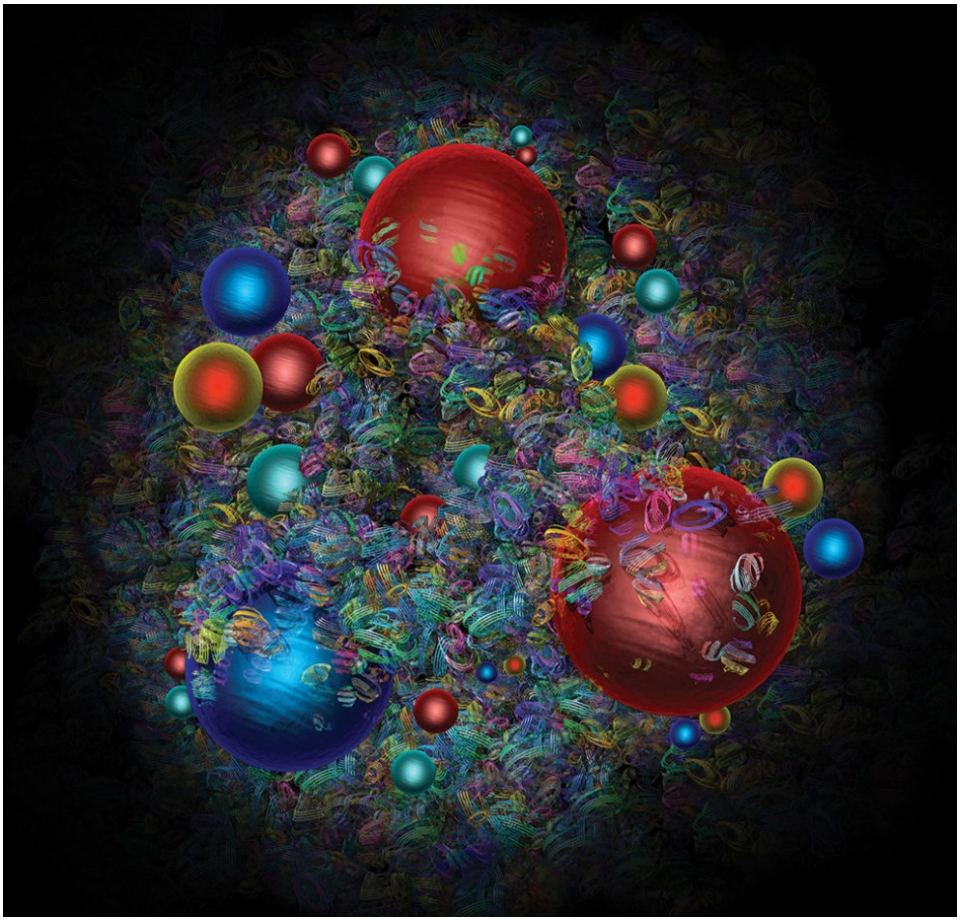


Figure 1.7: Artistic visualization of the internal structure of the proton. Figure adapted from [37].

is driven by the formation of a quark-antiquark condensate, $\langle \bar{q}q \rangle \neq 0$, meaning that the physical QCD vacuum is not empty but rather filled with a non-zero expectation value of quark fields.

Intuitively, this spontaneous symmetry breaking mechanism "dresses" the fundamental, nearly massless current quarks with a dense, co-moving cloud of virtual gluons and sea quarks. This dynamic dressing process effectively transforms the bare current quarks into massive quasi-particles known as constituent quarks, which acquire a dynamical mass on the order of $m_{u,d}^{\text{const}} \approx 300$ MeV. These massive effective degrees of freedom form the foundation of the Constituent Quark Model (CQM) [39]. By absorbing the intractable, non-perturbative complexity of the gluon sea into these quasi-particles, the model brilliantly simplifies the relativistic many-body problem of QCD into a solvable quantum mechanical system.

Consequently, the CQM provides an intuitive description of many qualitative features of the hadron mass spectrum. For baryons (fermionic bound states of three constituent quarks, qqq), the simple scalar sum of three light constituent quarks (u or d) yields a characteristic mass scale of approximately 900 MeV, close to the mass of the lightest baryon doublet, the nucleon (~ 938 MeV). Heavier baryons can be described by incorporating heavier constituent flavors (e.g., the strange quark

with $m_s^{\text{const}} \approx 500$ MeV) together with additional interactions between the constituent quarks. In particular, a color-magnetic spin-spin interaction provides a natural explanation for the substantial mass splitting between the spin-$\frac{1}{2}$ nucleon and the spin-$\frac{3}{2}$ Δ baryon (~ 1232 MeV), despite their similar light-quark content. Analogous spin-dependent interactions also contribute to the mass splittings between mesons with the same flavor content but different total spin. However, the exceptionally small mass of the pion (~ 140 MeV) compared with that of the ρ meson (~ 770 MeV) cannot be understood solely as a conventional hyperfine splitting within the CQM. As discussed above, the pion occupies a special role in QCD as a pseudo-Goldstone boson associated with the spontaneous breaking of chiral symmetry. Its unusually small mass therefore reflects the approximate chiral symmetry of QCD and its explicit breaking by the small but finite current masses of the light quarks, whereas the ρ meson remains at the characteristic hadronic mass scale [40].

1.3 Hadron Physics

The study of hadron physics is fundamentally driven by the quest to understand the strong interaction in the non-perturbative regime of QCD. While the Standard Model has successfully described interactions at high energy scales through perturbation theory, the dynamics at the energy scale of hadron masses (typically in the GeV range) remain one of the most challenging frontiers in modern physics.

The QCD Spectrum and Exotic States

According to the foundational principles of QCD, physical hadrons must exist as color-singlet bound states. Historically, within the framework of the CQM, these states have been predominantly classified into two conventional categories: mesons ($q\bar{q}$) and baryons (qqq). Before the fundamental dynamic theory of QCD was firmly established, the proliferation of newly discovered hadrons in the 1960s was elegantly systematized by Murray Gell-Mann and Yuval Ne'eman through a classification scheme known as the Eightfold Way [41, 42]. This scheme is rooted in the approximate $SU(3)_F$ flavor symmetry of the three lightest quarks (u, d, s). Within this mathematical framework, hadrons are organized into highly symmetric geometric multiplets by plotting their third component of isospin (I_3) against their hypercharge (Y). These quantum numbers are fundamentally linked to the observable electric charge (Q) of the particle via the Gell-Mann-Nishijima formula:

$$Q = I_3 + \frac{Y}{2}, \tag{1.17}$$

where $Y = B+S$ is the sum of the baryon number (B) and strangeness (S).

For mesons ($q\bar{q}$), the combination of a quark (flavor triplet, $\mathbf{3}$) and an antiquark (anti-triplet, $\bar{\mathbf{3}}$) decomposes according to the group-theoretical relation $\mathbf{3}\otimes\bar{\mathbf{3}} = \mathbf{8}\oplus\mathbf{1}$. Consequently, mesons naturally populate nonets, consisting of an $SU(3)$ octet and a flavor-singlet. In their ground state, the orbital angular momentum between the

constituent quarks is zero ($L = 0$). The parity and charge conjugation quantum numbers are given by $P = (-1)^{L+1}$ and $C = (-1)^{L+S}$ (for neutral states), respectively. This naturally leads to two primary nonets depending on the total quark spin S as shown in Figure 1.8. When the spins are anti-parallel ($S = 0$), the system forms the **pseudoscalar meson nonet** with quantum numbers $J^{PC} = 0^{-+}$. This nonet comprises an isospin triplet (π^+, π^0, π^-), two isospin doublets (K^+, K^0 and $K^-, \bar{K}^0$), and two isospin singlets (η_8 and η_1). Here, η_8 belongs to the octet, while η_1 represents the fully symmetric $SU(3)$ singlet state. However, because the strange quark is significantly heavier than the up and down quarks, the $SU(3)_F$ symmetry is explicitly broken. Consequently, the physical mass eigenstates observed in experiments, the η and η' mesons, are not the pure $SU(3)$ states, but rather linear superpositions (mixtures) of the η_8 and η_1 basis states. Similarly, when the quark spins are parallel ($S = 1$), they form the heavier **vector meson nonet** ($J^{PC} = 1^{--}$). This includes the ρ triplet, the K^* doublets, and the $SU(3)$ singlets ω_8 and ω_1. In strong contrast to the pseudoscalar sector, the state mixing in the vector nonet is nearly ideal. This specific mixing angle results in the physical ϕ meson being an almost pure $s\bar{s}$ state, while the physical ω meson is predominantly a symmetric combination of the lighter up and down quarks ($(u\bar{u} + d\bar{d})/\sqrt{2}$).

For baryons (qqq), the combination of three valence quarks decomposes as $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$. The physical realization of these states is strictly governed by the Pauli exclusion principle: the overall wavefunction of a baryon (comprising color, space, flavor, and spin parts) must be totally antisymmetric under the exchange of any two constituent quarks. Since all physical hadrons are color singlets, their color wavefunction is totally antisymmetric. For ground state baryons ($L = 0$), the spatial wavefunction is symmetric. Therefore, the combined flavor-spin wavefunction is compelled to be totally symmetric. This constraint dictates the baryon spectrum as shown in Figure 1.9. When the total spin is $S = 3/2$ (a symmetric spin state), the flavor wavefunction must also be totally symmetric, uniquely selecting the **baryon decuplet** ($J^P = 3/2^+$). This decuplet forms an inverted triangle in the $I_3 - Y$ plane, housing the Δ resonances, the Σ^* and Ξ^* hyperons, and the purely strange Ω^- hyperon. Conversely, when the total spin is $S = 1/2$ (mixed symmetry), it must pair with the mixed-symmetry flavor octet to form a totally symmetric flavor-spin combination. This forms the **baryon octet** ($J^P = 1/2^+$), a hexagonal structure that accommodates the nucleons (p, n) at the top, alongside the heavier Λ, Σ, and Ξ hyperons.

However, the non-Abelian nature of QCD, characterized notably by the self-interaction of gluons, theoretically permits a much richer spectrum of color-singlet configurations possessing active gluonic or multi-quark degrees of freedom. These are collectively referred to as exotic hadrons which are shown in Figure 1.10.

Among these, **glueballs** represent the most direct manifestation of gluon self-coupling, being bound states composed entirely of valence gluons (gg or ggg)[45]. First-principles calculations from Lattice QCD (LQCD) predict a robust mass spectrum for these states as shown in Figure 1.11. The LQCD spectra indicate that the lightest scalar glueball ($J^{PC} = 0^{++}$) is expected to lie in the mass region of $1.5 - 1.7$ GeV/c^2 , followed by heavier tensor (2^{++}) and pseudoscalar (0^{-+}) states.

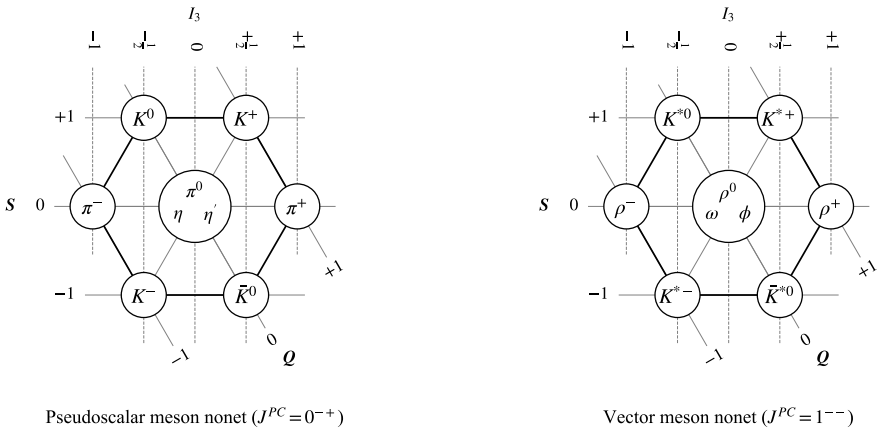


Figure 1.8: The ground state meson nonets.

Another class of gluonic excitations is the **hybrids**, which consist of a conventional quark configuration coupled to an actively excited valence gluon field. These can manifest either as hybrid mesons ($q\bar{q}g$) or hybrid baryons ($qqqg$) [47]. For hybrid mesons, physicists can unambiguously identify these states in experiments by searching for exotic quantum numbers, such as $J^{PC} = 0^{--}, 0^{+-}, 1^{-+}$, and 2^{+-}, which are strictly forbidden for any standard fermion-antifermion ($q\bar{q}$) configuration. Conversely, hybrid baryons do not exhibit such unique exotic signatures, they share the exact same quantum numbers as conventional excited baryons (qqq^*). This kinematic overlap makes their experimental isolation significantly more challenging, often requiring sophisticated partial-wave analyses of high-statistics data sets to disentangle them from the conventional background. The physical origin of these exotic signatures is beautifully elucidated by LQCD calculations. In the charmonium sector ($c\bar{c}g$), LQCD consistently predicts a fundamental supermultiplet comprising exactly eight lowest-lying hybrid states. The generation of these eight states can be intuitively understood through the constituent gluon picture [48]. The hybrid is formed by coupling a ground-state S-wave $c\bar{c}$ core, which can exist either as a spin-singlet ($S = 0$, $J^{PC} = 0^{-+}$) or a spin-triplet ($S = 1$, $J^{PC} = 1^{--}$), to the lightest gluonic excitations. The gluonic field itself behaves as an effective constituent, primarily manifesting in a chromomagnetic mode ($J_g^{P_g C_g} = 1^{+-}$) or a chromoelectric mode ($J_g^{P_g C_g} = 1^{--}$). The angular momentum addition of the $c\bar{c}$ core and the gluonic field naturally generates two distinct multiplets. Coupling the core to the 1^{+-} gluonic mode yields the four states $\{1^{--}, 0^{-+}, 1^{-+}, 2^{-+}\}$, while coupling to the 1^{--} mode yields another four states $\{1^{+-}, 0^{+-}, 1^{+-}, 2^{+-}\}$, perfectly accounting for the supermultiplet of eight states. Crucially, among these eight configurations, three states possess the exotic quantum numbers 1^{-+}, 0^{+-}, and 2^{+-}. Since conventional $q\bar{q}$ models can never reproduce these specific J^{PC} combinations, their experimental observation serves as a conclusive evidence for the hybrid nature of the resonance. Currently, the lightest of these exotic states, the 1^{-+} charmonium hybrid, is predicted by LQCD to lie in the mass region of $4.2 - 4.3$ GeV/c^2. This specific mass window makes it a prime theoretical target for ongoing and future high-precision

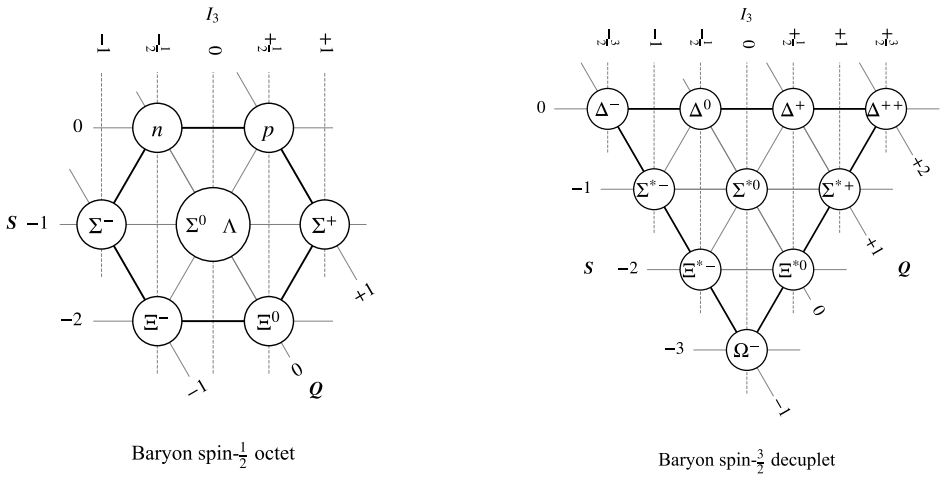


Figure 1.9: The ground state baryon octet and decuplet.

experiments, as its discovery would definitively confirm the active dynamic role of gluons in hadron spectroscopy.

Furthermore, the fundamental Lagrangian of QCD does not prohibit **multi-quark systems**, such as tetraquarks ($qq\bar{q}\bar{q}$) and pentaquarks ($qqqq\bar{q}$). Theoretical models and a plethora of recent experimental discoveries (such as the XYZ states [49]) suggest that these complex systems can manifest in various distinct internal configurations. They may exist as *compact (deeply bound) states*, where all constituent quarks share a single, tightly bound confinement volume, often modeled via diquark-antidiquark interactions [50, 51]. Alternatively, they may form *hadronic molecules*, which are extended, loosely bound states of two distinct color-singlet hadrons (e.g., a meson-meson or meson-baryon pair) interacting via residual nuclear forces, such as pion exchange [52, 53]. A third proposed configuration is the *hadrocharmonium* [54, 55], featuring a compact, color-singlet heavy quarkonium core ($c\bar{c}$) embedded within a diffuse, loosely interacting cloud of light hadronic matter. Closely related to this is the *adjoint charmonium* model [56, 57]. In this configuration, the heavy $c\bar{c}$ pair exists in a color-octet (adjoint) representation rather than a color-singlet. To satisfy the fundamental QCD requirement of overall color confinement, this compact color-octet core must be neutralized by binding with a surrounding cloud of light quarks or gluonic degrees of freedom that also carry a complementary color-octet charge.

The unambiguous experimental identification and detailed phenomenological study of these exotic states are crucial for validating LQCD predictions, deciphering the exact mechanisms of color confinement, and understanding the complete dynamics of hadron mass generation.

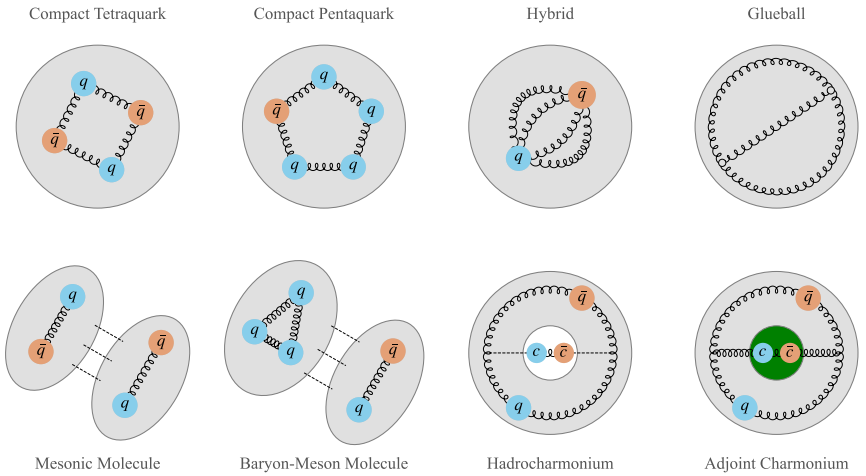


Figure 1.10: Schematic illustration of exotic hadrons. Figure inspired by [43, 44].

Charmonium and the XYZ States

The goal of experimental hadron spectroscopy is to identify the states of hardronic spectra which belong to a certain potential by measuring the mass, width, and quantum numbers (J^{PC}) of these states. The charm quark sector ($c\bar{c}$), in particular, offers a unique laboratory where non-relativistic potential models can be tested against experimental data.

The history of precision studies in this sector began with the discovery of the J/ψ meson by the SLAC-SP-017 collaboration [58] and the E598 collaboration at Brookhaven National Laboratory [59], provided the first compelling experimental evidence for the existence of the charm quark. The charmonium spectrum (see Figure 1.12) is well described by the Cornell potential [60]:

$$V(r) = -\frac{4}{3}\frac{\alpha_s}{r} + \kappa r, \tag{1.18}$$

incorporating both the one-gluon exchange (Coulomb like) and confinement (linear) terms.

A crucial feature of the spectrum is the open-charm threshold ($2m_D \approx 3.73$ GeV). States lying below this threshold (such as J/ψ and η_c) cannot decay into a pair of charmed mesons ($D\bar{D}$). Their decay into light hadrons requires the annihilation of the heavy quark pair into gluons, a process with disconnected quark lines. According to the OZI rule (Okubo-Zweig-Iizuka) [61], such disconnected diagrams are strongly suppressed by the running coupling constant α_s as shown in Figure 1.13. Consequently, these sub-threshold states exhibit extremely narrow widths ($\Gamma \sim$ keV), making them metastable compared to states above the threshold which fall apart rapidly into D mesons.

However, the traditional quarkonium paradigm has been profoundly challenged by the recent discovery of numerous unexpected resonances in the charmonium mass region. As explicitly detailed in the charmonium spectrum (Figure 1.12), the conventional $c\bar{c}$ states (highlighted in yellow) are predominantly located below the open-charm threshold ($2M_D$, indicated by the lower dashed line) and align remarkably well

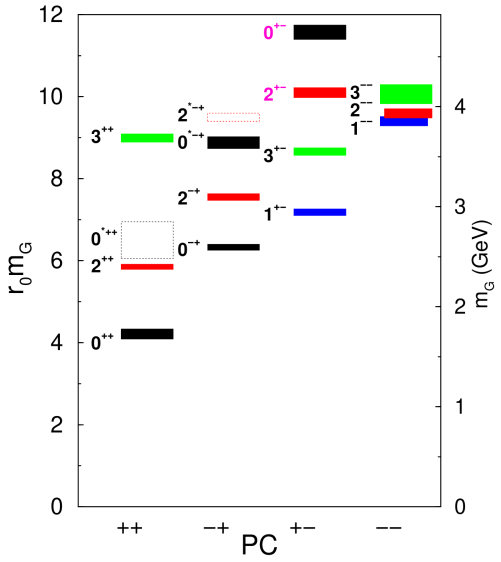


Figure 1.11: The glueball spectrum predicted by Lattice QCD. The lightest scalar glueball is expected around $1.5-1.7$ GeV/c^2. Figure adapted from [46].

with potential model predictions. In strong contrast, a new class of enigmatic particles, collectively known as the XYZ states, populates the higher-mass region. These include the neutral XY mesons (highlighted in red) and the electrically charged Z mesons (highlighted in purple). The observation of these charged states (such as the $Z_c(3900)^+$) is particularly striking, as their non-zero electric charge unequivocally dictates that they must contain at least four valence quarks (e.g., $c\bar{c}u\bar{d}$). Furthermore, these newly discovered states exhibit properties that strongly deviate from standard $c\bar{c}$ assignments. For instance, the renowned $X(3872)$ lies extremely close to the $D^{*0}\bar{D}^0$ + c.c. threshold (the upper dashed line), strongly hinting at exotic internal configurations, such as a loosely bound hadronic molecule or a compact tetraquark [62]. To definitively resolve the exact nature of these resonances, high-precision spectroscopic data are essential for mapping their exact line shapes and deciphering their complex decay patterns.

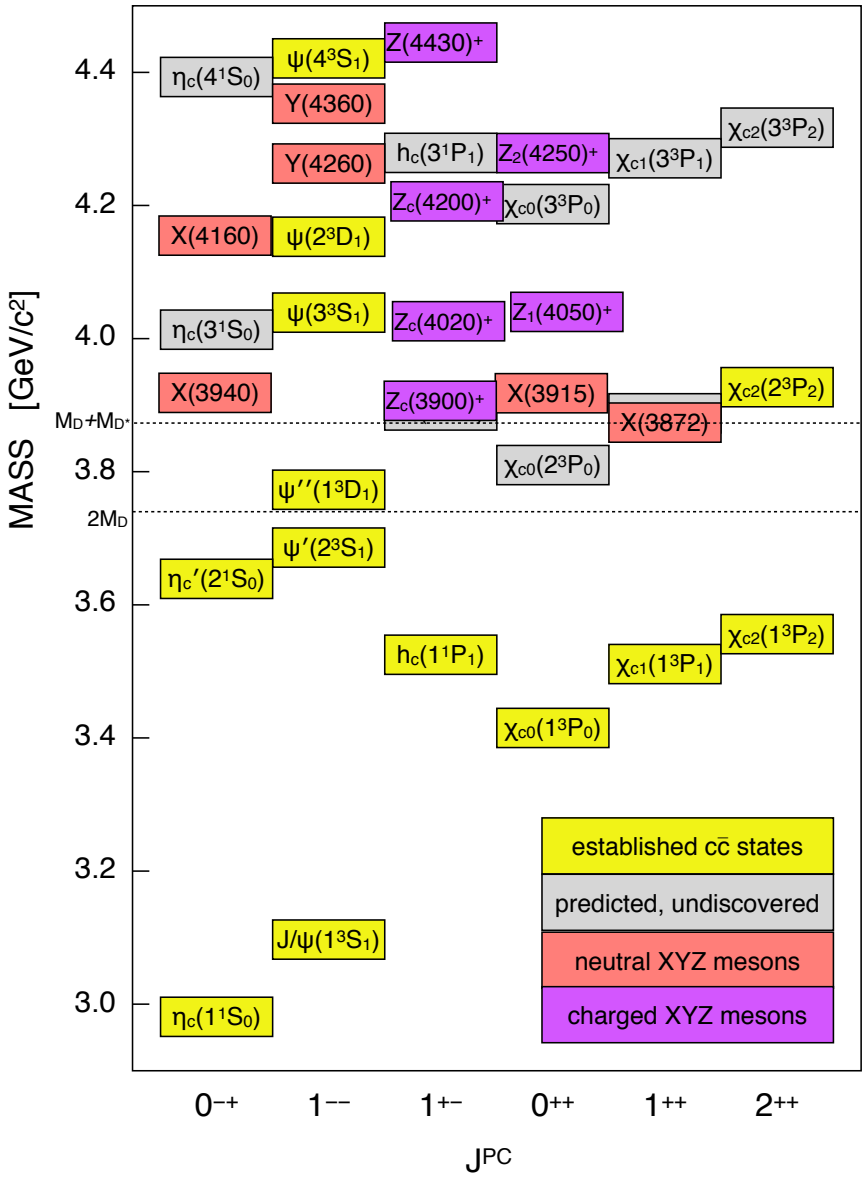


Figure 1.12: The charmonium spectrum. The conventional states ($J/\psi, \chi_c$, etc.) are shown in yellow. The predicted but not discovered chamonium states are shown in gray. The newly discovered XYZ states are shown in red (neutral) and purple (charged), respectively. Figure adapted from [49].

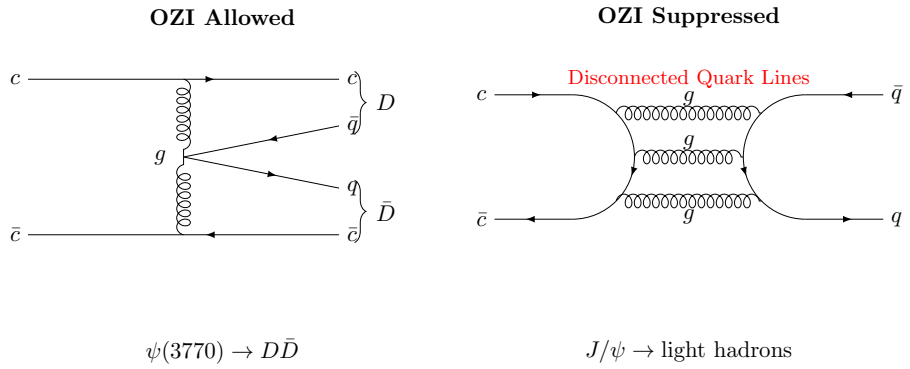


Figure 1.13: Visual representation of the OZI rule. Left: An OZI-allowed decay (e.g., $\psi(3770) \to D\bar{D}$) where quark lines are connected throughout the process. Right: An OZI-suppressed decay (e.g., $J/\psi \to$ light hadrons) requiring the annihilation of the heavy quark pair into hard gluons (disconnected quark lines).

Experiment Methods

To map the hadron spectrum with high precision, the choice of the production mechanism is decisive. Current experiments primarily rely on proton-proton (pp) collisions, electron-positron (e^+e^-) annihilation, or antiproton-proton ($\bar{p}p$) annihilation, as shown in Figure 1.14.

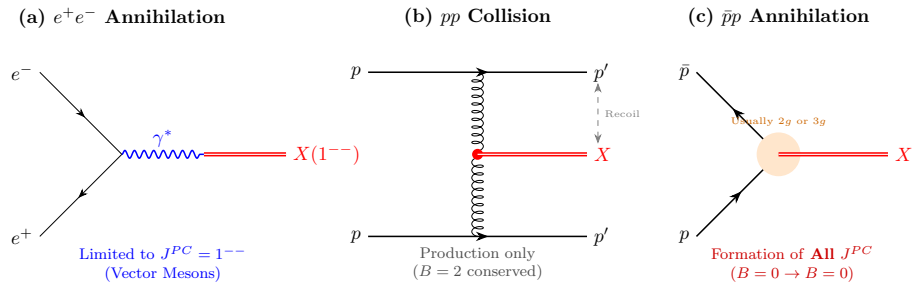


Figure 1.14: Comparison of hadron production mechanisms. **(a)** e^+e^- annihilation proceeds via a virtual photon, restricting formation to vector states (1^{--}). **(b)** In pp collisions, baryon number conservation ($B = 2$) requires the protons (or their remnants) to exist in the final state, making it a production process with recoil particles. **(c)** In $\bar{p}p$ annihilation ($B = 0$), the entrance channel can fuse completely to form a resonance X. This allows direct formation of all non-exotic quantum numbers.

Electron-positron annihilation (e.g., at BESIII [63]) provides a markedly cleaner experimental environment but suffers from stringent quantum number selection rules. As shown in Figure 1.14(a), this process predominantly proceeds via the s-channel exchange of a single virtual photon. This mechanism restricts the directly formed states exclusively to those sharing the quantum numbers of the photon ($J^{PC} = 1^{--}$, vector states). Other configurations, such as the spin-singlet η_c or h_c, cannot be formed directly; they can only be accessed indirectly via the radiative or hadronic decays of heavier vector states. This indirect production mechanism significantly restricts the available phase space and cross sections, severely limiting the achievable statistics for non-vector states. Furthermore, similar to pp collisions, the mass resolution for these indirectly produced states remains bottlenecked by the detector resolution.

In high energy pp collisions (e.g., at LHCb [64]), hadrons are produced with copious cross sections. However, as illustrated in Figure 1.14(b), baryon number conservation ($B = 2$ in the initial state) strictly requires the original protons or their baryonic remnants to exist in the final state. Consequently, the resonances of interest are inextricably generated via a *production* mechanism alongside numerous recoil particles in a high-multiplicity background environment. In this mode, the mass of the resonance must be kinematically reconstructed from the four-momenta of its decay products, inherently limiting the mass resolution to the tracking and calorimetric resolution of the detector (typically $5 - 10$ MeV/c^2).

The antiproton-proton ($\bar{p}p$) system offers a uniquely powerful alternative: *Resonant Formation* [1]. Because the initial state has a net baryon number of zero

($B = 0$), the entrance channel can fuse completely to form a single intermediate resonance X, as depicted in Figure 1.14(c). Since the strong interaction, mediated by gluons, can couple to any quark-antiquark pair configuration, $\bar{p}p$ annihilation allows for the direct formation of coherent states with all non-exotic quantum numbers:

$$\bar{p}p \to X. \tag{1.19}$$

This extraordinary capability permits the direct study of states that are kinematically or strictly inaccessible in e^+e^- collisions.

Most importantly, in such formation experiments, the experimental mass resolution is no longer dictated by the detector's decay-product reconstruction limits. Instead, it is defined solely by the precision of the initial state kinematics, specifically the momentum spread of the antiproton beam ($\Delta p/p$). By conducting a systematic energy scan across the resonance region using phase-space cooled antiproton beams, physicists can map out the exact resonance line shape directly from the production cross section. This scan technique can achieve a phenomenal mass resolution on the order of ~ 50 keV/c^2. Such sub-MeV precision is absolutely essential for resolving the extremely narrow natural widths of charmonium states and definitively separating closely spaced or overlapping exotic XYZ resonances.

As a striking illustration of this technique, Figure 1.15 displays a scan of the χ_{c1} resonance performed at the Fermilab antiproton accumulator by the E835 experiment, utilizing the formation process $\bar{p}p \to \chi_{c1} \to J/\psi\gamma$ [65]. In this measurement, the horizontal error bar for each cross-section data point in Figure 1.15(a) corresponds precisely to the remarkably narrow width of the incident beam energy distribution shown in Figure 1.15(b). This exquisite beam control allowed the E835 experiment to carry out the most precise measurement to date of the χ_{c1} mass ($3510.719 \pm 0.051 \pm 0.019$ MeV/c^2) and its extremely narrow total width ($0.876 \pm 0.045 \pm 0.026$ MeV).

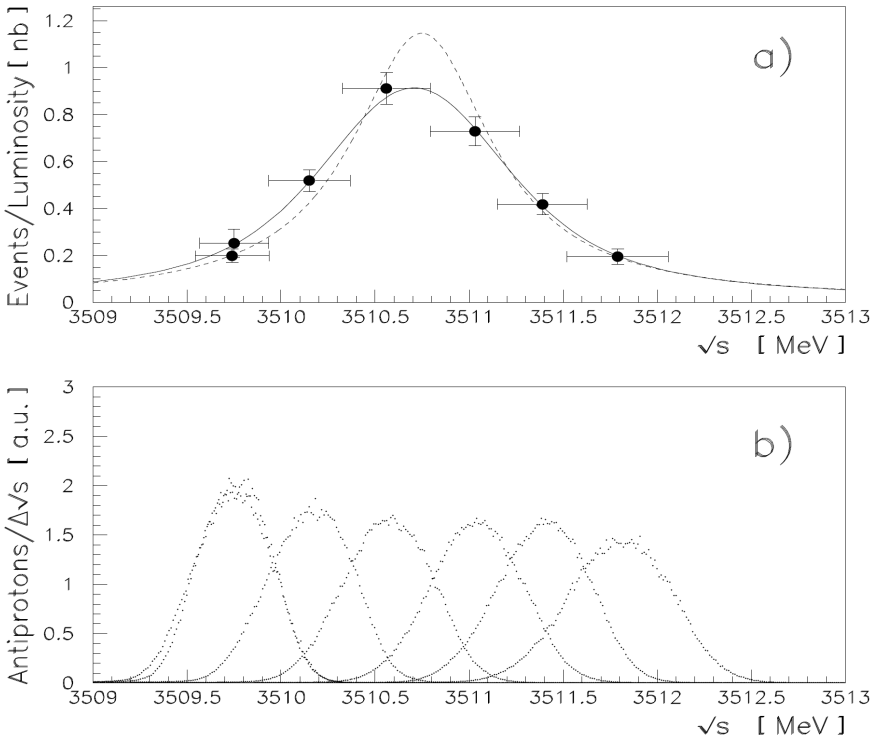


Figure 1.15: (a) The resonance scan of χ_{c1} performed by the E835 experiment at Fermilab. (b) The beam energy distribution for each scan point. Figure adapted from [65].

Chapter 2

The $\overline{\text{P}}$ANDA Experiment at FAIR

As discussed in Chapter 1, the non-perturbative regime of QCD poses some of the most profound questions in modern particle physics, particularly concerning the origin of hadron mass and the existence of gluonic excitations. Addressing these questions requires an experimental facility capable of combining high precision spectroscopy with high interaction rates to access rare processes [1].

The $\overline{\text{P}}$ANDA (AntiProton ANnihilation at DArmstadt) experiment is designed to meet these challenges. Situated at the future Facility for Antiproton and Ion Research (FAIR) [66], $\overline{\text{P}}$ANDA will utilize an intense, phase-space cooled antiproton beam colliding with a fixed proton target. This unique setup allows for the direct formation of charmonium and exotic states with unprecedented mass resolution, overcoming the limitations of electron-positron colliders.

2.1 Physics at $\overline{\text{P}}$ANDA

The $\overline{\text{P}}$ANDA experiment features a diverse and comprehensive physics program dedicated to the study of the hadron physics. While the theoretical motivations were established in Section 1.3, this section focuses on the specific experimental objectives and the unique capabilities of $\overline{\text{P}}$ANDA to address them. The primary goals include high precision charmonium spectroscopy, the search for gluonic excitations and exotic states, and the investigation of hadrons in nuclear matter.

Charmonium Spectroscopy

The charmonium system ($c\bar{c}$) serves as a precision calibration tool for the strong force. Figure 2.1 illustrates the spectrum for the charmonium states. While the states below the open charm threshold ($D\bar{D}$) are relatively well known, many questions remain regarding their precise widths and decay modes. Above the open charm threshold, the landscape is less understood. The identification of higher angular momentum states, specifically the D and F waves, remains elusive, and

the nature of the recently discovered XYZ states requires further measurements [62].

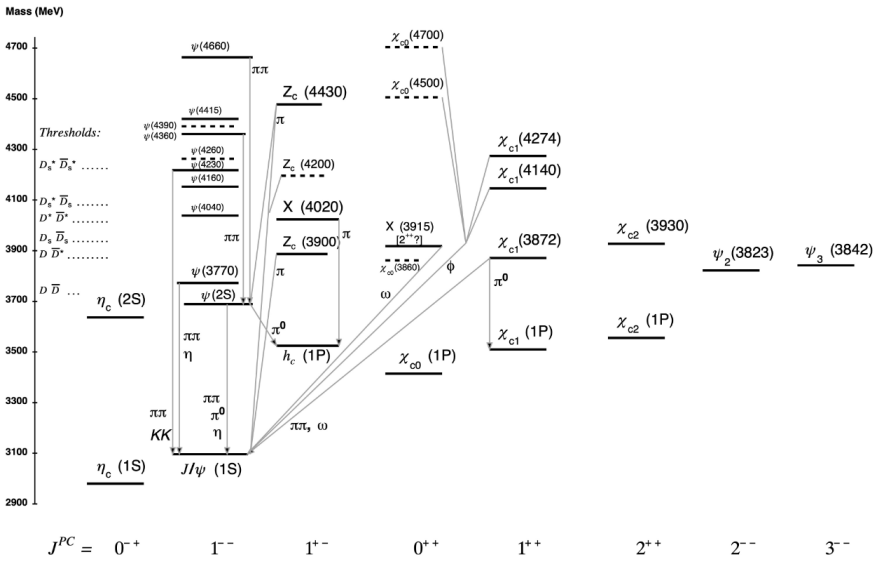


Figure 2.1: The level scheme of meson states containing a minimal quark content of $|c\bar{c}\rangle$. Figure adapted from [33].

Unlike e^+e^- experiments (e.g., BESIII, Belle II) which are restricted to forming vector states ($J^{PC} = 1^{--}$) via a virtual photon, $\overline{P}$ANDA utilizes antiproton-proton annihilation ($\bar{p}p \rightarrow X$). This allows for the direct formation of all non-exotic quantum numbers. By performing fine energy scans with the High Energy Storage Ring (HESR) beam, $\overline{P}$ANDA can measure the mass and width of resonances with an accuracy of ~ 50 keV/c^2, solely determined by the beam momentum spread ($\Delta p/p \sim 4 \times 10^{-5}$) rather than the detector resolution [67]. This precision is crucial for resolving the hyperfine splitting of the charmonium spectrum, identifying narrow states (such as D and F wave states) that may have been missed by previous experiments, and systematically mapping the properties of the XYZ states.

Open Charm and Baryon Spectroscopy

$\overline{P}$ANDA is expected to make significant contributions to the study of open charm mesons (D mesons) as well as charmed and strange baryons.

Open Charm Mesons A meson composed of a heavy charm quark and a lighter quark (u, d, s) provides a unique testing ground for QCD. Such a system can be analogized to a "Hydrogen atom" of hadron physics, where the heavy charm quark serves as a static nucleus and the light quark fills the energy levels. This picture is formalized in the Heavy Quark Effective Theory [68]. While this model successfully described early D meson data, the discoveries of narrow D_{sJ} states at BaBar, Belle, and CLEO have revealed discrepancies that do not conform well to standard potential model predictions [69]. Crucial to understanding the physical nature of these

states is the measurement of their decay widths. Thanks to the exceptional momentum definition of the antiproton beam from HESR, $\overline{\text{P}}$ANDA can perform threshold scans to accurately measure these widths down to values of approximately 100 keV, providing critical constraints for theoretical models [1].

Baryon Spectroscopy In the baryon sector, $\overline{\text{P}}$ANDA allows for the study of multi-strange and charmed baryons ($\Lambda_c, \Sigma_c, \Xi_c, \Omega_c$). The production cross-sections for baryon-antibaryon pairs ($\bar{p}p \rightarrow Y\bar{Y}$) peak at $\overline{\text{P}}$ANDA energies, enabling high-statistics studies. Understanding the excitation spectrum of these baryons provides complementary insight into the constituent quark model and the role of diquark correlations within hadrons [1].

Search for Gluonic Excitations

QCD explicitly predicts the existence of hadrons with active gluonic degrees of freedom, as color confinement does not prohibit color-singlet states formed by gluons:

- **Glueballs:** These are states composed solely of gluons (gg, ggg). Lattice QCD predicts the lightest scalar glueball (0^{++}) to have a mass between 1.5 and 1.7 GeV/c^2, and a tensor glueball (2^{++}) around 2.4 GeV/c^2 [70]. This mass region is perfectly covered by the $\overline{\text{P}}$ANDA center-of-mass energy range ($\sqrt{s} \approx 2.2 - 5.5$ GeV).
- **Hybrids:** Charmonium hybrids ($c\bar{c}g$) consist of a quark-antiquark pair coupled to an excited gluon. They are expected to appear in the mass range of $4.0 - 4.5$ GeV/c^2. Crucially, some hybrids may carry spin-exotic quantum numbers (e.g., $J^{PC} = 1^{-+}$), which are forbidden for conventional $q\bar{q}$ mesons. The observation of such a state would provide conclusive evidence for gluonic excitation and non-perturbative QCD dynamics.

The gluon-rich environment of $\bar{p}p$ annihilation is ideal for producing these states [71]. Unlike e^+e^- collisions or pp scattering, the annihilation process allows gluons to be produced directly from the annihilarion of the initial quarks, giving $\overline{\text{P}}$ANDA a distinct discovery potential.

Other Topics

Beyond spectroscopy, $\overline{\text{P}}$ANDA will investigate the structure of the nucleon itself via electromagnetic processes. The Drell-Yan process ($\bar{p}p \rightarrow \mu^+\mu^- X$) allows for the measurement of nucleon structure functions in the time-like region.

Furthermore, $\overline{\text{P}}$ANDA seeks to examine the effect of matter on hadrons. Specifically, the experiment aims to understand hadron masses in the context of spontaneous chiral symmetry breaking in QCD and its partial restoration in a hadronic environment. In addition, production cross sections of J/ψ and D mesons will also be measured with several nuclear targets.

The introduction of a strange quark in hypernuclei gives rise to the quantum number of strangeness. To date, only six double Λ-hypernuclei have been discovered. Therefore, $\overline{\text{P}}$ANDA will utilize dedicated experimental techniques to identify and investigate such new states in the nucleus with abundant production.

2.2 Facility for Antiproton and Ion Research (FAIR)

The $\overline{\text{P}}$ANDA experiment is one of the key pillars of the Facility for Antiproton and Ion Research (FAIR), currently under construction at the GSI Helmholtz Centre for Heavy Ion Research in Darmstadt, Germany. FAIR represents a substantial upgrade to the existing GSI facility, designed to provide high-intensity primary and secondary beams of ions and antiprotons for a wide range of physics research [66].

The accelerator complex (Figure 2.2) is designed to operate with parallel beams, allowing multiple experiments to run simultaneously. The primary acceleration chain for $\overline{\text{P}}$ANDA begins with the existing UNILAC (Universal Linear Accelerator) and SIS18 (synchrotron), which serve as injectors. The beams are then transferred to the new SIS100 synchrotron, a superconducting ring with a circumference of 1100 m [72].

For the $\overline{\text{P}}$ANDA physics program, the SIS100 accelerates protons up to a momentum of 29 GeV/c. These protons are extracted and focused onto a metal antiproton production target (nickel or iridium) to generate antiprotons via the reaction $p+A \rightarrow \bar{p}+X$. The produced antiprotons, with a momentum centered around 3 GeV/c, are collected and pre-cooled in the Collector Ring (CR) before being injected into the High Energy Storage Ring (HESR) [73].

2.3 High Energy Storage Ring (HESR)

The High Energy Storage Ring (HESR) is the central accelerator component for the $\overline{\text{P}}$ANDA experiment and it also hosts the KOALA [75] and SPARC [76] experiment (see Figure 2.3). Designed as a synchrotron and storage ring, it is capable of accumulating, accelerating, and storing antiprotons within a momentum range of 1.5 to 15 GeV/c [67]. The ring has a racetrack-shaped geometry with a circumference of 575 m, consisting of two 180° arcs and two 132 m long straight sections. One straight section houses the $\overline{\text{P}}$ANDA detector and the internal target.

The interaction of the antiproton beam with the internal target, as well as intra-beam scattering, leads to a continuous increase in the beam's emittance and momentum spread. To counteract these effects and maintain the beam quality essential for the physics goals, HESR is equipped with the **stochastic cooling system** [78]. This system is effective across the entire momentum range of HESR ($1.5 - 15$ GeV/c). It operates by detecting fluctuations in the beam's position and momentum using pickup electrodes, amplifying these signals, and applying corrective kicks via kicker electrodes to reduce the beam's phase space volume.

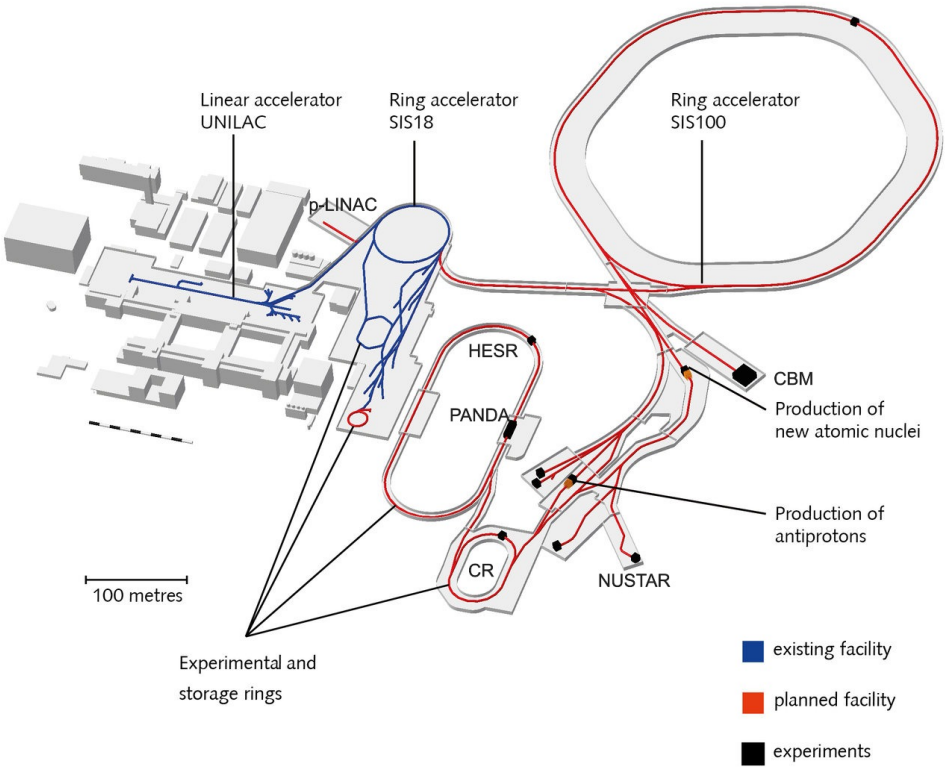


Figure 2.2: Layout of the FAIR facility. Existing GSI facilities are shown in blue, while the new FAIR installations are indicated in red. The acceleration chain for $\overline{\text{P}}$ANDA involves the UNILAC, SIS18, and SIS100, followed by the antiproton production target and the experimental storage rings (including CR and HESR). Figure adapted from [74].

The HESR receives antiprotons from the Collector Ring (CR) at a fixed injection momentum of 3.8 GeV/c. During the initial operational phase, the accumulation of antiprotons is performed directly within the HESR to reach the requisite beam intensities. In a future upgrade, the dedicated Recycled Experimental Storage Ring (RESR) is foreseen to act as an accumulator, enabling significantly higher beam intensities [79]. To fulfill the diverse physics requirements of the $\overline{\text{P}}$ANDA program, the HESR is designed to operate in two distinct modes, characterized by their respective trade-offs between luminosity and momentum resolution, as summarized in Table 2.1 [80]. The **High Resolution (HR) mode** is optimized for high-precision spectroscopy, particularly for conducting fine energy scans of extremely narrow resonances, such as charmonium states with natural widths below 1 MeV. By fully exploiting the beam cooling systems, this mode achieves an exceptionally sharp relative momentum spread down to $\Delta p/p \approx 4 \times 10^{-5}$. In this configuration, the stored beam intensity is typically maintained at 10^{10} antiprotons, which yields a peak luminosity of up to 2×10^{31} cm^{-2}s^{-1}. Conversely, the **High Luminosity (HL) mode** is specifically tailored to maximize the event interaction rate, a crucial requirement for studying extremely rare processes. By utilizing the RESR to accumulate a vastly larger number of antiprotons ($N_{\text{stored}} \approx 10^{11}$), the system can reach a peak luminosity of 2×10^{32} cm^{-2}s^{-1}. To sustain this high-density beam, the stochastic cooling

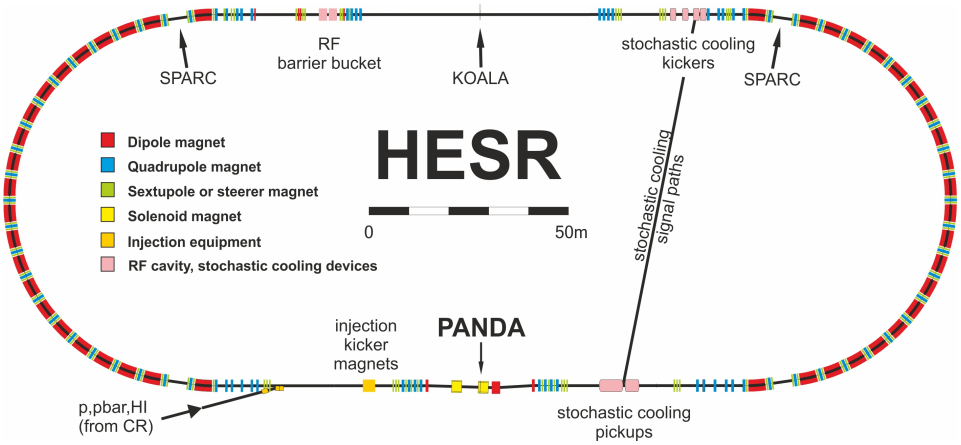


Figure 2.3: Schematic layout of the High Energy Storage Ring (HESR). The ring features a racetrack geometry with a circumference of 575 m. The $\overline{\text{P}}$ANDA detector is situated in one of the straight sections (bottom in this view), operating with an internal fixed target. Stochastic cooling components (pickups and kickers) are distributed along the ring to counteract beam heating. Antiprotons are injected into the ring from the Collector Ring (CR). Additionally, the ring accommodates the experimental setups for the SPARC and KOALA. Figure adapted from [77].

system is continuously employed to compensate for intra-beam scattering and beam heating, thereby preserving a broader but perfectly stable momentum spread of $\Delta p/p \approx 10^{-4}$.

Table 2.1: Comparison of the two operation modes of the HESR. The High Resolution (HR) mode and the High Luminosity (HL) mode.

Beam Parameters	High Resolution (HR)	High Luminosity (HL)
p (GeV/c)	$1.5 - 15$	$1.5 - 15$
Stored Antiprotons ($N_{\bar{p}}$)	10^{10}	10^{11}
Momentum Resolution ($\Delta p/p$)	$\geq 4 \times 10^{-5}$	$\sim 1 \times 10^{-4}$
Peak Luminosity [cm^{-2}s^{-1}]	2×10^{31}	2×10^{32}

Beam Parameters at the Interaction Point

The experimental performance of $\overline{\text{P}}$ANDA is critically dependent on the precise control of the antiproton beam parameters at the interaction point (IP). The effective luminosity and the quality of physics data are governed by the beam spot size, the angular divergence, the beam offset and the beam tilt.

The beam spot size $\sigma_{x,y}$ at a specific interaction point (x, y) is determined by the transverse emittance $\epsilon_{x,y}$ and the betatron amplitude function (beta function) $\beta_{x,y}$ at that point:

$$\sigma_{x,y} = \sqrt{\epsilon_{x,y} \cdot \beta_{x,y}}. \tag{2.1}$$

To achieve a high interaction rate, the root-mean-square (rms) of the beam spot size is required to be roughly 0.8 mm [81], and is assumed to be a fixed parameter in the experimental design. The angular divergence of the beam $\chi_{x,y}$ at the point (x, y) is inversely related to the beta function:

$$\chi_{x,y} = \sqrt{\frac{\epsilon_{x,y}}{\beta_{x,y}}}. \tag{2.2}$$

Alternatively, by substituting the beta function using Equation 2.1, the divergence can be expressed directly in terms of the emittance and the fixed beam spot size [82]. These kinematic parameters are summarized in Table 2.2 for both the High Resolution (HR) and High Luminosity (HL) operation modes of the HESR. As detailed in the table, while the rms beam spot size is kept strictly constant at 0.8 mm across all configurations to maintain a consistent interaction vertex, the emittance and angular divergence exhibit strong dependencies on both the incident beam momentum and the chosen operation mode. Specifically, as the beam momentum (p_{beam}) increases from 1.50 GeV/c to 15.0 GeV/c, both the emittance and the angular divergence decrease significantly. Furthermore, when comparing the two operation modes, the HR mode consistently features a much smaller emittance and angular divergence than the HL mode. For instance, at maximum momentum (15.0 GeV/c), the HR mode achieves an exceptionally tight emittance of just 0.023 mm$\cdot$mrad. This highly constrained beam phase space in HR mode is absolutely essential for minimizing the momentum spread ($\Delta p/p$), thereby achieving the sub-MeV mass resolution required for precision resonance scanning, albeit at the expense of the maximum achievable luminosity.

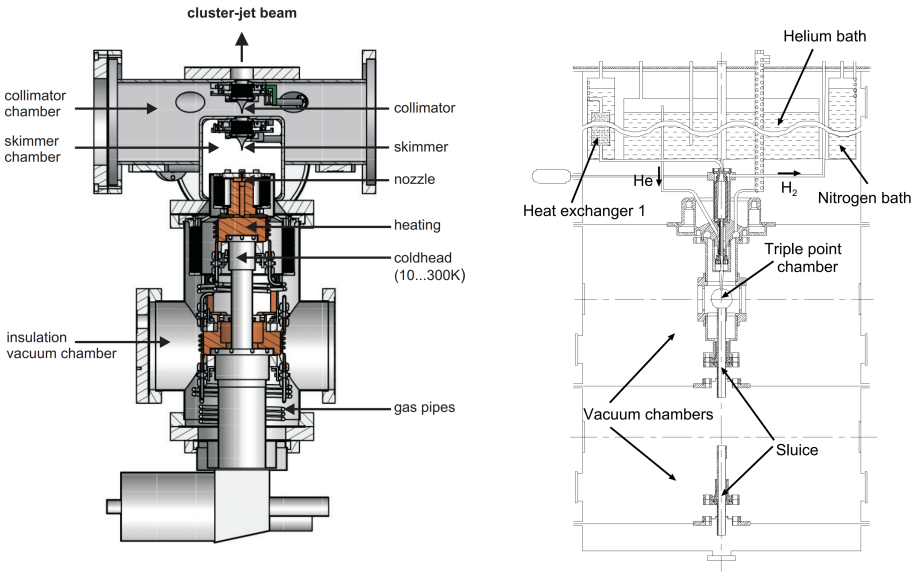
Table 2.2: HESR beam parameters for the High Resolution (HR) and High Luminosity (HL) modes at various beam momentum.

p_{beam} [GeV/c]	Spot Size (rms) [mm] (All Modes)	Emittance (rms) [mm · mrad]		Divergence (rms) [mrad]	
		HR	HL	HR	HL
1.50	0.8	0.23	1.45	0.29	1.81
4.06	0.8	0.08	0.53	0.11	0.67
8.90	0.8	0.04	0.244	0.05	0.31
15.0	0.8	0.023	0.145	0.03	0.18

Besides the beam spot size and angular divergence, the beam offset and tilt are also critical parameters. The beam offset denotes the transverse displacement of the beam centroid from the nominal trajectory in both the horizontal (x) and vertical (y) planes. The beam tilt represents the mean angular deviation of the beam axis relative to this ideal path. These two parameters are measured with Beam Position Monitors located upstream and downstream of the $\overline{\text{P}}$ANDA experiment, separated by approximately 20 m, resulting in uncertainties of ~ 200 μm in beam offset and ~ 0.2 mrad in beam tilt, respectively [83].

2.4 The $\overline{\text{P}}$ANDA Target

To achieve the design peak luminosity of 2×10^{32} cm^{-2}s^{-1} with the HESR antiproton beam, the $\overline{\text{P}}$ANDA experiment utilizes internal targets. Unlike external targets used in other experiments, the internal target system must operate within the ultra-high vacuum of the storage ring while providing sufficient areal density for collisions [84]. Two primary target technologies are foreseen for the different operation modes of $\overline{\text{P}}$ANDA: the Cluster-Jet Target and the Pellet Target (see Figure 2.4) [85].



(a) Cluster-Jet Target.

(b) Pellet Target.

Figure 2.4: Sketch of the cluster-jet (*left*) and pellet target (*right*). Figure adapted from [85].

The **Cluster-Jet Target** produces a continuous stream of supersonic hydrogen clusters. By expanding pressurized hydrogen gas through a fine nozzle at cryogenic temperatures, a directed jet is formed. This target type offers a homogeneous density profile and allows for a precise definition of the interaction region. The density can be adjusted continuously, making it particularly suitable for the initial commissioning phase and luminosity scans [86, 87].

The **Pellet Target** generates a stream of frozen hydrogen microspheres (pellets) with a typical diameter of ~ 20 μm [88]. These pellets are injected vertically into the beam line, which will provide accurate position coordinates in the accelerator beam region. While this system can achieve higher effective areal densities compared to the cluster jet, the discrete nature of the pellets introduces time-dependent variations in the instantaneous luminosity [89, 90]. Tracking of individual pellets is required to reconstruct the exact vertex position for each event.

Implications for the Interaction Region

Crucially for the luminosity determination discussed in this thesis, is that neither target system creates a point-like interaction vertex. The interaction region is defined by the spatial overlap of the circulating antiproton beam (with a finite transverse width) and the target stream (with a transverse diameter of several millimeters). This results in an *extended interaction volume* along the beam axis (z). Furthermore, the unavoidable diffusion of target material into the beam pipe creates a distributed profile of residual gas (Restgas) extending several meters upstream and downstream of the nominal interaction point [85, 5]. The precise reconstruction of this extended profile is a central challenge addressed in this work.

2.5 The $\overline{P}$ANDA Detector

The $\overline{P}$ANDA detector is designed to achieve nearly 4π solid angle coverage for both charged particles and photons. An overview of the spectrometer is shown in Figure 2.5. Given the fixed-target kinematics, where reaction products are boosted forward, the detector is split into two magnetically independent spectrometers: the Target Spectrometer (TS), surrounding the interaction point, and the Forward Spectrometer (FS) covering the downstream region [3].

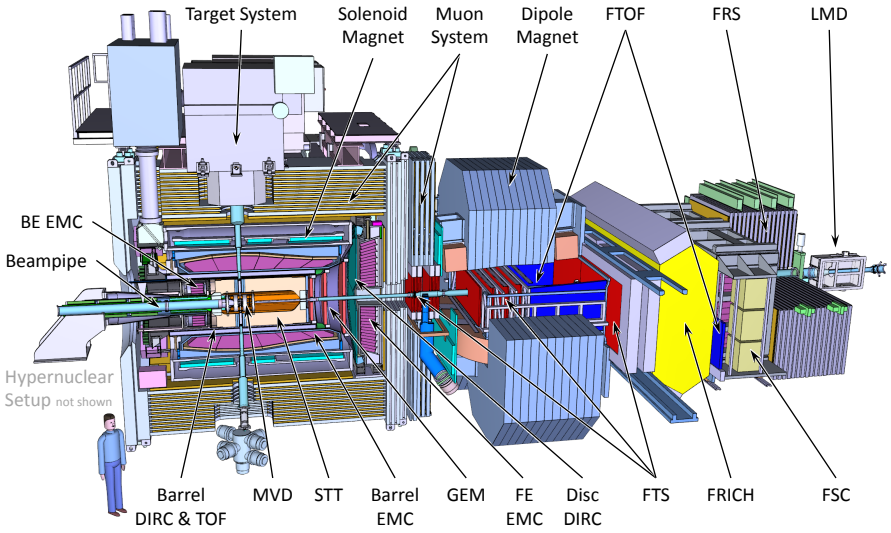


Figure 2.5: Schematic side view of the $\overline{P}$ANDA detector. The setup is divided into the Target Spectrometer, inserted in the solenoid magnet (left), and the Forward Spectrometer, utilizing a dipole magnet (right). Figure adapted from [3].

$\overline{P}$ANDA Magnets

The magnetic field configuration is defined by two separate magnets, which determine the particle bending for momentum analysis. The Target Spectrometer is housed within a superconducting **Solenoid Magnet** with a length of 2.5 m and inner radius of 0.9 m. It provides a highly homogeneous magnetic field (better than 2%) of 2 T parallel to the beam axis, ensuring fast and high precision charged particle tracking in the central region [91]. The Forward Spectrometer utilizes a large **Dipole Magnet** which provides a field integral of 2 Tm. It deflects high momentum particles passing through the forward angular acceptance (horizontal $< 10°$, vertical $< 5°$) to achieve the required momentum resolution for charged particles [91].

Tracking System

Precise reconstruction of charged particle trajectories and vertices is performed by a combination of detectors in the TS (the Micro Vertex Detector, the Straw Tube Tracker and the Gas Electron Multipliers) and FS (the Forward Tracker), as shown in Figure 2.6.

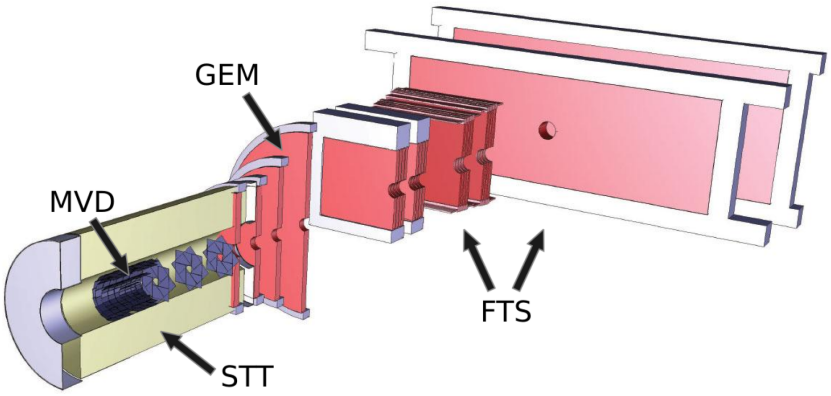


Figure 2.6: Tracking system of the $\overline{P}$ANDA detector, which contains the Micro Vertex Detector (MVD), Straw Tube Tracker (STT), Gas Electron Multipliers (GEM), and Forward Tracker (FTS). Figure adapted from [92].

The **Micro Vertex Detector (MVD)** is the innermost tracking device which consists of four barrel layers and six disks. It is crucial for precise determination of charged tracks and vertices of charged particles. (Detailed in Section 2.6).

Surrounding the MVD, the **Straw Tube Tracker (STT)** consists of 4636 individual straws, each with a length of 150 cm and spanning an outer radial region of 15 to 41.8 cm. The self-supporting tubes are made of aluminized Mylar and filled with an Ar/CO_2 gas mixture. When charged particles traverse the straws, they ionize the gas; the resulting electrons drift toward a central tungsten anode wire under a high-voltage electric field. By accurately measuring this drift time, the STT achieves

a spatial resolution of approximately 150 μm in the $r - \phi$ plane. Consequently, the STT serves as the primary central tracker, providing robust reconstruction of helical trajectories over a broad momentum range ($\sim$ 100 MeV/c to 8 GeV/c) and offering vital particle identification (PID) capabilities via specific energy loss (dE/dx) measurements [93].

Further downstream, three planar **Gas Electron Multiplier (GEM)** stations are positioned in the forward end-cap region (approximately 1.1 m, 1.4 m, and 1.9 m downstream of the interaction point), covering polar angles below 22°. These detectors operate using thin, Cu-coated polymide foils perforated with a high density holes. A strong electric field applied across these holes induces a localized electron avalanche, providing massive gas amplification. This unique multiplication principle is specifically chosen for its exceptional ability to handle the enormous particle rates in the forward direction while simultaneously maintaining an excellent spatial resolution of below 100 μm [94].

The track reconstruction and momentum analysis of forward-going particles, which are deflected by a large dipole magnet (covering polar angles below 5° vertically and 10° horizontally), are performed by the **Forward Tracker (FTS)**. It consists of three pairs of tracking stations strategically placed upstream, inside, and downstream of the dipole magnet. Each station comprises four double-layers of pressurized straw tubes (10 mm in diameter). Operating on the same drift-time measurement principle as the STT, the FTS yields a spatial resolution of roughly 150 μm per tube, which translates to a highly precise momentum resolution of $\Delta p/p \approx 1 - 2\%$ for forward tracks. To efficiently resolve geometrical ambiguities (ghost hits) and reconstruct full 3D coordinates, the first and fourth layers of each station contain vertical straws (0°), while the intermediate layers contain stereo straws inclined at +5° and −5°, respectively [95].

Located ∼11 m downstream, the **Luminosity Detector (LMD)** is a specialized tracking detector for elastic scattering recoil protons. (Detailed in Section 2.7).

Calorimeters

Measurements for photons and electrons is provided by two calorimeter systems designed to operate in high rate environments, as shown in Figure 2.7.

The **Electromagnetic Calorimeter (EMC)** of the Target Spectrometer is a homogeneous calorimeter composed of approximately 15,744 fast scintillating crystals made of $PbWO_4$, organized into a barrel section and two end-caps. The barrel covers a solid angle range of 22° to 140° for the barrel part, while the end caps cover the remaining angular range close to the beam axis, except for 5° in the vertical plane and 10° in the horizontal plane, which are necessary for the beam. Electromagnetic showers are created in fast scintillating crystals which are cooled to −25°C to maximize light yield and the total energy deposit is measured through the scintillating light. Readout is performed by large area Avalanche Photodiodes (APDs) in the barrel, the backward end-cap and the outer forward end-cap and Vacuum Phototetrodes (VPTTs) in the inner forward end-cap. The EMC achieves

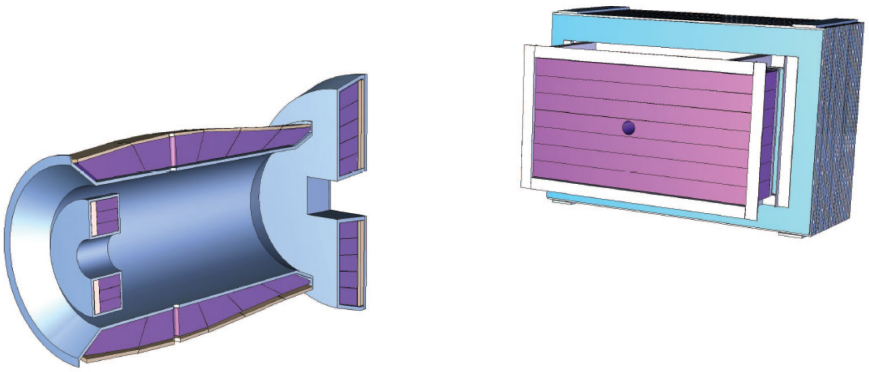


Figure 2.7: Calorimeter system of the $\overline{\text{P}}$ANDA detector, which contains the Electromagnetic Calorimeter (EMC) in the Target Spectrometer and the Forward Spectrometer Calorimeter (FSC). Figure adapted from [92].

a superior energy resolution of $1.5\%/\sqrt{E/\text{GeV}}$ and a spatial resolution of 1.1 mm [96].

Situated approximately 7.5 m downstream, the **Forward Spectrometer Calorimeter (FSC)** covers the acceptance of the dipole magnet window. To handle the high radiation dose in the forward region, it utilizes a Shashlyk-type sampling calorimeter design. This consists of 378 modules of lead absorber plates interleaved with scintillator tiles, read out by wavelength-shifting fibers coupled to photomultipliers, ensuring robust energy measurement for high energy particles with resolution of $4\%/\sqrt{E/\text{GeV}}$ [97].

Particle Identification (PID)

The physics program of $\overline{\text{P}}$ANDA requires the measurement of the four-momentum, which is not provided by tracking measurements alone, however can be determined by measuring the energy, time of flight, energy loss (dE/dx) or the velocity of the particle. $\overline{\text{P}}$ANDA employs a comprehensive PID scheme to identify different particle species over a wide momentum range, utilizing several various particle identification systems. The main PID systems are illustrated in Figure 2.8.

To identify charged hadrons (π, K, p) in the medium to high momentum range, $\overline{\text{P}}$ANDA employs two distinct Cherenkov detectors based on the **Detection of Internally Reflected Cherenkov light (DIRC)** principle. Both systems use synthetic fused silica as radiator material and Micro-Channel Plate Photomultiplier Tubes (MCP-PMTs) for readout, enabling operation within the strong magnetic field of the solenoid [98, 99]. The *Barrel DIRC* surrounds the STT at a radius of 47.6 cm and covers the polar angular range of $22° < \theta < 140°$. It consists of 48 narrow radiator bars ($17 \times 35 \times 2400$ mm^3) grouped into 16 sectors. The Cherenkov photons produced by traversing particles are guided by total internal reflection to the backward end, where they enter a compact fused silica expansion volume (prism). The photons are focused by a 3-layer compound lens system and detected by an array of MCP-PMTs with excellent timing resolution ($\sigma_t \approx 100$ ps).

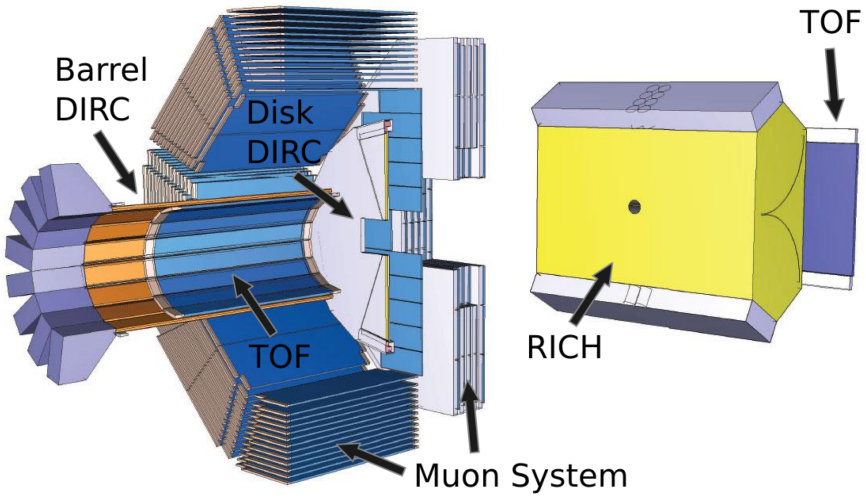


Figure 2.8: Particle Identification (PID) systems of the $\overline{P}$ANDA detector, which contains the Detection of Internally Reflected Cherenkov radiation (DIRC), the Ring Imaging Cherenkov Detector (RICH), the Time of Flight System (TOF) and the Muon Detection System. Figure adapted from [92].

This design provides π/K separation with a separation power of $> 3\sigma$ for momenta up to 3.5 GeV/c [98]. The forward end-cap region ($5° < \theta < 22°$) is instrumented with the *Endcap Disc DIRC*. Unlike the barrel, this detector is constructed from four physically independent quadrants, forming a dodecagonal disc structure. Each quadrant consists of a large fused silica radiator plate and dedicated Focusing Elements (FEL) that guide the photons to the readout modules attached to the outer rim. This novel design is optimized to separate pions from kaons up to momenta of 4 GeV/c, extending the PID capabilities into the forward spectrometer acceptance [99].

For high-momentum particles entering the forward spectrometer, a dual radiator aerogel **Ring Imaging Cherenkov Counter (RICH)** detector is foreseen. It uses silica aerogel and C_4F_{10} gas as radiators to provide $\pi/K/p$ separation in the momentum range from 2 GeV/c up to 15 GeV/c at polar angles between 5° and 22° [100].

Precise timing information is essential for particle identification (PID) in the low-to-intermediate momentum region, as well as for defining the relative timing between all tracks and photons. $\overline{P}$ANDA employs two dedicated **Time of Flight (TOF)** systems covering the target and forward spectrometers. Located in the Target Spectrometer, the *Barrel TOF* detector occupies the minimal radial gap between the STT and the DIRC. Its primary design goal is to provide a fast timing signal with a resolution of $\sigma_t < 100$ ps for event building and software triggering. The system consists of 5,760 small plastic scintillator tiles, each measuring $28.5 \times 28.5 \times 5$ mm^3, arranged in a cylindrical barrel geometry. The high granularity ensures low occupancy even at maximum interaction rates. To operate within the 2 T magnetic field and limited space, each tile is read out by Silicon Photomultipliers (SiPMs). The signals from multiple SiPMs are connected in series/parallel configurations to

optimize the signal-to-noise ratio and impedance matching. Prototype tests have demonstrated a time resolution of approximately 80 ps, sufficient to separate pions from kaons up to momenta of 700 MeV/c and to determine the event t_0 with high precision [101]. The Forward Spectrometer is equipped with the *Forward TOF (FTOF)* wall, positioned approximately 7.5 m downstream of the interaction point. It serves as the primary PID device for charged particles emitted at small polar angles (vertical $< 10°$, horizontal $< 5°$). The detector is constructed as a "scintillation wall" composed of 66 vertical plastic scintillator slabs (Bicron BC-408). The central slabs are 140 mm wide, while the outer ones are 200 mm wide, all having a thickness of 20 mm. Each slab is read out at both ends by fast Photomultiplier Tubes (PMTs) via twisted light guides. This dual-ended readout allows for position-independent time reconstruction and improves resolution. The system is designed to achieve a time resolution better than 100 ps. Experimental validation with 920 MeV protons yielded a resolution of $\sigma \approx 70$ ps, enabling π/K separation up to 2.8 GeV/c and p/K separation up to 4.7 GeV/c [102].

To distinguish between pions and muons, which cannot be differentiated by Cherenkov detectors or TOF systems, a **Muon Detection System** is implemented. It relies on the range system technique implemented in the instrumented flux return yokes of the magnets. The system is designed to discriminate muons from the abundant background of pions by exploiting their different interaction mechanisms: while hadrons (mostly pions) initiate hadronic showers and are absorbed within the iron, muons interact primarily via ionization and penetrate deep into the material. The Mini-Drift Tube (MDT), a robust gas-filled drift cell capable of operating in high magnetic fields, will be used as the basic detection element. The digital hit information from the MDTs is used to reconstruct the range of the particle within the absorber block. The yoke is segmented, with the gaps instrumented with rectangular aluminum Mini-Drift Tubes (MDTs). This interleaved structure (Iron-MDT-Iron) allows for sampling the particle trajectory at various depths, achieving an efficiency above 90% for muon identification against pion background in the energy range from 500 MeV/c to 1200 MeV/c [103].

Among these systems, the **MVD** (tracking precision) and the **Luminosity Detector** (luminosity reference) form the hardware basis for the research presented in this thesis.

2.6 Micro Vertex Detector (MVD)

The MVD is the innermost tracking device of the Target Spectrometer. It is designed to detect charged particles with high spatial resolution immediately after their production at the interaction point. The primary physics goals requiring such precision include the identification of open charm and strangeness production via the reconstruction of secondary decay vertices (e.g., D mesons, hyperons) displaced from the primary interaction point by only a few hundred micrometers [104].

The Micro Vertex Detector (MVD) tightly encloses the beam pipe and the target interaction region. Its geometrical layout, optimized to maximize angular acceptance while minimizing the material budget, consists of two primary intersecting subsystems (see Figure 2.9).

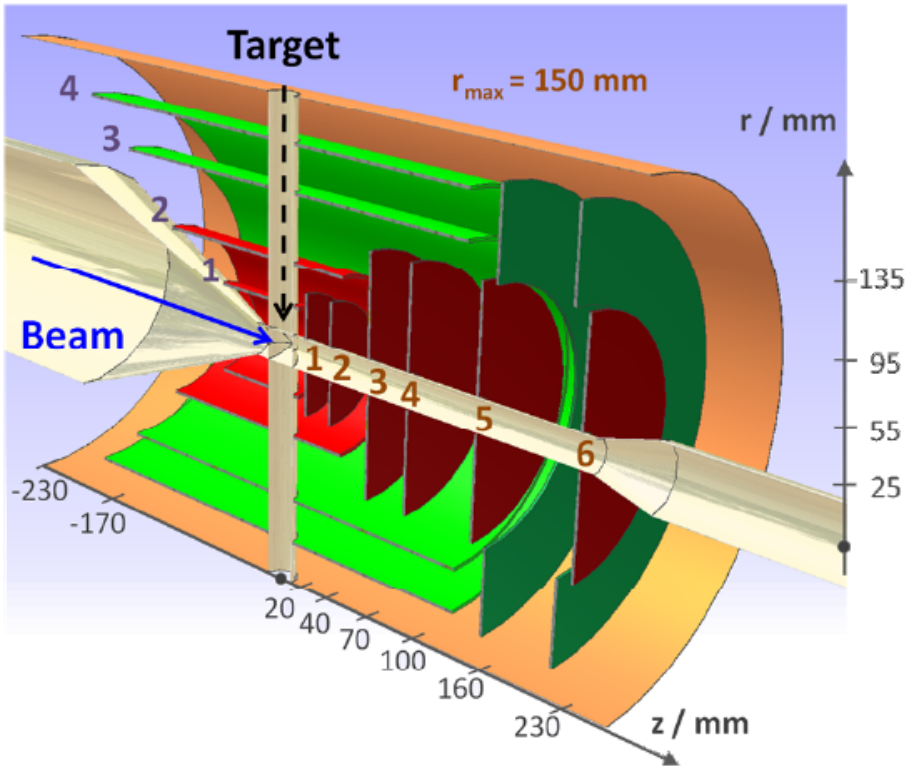


Figure 2.9: Schematic layout of the PANDA MVD showing the barrel and disk sections, the green parts are strip sensors and the red parts are pixel sensors. Figure adapted from [104].

The **barrel section** comprises four concentric cylindrical layers arranged around the beam axis at radial distances of approximately 25, 55, 95, and 135 mm. To achieve extreme spatial resolution near the primary vertex, the two innermost layers are equipped with high-precision pixel sensors, whereas the two outer layers utilize double-sided silicon strip detectors. Complementing this cylindrical geometry is the forward **disk section**, which consists of six planar tracking disks oriented perpendicular to the beam axis. To handle the high track multiplicity in the forward direction, the first four disks, located at distances of 2, 4, 7, and 10 cm downstream from the interaction point, are instrumented entirely with pixel sensors. The remaining two disks, positioned further downstream at 16 and 23 cm, employ a combination of both pixel and strip detector technologies. Together, this hybrid geometric configuration ensures a broad polar angular coverage ranging from 3° to 150°. Consequently, the MVD provides highly accurate decay vertex information for almost all charged tracks falling within the acceptance of the Target Spectrometer, as illustrated in Figure 2.10. This configuration ensures a polar angular coverage from 3° to 150°, providing vertex information for almost all tracks within the acceptance of the Tar-

get Spectrometer, as shown in Figure 2.10.

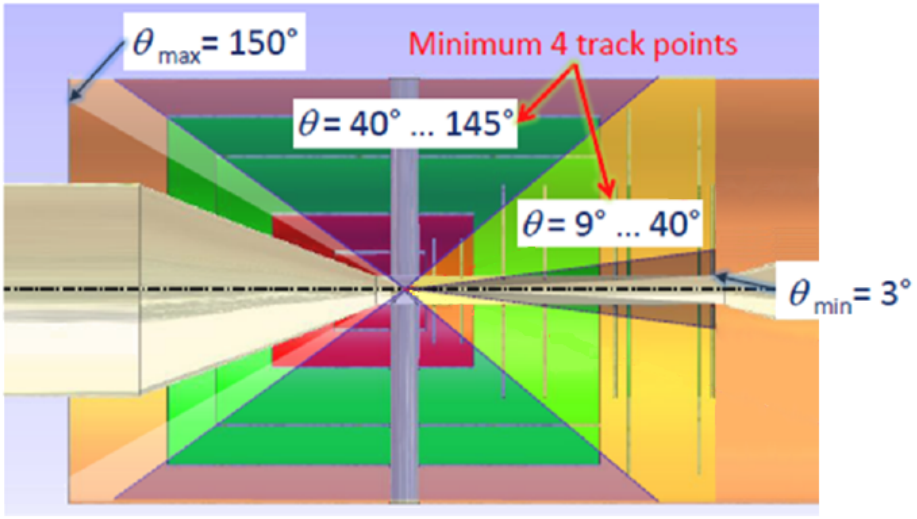
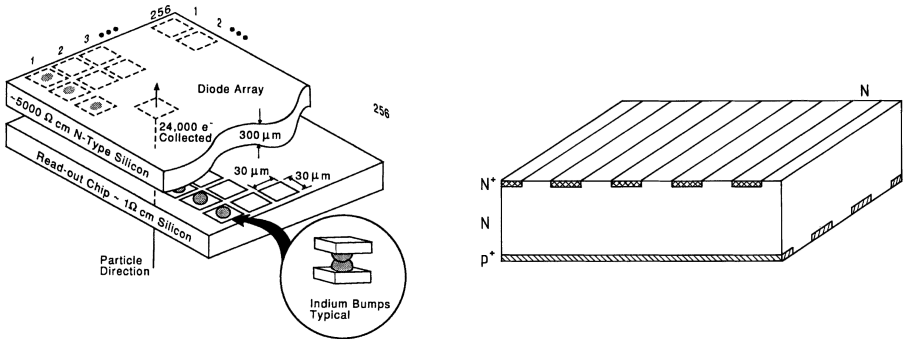


Figure 2.10: Polar angle coverage of the MVD for charged particles originating from the interaction point, at least four hits per track are guaranteed in the range of $9° < \theta < 145°$. Figure adapted from [104].

To optimally balance the stringent requirements of high-rate capability, exquisite spatial resolution, and a minimized material budget, the original design of the MVD employs two distinct silicon technologies based on their radial distance from the interaction point.

In regions subjected to the highest particle fluxes, specifically the two innermost barrel layers and the inner rings of all six forward disks, are instrumented with **Hybrid Pixel Sensors**. The term "hybrid" refers to the physical separation of the sensitive detection volume and the readout electronics, as shown in Figure 2.11a. The sensor material consists of oxygenated epitaxial silicon with a thickness of 200 μm, chosen for its high radiation hardness against non-ionizing energy loss. Each sensor is bump-bonded to a custom-developed readout ASIC named ToPix, fabricated in 0.13 μm CMOS technology. The pixel cell size is $100 \times 100\,\mu\text{m}^2$ (see Figure 2.12), providing excellent spatial resolution. Unlike frame-based imagers, ToPix operates in a triggerless, continuous readout mode. It registers the time of arrival (timestamp) and the energy deposition via the Time-over-Threshold (ToT) method for every hit, which also enables particle identification via dE/dx.

Moving outward, to instrument the larger surface areas of the outer regions (Barrel Layers 3-4 and the outer rings of Disk Layers 5-6), the MVD utilizes **Double-Sided Silicon Strip Detectors**. This technology is selected to drastically minimize the total number of readout channels while maintaining high spatial precision. These sensors comprise orthogonal strips on opposing surfaces, which are interconnected to the electronics, as illustrated in Figure 2.11b. The barrel utilizes rectangular sensors with orthogonal strips (90° stereo angle). The intrinsic strip pitch is 65 μm. To improve spatial resolution while reducing the number of readout channels, intermediate floating strips are used, resulting in a readout pitch of 130 μm. The forward disks



(a) Schematics of the hybrid pixel sensor, showing the attachment of the detector and readout electronics. Figure adapted from [105].

(b) Schematics of the Double-Sided Silicon Strip Detector, showing the orthogonal strips on opposing surfaces. Figure adapted from [106].

Figure 2.11: Hybrid pixel sensor and its readout ASIC used in the MVD.

employ trapezoidal sensors. To reduce ghost hits in this high-occupancy region, a small stereo angle of 15° is implemented between the top and bottom strips. The strip pitch is adapted to 65 μm to maintain consistent spatial precision (see Figure 2.12). The strips are read out by the PASTA (PAnda STrip Asic) chip. Similar to ToPix, PASTA utilizes the ToT technique to determine the hit time, ensuring compatibility with the experiment's free-streaming data acquisition architecture to measure energy loss with a high dynamic range.

While the hybrid pixel technology offers excellent time resolution and radiation hardness, it possesses a significant drawback concerning the material budget. A single tracking station, consisting of the sensor (200 μm), the readout chip (150 μm), the flex-print cable, and the necessary cooling structures (carbon foam/pipes), introduces a material thickness of approximately 1.1% X_0 (radiation length) [107]. For high-momentum particles, this material budget is acceptable. However, for low-momentum particles ($p < 1$ GeV/c), which are abundant in $\bar{p}p$ annihilation, the *multiple Coulomb scattering* induced by this material becomes the dominant factor limiting the vertex resolution. As reducing the material budget is the key to improving the tracking performance, this limitation motivates the investigation of **High Voltage Monolithic Active Pixel Sensors (HV-MAPS)** (see section 2.7) as a potential upgrade technology with a reduced material budget to $\sim 0.3\%\, X_0$, which is a central topic of this thesis (discussed in Chapter 6).

2.7 Luminosity Detector (LMD)

The luminosity determination with high accuracy is crucial for many aspects in $\overline{\text{P}}$ANDA experiment. The primary dedicated device for this task is the Luminosity Detector (LMD). Its design goal is to determine the absolute time-integrated luminosity with a systematic uncertainty of less than 1% by measuring the rate of antiproton-proton elastic scattering ($\bar{p}p \to \bar{p}p$) close to the Coulomb-Nuclear Interference region [108].

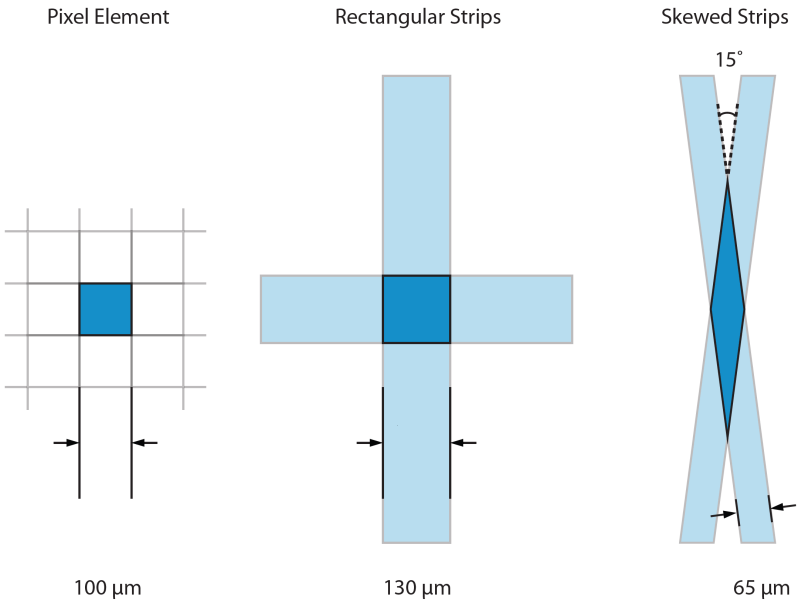


Figure 2.12: Schematic view of readout cells, hybrid pixel sensors (*left*), rectangular strip sensors (*center*) and 15° skewed strip sensors (*right*). Figure adapted from [107].

Since the differential cross-section of elastic scattering rises steeply towards small momentum transfers (t), the LMD is positioned far downstream in the Forward Spectrometer to detect scattered antiprotons at very small polar angles. The detector is located approximately 10.5 m to 12.5 m from the interaction point. The complete setup of the LMD is shown in the Figure 2.13. It consists of four planar tracking stations situated in the vaccum box separated by distances of 20 cm between the first two planes and 10 cm between the last three planes. The tracks of the scattered antiprotons will be measured by High Voltage Monolithic Active Pixel Sensors (HV-MAPS) with very high spatial resolution and low material budget. There is a transition cone providing the Electromagnetic shielding for the beam, allowing the forward scattered antiprotons to enter the LMD volume. The linear shift mechanism as well as the half plane design allows the LMD to be retracted. Consequently, this geometry of LMD defines a polar angular acceptance of $3.5\,\text{mrad} \leq \theta \leq 10\,\text{mrad}$ with full azimuthal ($0 \leq \phi < 2\pi$) coverage [7].

Figure 2.14 shows the detailed design of one half-plane of the LMD. Each half plane consists of 5 sensor modules. Each module contains a chemical vapor deposition (CVD) diamond wafer that has 5 HV-MAPS glued on each side. The heat generated by the sensors is conducted through the CVD diamond to liquid-cooled aluminum heat sink support (operated with ethanol at $-20°$C) located outside the active acceptance. This design ensures efficient cooling while maintaining an extremely low material budget.

The LMD is the first system in $\overline{P}$ANDA, and one of the first in high-energy physics, designed to employ HV-MAPS. HV-MAPS are fabricated in a high-voltage complementary metal oxide semiconductor (CMOS) process. Unlike standard CMOS sensors where charge collection relies on slow thermal diffusion, HV-MAPS apply a

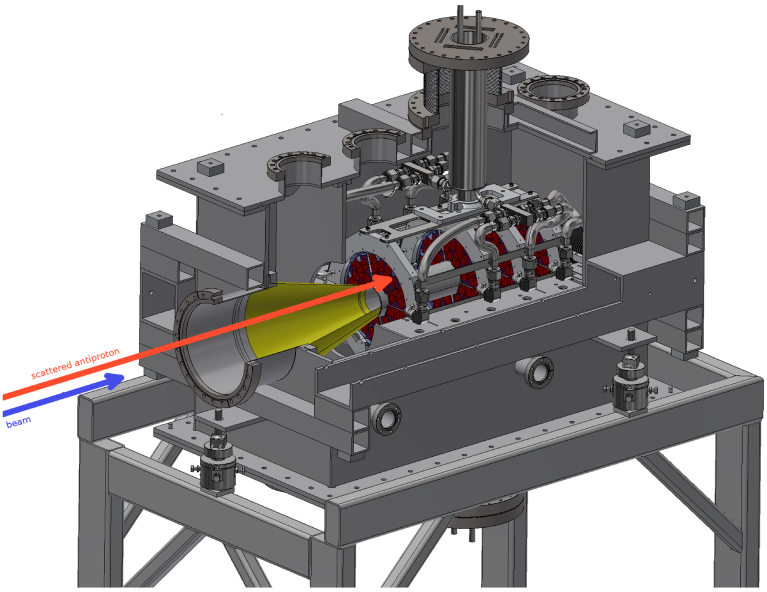


Figure 2.13: CAD drawing of the LMD. Figure adapted from [109].

high bias voltage (typically $\geq 60\,\mathrm{V}$) between the deep n-well and the p-substrate. This creates a wide depletion zone where signal charge is collected via **drift** in a strong electric field, as shown in Figure 2.15. The drift mechanism ensures very fast charge collection ($< 10\,\mathrm{ns}$), enabling high time resolution and the ability to assign timestamps to individual hits. The depletion and fast collection make the sensors inherently more robust against radiation damage compared to standard MAPS.

The monolithic architecture of HV-MAPS ensures that the readout electronics, including the charge-sensitive amplifier and digital logic, are integrated directly into the pixel cell or the sensor periphery on the same silicon die. This eliminates the need for an external readout chip and the corresponding bump-bonding interconnections required by hybrid pixels in the original MVD design.

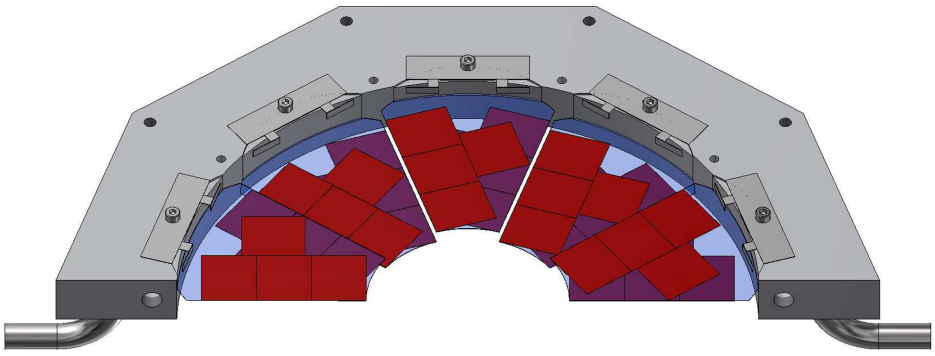


Figure 2.14: Detailed design of one half-plane acting as a cooling support of 5 sensor modules. The aluminum structure contains a stainless steel cooling pipe. Figure adapted from [109].

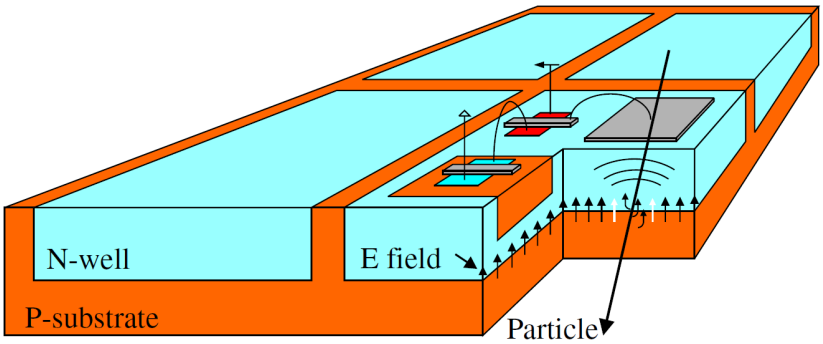


Figure 2.15: Schematic view of HV-MAPS with four pixels. The high voltage applied between the deep n-well and the p-substrate creates a depletion zone where signal charge is collected via drift in a strong electric field. Figure adapted from [110].

HV-MAPS technology is now implemented using the MuPix sensor family, which has been developed and optimized over the past decade within the Mu3e experiment [111]. The MuPix sensors feature a pixel size of $80 \times 80\,\mu\text{m}^2$, an overall dimension of $2 \times 2\,\text{cm}^2$ and a thickness of only $50\,\mu\text{m}$. Combined with the CVD diamond support, the total material budget is reduced to approximately 0.3% X_0 per station. This minimization of material is crucial for the LMD to achieve the required angular resolution [108].

2.8 The PandaRoot Software

The PandaRoot framework [112] is a specific implementation derived from the FairRoot project [113]. The FairRoot framework is an object-oriented simulation, reconstruction, and data analysis framework based on ROOT [114]. It was explicitly developed to serve multiple experiments at the FAIR facility, including CBM, $\overline{\text{P}}$ANDA, and R3B. By leveraging ROOT's tree and branch-based data structures, the framework delivers base classes which enable the users to easily construct their experimental setup in a fast and convenient way.

A comprehensive schematic structure of FairRoot is presented in Figure 2.16. As explicitly illustrated, the software architecture is intrinsically divided into three hierarchical layers: the underlying dependencies (top), the core framework (middle), and the experiment-specific implementations (bottom). The **top layer** comprises the foundational ROOT libraries and the Virtual Monte Carlo (VMC) [115] interface. Developed at CERN, ROOT provides a vast array of tools essential for high-energy physics, encompassing robust data analysis and performance simulation functionalities, thereby serving as the absolute bedrock of FairRoot. Working together with ROOT is the VMC, a widely utilized abstraction tool in high-energy physics. The VMC enables seamless access to various Monte Carlo transport engines without necessitating any alterations to the user's code, geometry, or detector descriptions. Presently, VMC supports the use of both GEANT3 [116] and GEANT4 [117], with

support for FLUKA [118] currently under development. GEANT itself is the standard application suite for simulating the passage of particles through matter. It exists in two major iterations: GEANT3, which is a legacy system coded in FORTRAN, and GEANT4, a modern evolution written in C++ that employs an object-oriented design to implement physics processes more generally. Because the simulation results produced by GEANT3 and GEANT4 are nearly identical for many standard applications, both versions remain extensively employed and validated within the high-energy physics community. The **middle layer** constitutes the FairRoot core, which centrally manages the execution flow through the Run Manager. On one hand, the framework delivers managing classes that drive the execution of a simulation or reconstruction session, handle the Input/Output (I/O) to files, process parameter I/O, and build the geometry for a simulation from various input files. On the other hand, it provides essential base classes for detector handling, magnetic field definition, event generators for simulation sessions, and base task structures used for analysis sessions. Finally, the **bottom layer** contains the experiment-specific code. Historically, the code base was split into the base part called FairRoot and the experiment-specific parts, such as CbmRoot and PandaRoot. In this bottom tier, individual $\overline{\text{P}}$ANDA sub-detectors (e.g., MVD, STT, EMC) and their distinct digitization or track-finding algorithms are securely implemented by inheriting from the middle-layer core base classes. Using and extending the task mechanism of ROOT, it is possible to implement complex analysis tasks in a highly structured manner. Within this framework, `FairTask` serves as the fundamental class that facilitates the construction of a hierarchical tree of processing tasks. Derived from ROOT's base `TTask` class, it can elegantly contain multiple sub-tasks that are executed recursively. Consequently, highly sophisticated simulation and reconstruction programs can be dynamically assembled and executed simply by invoking the services of the top-level task or its specific sub-tasks.

Figure 2.17 depicts the data flow of the PandaRoot framework. The simulation process involves the generation of physics and background events using several event generators such as "Particle Gun", EvtGen, DPM, PYTHIA, and UrQMD. The VMC is utilized for the transportation of the generated particles through the detectors, where they interact with the sensitive elements. The energy deposit resulting from these interactions is digitized to simulate future measurements of the experiments. Using the digitized output of the sinulation or experiment data, the hits can be reconstructed, followed by track finding, fitting, and particle identification to perform the physics analysis.

The PandaRoot framework incorporates a modular design approach, where each subsystem is allocated its own package for Monte Carlo processing, digitization, reconstruction, tracking and particle identification (PID). The detector components within each subsystem employ unique algorithms to compute the probability density function (PDF) for each particle hypothesis. The combination of the PDFs from all detector components is accomplished through the use of a Bayesian approach, resulting in a global likelihood for PID.

Upon completion of data flow, the PandaRoot framework offers a range of algorithms for the tracking and particle identification, in addition to comprehensive analysis tools including the *Rho* package and the kinematic fitter package. The latter package facilitates a range of fitting techniques, including vertex fitting, mass

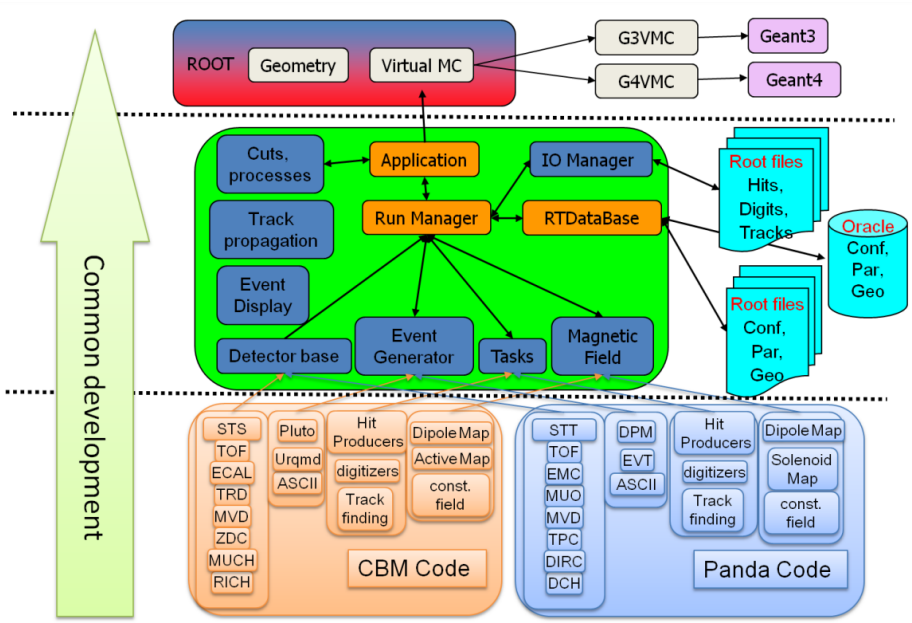


Figure 2.16: Structure of FairRoot. External libraries at the top, the framework in the middle and the special experiments at the bottom. Figure adapted from [112].

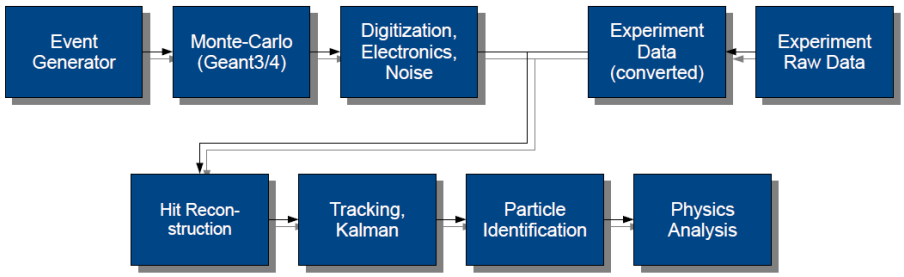


Figure 2.17: Scheme of PandaRoot data flow. Figure adapted from [107].

constraint fitting, 4C Kinematic fitting, and decay tree fitting. The Rho package provides a unique mechanism for consolidating particle candidate lists, which effectively eliminates the potential for double counting through the use of indexed candidates. Additionally, the package includes support for PID, tracking, and generated Monte Carlo information, further enhancing the efficiency and accuracy of the physics analysis process.

The **Silicon Detector Software (SDS)** is a specialized, highly integrated software package designed for all silicon detector components within the $\overline{\text{P}}$ANDA experiment, including MVD and LMD. By utilizing C++ inheritance mechanisms, the SDS significantly simplifies the development and maintenance of software packages intended for individual silicon sub-detectors. The SDS interfaces seamlessly with PandaRoot through the `FairTask` mechanism, which orchestrates the entire data processing and analysis workflow. Conceptually, the SDS workflow proceeds sequentially through two principal phases: Monte-Carlo transport coupled with electronic digitization,

followed by hit reconstruction and clustering.

To accurately simulate the physical processes, the detector geometry is first meticulously modeled based on the ROOT framework, organized as a hierarchical nesting of volumes called assemblies. The basic physical volumes are translated and rotated to form these higher-level assemblies. Because the tree of assemblies is accessed via a directory of their names, identifying specific sensors requires a full structural path. To streamline this process during data processing, the `PndGeoHandling` class is utilized to dynamically map the sensitive physical volumes to unique integer identifiers known as sensor IDs. Furthermore, to optimize computational performance, only interactions within predefined active volumes (e.g., designated by names like "PixelActive") are evaluated and stored.

The data processing pipeline, illustrated in detail in Figure 2.18, begins with **Monte-Carlo Transport and Digitization**. Primary particles generated by event generators are propagated through the detector geometry and magnetic field via the Virtual Monte Carlo (VMC) interface (utilizing engines like GEANT3 or GEANT4). Whenever a simulated particle traverses an active silicon volume managed by `PndSdsDetector`, a Monte-Carlo point (`SdsMCPoint`) is instantiated. This point records crucial exact physical quantities, including the entry and exit coordinates, momentum, energy loss within the volume, and the precise time of flight.

Subsequently, the `SdsHitProducer` executes the digitization process. Digitization simulates the specific electronic response of the detector, converting the continuous energy loss of the `SdsMCPoint` into discrete digital signals (`SdsDigi`, specialized as `PndSdsDigiPixel` or `PndSdsDigiStrip`). Depending on the sensor thickness and technological properties, the SDS package employs different physical models to distribute the deposited charge across the readout channels:

Linear Model: For ultra-thin silicon sensors, such as the 50 μm thick HV-MAPS, the particle trajectory within the sensor bulk is sufficiently short to be approximated as a straight line. The deposited charge is assumed to be uniformly distributed along this projected track. Therefore, the charge Q_i measured by each strip i, or $Q_{i,j}$ measured by each pixel (i, j), is determined geometrically:

$$Q_i = Q \cdot \frac{|x_{i,out} - x_{i,in}|}{|x_{out} - x_{in}|} \qquad \text{(Strips)} \tag{2.3}$$

$$Q_{i,j} = Q \cdot \frac{|x_{i,out} - x_{i,in}|}{|x_{out} - x_{in}|} \cdot \frac{|y_{j,out} - y_{j,in}|}{|y_{out} - y_{in}|} \qquad \text{(Pixels)} \tag{2.4}$$

where Q is the total deposited charge, (x_{in}, y_{in}) and (x_{out}, y_{out}) are the entry and exit coordinates of the track inside the sensor volume, and the subscripted coordinates represent the entry and exit points within the specific pitch or pixel boundaries.

Static Charge Cloud Model: Conversely, for thicker sensors, such as the 200 μm hybrid pixel sensors, the drifting charge carriers experience significant lateral diffusion, rendering the straight-line approximation inadequate. This phenomenon is modeled by projecting a Gaussian probability density function of the charge cloud onto the sensor surface. The charge Q_i measured by a strip i is calculated

as:

$$Q_i = \frac{Q}{2p \cdot |x_{out} - x_{in}|} \cdot \Big(F_i(x_{out}) - F_i(x_{in}) - F_{i+1}(x_{out}) + F_{i+1}(x_{in}) \Big) \tag{2.5}$$

where $p = |x_i - x_{i+1}|$ is the pitch size, and $F_k(x)$ is the primitive of the error function $\mathrm{erf}(x)$ for strip k in the x-direction:

$$F_k(x) = (x_k - x) \cdot \mathrm{erf}\left(\frac{x_k - x}{\sqrt{2}\sigma} \right) + \sqrt{\frac{2\sigma^2}{\pi}} \cdot e^{-\frac{(x_k - x)^2}{2\sigma^2}}. \tag{2.6}$$

The charge cloud width σ for 300 μm thick silicon is theoretically parameterized as [105]:

$$\sigma = \sqrt{\frac{kT}{q} \cdot \frac{2\epsilon\epsilon_0}{qN_0}} = 5.81\,\mu\mathrm{m} \tag{2.7}$$

where k is the Boltzmann constant, T is the absolute temperature, q is the elementary charge, ϵ and ϵ_0 are the relative and vacuum permittivities, and N_0 is the effective donor concentration. For pixel sensors featuring a second dimension of readout channels, the formula is naturally extended to integrate over both x and y distributions:

$$\begin{aligned} Q_{i,j} = & \frac{Q}{4p \cdot |x_{out} - x_{in}| \cdot |y_{out} - y_{in}|} \\ & \cdot (F_i(x_{out}) - F_i(x_{in}) - F_{i+1}(x_{out}) + F_{i+1}(x_{in})) \\ & \cdot (F_i(y_{out}) - F_i(y_{in}) - F_{i+1}(y_{out}) + F_{i+1}(y_{in})). \end{aligned} \tag{2.8}$$

This digitally synthesized charge information is then stored alongside its corresponding pixel or strip hardware address.

The final phase in the SDS pipeline is **Hit Reconstruction and Clustering**, executed by the `SdsClusterTask`. Clustering is the pattern recognition process that aggregates adjacent fired pixels or strips (digis) that are statistically likely to originate from the passage of the same incident particle. To belong to a cluster, a digital entry must lie within a predefined search radius (set by default to include immediate neighbors). To aggressively suppress electronic noise and reject "ghost" clusters, the summed charge of the cluster must exceed a rigorous predefined threshold. Once a valid cluster is formed, its spatial centroid is determined to extract the precise spatial hit information in the $x-y$ plane of the local sensor coordinate system (with $z = 0$ defining the sensor mid-plane). Finally, `PndGeoHandling` performs the necessary mathematical translations and rotations to transform this local hit into the global $\overline{P}$ANDA laboratory frame. The calculation of the cluster centroid relies on several specific algorithms, chosen dynamically based on the cluster size and the underlying sensor technology:

Binary Center: This algorithm places the hit exactly in the geometric center of the fired channel with the highest charge. It is the only viable method for single-channel clusters, yielding an intrinsic spatial variance of [119]

$$\Delta x^2 = \frac{p^2}{12} \tag{2.9}$$

where p is the pixel or strip pitch.

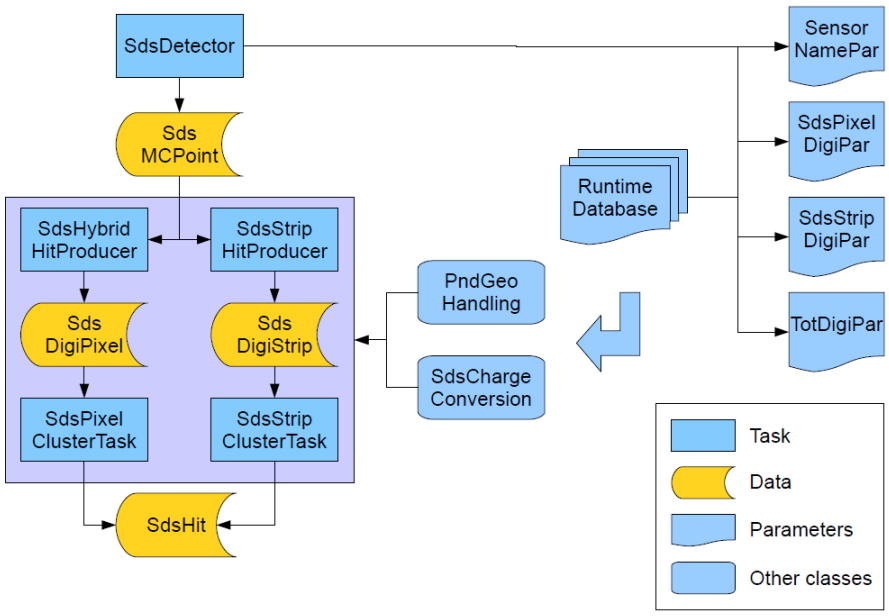


Figure 2.18: Schematic layout of the SDS package illustrating the sequential data flow from Monte-Carlo generation to hit reconstruction. Figure adapted from [107].

Center of Gravity: For clusters spanning two or more channels ($n \geq 2$), the Center of Gravity method utilizes the individual charge measurements q_i as statistical weights to calculate a refined barycenter of the fired pixels or strips x_i as follows:

$$\bar{x} = \frac{\Sigma_i q_i x_i}{\Sigma_i q_i}. \tag{2.10}$$

The uncertainty of this method is:

$$\Delta x^2 = \frac{\sigma^2_{\text{Noise}}}{(\Sigma_i q_i)^2} \cdot \sum_{i=1}^{n} (x_i - \bar{x})^2 + \frac{Q^2_{\text{Thr}}}{12 \cdot (\Sigma_i q_i)^2} \left[(x_0 - \bar{x})^2 + (x_{n+1} - \bar{x})^2 \right] \tag{2.11}$$

where Q_{Thr} is the charge threshold and σ_{Noise} is the electronic noise.

Median: This algorithm approximates the hit position using the geometric middle of the clustered channels, which is highly effective if the track produces a statistically flat charge density profile. The uncertainty of this method is the same as the binary center method.

Head-Tail Algorithm: For extensively wide clusters ($n > 3$), the Head-Tail Algorithm offers superior precision. By evaluating only the first (head x_h) and last (tail x_t) channels of the cluster, and modifying their geometric midpoint based on the charge asymmetry at the cluster edges (Q_h and Q_t), it mitigates fluctuations within the cluster core. The hit position is calculated as:

$$x = \frac{x_h + x_t}{2} + p \cdot \frac{Q_h - Q_t}{2Q_0} \tag{2.12}$$

where p is the pixel size or strip pitch, Q_0 is the charge the particle leaves in one readout channel which is approximated by the average charge deposit in the channel

between the head and tail ($n = t - h$):

$$Q_0 = \frac{1}{n-2} \sum_{i=h+1}^{t-1} Q_i. \tag{2.13}$$

The uncertainty of this method is:

$$\Delta x^2 = 2 \cdot \sigma^2_{\text{Noise}} \cdot \frac{p^2}{4Q_0^2}. \tag{2.14}$$

In practice, the application of these reconstruction algorithms is strictly dictated by the sensor technology. For HV-MAPS, charge sharing information cannot be reliably exploited for cluster-position reconstruction because the charge is collected within a very thin depletion region of approximately 15 μm. In such a thin sensitive layer, the energy deposited by traversing particles exhibits large relative Landau fluctuations, resulting in substantial variations in the generated and collected charge and thereby making charge-based interpolation unreliable. Consequently, single-pixel clusters are reconstructed using the binary center method, while multiple-hit clusters rely on the median method. In contrast, the fully analog hybrid pixel sensors provide sufficiently precise information on the energy deposited in individual pixels and can therefore exploit charge sharing for position reconstruction. They use the binary method for single-pixel clusters, switch to the center-of-gravity method for clusters of two or three pixels, and employ the highly precise head-tail algorithm for larger clusters ($n > 3$).

Chapter 3

Luminosity Determination

3.1 The Concept of Luminosity

In high-energy physics experiments, the Luminosity ($\mathcal{L}$) is the fundamental parameter that relates the physical cross section (σ) of a specific reaction to the experimentally observed interaction rate (R). It serves as the absolute overall normalization factor for virtually all physics analyses.

The instantaneous luminosity characterizes the interaction potential of the colliding system and is quantified as the number of particles per unit area and unit time that can, in principle, participate in an interaction. For a fixed-target experimental setup like $\overline{\text{P}}$ANDA, the luminosity is determined by the specific beam and target parameters. It is defined as [2]:

$$\mathcal{L} = f \cdot n_b \cdot N_{beam} \cdot \rho_{target} \tag{3.1}$$

where f is the revolution frequency of the beam in the storage ring, n_b is the number of circulating bunches, N_{beam} is the number of particles per bunch, and ρ_{target} is the areal density of the target (number of target atoms per unit area). The total interaction rate for a process is simply given by the relation $R = \mathcal{L} \cdot \sigma_{tot}$.

3.2 Motivation for Precise Luminosity Determination

The primary objective of the $\overline{\text{P}}$ANDA experiment is to study the strong interaction in the transitional region between perturbative and non-perturbative QCD. This involves high-precision spectroscopy of charmonium and open-charm states, alongside searches for exotic hadrons such as glueballs and hybrids. To successfully accomplish these physics goals, a precise determination of the luminosity is mandatory.

Absolute Cross Section Measurement

The most fundamental quantity in scattering experiments is the cross section σ, which reflects the probability of a specific physical process occurring. In an experiment, the number of observed events N_{obs} for a given reaction channel is related to the physical cross section by:

$$N_{obs} = \mathcal{L}_{int} \cdot \sigma \cdot \epsilon_{det} \tag{3.2}$$

where ϵ_{det} is the total detection efficiency (determined via simulations) and $\mathcal{L}_{int}$ is the time-integrated luminosity, defined as the integral of the instantaneous luminosity $\mathcal{L}(t)$ over the data taking period from t_{start} to t_{end}:

$$\mathcal{L}_{int} = \int_{t_{start}}^{t_{end}} \mathcal{L}(t)\, dt \tag{3.3}$$

Consequently, the experimental determination of the cross section is given by $\sigma = N_{obs}/(\mathcal{L}_{int} \cdot \epsilon_{det})$. Any uncertainty in the luminosity determination, $\Delta\mathcal{L}/\mathcal{L}$, translates directly into a systematic uncertainty in the measured cross section. For $\overline{\mathrm{P}}$ANDA, the design goal is to achieve a systematic precision of better than 1% for the absolute luminosity. This precision is essential to distinguish between different theoretical models that often predict cross sections differing by only a few percent.

Resonance Energy Scanning in PANDA

The primary advantage of $\overline{\mathrm{P}}$ANDA as a fixed-target experiment is the ability to perform Resonance Energy Scans. In this mode, the center-of-mass energy $\sqrt{s}$ is varied in small, discrete steps across the expected mass region of the resonance. The goal is to reconstruct the resonance lineshape and extract its properties (mass M_R and width Γ_R) with a precision limited only by the HESR beam momentum resolution ($\Delta p/p \sim 10^{-5}$), rather than the detector resolution.

The principle of this method, and the critical role of luminosity normalization, is illustrated by Figure 3.1. The scanning procedure involves systematically adjusting the HESR beam momentum $p_{beam,i}$, meaning the center-of-mass energy $E_{cm,i}$ is varied accordingly. At each step i, the number of observed events $N_{obs,i}$ is counted. However, as shown in Figure 3.1 *(Left)*, the raw measured rates do not directly reveal the resonance shape. In a real experiment, the time for each measurement varies and thus the area below the beam profile changes from point to point. Consequently, the raw measured rates appear chaotic and scattered. To recover the physical observable, all data points must be normalized. The measured rates and beam profiles are normalized by the precise relative time-integrated luminosity $\mathcal{L}^{rel}_{int,i}$. As illustrated in Figure 3.1 *(Middle)*, the beam profiles are normalized to be the same at every point, and after normalization the measured rates align to form a smooth curve. This curve represents the convolution of the intrinsic resonance lineshape and the specific beam energy resolution function. The final step is the extraction of the resonance parameters, depicted in Figure 3.1 *(Right)*. By deconvoluting the normalized scan curve with known beam profiles, the true intrinsic parameters of the resonance (its

mass M_R and width Γ_R) are revealed. This allows $\overline{P}$ANDA to measure widths of very narrow states (e.g., < 100 keV) which are otherwise inaccessible to conventional spectrometers.

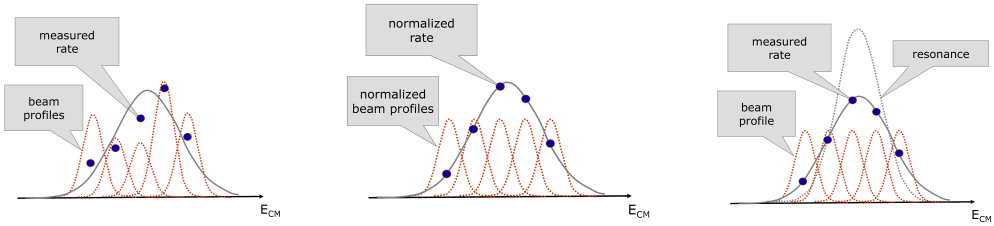


Figure 3.1: The principle of the resonance energy scan method. *(Left)* The raw measured rates (blue dots) at different energy steps appear chaotic due to varying measurement durations (red dashed lines) and luminosity fluctuations. *(Middle)* After normalizing each point by its precise luminosity $\mathcal{L}^{rel}_{int,i}$, the data points align to the scan curve (black solid line), while the beam profiles are normalized to the same entries (red dashed lines). *(Right)* The intrinsic resonance lineshape (black dashed line) is deconvoluted from the scan curve with beam profiles, yielding the precise mass and width.

3.3 Method at $\overline{P}$ANDA

The reference method for the absolute luminosity determination in $\overline{P}$ANDA is based on the measurement of events of antiproton-proton elastic scattering ($\bar{p}p \to \bar{p}p$) at small polar angles. This channel is chosen because its differential cross section is high and, more importantly, the theoretical model of this channel is well-known.

Antiproton-Proton Elastic Scattering

The physics of elastic scattering at low momentum transfer squared (t) is governed by the interference between the electromagnetic (Coulomb) interaction and the strong (Nuclear) interaction. The differential cross section is proportional to the modulus squared of the sum of the scattering amplitudes [120]:

$$\frac{d\sigma}{dt} = \frac{\pi}{k^2}|f_C + f_N|^2, \tag{3.4}$$

where k is the particle momentum in the center-of-mass system, f_C is the Coulomb amplitude, and f_N is the Nuclear amplitude.

The Coulomb Amplitude (f_C) dominates at very small values of $|t|$ (near-forward scattering). It is precisely calculable from Quantum Electrodynamics (QED) and

describes the scattering of two point-like charges. To first order, it follows the Rutherford scattering law:

$$f_C(t) = -\frac{2\alpha_{EM} G^2(t)}{|t|} e^{i\alpha_{EM}\phi}, \tag{3.5}$$

where α_{EM} is the fine-structure constant, $G(t) = (1+\Delta)^{-2}$ is the proton electromagnetic form factor with $\Delta = |t|/0.71\,\mathrm{GeV}^2$, and ϕ is the Coulomb-Nuclear interference phase [121].

The Nuclear Amplitude (f_N) describes the strong interaction which dominates at larger $|t|$. Since Quantum Chromodynamics (QCD) cannot be solved perturbatively in this regime, f_N is typically described using phenomenology based on the optical theorem. A common parameterization is:

$$f_N(t) = \frac{\sigma_{tot}}{4\pi}(\rho + i)e^{-b|t|/2}. \tag{3.6}$$

Here, σ_{tot} is the total hadronic cross-section, ρ is the ratio of the real to the imaginary part of the forward scattering amplitude, and b is the nuclear slope parameter [122].

The Coulomb-Nuclear Interference (CNI) Region lies between the Coulomb-dominated region and the Nuclear-dominated region. In this specific angular range, the Coulomb and Nuclear amplitudes are of comparable magnitude [123].

The differential cross section hence is explicitly decomposed into three components:

$$\frac{d\sigma}{dt} = \underbrace{\left.\frac{d\sigma}{dt}\right|_C}_{\text{Coulomb}} + \underbrace{\left.\frac{d\sigma}{dt}\right|_{Had}}_{\text{Hadronic}} + \underbrace{\left.\frac{d\sigma}{dt}\right|_{Int}}_{\text{Interference}}. \tag{3.7}$$

Each term can be expressed as:

$$\begin{aligned}
\left.\frac{d\sigma}{dt}\right|_C &= \frac{4\pi\alpha_{EM}^2(\hbar c)^2 G^4(t)}{\beta^2 t^2}, \\
\left.\frac{d\sigma}{dt}\right|_{Had} &= \frac{\sigma_{tot}^2(1+\rho^2)}{16\pi(\hbar c)^2} e^{-b|t|}, \\
\left.\frac{d\sigma}{dt}\right|_{Int} &= \frac{\alpha_{EM}\sigma_{tot}}{\beta|t|} G^2(t) e^{-b|t|/2} \left(\rho\cos(\alpha_{EM}\phi) - \sin(\alpha_{EM}\phi)\right),
\end{aligned} \tag{3.8}$$

where β is the beam velocity. The differential cross section of Coulomb part can be calculated precisely, while the Hadronic and Interference parts depend on the parameters σ_{tot}, ρ, and b, which must be determined experimentally.

Figure 3.2 illustrates the differential cross section of $\bar{p}p$ elastic scattering in dependence on the four-momentum transfer $|t|$ according to Eq. 3.8 at a beam momentum of $5.7\,\mathrm{GeV}/c$, highlighting the contributions from the Coulomb, Hadronic, and Interference components. At a small momentum transfer $|t|$, the Coulomb term dominates, while at larger $|t|$ values, the Hadronic term becomes significant. The CNI region lies in between, where both contributions are comparable. The small $|t|$ region is preferred for the luminosity determination due to the precise calculable nature of the Coulomb term independent of experimental parameters like σ_{tot},

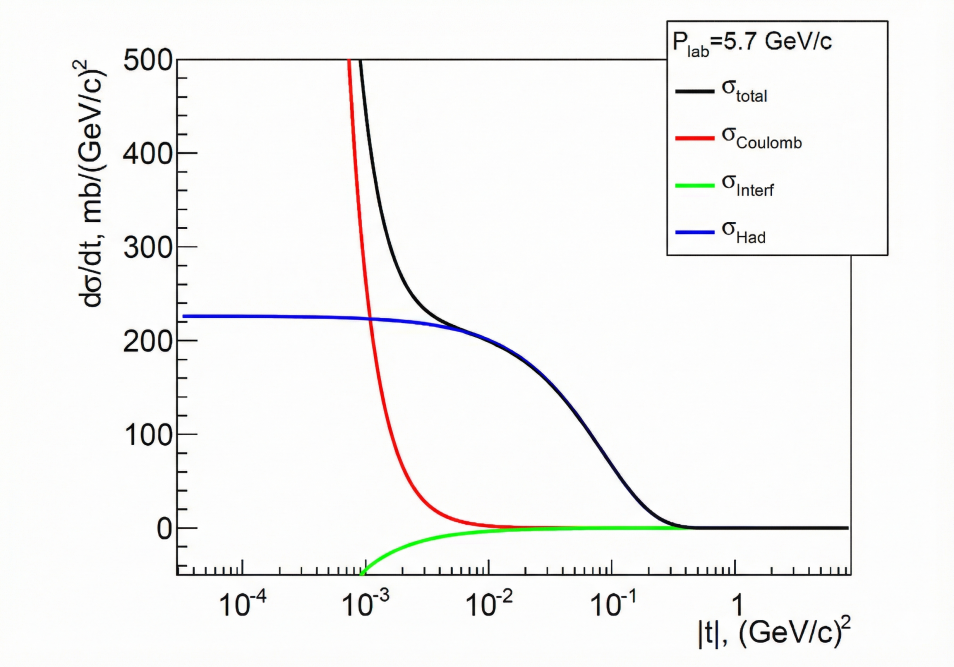


Figure 3.2: Differential cross section of the $\bar{p}p$ elastic scattering in dependence on the four-momentum transfer $|t|$ at a beam momentum of 5.7 GeV/c. The contributions from the Coulomb, Hadronic, and Interference components are shown. Figure adapted from [92].

ρ, and b. Therefore, the LMD is located far downstream ($z \approx 11\,\mathrm{m}$) to capture the scattered antiprotons. The recoil protons are not used, because their energy is too low to leave the beam pipe. The high momentum transfer protons are able to reach the LMD, but the Coulomb term is negligible in this region. Therefore, only the scattered antiprotons are detected by the LMD for the luminosity determination.

For the elastic scattering at a fixed target experiment, t is directly related to the scattering angle θ_{cms}:

$$|t| = -2p_{cms}^2(1 - \cos\theta_{cms}), \tag{3.9}$$

where p_{cms} is the momentum of the particles in the center-of-mass system. Boosted from the laboratory system, the scattering angle θ_{cms} can be expressed with the laboratory angle θ_{lab} as:

$$\tan\theta_{cms} = \frac{\sin\theta_{lab}}{\gamma\left(\cos\theta_{lab} - \beta\frac{E_{lab}}{p_{lab}}\right)}, \tag{3.10}$$

where γ is the Lorentz factor, β is the velocity of the beam divided by the speed of light, and E_{lab} and p_{lab} are the energy and momentum of the antiproton in the laboratory system, respectively.

The LMD detects antiprotons scattered at very small angles ($3 - 9\,\mathrm{mrad}$). Using Eqs. 3.9 and 3.10, the corresponding momentum transfer $|t|$ can be calculated, as shown in Figure 3.3. Unfortunately, $|t|$ is not always at very small values where the Coulomb term dominates. At the beam momentum of 3.5 GeV/c, the hadronic

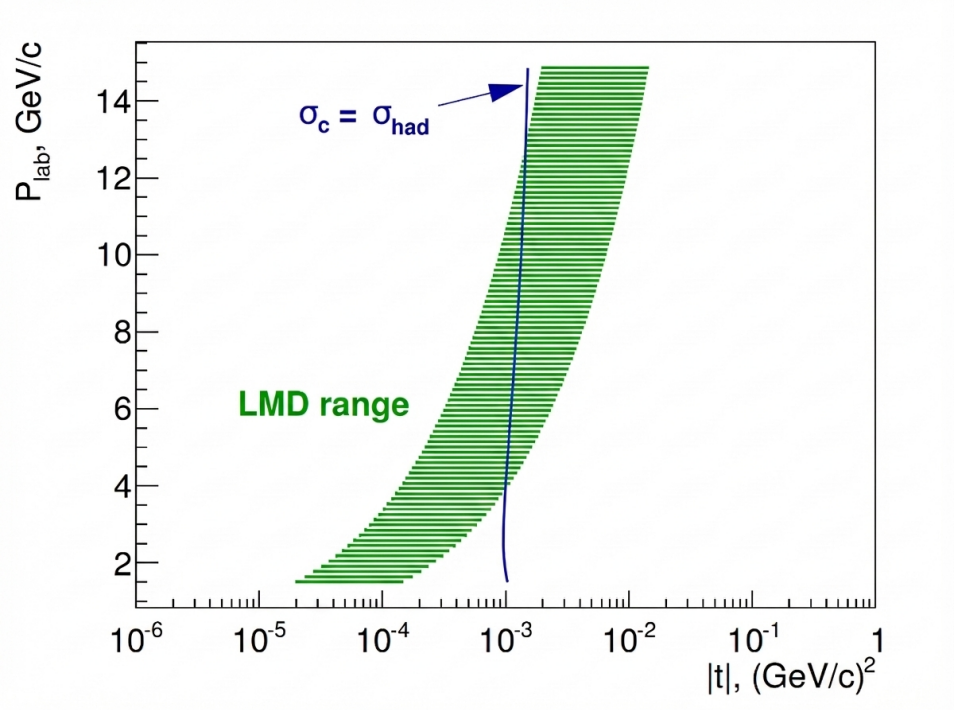


Figure 3.3: Momentum transfer $|t|$ range covered by the LMD ($\theta_{lab} : 3 - 9\,\text{mrad}$) for the momentum range covered by $\overline{\text{P}}$ANDA. The blue line indicates the $|t|$ where the Coulomb term equals to the Hadronic term. Figure adapted from [108].

term begins to take great effects, while at $12\,\text{GeV}/c$, the hadronic term dominates. Therefore, the full expression of Eq. 3.8 must be used to fit the measured differential cross section to extract the luminosity.

The Dual Parton Model (DPM)

To overcome the severe limitations of simple analytical parameterizations, the Dual Parton Model (DPM) is employed as a robust phenomenological framework to describe high-energy $\bar{p}p$ elastic scattering [124, 125]. For the specific purpose of luminosity determination at $\overline{\text{P}}$ANDA, the primary practical advantage of the DPM lies in its ability to accurately model the hadronic amplitude across a wide range of momentum transfers ($|t|$) without relying on unstable free parameters. Simple exponential parameterizations of the hadronic amplitude (e.g., $e^{-b|t|}$) via Eq. 3.6 are valid only at small momentum transfers [126, 127]. They completely fail to describe the complex diffractive structures, such as pronounced dips and secondary maxima, that inherently appear at larger $|t|$ values in $\bar{p}p$ scattering. Furthermore, attempting to extract the luminosity by treating fundamental variables like σ_{tot}, b, and ρ as independent free parameters for every single energy scan point can easily lead to unphysical mathematical fluctuations in the fit results. To overcome these experimental and phenomenological limitations, a more complicated DPM-based model is

employed [128]:

$$\left.\frac{d\sigma}{dt}\right|_{Had} = A_1 \cdot [e^{t/2t_1} - A_2 \cdot e^{t/2t_2}]^2 + A_3 \cdot e^{t/t_2}, \tag{3.11}$$

where A_1, A_2, A_3, t_1, and t_2 are parameters determined by fitting existing experimental data on $\bar{p}p$ elastic scattering over a wide range of beam momenta. Here t_1 is fixed at value of 0.0899 for all momenta, while all other parameters are fitted via smooth functions of the beam momentum p_{lab}:

$$x = a + c \cdot e^{-p_{lab}/d}, \tag{3.12}$$

where x represents the DPM parameters (A_1, A_2, A_3, or t_2), and a, c, and d are constants determined from the fit. For beam momenta in the range of $2 \leq p_{lab} \leq 16\,\mathrm{GeV}/c$, the fitted parameters are listed in Table 3.1. This model captures the

Table 3.1: Fitted parameters for the DPM model in Eq. 3.12.

Parameter	a	c	d
A_1	115.0	650.0	4.08
A_2	0.0687	0.307	2.367
A_3	0.8372	39.53	0.765
t_2	-2.979	3.353	0.670

essential features of the hadronic amplitude, including the diffractive structures, and provides a more accurate description of the differential cross section across the relevant $|t|$ range in $\overline{\mathrm{P}}$ANDA LMD acceptance. Then the hadronic amplitude derived from the DPM model is used to form the full interference model with the Coulomb amplitude. By using this robust description, $\overline{\mathrm{P}}$ANDA can reliably determine the luminosity even in large $|t|$ regions, where the simple exponential ansatz breaks down, ensuring the stability required for high-precision resonance scans. Figure 3.4 shows the Coulomb, Hadronic (from DPM), and Interference components of the differential cross section at four different beam momenta, where the accurate magnetic field maps exists in $\overline{\mathrm{P}}$ANDA simulation software. These differential cross sections will then be used for the luminosity determination in the following sections.

The DPM generator is also used to produce the simulated events for comparison with events measured with the future LMD. These events are tracked through the detector and the information is provided in the same style as the real experiment would produce.

3.4 Track Reconstruction and Event Preparation

The raw data acquired by the Luminosity Detector consists of digitized pixel hits distributed across four detector planes. To perform a precision luminosity analysis, these discrete electronic signals must be transformed into physical particle trajectories and, crucially, propagated back to the interaction point (IP) through the

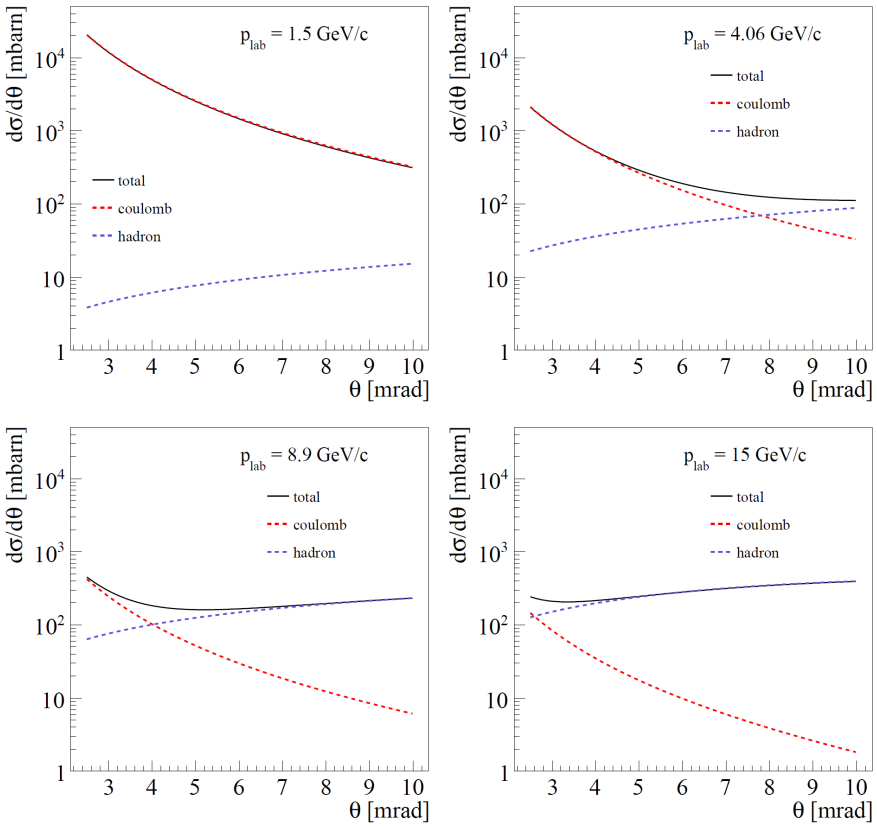


Figure 3.4: Differential cross section of $\bar{p}p$ elastic scattering in dependence on the scattering angle θ_{lab} in the laboratory rest frame at beam momenta of 1.5 GeV/c, 4.06 GeV/c, 8.9 GeV/c, and 15.0 GeV/c. The contributions from the Coulomb, Hadronic (from DPM), and Interference components are shown. Figure adapted from [7].

complex magnetic field to determine the scattering angles. The reconstruction chain is fully implemented within the **PandaRoot** framework as discussed before. The procedure consists of four sequential stages: Digitization and Hit Reconstruction, Track Finding, Track Fitting, Track Filtering and Back-Propagation, as illustrated in Figure 3.5.

Digitization and Hit Reconstruction

The first stage processes the data from the HVMAPS, which simulates the detector response to the traversing particles (in Monte Carlo) or processes the raw readout data (in Experiment). For the HVMAPS of the LMD, the SDS package handles the digitization and hit reconstruction.

Pixel Firing A particle traversing a sensor pixel generates a charge signal. The digitization step converts these into digital information (*"Digis"*) by applying a

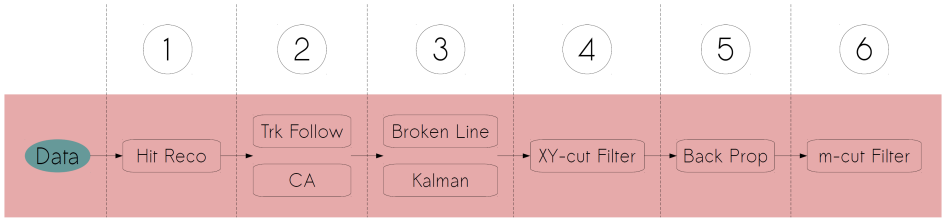


Figure 3.5: Schematic overview of the track reconstruction chain for the Luminosity Detector in $\overline{\text{P}}$ANDA. Figure adapted from [7].

threshold to simulate the sensor readout electronics. If this charge exceeds the threshold, the pixel is marked as *hit*.

Clustering A single traversing antiproton often deposits charge in multiple adjacent pixels due to charge sharing. A dedicated clustering algorithm groups these contiguous fired pixels into a *Cluster*.

Position Estimation The spatial position of the hit $(x_{hit}, y_{hit}, z_{hit})$ is calculated as the binary center (single cluster) or the median (multiple clusters) of the cluster. The spatial resolution is $\sigma_{spatial} = p/\sqrt{12} = 23\,\mu$m with pixel size $p = 80\,\mu$m of the HVMAPS. Since we can't use charge sharing for clustering in HVMAPS, center of gravity method is not applicable for multiple clusters to achieve better resolution. The simulation studies show that the fraction of multiple clusters is less than 40% due to the thin sensor thickness of $50\,\mu$m. Therefore, the overall spatial resolution is not significantly improved even if center of gravity method for multiple clusters would be used.

Track Finding Algorithms

For the LMD, track finding serves as the pattern recognition step to associate individual hits into trajectory candidates. The **PandaRoot** framework provides two algorithms for this task: **Track Following**, which represents a sequential local method, and **Cellular Automaton**, which acts as a global parallel approach.

Track Following Sequential "Track Follower" algorithms extrapolate a seed track segment layer-by-layer, searching for compatible hits in the next detector plane. While intuitive and effective in low-occupancy environments, this approach can suffer from combinatorial ambiguities in high-rate scenarios.

Cellular Automaton (CA) Unlike sequential followers, the CA creates short hit-to-hit segments (*cells*) in parallel across all layers and evolves them into continuous tracks based on neighborhood rules. This global approach is inherently robust against local inefficiencies and is well-suited for the planar geometry of the LMD [129].

For the LMD analysis, the **Cellular Automaton** algorithm is the preferred choice [92]. The implementation used in this work creates track candidates requiring hits in at least 3 out of 4 stations to ensure high efficiency while suppressing ghost tracks.

Track Fitting with GenFit

Once track candidates are found, their kinematic parameters must be estimated. The reconstruction utilizes the **GenFit** toolkit [130], which supports multiple fitting algorithms, most notably the **General Broken Lines (GBL)** [131, 132] fit and the **Kalman Filter** [133, 134].

General Broken Lines (GBL) The GBL algorithm constructs a trajectory from a sequence of thin scatterers. It is particularly powerful for detector alignment as it provides explicit access to the residuals of all measurements simultaneously.

Kalman Filter The Kalman filter is a recursive estimator that updates the track state vector $\vec{q} = (x, y, z, p_x, p_y, p_z)$ at each measurement plane. It is the preferred algorithm for track reconstruction due to its superior handling of complex material effects. Specifically, it incorporates the mean energy loss described by the Bethe-Bloch formula when particles traverse the detector sensors and support structures. Furthermore, the filter elegantly accounts for Multiple Coulomb Scattering (MCS) by treating it as stochastic process noise. This approach appropriately inflates the track's covariance matrix, accurately reflecting the accumulated spatial uncertainties induced by the detector's material budget.

For the determination of the luminosity, the **Kalman Filter** is selected as the primary fitting algorithm. This ensures that the final error estimates at the first detector layer are statistically rigorous, which is a prerequisite for the accurate back-propagation to the interaction point.

Back-Propagation to the Interaction Point

For luminosity determination, the physics observable is the scattering angle θ at the Interaction Point ($z = 0$), not at the detector ($z \approx 11\,\mathrm{m}$). Between the IP and the LMD, the antiprotons traverse the stray field of the solenoid and the strong dipole magnet field, which deflects trajectories horizontally. To recover the true scattering angle, the fitted tracks must be propagated backwards using the **GEANE** track propagator [135, 136]. This step involves a subtle but critical assumption regarding the particle momentum.

Momentum Assumption The LMD, being a forward spectrometer without magnetic field in the tracking volume, cannot measure the momentum of the particles. However, for elastic scattering, the momentum loss of the forward scattered antiproton is negligible ($\Delta p \approx 100\,\mathrm{keV}/c$ at $1.5\,\mathrm{GeV}/c$ beam momentum and $\Delta p \approx 10\,\mathrm{MeV}/c$ at $15\,\mathrm{GeV}/c$). Therefore, the track momentum for the back-propagation is fixed to the nominal beam momentum p_{beam}.

Track Transport GEANE numerically integrates the equation of motion through the detailed magnetic field map $\vec{B}(\vec{r})$, as illustrated in Figure 3.6. This transforms the track state $\vec{q}_{LMD}$ to the vertex state $\vec{q}_{IP}$.

The reconstructed scattering angle θ_{rec} is then defined as the angle between the back-propagated momentum vector $\vec{p}_{IP}$ and the incoming beam axis (typically the z-axis, or the tilted beam axis as discussed in Section 3.6).

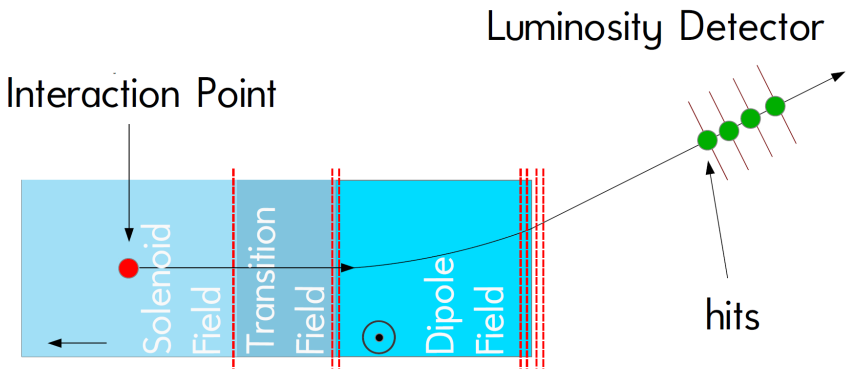


Figure 3.6: Back-propagation of tracks from the LMD to the Interaction Point through the dipole, transition and solenoid fields using the GEANE track propagator. Figure adapted from [92].

Track Filtering and Selection

During the whole reconstruction process, several additional filtering stages are applied to ensure data quality:

- **Cleaning:** "Split Tracks" (a single particle reconstructed as two tracks) and "Ghost Tracks" are removed based on shared hit criteria and χ^2 quality cuts at the track finding stage.
- **Box Cut:** Secondary particles produced in the beam pipe material or vacuum box frames typically originate from vertices far from the IP. By applying cuts on the reconstructed azimuthal (ϕ) and polar (θ) angles, a significant portion of this background is suppressed while maintaining high efficiency for the elastically scattered antiprotons. This filter is optimized before back-propagation and labeled as "XY-Cut Filter" in Figure 3.5.

- **m-cut:** After back-propagation, an additional momentum consistency cut is applied for further background suppression.

Figure 3.5 shows the two-dimensional distribution of DPM generated events. The reconstructed scattering angle θ_{rec} at the IP in x- and y-directions is shown as θ_x and θ_y after the full reconstruction chain.

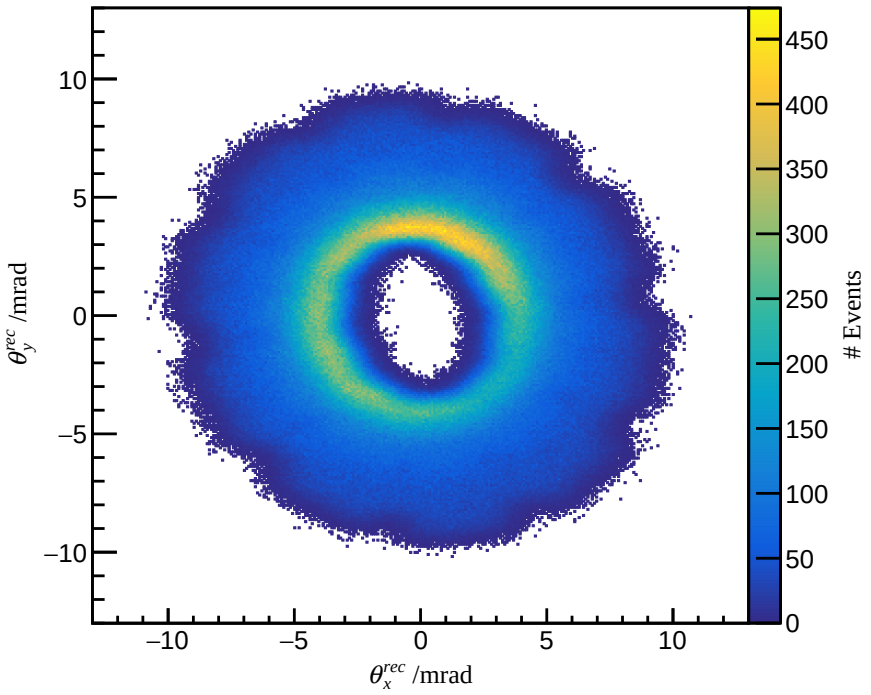


Figure 3.7: Distribution of the reconstructed track direction of simulated DPM events at the IP at 4.06 GeV/c. The angular distributions are plotted versus the scattering angle θ_{rec}, separated into its x- and y-components (θ_x and θ_y).

3.5 The Luminosity Fit Framework

The extraction of the time-integrated luminosity $\mathcal{L}_{int}$ constitutes a parameter estimation problem, where a theoretical model $\frac{d\sigma(\theta)}{d\theta}$ is optimized to match the experimental angular distribution $\frac{dN(\theta)}{d\theta}$, as illustrated in Eq. 3.13, where the theoretical model is corrected with detector acceptance $\epsilon(\theta)$ and resolution $R(\theta)$. To address the specific challenges of the $\overline{\text{P}}$ANDA environment—such as extended target profiles, beam divergence, and complex detector acceptance—a dedicated software framework, *LuminosityFit*, was developed [7].

$$\frac{dN(\theta)}{d\theta} = \mathcal{L}_{int} \cdot \left[\frac{d\sigma(\theta)}{d\theta} \cdot \epsilon(\theta)\right] \otimes R(\theta) \tag{3.13}$$

Designed with a strict Object-Oriented architecture, the framework prioritizes modularity and flexibility. It decouples the physics implementation from the numerical optimization engine and the data storage backend. This separation of concerns allows for the seamless interchange of interaction models (e.g., different cross section parametrizations) or minimization algorithms without modifying the core analysis logic.

Architectural Design

The framework is structured around a central orchestration layer, utilizing the *Facade Design Pattern* to manage the interaction between four abstract components:

- **Data Representation Layer:** Experimental data and Monte Carlo simulations after the full track reconstruction procedure in Section 3.4 are encapsulated into a unified histogram interface. This abstraction layer handles the binning of the angular phase space in the projected angular coordinates (θ_x and θ_y, defined as $\theta_x \approx p_x/p_z$ and $\theta_y \approx p_y/p_z$ for small angles) and provides coherent access to bin contents and statistical errors. By isolating the analysis from the underlying file format (e.g., ROOT Trees), the fit performance is optimized for high-statistics datasets.
- **Abstract Model Interface:** The physics of the scattering process is defined through a generalized model interface. Concrete implementations (e.g., the DPM) provide the probability density functions (PDF). Crucially, this layer is responsible for the "Forward-Folding" process: it accounts for the detector acceptance, which incorporates the beam shift effects, and convolves the theoretical cross section (as shown in Figure 3.4) with the detector resolution and beam smearing effects during the fit.
- **Estimator Strategy:** The quantification of the agreement between the model and the data is handled by an interchangeable estimator component. While standard χ^2 metrics are available for diagnostics, the primary analysis utilizes a **Binned Log-Likelihood** estimator. This approach is statistically required to correctly treat low-statistics bins in the high-$|t|$ tail of the distribution.
- **Minimization Backend:** The numerical optimization of the likelihood function is delegated to an external engine wrapper. This design allows the framework to utilize robust third-party libraries (such as *Minuit2*) to find the global minimum in the multi-dimensional parameter space efficiently.

The fit determines the optimal set of physics parameters $\vec{\alpha}$ (including $\mathcal{L}_{int}$) by minimizing the negative log-likelihood function:

$$-\ln L(\vec{\alpha}) = \sum_{i \in \text{bins}} \left[\nu_i(\vec{\alpha}) - n_i \ln \nu_i(\vec{\alpha}) + \ln(n_i!)\right], \tag{3.14}$$

where n_i is the observed number of events in angular bin i, and ν_i is the expected number of events predicted by the model. A key feature of the software is its handling of detector effects. Instead of unfolding the experimental data—a process

known to introduce correlations and instabilities—the framework applies a forward-folding technique. The expected number of events ν_i in a generic angular bin i is constructed as:

$$\nu_i(\vec{\alpha}) = \mathcal{L}_{int} \cdot \epsilon_i \cdot \int_{\Delta\Omega_i} \left[\frac{d\sigma}{d\Omega}(\vec{\alpha}, \Omega') \otimes R(\Omega, \Omega') \right] d\Omega + B_i \tag{3.15}$$

Here, ϵ_i represents the geometrical acceptance map, B_i is the background term, and R is the resolution function. The symbol $\otimes$ denotes the numerical convolution of the theoretical cross section with the detector resolution. By modeling these effects explicitly, the software ensures that the extracted parameters are free from detector-induced biases.

IP Alignment and Coordinate Transformation

A critical pre-processing step implemented in the framework is the alignment of the coordinate system. The standard laboratory frame assumes the interaction point is at $(0, 0, 0)$. However, real beam operations imply a transverse offset (x_0, y_0).

The framework utilizes the full 3D tracking information available after the back-propagation step to determine this offset geometrically. For each back-propagated track, the Point of Closest Approach (POCA) relative to the nominal coordinate origin $(0, 0, 0)$ (or given point $(x_i, y_i, 0)$ taken from other detectors) is calculated. This results in a distribution of interaction vertices in the transverse plane. The arithmetic mean of these POCA coordinates is identified as the measurement of the mean interaction point $(x_0, y_0, 0)$. Before the luminosity fit, a global coordinate transformation is applied to all track data:

$$\vec{r}' = \vec{r}_{lab} - \vec{r}_{IP}. \tag{3.16}$$

This virtually shifts the detector system such that the interaction point coincides with the origin, restoring the azimuthal symmetry required for the 2D luminosity fit model.

Detector Response Modeling

The raw angular distribution measured by the LMD differs from the theoretical cross-section due to two distinct instrumental effects: the limited geometrical acceptance and the finite angular resolution. In the *LuminosityFit* framework, these effects are decoupled and treated sequentially using a forward-folding approach.

The **Acceptance** ϵ represents the probability that an elastic scattering event is successfully detected and reconstructed. Due to the planar geometry of the HV-MAPS, the acceptance structures (such as sensor edges, overlaps, and module gaps) are most naturally described in the projected angular coordinates (θ_x, θ_y).

The acceptance map $\epsilon(\theta_x, \theta_y)$ is generated via extensive Monte Carlo simulations. A sample of N_{gen} single track events is generated utilizing a Box Generator and processed through the full Geant4 detector simulation and tracking chain to obtain the number of reconstructed tracks (N_{rec}) independence on the coordinates

(θ_x, θ_y). The acceptance is then defined as the bin-wise ratio in the 2D projection plane:

$$\epsilon_{ij}(\theta_{x,i}, \theta_{y,j}) = \frac{N_{rec}(\theta_{x,i}, \theta_{y,j})}{N_{gen}(\theta_{x,i}, \theta_{y,j})} \tag{3.17}$$

This 2D map captures the rectangular footprint of the sensors and the beam pipe aperture accurately as shown in Figure 3.8. In the fit, for any generated physics event with angle θ, the corresponding (θ_x, θ_y) coordinates are calculated to look up the acceptance factor $\epsilon(\theta_x, \theta_y)$, which is then applied as a multiplicative weight to the event rate.

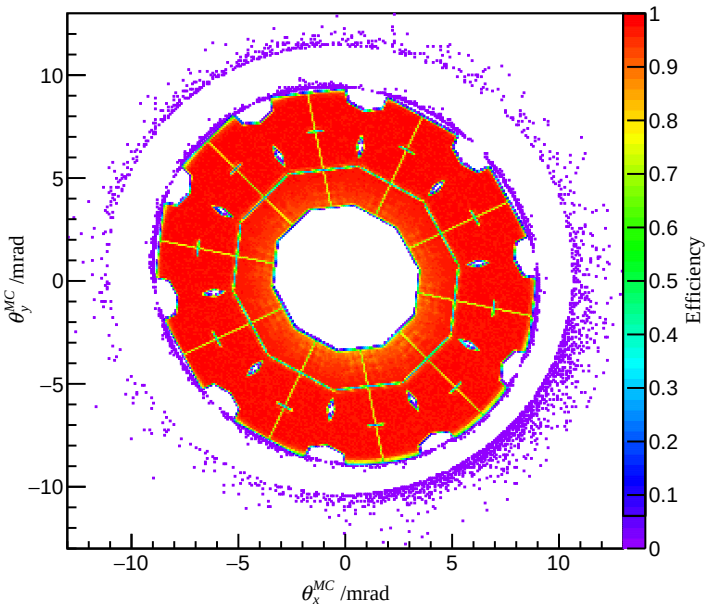


Figure 3.8: Example of a 2D acceptance map $\epsilon(\theta_x, \theta_y)$ derived from Monte Carlo simulations at $p_{beam} = 4.06\,\text{GeV}/c$. Regions of reduced efficiency correspond to sensor edges and beam pipe shadowing.

The **Angular Resolution** describes the statistical deviation of the reconstructed angle θ_{rec} from the true physical angle θ_{true}. This smearing is primarily dominated by Multiple Coulomb Scattering (MCS) in the detector material. To model this effect, the distribution of residuals $\Delta\theta = \theta_{rec} - \theta_{true}$ is analyzed. As shown in previous studies [7], a single Gaussian is often insufficient to describe the resolution function $R(\Delta\theta)$, as it fails to capture the tails arising from non-Gaussian scattering processes. Therefore, a Double Gaussian parameterization is employed:

$$R(\Delta\theta; \sigma_1, \sigma_2, \eta) = \eta \cdot G(\Delta\theta; 0, \sigma_1) + (1 - \eta) \cdot G(\Delta\theta; 0, \sigma_2), \tag{3.18}$$

where G is a normalized Gaussian, σ_1 represents the width of the core (signal), σ_2 the width of the tails, and η the relative fraction. Figure 3.9 illustrates the resolution as a function of the two beam momenta, whose σ_1 and σ_2 are shown. As expected, the MCS contribution is more prominent at lower momenta.

In the fit, the theoretical cross-section is convolved with this resolution function:

$$\frac{d\sigma^{smeared}}{d\Omega}(\theta) = \int \frac{d\sigma}{d\Omega}(\theta') \cdot R(\theta - \theta')\, d\theta' \tag{3.19}$$

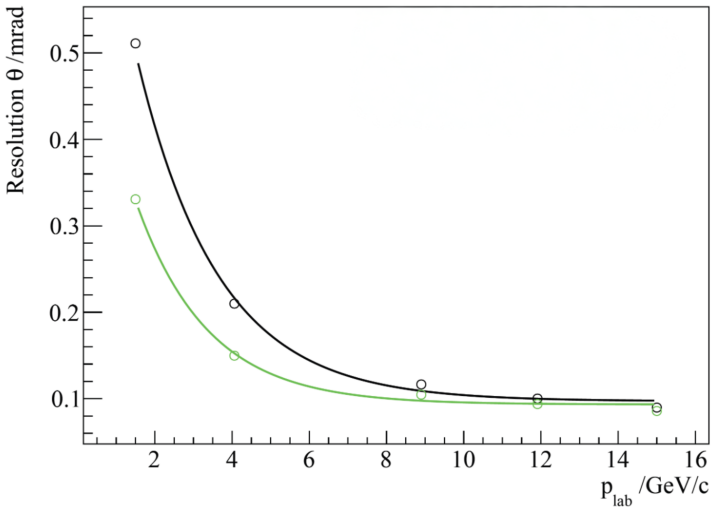


Figure 3.9: Angular resolution parameters σ_1 (green curve) and σ_2 (black curve) as a function of beam momentum, derived from Monte Carlo simulations. The increasing trend at lower momenta reflects the enhanced Multiple Coulomb Scattering effects. Figure adapted from [7].

The framework explicitly separates the geometrical masking (in θ_x, θ_y) from the physics smearing (in θ), ensuring a rigorous description of the detector response.

3.6 Effects of Beam Properties

In the ideal mathematical description of elastic scattering, the interaction occurs at a singular point $(0, 0, 0)$ and the antiproton beam vector is perfectly aligned with the z-axis. However, the realistic $\overline{\text{P}}$ANDA beam conditions at the HESR introduce geometric and kinematic deviations. To achieve the target precision for the luminosity determination, these effects must be rigorously modeled to avoid systematic biases. They can be categorized into deterministic mean shifts, shift and tilt as illustrated in Figure 3.10, and stochastic broadening (Divergence). The **IP Offset**, shown in green, shifts the primary vertex position away from the nominal detector center, thereby breaking the azimuthal symmetry of the tracks. Furthermore, the **Beam Tilt**, shown in blue, causes a systematic angular deviation that strongly depends on the azimuthal angle ϕ.

IP Offset and Distribution

The interaction zone is defined by the spatial overlap of the antiproton beam and the target stream. This creates a distributed source rather than a point source. The framework treats the mean position (Offset) and the profile width of this distribution differently based on the detector's sensitivity. A transverse displacement (x_0, y_0) of the beam breaks the azimuthal symmetry of the detector acceptance. To determine this offset precisely, the **Back-Propagation** procedure is exploited as described

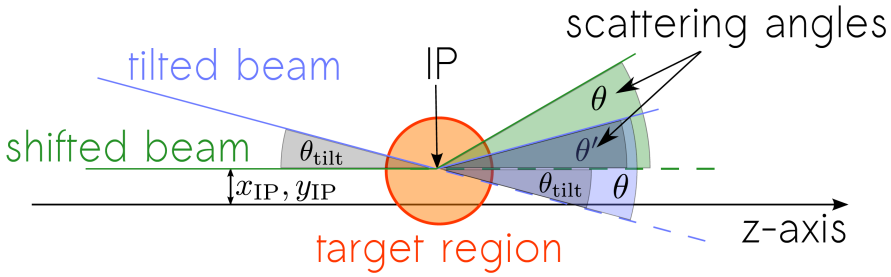


Figure 3.10: Schematic illustration of IP offset and beam tilt. Figure adapted from [108].

in Section 3.4. After extrapolating the reconstructed tracks through the magnetic field back to the target region, the geometrical origin of each track is determined by calculating its **Point of Closest Approach (POCA)** to the nominal coordinate origin $(0, 0, 0)$. The spatial distribution of these POCA points reflects the beam spot profile smeared by the resolution from reconstruction precision. Nevertheless, the **Mean** of this distribution provides the mean interaction vertex $(x_0, y_0, 0)$ with high precision ($\mathcal{O}(10\,\mu\text{m})$). This measured position is then used to define the "Beam Reference Frame". All reconstructed quantities are calculated relative to this physically determined center, effectively removing the systematic bias induced by the beam offset.

The transverse beam width (typically $\sigma_{x,y} \approx 1\,\text{mm}$) introduces a spatial smearing. However, due to the long lever arm of the LMD setup, the resulting angular uncertainty is small:

$$\sigma_{\theta,beam} \approx \frac{\sigma_{x,y}}{z_{LMD}} \approx \frac{1\,\text{mm}}{11000\,\text{mm}} \approx 0.09\,\text{mrad} \tag{3.20}$$

This contribution is significantly smaller than the resolution limit imposed by Multiple Coulomb Scattering in the MVD. Consequently, the LMD lacks the sensitivity to precisely disentangle the beam width from the detector resolution in a combined fit. Therefore, the beam width parameters (σ_x, σ_y) are fixed to the values provided by the HESR beam group ($\sigma_{x,y} = 0.8\,\text{mm}$) instead of using the POCA distribution. These external values are explicitly incorporated into the generation of the Acceptance Map $\epsilon(\theta_x, \theta_y)$ to correctly account for edge effects in the sensor acceptance.

The distribution along the beam axis (z) is a combination of the flat-distributed target and the extended residual gas background as discussed later.

Beam Tilt

The mean direction of the antiproton beam may not be perfectly parallel to the mechanical axis of the LMD. A **Beam Tilt** is defined by two small angles: t_x (tilt in the $x - z$ plane) and t_y (tilt in the $y - z$ plane). A non-zero tilt introduces a systematic shift in the scattering angles, the relationship between the true scattering angle θ^* and the angle θ_{lab} measured in the detector frame is approximated

by:

$$\theta_{lab} \approx \theta^* + t_x \cos\phi + t_y \sin\phi. \tag{3.21}$$

Neglecting this tilt induces a spurious ϕ-dependence in the data. The measured polar angle θ_{lab} would be systematically shifted depending on ϕ, distorting the t-distribution and ruining the fit quality. The *LuminosityFit* framework treats the Beam Tilt as a coordinate transformation. The reconstructed momentum vectors of all tracks are rotated from the "Laboratory Frame" into the "Beam Reference Frame" using a rotation matrix $R(t_x, t_y)$, effectively aligning the effective beam axis with the z-axis for the analysis, which define the variables θ_x and θ_y:

$$\theta_x = \theta_x^{lab} - t_x, \quad \theta_y = \theta_y^{lab} - t_y. \tag{3.22}$$

Beam Divergence

In contrast to the coherent tilt, the **Beam Divergence** represents the intrinsic statistical spread of the particle directions within the beam bunch. It follows a Gaussian distribution with width σ_{div}. Since the initial direction of a specific antiproton is unknown, this effect cannot be corrected event-by-event. Instead, it acts as an irreducible angular smearing.

In the luminosity fit model, the beam divergence is treated by convolution. It adds in quadrature to the detector's intrinsic resolution R_{det}. The total effective resolution kernel R_{eff} is given by:

$$R_{eff}(\theta) = R_{det}(\theta) \otimes G(\theta; 0, \sigma_{div}) \tag{3.23}$$

where G denotes the normalized Gaussian distribution describing the beam divergence:

$$G(\theta; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\theta-\mu)^2}{2\sigma^2}\right) \tag{3.24}$$

Here, the mean $\mu = 0$ reflects the fact that the systematic mean direction (Tilt) has already been corrected, leaving only the stochastic spread $\sigma = \sigma_{div}$ to be convolved. While the transverse width is handled in the *Acceptance* (multiplicative term), the divergence is handled in the *Resolution* (convolution kernel), ensuring a mathematically consistent treatment of all beam properties.

3.7 Integrated Fit Workflow

To rigorously account for the beam properties discussed above, the luminosity determination follows a sequential five-step workflow as illustrated in Figure 3.11. This procedure ensures that the physics model used for the fit matches the specific experimental conditions of the data taking period.

1. **IP Determination:** Using the experimental data taken by the LMD, the full track reconstruction is performed to determine the interaction vertex.

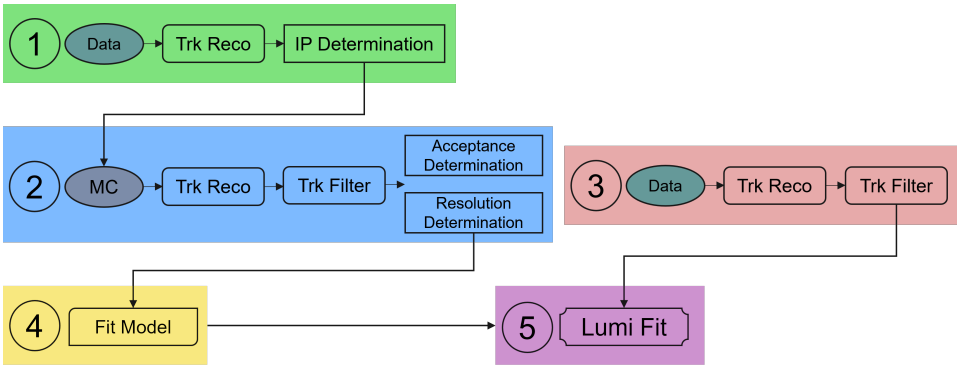


Figure 3.11: Schematic overview of the integrated luminosity fit workflow incorporating beam property corrections.

2. **Condition-Specific MC Production:** The determined IP parameters (mean offset and beam width from HESR) serve as input for a dedicated Monte Carlo simulation. These simulations with single track events generate the **Acceptance Map** $\epsilon(\theta_x, \theta_y)$ and determine the **Resolution Function** $R_{det}(\theta_x, \theta_y)$. This step is crucial: by customizing the MC samples to the observed beam parameters, the geometrical acceptance corrections implicitly include the effects of the beam offset and width.

3. **Track Filtering and Preparation:** The tracks reconstructed from the experimental data undergo a strict cleaning procedure to reject ghost tracks and secondary particles. The surviving data is binned in (θ_x, θ_y). The angular distributions is shown in Figure 3.12 *(left)*.

4. **Fit Model Construction:** Simultaneously, the theoretical differential cross-section $\frac{d\sigma}{d\Omega}$ is calculated using the DPM model for the relevant beam momentum. The fit model is constructed by convolving the theory with the resolution and multiplying by the acceptance. Figure 3.12 *(right)* shows the final 2D DPM model.

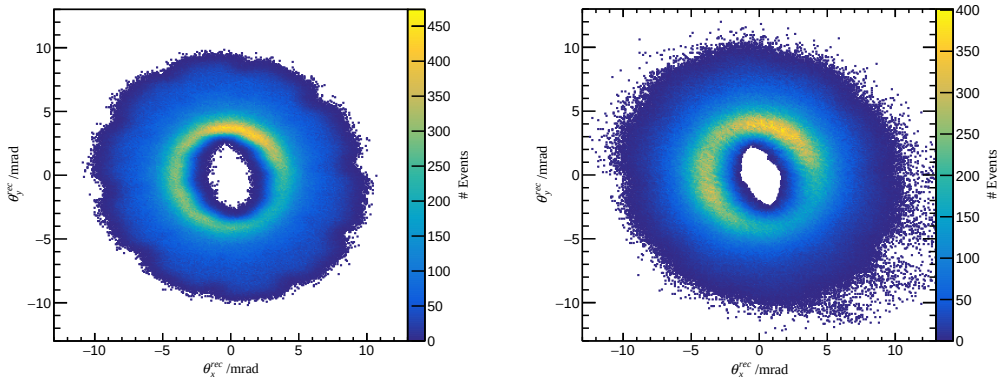


Figure 3.12: Experimental angular distribution after track selection *(left)* and 2D DPM fit model including acceptance and resolution effects *(right)* at a beam momentum of 4.06 GeV/c for Luminosity Fit.

5. **Global Luminosity Fit:** In this final step, the minimizer then optimizes the parameters to describe the data to fit the model. The Beam Tilt and Beam Divergence are typically treated as floating nuisance parameters alongside the Luminosity. This allows the fit to find the global minimum that best accommodates the residual angular distortions, yielding a precise and unbiased value for $\mathcal{L}_{int}$.

The final fit results for the example dataset at 4.06 GeV/c (shown in Figure 3.12) are listed in Table 3.2. The fitted IP offsets and beam tilts are in good agreement with the generated values, while the beam divergence parameters exhibit larger uncertainties due to their subtle influence on the angular distribution, but they remain consistent with the input values within errors. The extracted luminosity achieves a relative precision of $0.11 \pm 0.07\%$, demonstrating the effectiveness of the integrated fit workflow in accounting for beam property effects.

Table 3.2: Fit results for the example dataset at a beam momentum of 4.06 GeV/c.

Parameter	Generated Value	Fitted Value
IP_x [mm]	1.0	0.93 ± 0.002
IP_y [mm]	1.0	1.00 ± 0.002
tilt_x [mrad]	100	104.9 ± 1.3
tilt_y [mrad]	100	94.6 ± 1.4
div_x [mrad]	100	78.8 ± 26.4
div_y [mrad]	100	62.1 ± 35.7
$\Delta\mathcal{L}/\mathcal{L}$ [%]	0.11 ± 0.07	

Chapter 4

Luminosity Determination with Restgas Background

4.1 Realistic Target Distribution

While the beam properties discussed in the previous chapter primarily affect the angular resolution, the properties of the target introduce a spatial complexity that fundamentally challenges the point-like interaction assumption. In the $\overline{\text{P}}$ANDA fixed-target setup, the interaction region is not confined to a dimensionless point but follows the density distribution of the gas molecules along the beam axis.

Origin of the Restgas

The primary interaction region in $\overline{\text{P}}$ANDA is a stream of frozen hydrogen pellets or a cluster jet, intersecting the antiproton beam at the nominal interaction point (IP, $z = 0$). Ideally, the scattering process occurs exclusively within the geometric overlap of the beam and this target stream.

However, the injection of target material into the beam pipe inherently degrades the local vacuum quality. Despite the high-capacity pumping system, a fraction of the hydrogen target evaporates or scatters out of the main stream, creating a cloud of so-called Restgas. This gas accumulates within the scattering chamber and diffuses along the beam pipe until it is removed by the nearest vacuum pumps. Figure 4.1 shows the vacuum pump locations at $\overline{\text{P}}$ANDA close to the interaction point.

Consequently, the antiproton beam interacts not only with the dense target but also with this low-density Restgas. Crucially, these Beam-Restgas interactions ($\bar{p}p \to \bar{p}p$) are kinematically indistinguishable from the desired signal Beam-Target interactions, as they share the exact same final state particles and kinematics. The only spatial

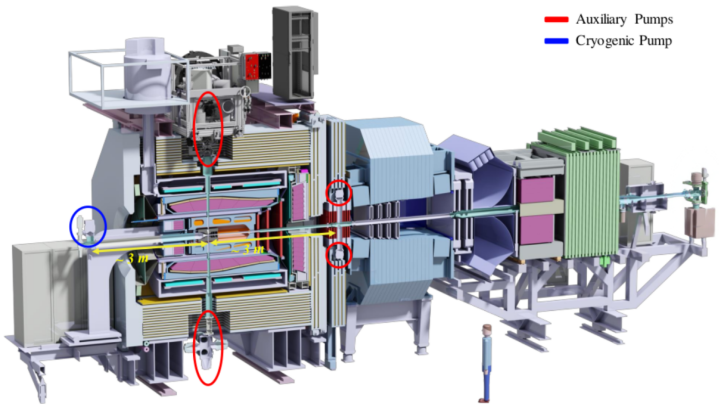


Figure 4.1: Schematic of the vacuum pump locations at $\overline{\text{P}}$ANDA close to the interaction point. Figure adapted from [137].

distinction between the two is the longitudinal position of the reconstructed interaction vertex, z_{vtx}. While the signal events are highly localized at the nominal interaction point ($z \approx 0$), the Restgas background events are continuously distributed along the entire z-axis of the beam pipe.

Spatial Density Distribution

The following study is based on the Cluster Jet Target developed by Münster Group, which is currently the default target option for $\overline{\text{P}}$ANDA. The $\overline{\text{P}}$ANDA Cluster Jet Target generates a directed stream of hydrogen clusters by expanding pre-cooled hydrogen gas through a fine Laval nozzle into the vacuum. The expansion creates a core jet of solidified hydrogen clusters. A system of skimmers and collimators is employed to geometrically define this beam, removing the majority of the uncondensed gas. Ideally, this results in a well-defined target density profile ρ_{jet} with a transverse diameter of few millimeters to centimeters at the interaction point ($z = 0$). Due to the constant flow of the jet, the time-averaged density $\rho_{jet}(z)$ is modeled as a flat distribution in the longitudinal direction, with a width σ_{jet} determined by the jet's geometrical intersection with the beam ($\sigma_{jet} = 2\,\text{cm}$) [138].

However, despite the skimmers, a fraction of the hydrogen gas inevitably enters the scattering chamber. This arises from two mechanisms:

- **Gas Halo:** Imperfect separation of the clusters from the carrier gas leads to a diffuse "curtain" of hydrogen molecules accompanying the main jet.
- **Evaporation and Scattering:** Clusters interacting with the beam pipe edges or evaporating during flight contribute to a continuous gas load.

The shape of the residual gas distribution $\rho_{gas}(z)$ is governed by the dynamic equilibrium between the gas load from the target and the effective pumping speed at the interaction region. Due to the limited conductance of the narrow beam pipe and the placement of the vacuum pumps the gas density is not uniform.

Figure 4.2 shows the simulation of the gas density distribution along the beam axis, detailing both the target itself and the gas load in the surrounding so-called Restgas. It's zoomed in the region around the target to illustrate the shape of the target and Restgas distributions. The Restgas density is maximal within the target stream ($z = \pm 1\,\mathrm{cm}$), where the evaporation occurs, and decreases linearly or exponentially towards the pumping ports, depending on the molecular flow regime.

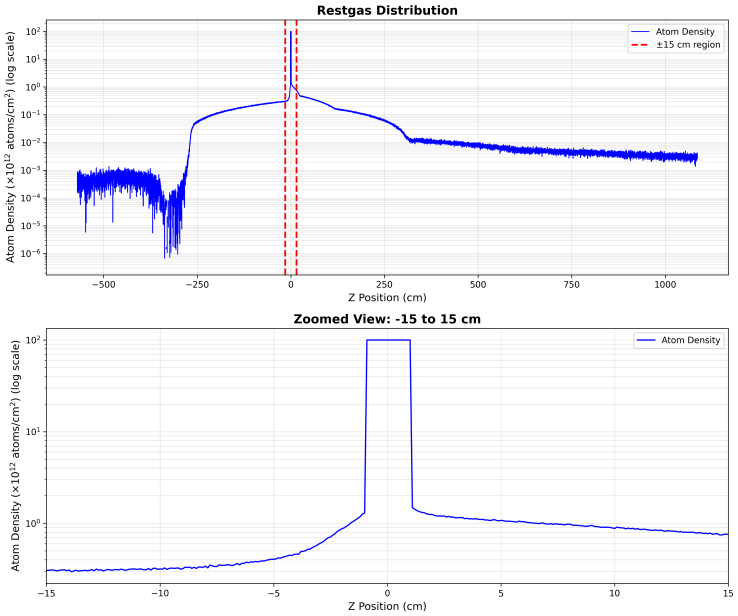


Figure 4.2: Target density profile along the beam axis by molecular flow simulation showing the combination of the Cluster Jet Target and the Restgas distribution. The dip shows the location of cryopump at $\sim 3\,\mathrm{m}$ upstream. Red lines indicate the zoomed region around the target, which is shown in the bottom. Data taken from [139].

4.2 Luminosity Determination with Restgas Distribution

As demonstrated in previous studies [92, 7], the precise track reconstruction and back-propagation algorithms (described in Section 3.4) effectively compensate for the kinematic smearing caused by longitudinal vertex displacements. Consequently, as long as the x and y coordinates of the IP are accurately known, the scattering angle θ is reconstructed correctly regardless of the z coordinate. That means the Restgas events do not distort the angular distribution shape used for luminosity extraction. However, the geometrical acceptance of the Luminosity Detector is highly sensitive to the longitudinal position of the interaction vertex.

Impact of Vertex Position on Acceptance

The LMD system has a fixed aperture defined by the sensor dimensions and the beam pipe constraints. An interaction occurring at a displaced position $z \neq 0$

projects the same scattering angle θ onto a different spatial coordinate on the detector plane compared to an interaction at the nominal IP ($z = 0$). This geometric effect implies that the geometrical acceptance map is a function of the vertex position: $\epsilon = \epsilon(\theta_x, \theta_y, z)$.

To characterize this effect, a systematic simulation study was performed. Elastic scattering events were generated at discrete vertex positions along the beam axis, ranging from $z = -10\,\text{cm}$ to $z = +10\,\text{cm}$ in steps of 2.5 cm. For each z-position, simulations ranging from $-1\,\text{cm}$ to $+1\,\text{cm}$ in steps of 0.1 cm in x and y directions were generated to get the acceptance maps as shown in Figure 4.3. While the abso-

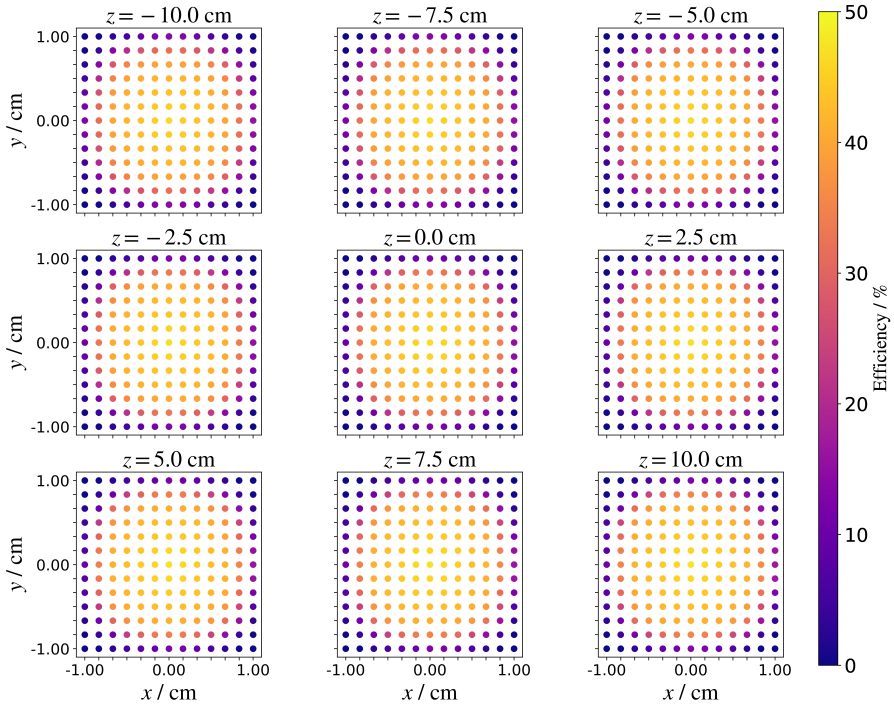


Figure 4.3: Acceptance maps of the LMD for different longitudinal vertex positions z. Each subplot corresponds to a fixed z value.

lute acceptance maps show a similar topology (dominated by the sensor geometry with $\sim$ 40% central efficiency), a differential analysis reveals a systematic gradient. Figure 4.4 compares the acceptance maps at two adjacent z positions (e.g., $z = -10\,\text{cm}$ and $z = -7.5\,\text{cm}$). The study yields that a longitudinal shift of just 2.5 cm results in an efficiency variation of approximately 0.5% between two planes. Although locally small, this gradient accumulates over the full extent of the Restgas distribution.

This confirms that the detector acceptance is not translationally invariant along z. Consequently, using a static map derived at $z = 0$ to also account for the Restgas distribution would introduce a significant systematic bias in the luminosity determination.

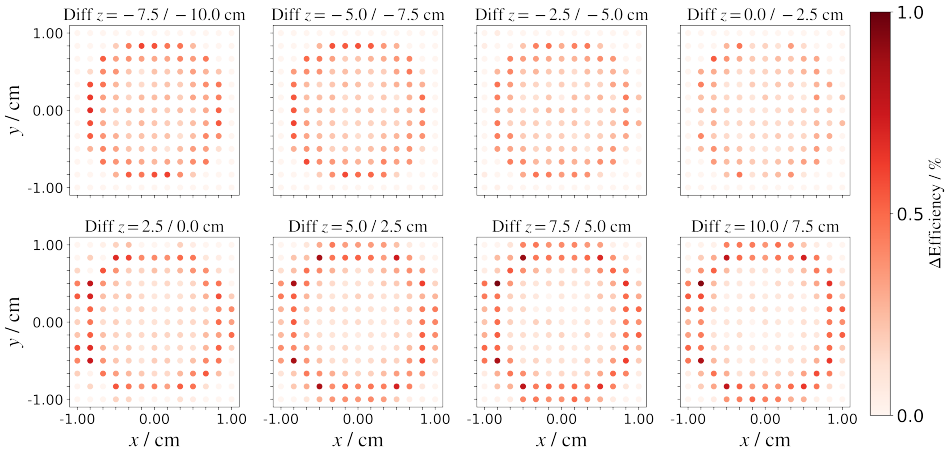


Figure 4.4: Differential acceptance maps showing the subtraction of two adjacent acceptance maps of Figure 4.3.

Generation of the Effective Acceptance including Restgas

Instead of performing a complex numerical integration of the acceptance $\epsilon(\theta, z)$ over the gas profile during the fit, a Monte Carlo Template approach is employed. That means, the acceptance, which is recalculated afterwards to get the effective acceptance in dependence on θ_x and θ_y, is directly derived from simulations incorporating the Restgas distribution. The step to get it calculated correctly is as follows:

1. **Gas Profile Configuration:** The event generator is configured to sample interaction vertices according to the extended target density profile as shown in Figure 4.2.

2. **Full Simulation:** These events undergo the complete detector simulation and reconstruction chain. This process naturally accounts for all geometric cuts, sensor edges, and efficiency variations experienced by particles originating from displaced vertices.

3. **Effective Map Extraction:** The effective acceptance $\epsilon_{eff}^{gas}(\theta_x, \theta_y)$ is then calculated as the ratio of reconstructed to generated events:

$$\epsilon_{eff}^{gas}(\theta_x, \theta_y) = \frac{N_{rec}^{gas}(\theta_x, \theta_y)}{N_{gen}^{gas}(\theta_x, \theta_y)} \tag{4.1}$$

Crucially, this map ϵ_{eff}^{gas} is distinct from the standard point-source acceptance ϵ_{point} used for the point-like interaction region. It contains the convolution of the extended target density distributions with the z-dependent detector response.

Figure 4.5 shows the effective acceptance map derived from this procedure together with the acceptance map for the point-like interaction region. Compared to the point-source acceptance, the effective acceptance exhibits significant deviations. The point-source acceptance exhibits sharp boundaries and distinct efficiency drops corresponding to the gaps between sensor modules. This reflects the precise geometric

projection of the detector as seen from the fixed center $(0, 0, 0)$. In contrast, the effective acceptance necessary for the Restgas distribution displays significantly softer gradients. As the Restgas events originate from a range of z positions, the projection of the sensor edges onto the angular plane (θ_x, θ_y) shifts continuously. The superposition of these shifted projections results in a blurring effect, effectively smoothing out the dead zones and module gaps. This visual comparison underscores the necessity of the usage of the Restgas background. If the sharp point-source acceptance map were used to model the background component, the fit would attempt to impose non-existent sharp structures onto the smooth background data. This mismatch would manifest as artificial residuals around the sensor edges, leading to a poor likelihood and, critically, a systematic bias in the extracted total luminosity. By employing the effective acceptance, the fit model correctly matches the smeared geometry of the experimental data, ensuring a robust and unbiased determination of $\mathcal{L}_{int}$. Therefore, the final fit model $\nu(\theta)$ is constructed using this effective

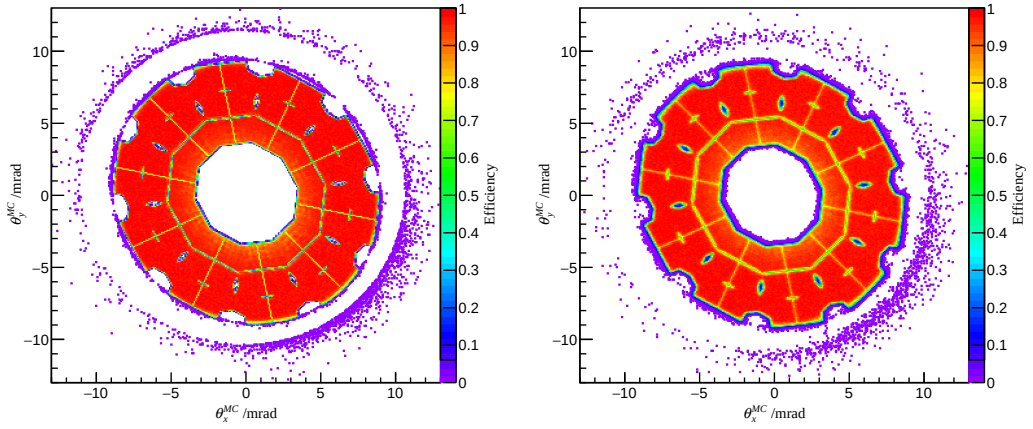


Figure 4.5: Acceptance map for point-like interaction region *(left)* in comparison with the effective acceptance map $\epsilon_{eff}^{gas}(\theta_x, \theta_y)$ *(right)*, derived from simulations incorporating the extended target profile.

acceptance. In Equation 3.15, the geometrical acceptance map ϵ_i is replaced by $\epsilon_{eff}^{gas}(\theta_x, \theta_y)$ to generate the new fit model that accurately accounts for the extended target effects:

$$\nu_i(\vec{\alpha}) = \mathcal{L}_{int} \cdot \epsilon_{eff}^{gas} \cdot \int_{\Delta\Omega_i} \left[\frac{d\sigma}{d\Omega}(\vec{\alpha}, \Omega') \otimes R(\Omega, \Omega') \right] d\Omega + B_i \tag{4.2}$$

Then the luminosity extraction proceeds as described in Section 3.7, ensuring that the final results are free from biases introduced by the Restgas background.

4.3 Handling of Spatial Vertex Cuts

In the final stage of the data analysis, it is often necessary to define a specific fiducial volume for the interaction point where the reaction took place. While the track reconstruction and back-propagation algorithms determine the interaction point (IP) for every event, applying explicit cuts on the reconstructed vertex coordinates

$(x_{vtx}, y_{vtx}, z_{vtx})$ is essential for two primary reasons. First it isolates the core physics region by restricting the analysis to the "Region of Physical Interest" where the beam-target overlap is maximal. Second, it provides robust background suppression by removing outliers and secondary interactions occurring far upstream or downstream or primary interactions outside the beam pipe.

The *LuminosityFit* framework supports the determination of the luminosity within any user-defined spatial range. However, applying geometric cuts on the experimental data fundamentally alters the reconstruction efficiency. If simple cuts are applied to the data but ignored in the fit model, the loss of signal events would be misinterpreted as a lower luminosity.

To ensure consistency, the framework introduces the concept of a Cut-Dependent Effective Acceptance Map. When a spatial cut condition Ω_{cut} (e.g., $|x_{vtx}, y_{vtx}| < 0.5\,\text{cm}$ and $|z_{vtx}| < 1\,\text{cm}$) is applied to the data, the effective acceptance map ϵ_{eff}^{gas} is replaced by a modified map ϵ_{cut}^{gas}.

The modified acceptance map is derived from the same high-statistics Monte Carlo simulations as described for the determination of the effective acceptance map ϵ_{eff}^{gas}, but with the cut condition applied to the reconstructed MC events. The effective acceptance map is defined as the ratio of the number of reconstructed events surviving the cut (N_{rec}^{cut}) to the total number of generated events (N_{gen}):

$$\epsilon_{cut}^{gas}(\theta_x, \theta_y) = \frac{N_{rec}^{cut}(\theta_x, \theta_y)}{N_{gen}(\theta_x, \theta_y)} \tag{4.3}$$

Here, $N_{gen}(\theta_x, \theta_y)$ represents events generated in the whole angular phase space, reflecting the full physical process (e.g., the complete extended target density profile including Restgas background). $N_{rec}^{cut}(\theta_x, \theta_y)$ is counted only by those MC tracks that are successfully reconstructed and whose back-propagated vertex falls within the defined cut limits.

By using ϵ_{cut}^{gas} in the fit function:

$$\nu_i(\vec{\alpha}) = \mathcal{L}_{int} \cdot \epsilon_{cut}^{gas} \cdot \int_{\Delta\Omega_i} \left[\frac{d\sigma}{d\Omega}(\vec{\alpha}, \Omega') \otimes R(\Omega, \Omega') \right] d\Omega + B_i, \tag{4.4}$$

the model correctly predicts the reduced event rate expected inside the cut region. Since the effective acceptance map ϵ_{cut}^{gas} implicitly includes the survival probability of the signal under the cut, the fitted parameter $\mathcal{L}_{int}$ correctly recovers the total integrated luminosity of the experiment, compensating for the signal fraction removed by the cuts. This feature allows for robust stability checks, where the luminosity is determined under varying cut conditions (e.g., varying the z-window) to verify the systematic stability of the measurement. Table 4.1 presents two representative vertex-cut scenarios at a beam momentum of $4.06\,\text{GeV}/c$, demonstrating the consistent recovery of the integrated luminosity. If the baseline effective acceptance map (ϵ_{eff}^{gas}) is naively utilized, applying a tight spatial window ($|x_{vtx}, y_{vtx}| < 0.1\,\text{cm}, |z_{vtx}| < 0.3\,\text{cm}$) induces an artificial luminosity deviation exceeding 85%. Similarly, a looser window ($|x_{vtx}, y_{vtx}| < 0.5\,\text{cm}, |z_{vtx}| < 1.0\,\text{cm}$) still yields a significant deviation of over 35%. However, these massive deviations completely vanish once the acceptance map is appropriately updated to incorporate the specific vertex cut conditions (ϵ_{cut}^{gas}).

Table 4.1: Deviation in the luminosity determination $\Delta\mathcal{L}/\mathcal{L}$ by using the effective acceptance map ϵ_{eff}^{gas} in comparison to ϵ_{cut}^{gas} where the cut is included into the determination under different vertex cut conditions at a beam momentum of 4.06 GeV/c.

Vertex Cut Condition	$\Delta\mathcal{L}/\mathcal{L}$ [%] with ϵ_{eff}^{gas}	$\Delta\mathcal{L}/\mathcal{L}$ [%] with ϵ_{cut}^{gas}
No Cut	-0.0002 ± 0.07	-0.0002 ± 0.07
$\lvert x_{vtx}, y_{vtx}\rvert < 0.1\,\text{cm}, \lvert z_{vtx}\rvert < 0.3\,\text{cm}$	-86.49 ± 0.01	-0.0001 ± 0.07
$\lvert x_{vtx}, y_{vtx}\rvert < 0.5\,\text{cm}, \lvert z_{vtx}\rvert < 1\,\text{cm}$	-37.61 ± 0.05	-0.0006 ± 0.08

Chapter 5

Performance Studies for Luminosity Determination with Restgas Background

Following the methodology established in Chapter 4, this chapter presents the comprehensive validation of the *LuminosityFit* software framework with Restgas background. The primary objective is to demonstrate that the extended target model allows for a precision luminosity determination ($\Delta\mathcal{L}/\mathcal{L} < 1\%$) even in the presence of significant Restgas background and realistic beam distortions.

The validation strategy follows a bottom-up approach:

1. **Single Parameter Validation:** Isolating individual beam effects (IP offset, Tilt, Divergence) to verify the robustness of the reconstruction against the extended background.
2. **Fit Strategy Optimization:** Establishing a stable iterative procedure to handle parameter correlations.
3. **System Validation:** Evaluating the final performance in realistic scenarios where all effects are superimposed, comparing the performance with and without the Restgas component.

To benchmark the algorithms, a series of Monte Carlo simulations were performed using the DPM Generator (to simulate realistic antiproton-proton elastic scattering events) and the Box Generator (to generate acceptance maps and resolution functions) in PandaRoot with the Cluster Jet target profile and the extended target model described in Section 4.1.

5.1 Single Parameter Performance

Before addressing complex scenarios, the stability of the reconstruction was evaluated for individual beam parameters when the others are ideal. This allows for a direct comparison with previous studies on point-like targets [7], serving as a baseline to ensure that the presence of the extended Restgas background does not degrade the intrinsic resolution of the algorithms

IP Shift

Scenarios with varying the IP were simulated (without tilt or divergence). The IP position was reconstructed using the standard POCA-based back-propagation algorithm. As shown in Figure 5.1a, the magnitudes of shifted IPs in x and y coordinates are either 1 mm or 2 mm (the black markers). The reconstructed IPs (red markers with error bars) closely match the simulated values, indicating accurate reconstruction of the IP. The accuracy of the measured offset in y direction is below 25 μm, while in x direction it is slightly worse (up to 80 μm). Overall, the IP reconstruction maintains an accuracy of tens of micrometers even in the presence of Restgas background. Applying these IP corrections in the luminosity fit procedure, the relative deviation of the luminosity are summarized in Figure 5.1b. The size of the markers indicates the magnitude of the deviation and the color (red/blue) represents the positive/negative sign of the deviation. The accuracy of the luminosity determination can be achieved within below 0.2% with the mean statistical error of 0.07% for all simulated IP offsets. These scenarios confirm that the Restgas does not influence the precision of the determination of the IP position nor the luminosity determination if applying the correction methods described in Section 4.1. The results are comparable to the point-like target case [7].

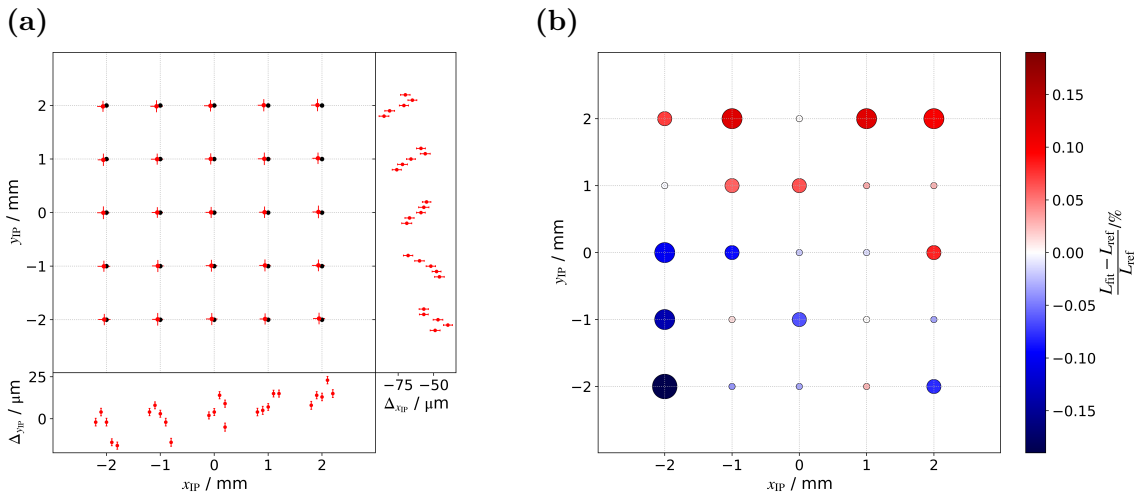


Figure 5.1: (a) Reconstructed IP positions (red markers with error bars) in comparison to the simulated IP (black markers) in x and y directions. At the edges the absolute deviations are shown. (b) Relative deviation of the luminosity for simulated IP shifts. The size of the markers indicates the magnitude of the deviation.

Beam Tilt

Similarly, scenarios with varying the beam tilt angles were simulated (ideal IP and divergence). The beam tilt is extracted as another free parameter alongside the luminosity in the log-likelihood fit from the model to data. The simulated tilt

angles range from -0.2 mrad to $+0.2$ mrad in both x and y directions (with a step of 0.1 mrad) as it is illustrated in Figure 5.2a (black markers). The resulting tilt values after the fit (red markers with error bars) closely follow the simulated values, demonstrating an accurate tilt determination of better than 10 μrad. Figure 5.2b shows the relative deviation of the luminosity for different tilt scenarios. The maximum relative deviation of the luminosity is below 0.1% with the mean statistical error of 0.07%, which is comparable to the results obtained for point-like scenarios [7]. These results prove that the beam tilt is determined with high accuracy together with the luminosity even when the signal is contaminated by Restgas background events.

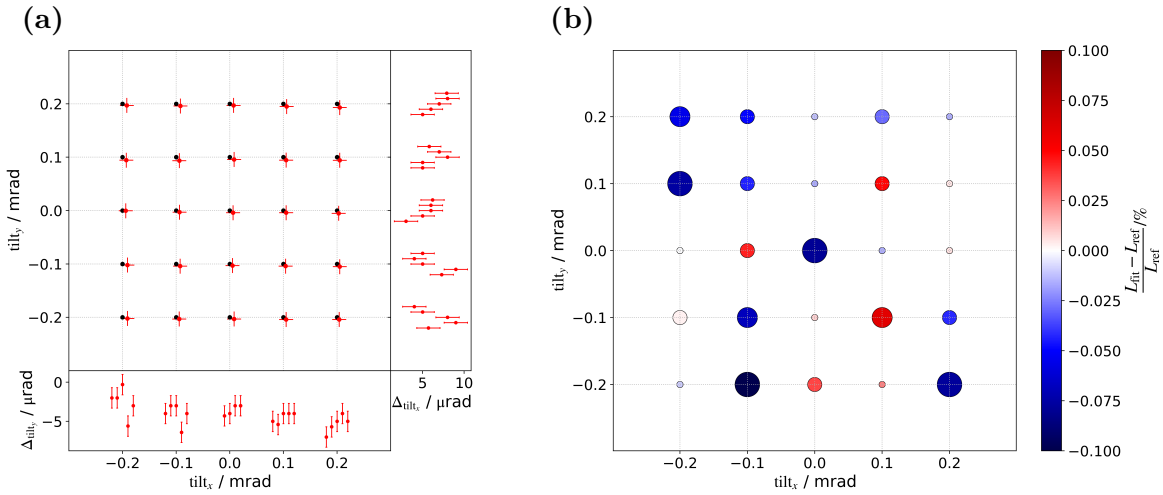


Figure 5.2: (a) Resulting beam tilt values after the fit (red markers with error bars) in comparison to the simulated tilt values (black markers) in x and y directions. At the edges the absolute deviations are shown. (b) Relative deviation of the luminosity for simulated beam tilt values. The size of the markers indicates the magnitude of the deviation.

5.1.1 Beam Divergence

Finally, scenarios with varying beam divergence values were simulated (ideal IP and zero tilt). The beam divergence is also extracted as a free parameter alongside the luminosity in the log-likelihood fit like the tilt. The simulated divergence values range from 0.0 mrad to 0.2 mrad in both x and y directions (with a step of 0.1 mrad) as shown in Figure 5.3a (black markers). The resulting divergence values determined by fitting the model to the data (red markers with error bars) do not really match the simulated values as closely as in the previous cases, but still within reasonable accuracy (better than 45 μrad). Also the beam divergence results show larger statistical fluctuations than the tilt, as it acts as a symmetric smearing term correlated with the detector resolution. However, the final luminosity remains minimal (below 0.2% with the mean statistical error of 0.14%) as shown in Figure 5.3b. The fit successfully disentangles the divergence smearing from the luminosity nor-

malization, yielding precise results with the presence of Restgas background, which is also comparable to the point-like target scenarios [7].

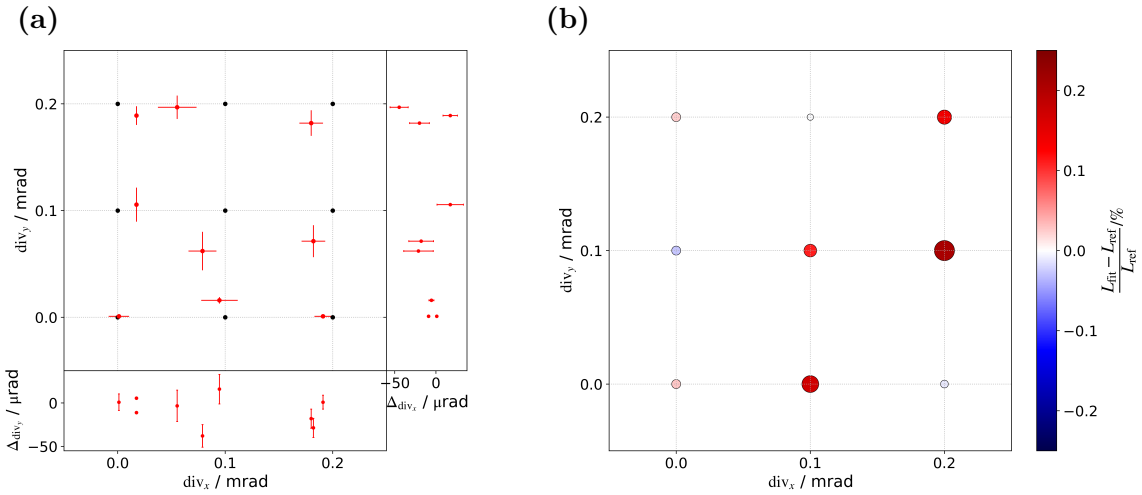


Figure 5.3: (a) Resulting divergence values determined by fitting the model to the data (red markers with error bars) in comparison to the simulated divergence values (black markers) in x and y directions. At the edges the absolute deviations are shown. (b) Relative deviation of the luminosity for simulated beam divergence values. The size of the markers indicates the magnitude of the deviation.

5.2 Optimization of the Fitting Strategy

When moving towards realistic scenarios where all three complications (IP offset, beam tilt and divergence) occur, a numerical challenge arises. IP offset can be reconstructed precisely alone before the fit procedure. However, attempting to fit the luminosity, the tilt, and the divergence simultaneously often leads to convergence failures or unstable results. This is attributed to the strong correlation between the beam tilt (mean shift) and beam divergence (width broadening), and too many floating parameters fitting at high dimensionality.

To address this, the sensitivity of the tilt determination was analyzed. A scenario with typical beam conditions ($\mathrm{tilt}_{x,y} = 100\,\mu\mathrm{rad}$, $\mathrm{div}_{x,y} = 100\,\mu\mathrm{rad}$) was simulated. The data was then fitted using three different assumptions for the beam divergence which were fixed to:

1. **0:** Ignoring divergence completely.
2. **100:** The ideal case where the model matches the simulation.
3. **200:** A case scenario where the divergence is significantly overestimated.

The results of the determination of the luminosity and the beam tilt in x and y directions are summarized in Table 5.1. Crucially, the determination of the beam

tilt is remarkably robust across all scenarios. In all cases the beam tilt values are about the same within 2 μrad and the deviation from the simulated value is about 5 μrad. The luminosity determination shows a larger dependence on the divergence assumption. In the case with the correct assumption, the luminosity value is close to the real value (0.05% deviation). In the two other cases, the deviation is 0.3% and even 1.05%, respectively.

Table 5.1: Results of the stability of the beam tilt determination under varying divergence assumptions. In all cases, the simulated parameters are fixed at $\text{tilt}_{x,y}^{sim} = 100\,\mu\text{rad}$ and $\text{div}_{x,y}^{sim} = 100\,\mu\text{rad}$. The fit assumes different fixed divergence values ($\text{div}_{x,y}^{fix}$) ranging from $0\,\mu$rad (neglected) to $200\,\mu$rad (overestimated).

Fit Assumption	**Fitted Tilt**		**Fitted Luminosity**
$\text{div}_{x,y}^{fix}$	tilt_x^{rec}	tilt_y^{rec}	$\Delta\mathcal{L}/\mathcal{L}$
[μrad]	[μrad]		[%]
0	104.9 ± 1.3	96.3 ± 1.4	0.30 ± 0.07
100	105.0 ± 1.2	95.9 ± 1.3	-0.05 ± 0.07
200	104.0 ± 1.3	94.5 ± 1.4	-1.05 ± 0.07

This evidence confirms that the determination of the beam tilt is decoupled from the divergence, enabling a stable two-step iterative fitting strategy:

- **Step 1 (Tilt Determination):** The fit is performed with the beam divergence fixed to 0 mrad. This stabilizes the minimizer and allows for a precise extraction of the beam tilt.
- **Step 2 (Divergence & Luminosity):** The beam tilt is fixed to the value determined in Step 1. The fit is repeated with beam divergence and luminosity as free parameters.

Both steps are more stable than a simultaneous fit of all floating parameters, ensuring reliable convergence. In principle, this two-step procedure can be iterated by feeding the fitted divergence back into Step 1 to refine the tilt. However, studies have shown that a single iteration is sufficient to achieve convergence within the desired precision.

5.3 Performance in Realistic Scenarios

Finally, the full performance was evaluated under realistic scenarios where IP offset, beam tilt, and divergence are all present simultaneously. To quantify the impact of the Restgas background, simulations were performed both with and without the Restgas component, allowing for a direct comparison of the fit performance. A set of 10 realistic scenarios at the beam momentum of 4.06 GeV/c were defined, each with different combinations of IP offsets (up to 2 mm), beam tilts (up to 0.2 mrad), and divergences (up to 0.2 mrad) based on expected beam conditions at $\overline{\text{P}}$ANDA. The fit was performed using the optimized two-step fitting strategy outlined in Section 5.2. The reconstructed IP positions, determined beam tilts and divergences are shown in

Figure 5.4 for all scenarios. Similar to the single parameter studies, the reconstructed IP positions and the fitted beam tilts (red markers with error bars) closely match the simulated values (black markers) for both with and without Restgas scenarios. The accuracy of the IP reconstruction remains within 10 μm in y and 70 μm in x for all scenarios, and the beam tilt determination maintains an accuracy of better than 10 μrad. The fitted beam divergences show larger statistical fluctuations but still reasonable agreement with the simulated values. The divergence accuracy is better than 25 μrad for all scenarios. The final luminosity results are summarized in Figure 5.5, showing that the maximum relative deviation of the luminosity remains below 0.25% with the mean statistical error of 0.09% for all scenarios without Restgas and 0.15% with the mean statistical error of 0.21% with Restgas. This demonstrates that the presence of Restgas background does not influence the luminosity precision compared to the ideal case, while the statistical error increases slightly due to the additional background.

Also scenarios at beam momenta of 1.5, 8.9 and 15.0 GeV/c (the beam momenta with the accurate magnetic field maps available in PandaRoot) were studied, yielding consistent results (see Table 5.2, 5.3 and 5.4). With these comprehensive studies, the following key conclusions can be drawn regarding the performance of the luminosity determination framework in realistic scenarios with Restgas background:

1. **Parameter Robustness:** The accuracy of the determination for IP, beam tilt and divergence is virtually identical for cases with and without Restgas. This proves that the effective acceptance map successfully describes the Restgas influence, preventing it from distorting the signal parameters.

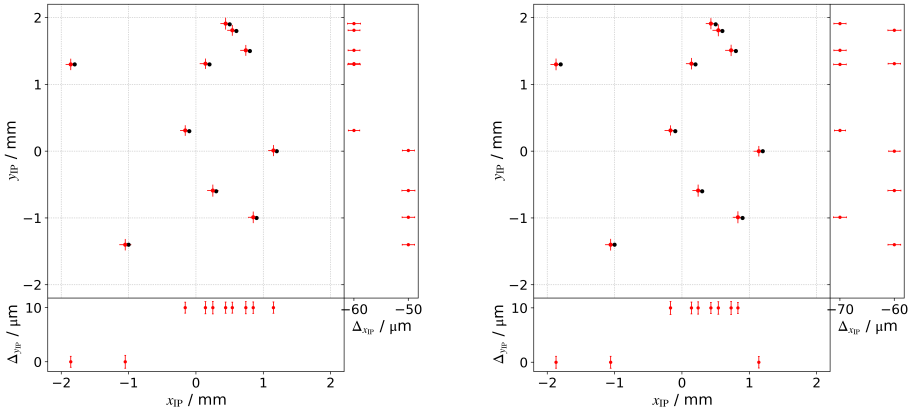
2. **Luminosity Precision:** Most critically, the luminosity determination shows an absolute relative deviation of:

$$\left|\frac{\Delta\mathcal{L}}{\mathcal{L}}\right| < 0.5\%$$

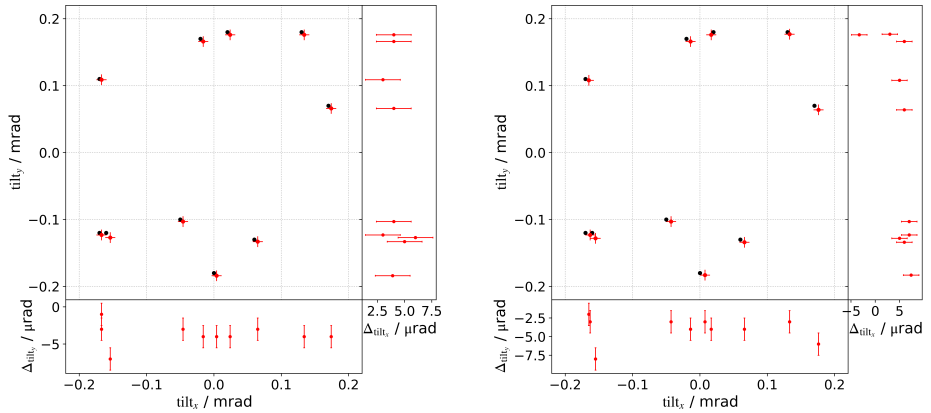
   for all investigated scenarios.

This result is a major milestone. It confirms that with the *LuminosityFit* framework, the presence of the Restgas background is no longer limiting the precision of the luminosity determination at $\overline{\mathrm{P}}$ANDA. The achieved precision surpasses the initial design goal of 1%, ensuring that $\overline{\mathrm{P}}$ANDA can perform high-precision physics measurements without being hindered by Restgas background.

(a) Reconstructed IP positions.



(b) Beam tilts after the fit.



(c) Beam divergences after the fit.

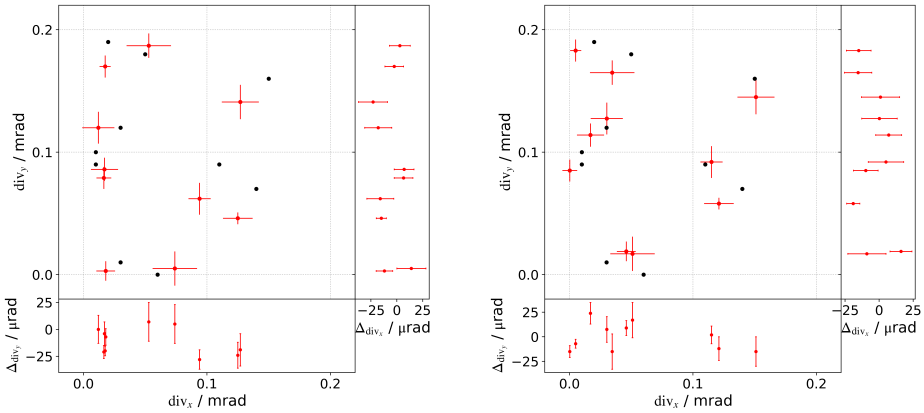


Figure 5.4: (a) Reconstructed IP positions, (b) beam tilts and (c) divergences after the fit for all realistic scenarios without (*left*) and with (*right*) Rest-gas background. The red markers with error bars represent the reconstructed/fitted values, while the black markers indicate the simulated values. At the edges the absolute deviations are shown.

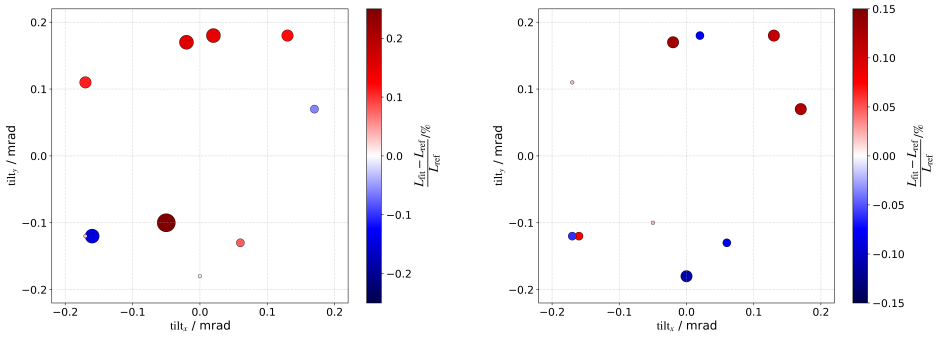


Figure 5.5: Relative deviation of the determined luminosity for all realistic scenarios without (*left*) and with (*right*) Restgas. The size and color of the markers indicate the magnitude and sign of the deviation, respectively. The results are shown in dependence on the beam tilt.

Table 5.2: Ten fit results of the luminosity determination for scenarios at beam momenta of 1.5 GeV/c. The first row shows the generated values, while the second row shows the values of the determined IP position, beam tilt and divergence. The last column shows the relative deviation of the luminosity for each scenario.

IP_x	IP_y	$tilt_x$	$tilt_y$	div_x	div_y	$\Delta\mathcal{L}/\mathcal{L}$
[mm]		[μrad]				[%]
−2.00	1.20	−150.0	−40.0	120.0	50.0	0.11 ± 0.11
−2.12	1.23	−144.1	−48.8	84.9	17.3	
−1.60	0.60	−180.0	−110.0	50.0	130.0	-0.18 ± 0.22
−1.69	0.62	−174.3	−115.9	74.2	137.4	
−0.50	−0.30	−80.0	0.0	80.0	100.0	0.16 ± 0.24
−0.62	−0.28	−73.3	−10.3	59.1	78.6	
1.30	0.60	−100.0	190.0	200.0	0.0	-0.09 ± 0.11
1.16	0.63	−90.2	177.7	199.1	3.3	
0.30	−1.80	−60.0	−130.0	120.0	20.0	0.13 ± 0.08
0.18	−1.77	−53.1	−139.8	139.3	17.7	
−0.50	−1.60	150.0	40.0	30.0	130.0	0.20 ± 0.17
−0.62	−1.57	158.1	30.0	5.2	146.0	
1.00	2.00	−170.0	− 90.0	160.0	60.0	-0.10 ± 0.11
0.87	2.04	−170.6	−104.1	162.3	12.3	
2.00	1.10	30.0	120.0	140.0	40.0	0.007 ± 0.15
1.86	1.13	39.3	107.8	106.9	10.8	
−1.50	−0.30	−110.0	160.0	40.0	170.0	-0.05 ± 0.11
−1.62	−0.28	−104.1	155.2	25.6	154.1	
1.40	−0.20	−10.0	−130.0	100.0	100.0	-0.54 ± 0.24
1.27	−0.16	− 2.0	−140.6	147.2	146.4	

Table 5.3: Ten fit results of the luminosity determination for scenarios at beam momenta of 8.9 GeV/c. The first row shows the generated values, while the second row shows the values of the determined IP position, beam tilt and divergence. The last column shows the relative deviation of the luminosity for each scenario.

IP_x	IP_y	$tilt_x$	$tilt_y$	div_x	div_y	$\Delta\mathcal{L}/\mathcal{L}$
[mm]		[μrad]				[%]
1.30	−1.90	−120.0	−120.0	90.0	90.0	-0.04 ± 0.17
1.23	−1.91	−111.3	−118.4	51.5	52.1	
0.90	0.70	−160.0	160.0	100.0	90.0	0.02 ± 0.09
0.83	0.71	−146.2	156.6	97.1	27.2	
1.20	0.80	−30.0	−100.0	100.0	110.0	0.03 ± 0.10
1.13	0.80	−20.3	−109.5	95.7	92.9	
1.80	−0.20	−20.0	80.0	180.0	140.0	0.08 ± 0.12
1.73	−0.20	− 6.3	81.6	157.8	135.6	
−0.50	0.50	−40.0	70.0	10.0	150.0	0.07 ± 0.09
−0.58	0.51	−34.1	74.5	54.4	86.8	
2.00	0.40	−160.0	−20.0	20.0	160.0	0.06 ± 0.14
1.82	0.40	−138.7	−18.2	8.9	174.1	
−0.50	0.70	20.0	−90.0	150.0	40.0	0.06 ± 0.09
−0.58	0.70	26.8	−91.7	173.3	34.6	
0.80	0.20	−120.0	−40.0	10.0	100.0	0.07 ± 0.11
0.73	0.20	−113.2	−41.8	12.2	113.9	
1.60	0.70	−160.0	−170.0	50.0	20.0	-0.01 ± 0.28
1.53	0.70	−148.1	−168.6	49.5	9.6	
−0.10	1.80	10.0	70.0	150.0	140.0	-0.03 ± 0.12
−0.17	1.72	8.3	58.2	121.8	106.1	

Table 5.4: Ten fit results of the luminosity determination for scenarios at beam momenta of 15.0 GeV/c. The first row shows the generated values, while the second row shows the values of the determined IP position, beam tilt and divergence. The last column shows the relative deviation of the luminosity for each scenario.

IP_x	IP_y	$tilt_x$	$tilt_y$	div_x	div_y	$\Delta\mathcal{L}/\mathcal{L}$
[mm]			[μrad]			[%]
1.80	1.00	50.0	0.0	80.0	80.0	0.09 ± 0.09
1.70	1.00	78.6	23.3	67.1	73.6	
0.80	0.10	60.0	20.0	60.0	10.0	0.01 ± 0.09
0.70	0.10	59.5	27.8	93.1	21.8	
−0.60	1.40	−140.0	−150.0	110.0	200.0	0.09 ± 0.09
−0.70	1.40	−128.8	−135.2	181.6	209.1	
1.20	−0.20	100.0	190.0	80.0	130.0	0.04 ± 0.09
1.10	−0.20	122.9	207.5	18.4	173.5	
−1.60	−0.10	150.0	140.0	190.0	90.0	-0.05 ± 0.09
−1.70	−0.10	147.2	156.1	173.4	76.5	
−1.70	−0.20	−130.0	−180.0	140.0	80.0	-0.09 ± 0.09
−1.79	−0.20	−146.0	−178.6	132.3	90.4	
0.50	−0.60	−80.0	120.0	170.0	180.0	0.15 ± 0.09
0.40	−0.60	−75.4	135.9	152.9	194.3	
1.50	−0.50	−120.0	−40.0	0.0	70.0	0.07 ± 0.09
1.36	−0.50	−123.6	−49.4	5.1	99.7	
0.80	1.70	110.0	90.0	160.0	90.0	0.21 ± 0.10
0.70	1.71	120.6	105.2	152.1	78.3	
0.20	−0.90	140.0	−50.0	70.0	160.0	-0.04 ± 0.09
0.10	−0.90	130.8	−43.0	110.7	169.1	

5.4 Impact of Target Profile Uncertainty

The results in the previous sections demonstrate the excellent performance of the luminosity determination, even if restgas background is present. Here, the assumption was made that the true Restgas distribution is perfectly known and correctly modeled in the Monte Carlo generation of the effective acceptance map. However, in the real experiment, Restgas may not be the same as the gas flow simulation of the target group predicts. Due to technical limitations or environmental changes, the vacuum conditions may fluctuate, leading to changes in the Restgas density or profile shape.

To quantify the systematic risk associated with these fluctuations, a sensitivity study was performed. Scenarios where the actual Restgas density deviates by $\pm 10\%$ from the nominal expectation are considered, while the density in the target area ($-1\,\text{cm} < z < +1\,\text{cm}$) remains constant, as shown in Figure 5.6.

Two distinct analysis approaches were applied to these modified datasets to evaluate the impact of model consistency:

1. **Consistent Model:** The modified profile of the Restgas is assumed to be known correctly and is incorporated into the analysis. The effective acceptance map ϵ_{eff}^{gas} used for the fit is updated to match the modified target density.
2. **Mismatched Model:** The more realistic case is chosen here where the change in the Restgas density is not noticed. The data contains the $\pm 10\%$ variation, but it is fitted using the original effective acceptance map, simulating a model mismatch.

The results of both scenarios are summarized in Table 5.5. As expected, when the effective acceptance map matches the physical reality, the framework adapts perfectly. The relative deviation of the determined luminosity remains negligible at $\left|\frac{\Delta\mathcal{L}}{\mathcal{L}}\right| \approx 0.02\%$. This confirms the mathematical validity of the method: if the correct profile is used as input, the correct result is obtained. However, when there is a mismatch between the actual Restgas profile and the one used for the acceptance map, a significant systematic bias emerges. The fit attempts to compensate for the incorrect model by shifting the total luminosity. The resulting deviation jumps to $\left|\frac{\Delta\mathcal{L}}{\mathcal{L}}\right| > 3.0\%$, which is well beyond the precision goal for the luminosity determination of $\overline{\text{P}}$ANDA, indicating that a blind reliance on a static Restgas profile is insufficient for high-precision luminosity determination.

Table 5.5: Sensitivity of the luminosity determination to Restgas profile variations. Two scenarios are considered: one where the modified profile is known and used in the fit (Consistent Model), and another where the original nominal profile is used despite the modification (Mismatched Model).

Target Profiles	$\Delta\mathcal{L}/\mathcal{L}$ with Consistent Model	$\Delta\mathcal{L}/\mathcal{L}$ with Mismatched Model
-10% Restgas	-0.02 ± 0.08	-3.39 ± 0.08
Original	0.02 ± 0.08	0.02 ± 0.08
+10% Restgas	0.02 ± 0.08	3.39 ± 0.08

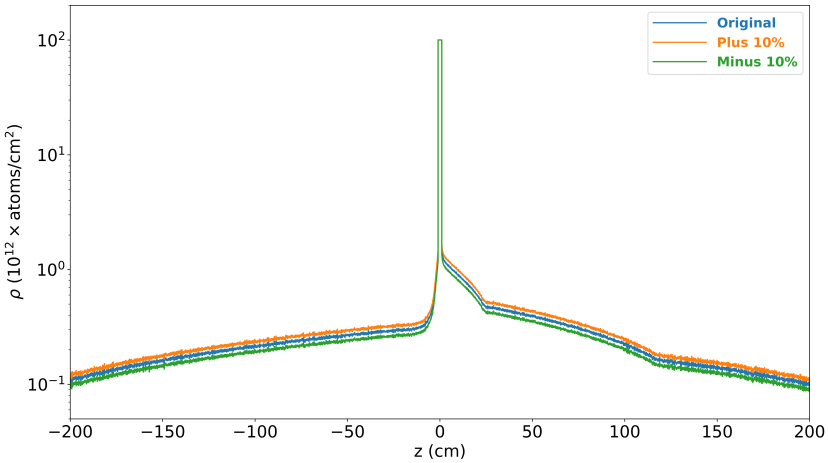


Figure 5.6: Modified Restgas profiles with ±10% density variations (orange and green curves) compared to the nominal profile (blue curve).

The contrast between the consistent and mismatched scenarios leads to a critical conclusion: The *LuminosityFit* framework is capable of high precision, but its accuracy is strictly limited by the knowledge of the underlying longitudinal target density profile $\rho(z)$.

A 3% systematic uncertainty induced by unnoticed Restgas variation is unacceptable. To mitigate this, one cannot rely on the LMD alone, as it cannot provide the target profile including the Restgas background during data taking. Therefore, an independent and precise measurement of the target and Restgas distribution is required. This necessitates the use of the tracking systems in $\overline{\text{P}}$ANDA, particularly a detector with high vertex resolution close to the interaction point.

This motivation leads directly to the next part of this thesis: the feasibility study of using the tracking system of the $\overline{\text{P}}$ANDA Target Spectrometer in order to precisely measure the target density profile along the beam axis, including the Restgas distribution.

Chapter 6

HV-MAPS for $\overline{\text{P}}$ANDA MVD

As demonstrated in Chapter 5, the precision of the luminosity determination at $\overline{\text{P}}$ANDA is strictly limited by the knowledge of the Restgas profile, $\rho(z)$. The sensitivity studies in Section 5.4 revealed that a discrepancy between the assumed and actual Restgas distribution can induce systematic biases in the luminosity measurement exceeding the 1% precision goal. To mitigate this, the Luminosity Detector (LMD) cannot rely on its own data alone, it requires an external determination on the primary interaction vertex distribution to get the precise Reatsgas profile.

This chapter investigates the feasibility of achieving this goal by using the tracking system in $\overline{\text{P}}$ANDA, specifically the Micro Vertex Detector (MVD). While the standard MVD design based on hybrid pixels offers robust tracking, the upgrade to High Voltage Monolithic Active Pixel Sensors (HV-MAPS) presents a unique opportunity. With their significantly reduced material budget and superior spatial resolution, HV-MAPS are expected to enhance the vertex reconstruction capability, thereby enabling a more accurate measurement of the Restgas profile. This chapter details the implementation of the HV-MAPS-based MVD in the PandaRoot simulation framework and evaluates its performance in reconstructing vertices.

6.1 The HV-MAPS

The HV-MAPS technology have been detailed in Section 2.7 for the Luminosity Detector. The advantage of MAPS over silicon hybrid pixel sensors as foreseen so far for the MVD pixel part is that the active sensor part and the readout electronics are combined on the same chip and the circuit is only located on one side of the substrate (see Figure 6.1). This allows the sensor to be thinned down to 50 μm. As a result, significantly lower mass occupancy can be achieved and undesired multiple scattering can be reduced. An additional advantage achieved by applying a high voltage is that the charge can be transported to the readout part of

the pixel much faster by drift than the diffusion process common in MAPS without high voltage applied. This leads to a better time resolution and higher rate capability.

For the specific application in $\overline{P}$ANDA, the *MuPix* sensor series, originally developed for the Mu3e experiment [140], is utilized. For this sensor at a high voltage of -60 V, the width of the depletion zone is about 15 μm. Figure 6.1 shows the layout of MuPix10, the first full scale prototype of this sensor series. The total size is $20.66 \times 23.18\,\mathrm{mm}^2$. The active area is about $20.48 \times 20.00\,\mathrm{mm}^2$ and has 256×250 quadratic pixels with a pixel size of 80 μm $\times$ 80 μm [141]. The MuPix11 is bug fixed version of the Mupix10 sensor and will eventually be used in $\overline{P}$ANDA [142]. Recent tests show that its time resolution is smaller than 5 ns and the power budget is smaller than 250 mW/cm^2 [143].

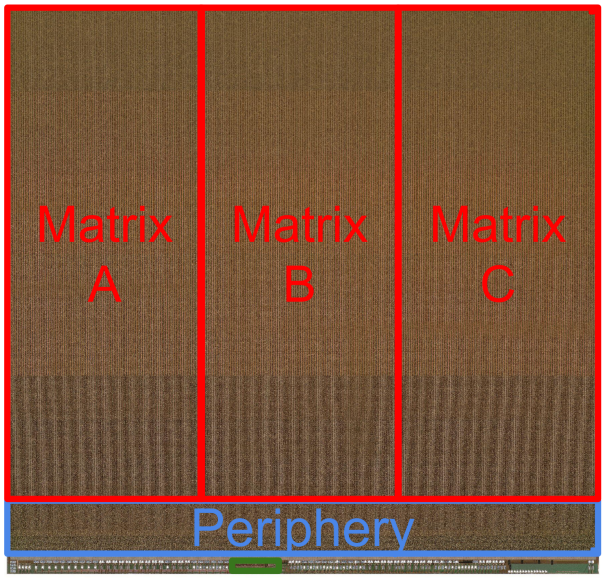


Figure 6.1: Layout of the MuPix10 sensor. Figure adapted from [144].

Compared with the hybrid pixel sensors, many advantages can be seen. Table 6.1 summarizes the parameters of MuPix sensors compared with hybrid pixel sensors. Since the readout electronics are embedded, the sensor can be thinned down to 50 μm, compared to the $\approx$ 330 μm combined thickness of the hybrid sensor and readout chip. This leads to a significant reduction of the material budget from approximately 1.0% X_0 per layer in hybrid pixels to approximately 0.3% X_0 per layer in HV-MAPS. This reduction is critical for minimizing multiple Coulomb scattering, which dominates the vertex resolution at low momenta. Also the pixel size is reduced from $100 \times 100\,\mu\mathrm{m}^2$ (Hybrid) to $80 \times 80\,\mu\mathrm{m}^2$ (MuPix), improving the intrinsic spatial resolution from $\sigma_{\mathrm{hit}} \approx 29\,\mu$m to $\approx 23\,\mu$m. Moreover, the applied high voltage leads to a depletion zone of $\sim 15\,\mu$m in comparison to 200 μm depletion zone of the whole bulk for the hybrid pixel sensors, ensuring fast charge collection (< 5 ns) for precise timing measurements and enabling a high signal-noise ratio, thereby achieving a superior signal and noise separation.

It should be noted that the current HV-MAPS implementation provides Time-over-

Threshold (ToT) information, rather than a precise measurement of energy deposition (dE/dx). Due to the thin depletion zone ($\sim 15\,\mu$m), the charge collection follows a broad Landau distribution with significant fluctuations. Consequently, charge-weighted clustering algorithms (e.g., center-of-gravity) are not applicable as the distribution of charge and therefore the ToT signal does not correlate linearly with the hit position since it is influenced by statistical fluctuations. Instead, a binary centroid method is employed, where all pixels in a cluster are assigned equal weight. This contrasts with hybrid pixel sensors, which utilize analog energy loss (dE/dx) measurements for precise charge-sharing interpolation, thereby achieving superior spatial resolution for multi-pixel clusters. The impact of this binary readout limitation is context-dependent. In the forward LMD configuration, particles typically traverse the sensors at near-perpendicular angles. Consequently, simulations show that in more than 90% of the cases, only one single pixel is hit per particle crossing. For single-pixel clusters, the binary geometric center is identical to the true hit position, making charge information redundant. However, the situation differs significantly for the MVD. Due to the proximity to the interaction point and the wide angular coverage, particles traverse the sensors at shallower angles. As demonstrated later, fewer than 50% of particle crossings result in single-pixel hits. For the remaining multi-pixel clusters, the binary centroid method, which assigns equal weight to all pixels, yields a lower spatial resolution compared to charge-weighted interpolation. This geometric effect represents the primary trade-off when adopting HV-MAPS for the inner tracking layers.

Table 6.1: Comparison of technical parameters between the standard Hybrid Pixel Sensors and the HV-MAPS used in this design.

Parameter	**Hybrid Pixel Sensor**	**HV-MAPS (MuPix)**
Material Budget ($/Layer$)	$\sim 1.0\%\ X_0$	$\sim 0.3\%\ X_0$
Sensor Thickness	$200\,\mu$m	$50\,\mu$m
Readout Chip	$130\,\mu$m	Integrated
Pixel Size	$100 \times 100\,\mu\text{m}^2$	$80 \times 80\,\mu\text{m}^2$
Intrinsic Resolution (σ_{xy})	$\sim 29\,\mu$m	$\sim 23\,\mu$m
Depletion Zone	Full Bulk ($200\,\mu$m)	$\sim 15\,\mu$m
Signal Information	dE/dx	ToT
Cluster Reconstruction	Charge Center of Gravity	Binary Centroid

6.2 Geometric Implementation in PandaRoot

To assess the performance of the upgraded tracking system, a dedicated MVD pixel geometry was implemented in the PandaRoot framework. This design strategy follows a "drop-in replacement" approach: the new HV-MAPS modules retain the spatial coverage of the original hybrid pixel sensors to ensure compatibility with the global mechanical structure. Figure 6.2 visualizes the placement of the new sensors without the mechanical support structures.

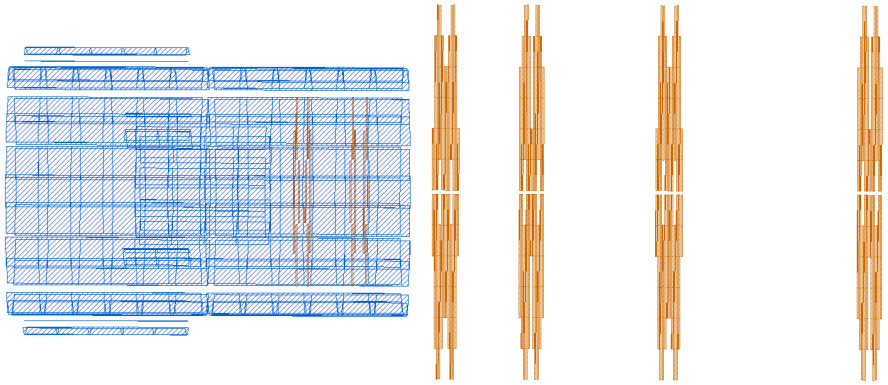


Figure 6.2: Layout of the HV-MAPS sensors in the MVD geometry within PandaRoot. The new pixel modules replace the original hybrid pixel sensors while maintaining the overall spatial coverage. Only the sensors are shown here.

Sensor Modules

The fundamental modification lies in the sensor material properties. While the active area dimensions are preserved, the sensor thickness is reduced from 200 μm to 50 μm. Crucially, the 130 μm thick external readout chips and the passive edges required by the hybrid pixel design are eliminated.

To replicate the elongated shape of the original staves, HV-MAPS modules were constructed by concatenating MuPix sensors. Four distinct module sizes were defined using combinations of $N = 2, 4, 5,$ and 6 sensors (see Figure 6.3). The module dimensions are defined as 1.16 cm [width] $\times$ (1.13 cm $\cdot N$) [length] $\times$ 50 μm [thickness]. A single sensor unit comprises 141 $\times$ 145 pixels with a pitch of 80 $\times$ 80 μm^2.

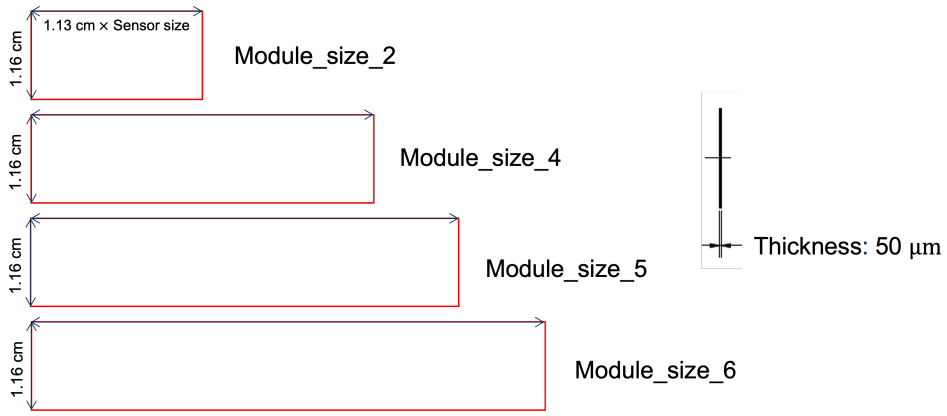


Figure 6.3: Technical configuration of the MVD HV-MAPS modules constructed from varying numbers of MuPix sensors.

Disk Configuration

The forward region is covered by two small disks and four large disks, all positioned orthogonal to the beam axis, as illustrated in Figure 6.4 *(right)*. Structurally, each disk comprises two half-disks mounted on opposite sides of a 0.4 cm thick support and cooling frame to ensure full hermetic coverage, which is depicted in the technical drawings of Figure 6.4 *(left)*.

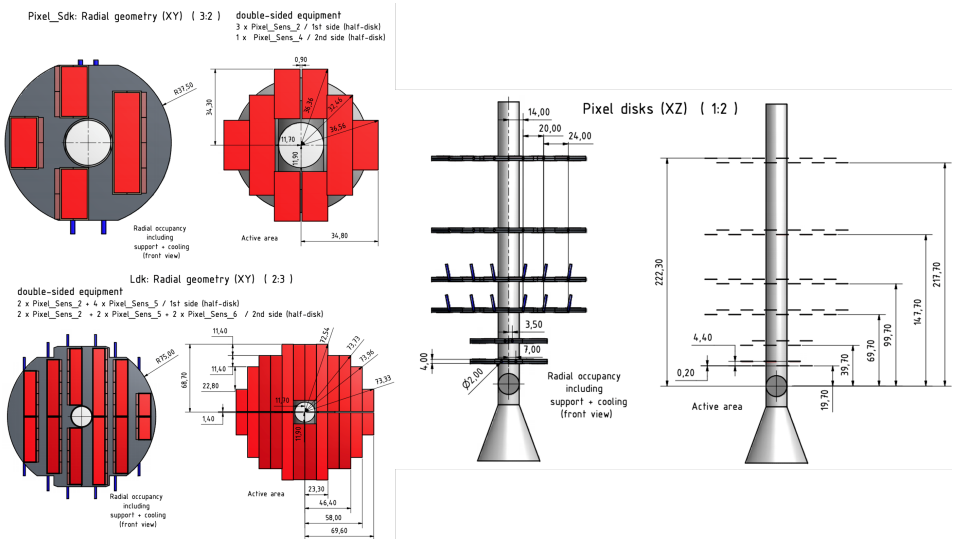


Figure 6.4: *(left)* Technical drawings of MVD half-disks and complete disks (Small disks *(top)* and large disks *(bottom)*) and *(right)* the position of each MVD disk along the beam axis.

The geometric composition is explicitly optimized for radial coverage:

- **Small Disks:** As detailed in Figure 6.4 (*left top*), each half-disk contains three Size-2 modules and one Size-4 module. The disks are positioned at $z \approx 2.0$ cm and 4.0 cm from the interaction point.
- **Large Disks:** As seen in Figure 6.4 (*left bottom*), each half-disk is populated with two Size-2, two Size-4, six Size-5, and two Size-6 modules. These larger disks are placed at $z \approx 7.0, 10.0, 14.8,$ and 21.8 cm.

Barrel Configuration

The barrel section, depicted in Figure 6.5 *(right)*, consists of two cylindrical layers surrounding the interaction point and covering the full azimuthal angle (360°). Both layers feature specific cutouts at the top and bottom to accommodate the vertical target pipe crossing.

The precise module arrangement for these layers is structured as follows:

- **Inner Layer ($r \approx 2.5$ cm):** Illustrated in Figure 6.5 *(left top)*, this layer consists of two cylindrical rows (upstream/downstream). Each row contains seven staves tilted by 6° in the xy-plane to ensure spatial hermeticity. The

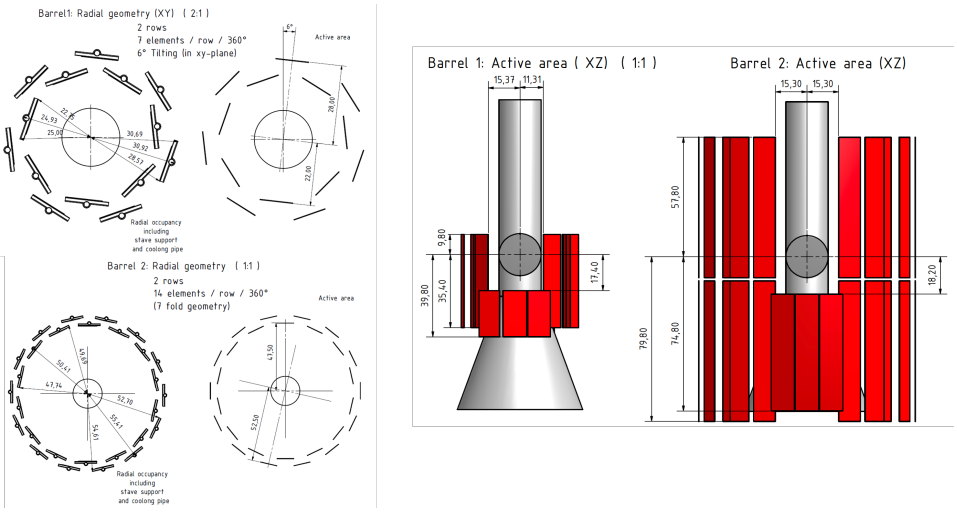


Figure 6.5: *(right)* Technical drawings of the MVD barrel layers and *(left)* detailed position of each barrel layer (Inner barrel layers *(top)* and outer barrel layers *(bottom)*).

staves near the target pipe utilize Size-2 modules, while the others use Size-4. The rows are positioned at $z \approx 2.2\,\text{cm}$ and $2.8\,\text{cm}$ (upstream edge).

- **Outer Layer ($r \approx 5.0\,\text{cm}$):** Shown in Figure 6.5 *(left bottom)*, this layer consists of two cylindrical rows with fourteen staves per row (without any tilt). The target region is covered by Size-5 modules, while the remaining sectors utilize two Size-6 modules per stave. The rows are positioned at $z \approx 4.75\,\text{cm}$ and $5.25\,\text{cm}$.

The outer part of the MVD with the double sided micro strip sensors as well as the passive components including the cooling system, support structures, and cabling frames, remain unchanged from the original TDR design [145]. The fully integrated geometry in PandaRoot, combining the new HV-MAPS silicon elements with the existing support structures, is shown in Figure 6.6.

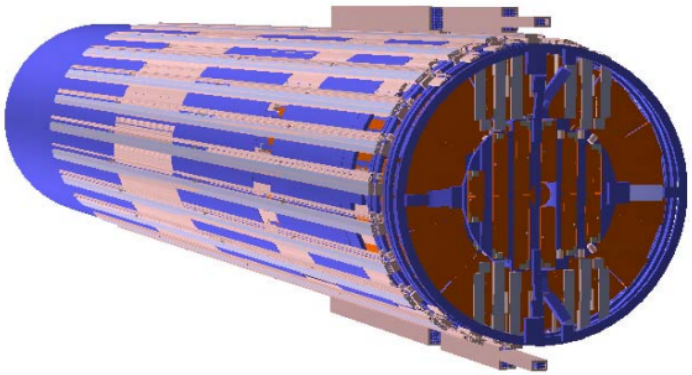


Figure 6.6: Full MVD geometry in PandaRoot framework. Figure adapted from [146].

The substitution of hybrid pixel sensors with HVMAPS fundamentally alters the tracking properties of the inner MVD layers. To quantify the impact on the vertex reconstruction precision, a comprehensive simulation study was performed using the PandaRoot framework. The study focuses on the vertex reconstruction precision, which is the decisive metric for resolving the distributed Restgas profile.

6.3 Hit resolution of the new MVD

For the first study only the MVD is used to investigate the performance of the two different sensor types, hybrid pixel sensors and HV-MAPS. The spatial hit resolution and the thickness of the material of the pixel sensors have impact on the momentum resolution of the particle tracks and on the resolution of the interaction point or the vertices of delayed decaying particles. So in the first step the hit resolution with the box generator was investigated. This generator produces π^+ with momenta of 0.3 or 1.0 GeV/c, covering the whole angular range of the MVD ($10°< \theta < 120°$ and $0°< \phi < 360°$).

The number of pixels which will be hit by a particle crossing is called the cluster size. The pixel size p as well as the sensor thickness d define the angle α_n where a track, approximated by a straight line inside the sensor, hits n readout zones:

$$\alpha_n = \arctan(\frac{n-1}{2} \cdot p/d) \tag{6.1}$$

In the case of the hybrid pixel sensors in the MVD, with a pitch $p = 100\ \mu m$ and a thickness of $d = 200\ \mu m$, these angles are $\alpha_2 = 14.0°$, $\alpha_3 = 26.6°$, $\alpha_4 = 36.9°$, $\alpha_5 = 45.0°$, $\alpha_6 = 51.3°$ and $\alpha_7 = 56.3°$. Then for HV-MAPS with a pitch $p = 80\ \mu m$ and a thickness of $d = 50\ \mu m$, these angles are $\alpha_2 = 38.7°$, $\alpha_3 = 58.0°$, $\alpha_4 = 67.4°$, $\alpha_5 = 72.6°$, $\alpha_6 = 76.0°$ and $\alpha_7 = 78.2°$. Therefore, smaller cluster sizes are expected for HV-MAPS.

Figure 6.7 shows the cluster size distribution of hybrid pixel sensors and HV-MAPS for π^+ with 0.3 GeV/c momentum. Cluster size 2 is the most frequent cluster size of hybrid pixel sensors and cluster size 1 and 3 has half of the entries of cluster size 2. While for the HV-MAPS, about 50% of the cases are cluster size 1 and about 50% are cluster size 2, as expected from the calculation above. By using HV-MAPS instead of hybrid pixel sensors, there will be slightly less hits (about 1%) but more hits with cluster size 1 (about twice) are observed.

The mean of all hits within one cluster is called hit. In Figure 6.8 the residual distributions of all hits of the two sensor types are separated by cluster size $n_{\text{cluster}} = 1$ and $n_{\text{cluster}} > 1$ for π^+ of 0.3 GeV/c beam momentum exemplarily for the projection in x direction. The coordinate system chosen here is always the system of the sensors with x and y describing the surface of the sensor and z the thickness of the sensor. Residuals for hits with cluster size 1 have the pixel resolution of 18.3 μm and 16.2 μm for hybrid pixel sensors and HV-MAPS. Residuals of hybrid pixel sensors for hits with cluster size larger than 1 have a double Gaussian structure with resolution of 3.48 μm. For HV-MAPS the resolution for clusters with size larger than 1 is 14.6 μm, which is close to the single pixel resolution. Table 6.2 also summarizes the hits

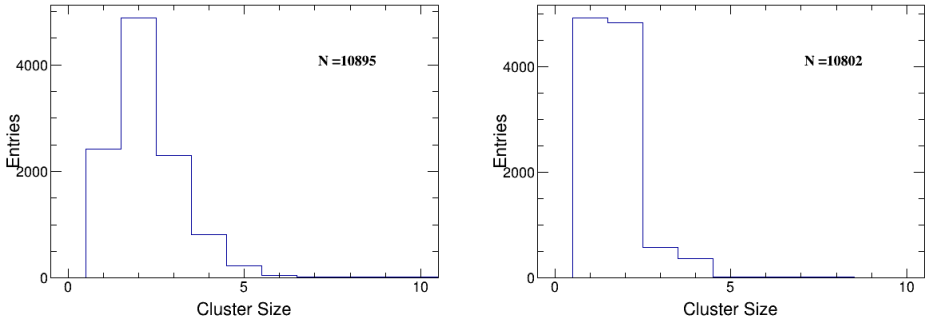


Figure 6.7: Cluster size distribution obtained from π^+ with 0.3 GeV/c momentum, covering the angular range $10°< \theta < 120°$ and $0°< \phi < 360°$. (*left*) Hybrid pixel sensors, (*right*) HV-MAPS.

resolution for the projections in x, y and z direction of 0.3 GeV/c pions as well as 1.0 GeV/c pions.

Table 6.2: Hits resolution of the projections in x, y and z direction for pions with 0.3/1.0 GeV/c beam momentum.

Cluster Size	Sensor Type	Beam Momentum	Hit Resolution		
			$x(\mu m)$	$y(\mu m)$	$z(\mu m)$
$n_{cluster} = 1$	Hybrid Pixel Sensor	0.3 GeV/c	18.3 ± 0.1	18.5 ± 0.2	8.36 ± 0.12
		1.0 GeV/c	18.4 ± 0.1	18.4 ± 0.1	8.43 ± 0.12
	HV-MAPS	0.3 GeV/c	16.2 ± 0.1	15.9 ± 0.1	6.63 ± 0.01
		1.0 GeV/c	16.3 ± 0.1	16.2 ± 0.1	6.64 ± 0.01
$n_{cluster} > 1$	Hybrid Pixel Sensor	0.3 GeV/c	3.5 ± 0.1	3.9 ± 0.1	4.75 ± 0.04
		1.0 GeV/c	3.9 ± 0.1	4.1 ± 0.1	4.87 ± 0.03
	HV-MAPS	0.3 GeV/c	14.6 ± 0.1	15.0 ± 0.1	4.10 ± 0.01
		1.0 GeV/c	14.9 ± 0.1	14.8 ± 0.1	4.18 ± 0.01

At the x and y projections, clearly the resolution of HV-MAPS for hits with cluster size 1 improves due to the pixel pitch in comparison to hybrid pixel sensors. For these hits, both hybrid pixel sensors and HV-MAPS use the binary center method for centroid calculation, which uncertainty is

$$\Delta x^2 = \frac{p^2}{12} \tag{6.2}$$

with p being the pixel size. So the calculated uncertainties of hybrid pixel sensors and HV-MAPS are 28.9μm and 23.1μm, which are a bit larger than the measured ones.

For hits with cluster size larger than 1, hybrid pixel sensors have a better resolution. Since HV-MAPS do not provide charge information for clustering, the median method which has the same uncertainty as the binary method is used for centroid calculation. For the calculation of the hit positions for cluster size larger than 1 of the hybrid pixel sensors the center of gravity method for cluster size 2 or 3

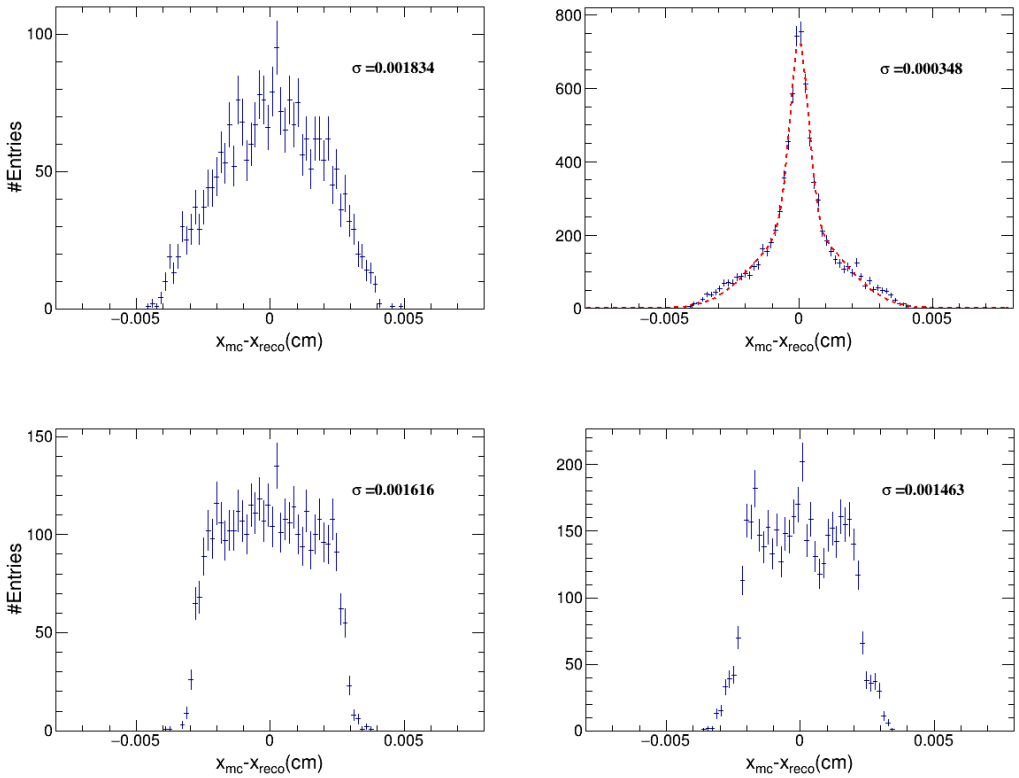


Figure 6.8: Residual distributions of the pixel sensors of hits with cluster size 1 (*left*) and larger than 1 (*right*) for pions of 0.3 GeV/c beam momentum in the x projection. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

is used and the head-tail algorithm for larger cluster sizes ($n > 3$). The residual distributions of cluster size larger than 1 of hybrid pixel sensors can be fitted with two Gaussian functions, leading to a better resolution. In the projection in z direction, the resolution of HV-MAPS is improved due to the much thinner sensor thickness. By using HV-MAPS instead of hybrid pixel sensors, there will be slightly less hits (about 1%) but more hits with cluster size 1 (about twice) are observed.

For hits with cluster size 1 HV-MAPS have a better resolution in x and y direction due to the smaller pixel size. Hits with cluster size larger than 1 hybrid pixel sensors have a better resolution in x and y direction due to the usage of dE/dx. HV-MAPS have a better resolution in z direction for hits with all cluster sizes due to the thinner sensors.

6.4 Reconstruction of D Mesons

To check the performance of the detector, a full analysis of a physics benchmark channel have been performed with the complete PANDA setup. The production of charged D^+D^- pairs in $\bar{p}p$ reactions have been chosen. At the interaction point

the $\bar{p}p$ system creates a D^+D^- pair, and the D mesons decay at the secondary vertices:

$$\bar{p}p \rightarrow D^+D^- \rightarrow K^-2\pi^+K^+2\pi^-.$$

This reaction channel was chosen to decide how precise the delayed decay of the long-living D mesons can be measured. An antiproton momentum of $p_{lab} = 8.9$ GeV/c has been used and 10,000 events have been produced for this simulation study. The tracks of these events were propagated through the detector system. Afterwards the reconstructed track candidates were further investigated.

There are 6 charged tracks per event, so the event selection criteria of $n_{\text{charged}} \geq 6$ is applied. For the PID, the global likelihood (P_i) of being a certain particle i was used. If the probability of being particle i is greater than that of any other particle j, this particle is assumed to be an i particle.

Reconstruction of the Final State Particles

Charged kaons and pions are the final state particles in the signal channel. For the investigation of the reconstruction performance of these particles, the generated MC information is used to ensure that the charged particles are correctly reconstructed kaons and pions. Figure 6.9 shows the momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of K^+ separated for hybrid pixel sensors and HV-MAPS in the center-of-mass system. Figure 6.10 shows the same distributions of π^+. These distributions look similar for hybrid pixel sensors and HV-MAPS, while there are more reconstructed kaons and pions and more low momentum pions with HV-MAPS.

The difference between the generated and the reconstructed information is the resolution. Figure 6.11 shows the resolution of the momentum (p) and spatial ($\cos\theta$ and ϕ) reconstruction of K^+ for hybrid pixel sensors and HV-MAPS in the center-of-mass system. Figure 6.12 shows the same resolutions of π^+. The standard deviation is calculated in the value range indicated with red arrows in Figures 6.11 and 6.12. All resolutions are improved with HV-MAPS. For distributions and resolutions of K^- and π^-, they look very similar to these from K^+ and π^+.

Table 6.3 summarized the spatial and momentum resolution, the number of tracks and efficiency of all pions and kaons reconstructed either with hybrid pixel sensors or HV-MAPS. For both kaons and pions, there are about 10% more tracks by using HV-MAPS instead of hybrid pixel sensors, which means tracks reconstructed with HV-MAPS have 10% higher efficiency for both kaons and pions. Also, more low momentum particle tracks were measured with HV-MAPS, so the track reconstruction with HV-MAPS is more sensitive to low momentum particles than hybrid pixel sensors. The momentum resolution of kaons and pions improves about 10% with HV-MAPS, while the spatial resolution improves about 20% with HV-MAPS in both direction $cos\theta$ and ϕ.

The pull distribution z for an observation i is defined as

$$z_i = \frac{\eta_{MC,i} - \eta_i}{\sigma(\eta_i)} \tag{6.3}$$

Table 6.3: Spatial and momentum resolution, quantity and efficiency of all pions and kaons with hybrid pixel sensors and HV-MAPS.

	Sensor Type	Resolution			Quantity	Efficiency
		p (MeV/c)	$\cos\theta$ ($\times 10^{-4}$)	ϕ ($\times 10^{-3}$)		
K^+	Hybrid Pixel Sensors	42.8 ± 0.4	10.02 ± 0.09	10.54 ± 0.09	7106 ± 47	71.1 ± 0.1 %
	HV-MAPS	38.8 ± 0.3	8.00 ± 0.07	8.60 ± 0.07	7868 ± 76	78.7 ± 0.1 %
K^-	Hybrid Pixel Sensors	44.2 ± 0.4	10.51 ± 0.10	10.92 ± 0.10	6940 ± 33	69.4 ± 0.3 %
	HV-MAPS	37.9 ± 0.3	7.80 ± 0.07	8.67 ± 0.08	7529 ± 65	75.3 ± 0.7 %
π^+	Hybrid Pixel Sensors	41.0 ± 0.3	12.54 ± 0.08	10.31 ± 0.06	15518 ± 21	77.6 ± 0.1 %
	HV-MAPS	35.5 ± 0.2	9.95 ± 0.06	7.88 ± 0.05	17063 ± 14	85.3 ± 0.1 %
π^-	Hybrid Pixel Sensors	40.0 ± 0.2	12.42 ± 0.08	10.03 ± 0.06	15585 ± 21	77.9 ± 0.1 %
	HV-MAPS	34.1 ± 0.2	9.96 ± 0.06	7.78 ± 0.05	17370 ± 11	86.9 ± 0.1 %

where $\eta_{MC,i}$ and η_i are MC and measured information of the variable η_i, $\sigma(\eta_i)$ is the standard deviation of the distribution of all η_i. It can be used to study the bias (systematic or non-random error) in measurements. In an ideal measurement, pull distributions should be normal distributed with mean equals to 0 and standard deviation is 1. If the normal distribution is narrower, the estimated uncertainties are over-estimated. If it is broader, the estimated uncertainties are under-estimated. Figure 6.13 shows pull distributions of K^+ (K^- and pions are similar to K^+), p_x, p_y and p_z represent the projections of the momentum in x, y and z direction. As can be seen in the plots, the uncertainties are determined in both cases a bit too large. The estimation for HV-MAPS is slightly better than hybrid pixel sensors.

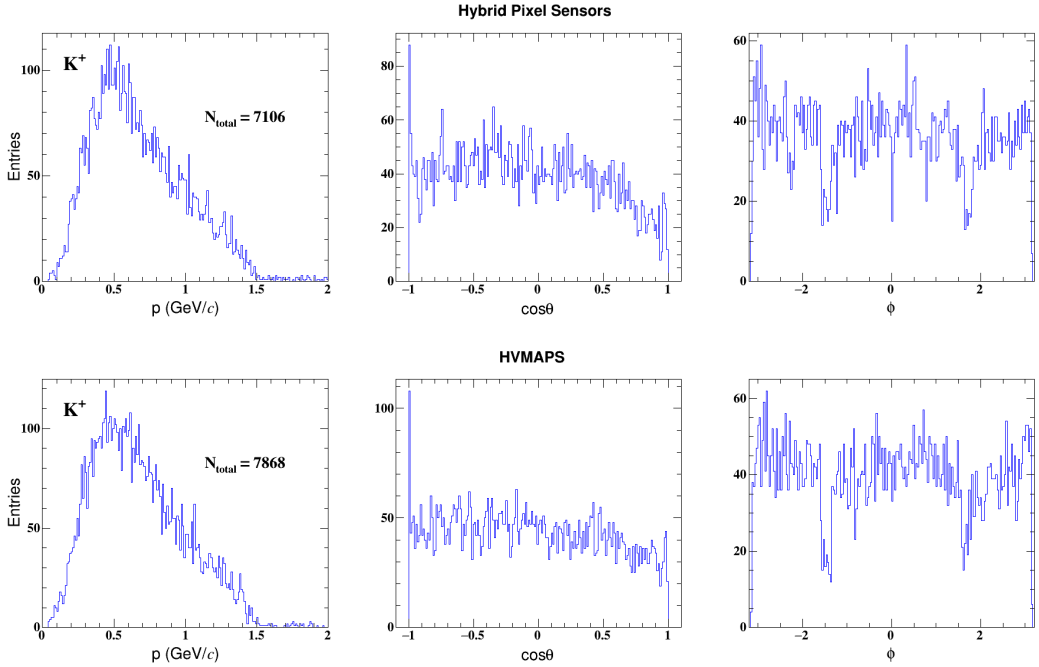


Figure 6.9: Momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of K^+ in the center of mass system. (*top*) hybrid pixel sensors, (*bottom*) HV-MAPS.

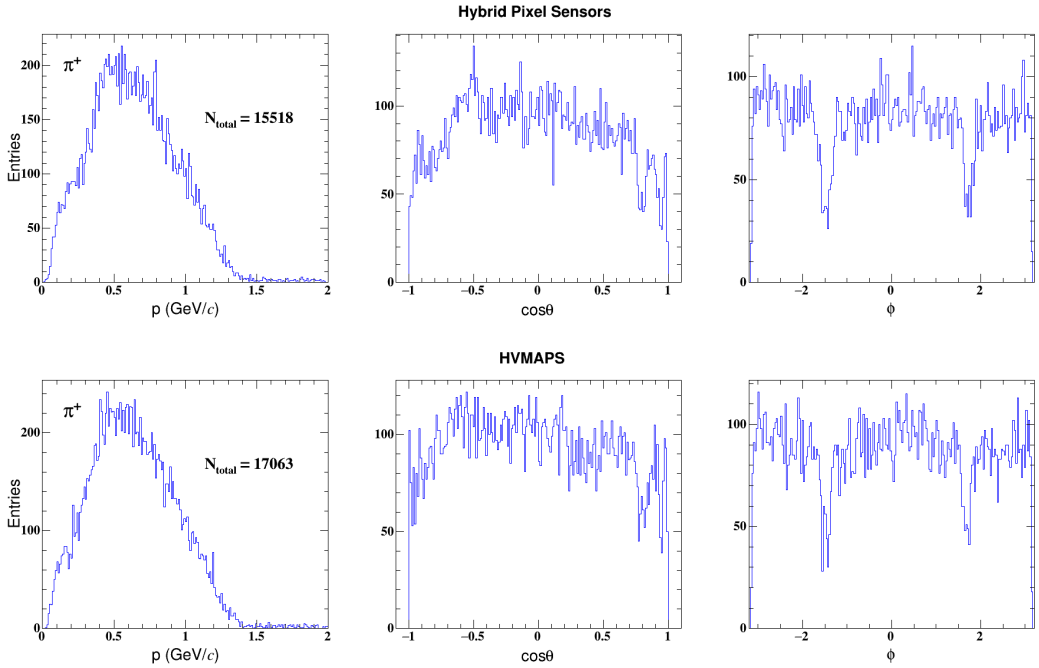


Figure 6.10: Momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of π^+ in the center of mass system. (*top*) hybrid pixel sensors, (*bottom*) HV-MAPS.

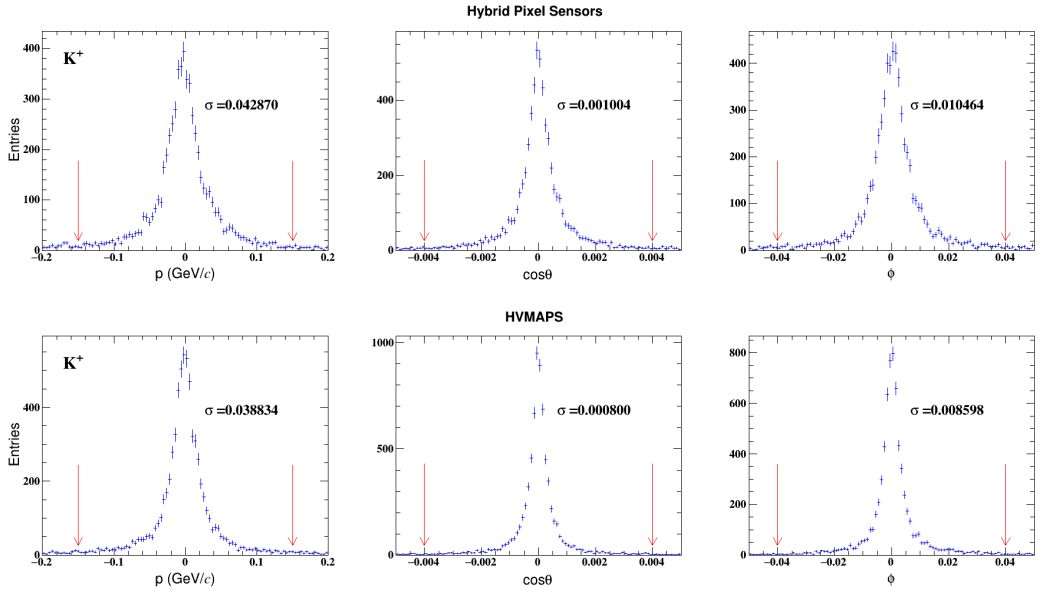


Figure 6.11: Distributions of the momentum (p) and spatial ($\cos\theta$ and ϕ) resolutions of K^+ in the center of mass system. (*top*) hybrid pixel sensors, (*bottom*) HV-MAPS.

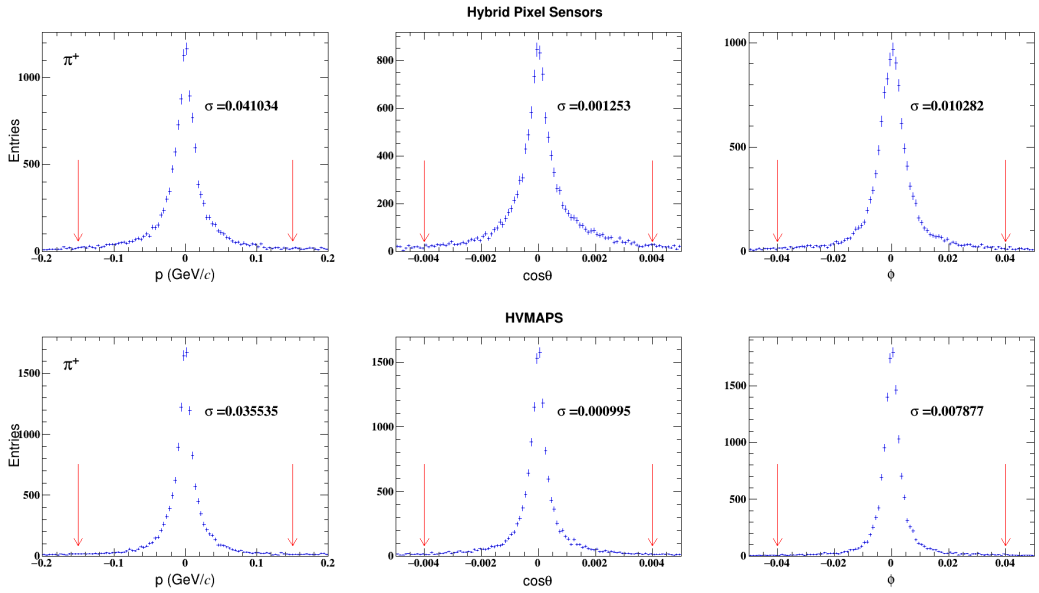


Figure 6.12: Distributions of the momentum (p) and spatial ($\cos\theta$ and ϕ) resolutions of π^+ in the center of mass system. (*top*) hybrid pixel sensors, (*bottom*) HV-MAPS.

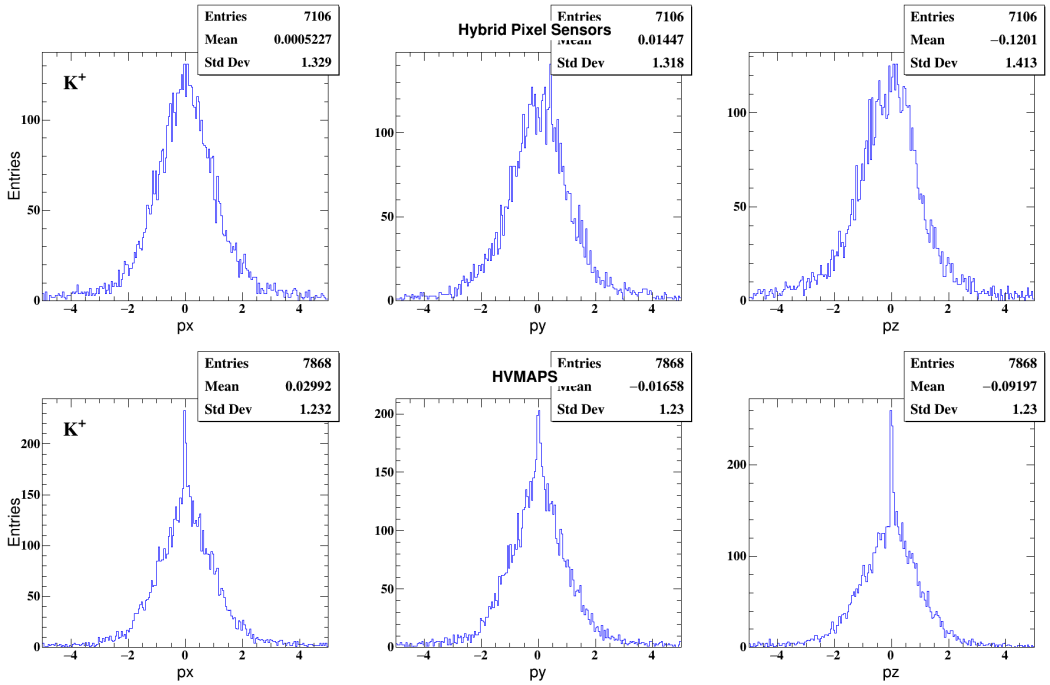


Figure 6.13: Pull distributions of the momentum components p_x, p_y and p_z of K^+, (*top*) hybrid pixel sensors, (*bottom*) HV-MAPS.

Invariant Mass of the D Candidates

The D candidates are constructed by combining two equal charged pion candidates and one oppositely charged kaon candidate:

$$K^- + 2\pi^+ \rightarrow D^+$$
$$K^+ + 2\pi^- \rightarrow D^-.$$

To check the quality of the reconstruction, also a matching with the generated values of each event is performed (truth match). In this step, truth match does not only require charged particles are kaons and pions, it is also checked if the tracks belong to the correct charged D mesons which should be produced from initial $\bar{p}p$ system. Figure 6.14 shows the distributions of the invariant $\pi\pi K$ mass (D candidates) measured with hybrid pixel sensors and HV-MAPS. Comparison of the invariant mass without truth match is also shown. In all cases a clean mass peak of the D mass is visible. The quantity of D candidates is much larger without truth match since wrong combinations are present due to the wrong PID and clone tracks. There are about 100% more truth matched D candidates with HV-MAPS, which means the efficiency of this benchmark channel improves about 100% at this step.

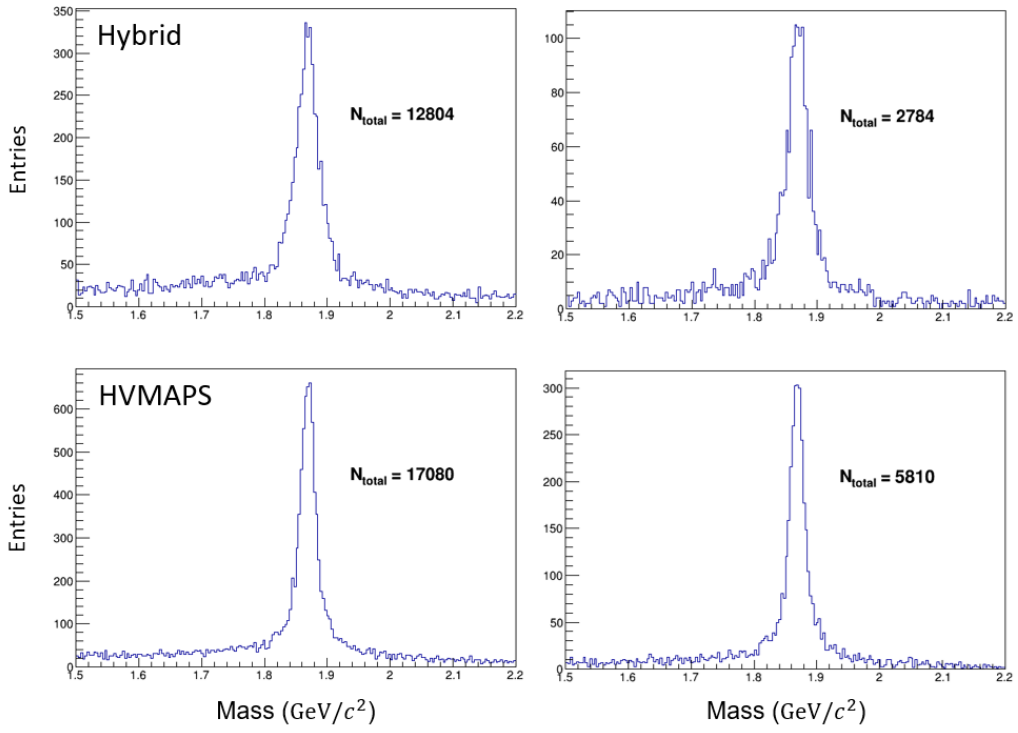


Figure 6.14: Distributions of the invariant $\pi\pi K$ mass (D mass) (*left*) without or (*right*) with truth match. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

To reduce the background distributions underneath the mass peaks, for the kinematic fit the decay tree fitter is used which takes into account the complete event.

Unlike the traditional method to kinematically fit a decay tree 'leaf-by-leaf', the decay tree fitter starts by fitting the vertices most downstream in the decay and builds up the tree by propagating information upstream. Therefore, the decay tree fitter fits the whole decay tree at once. Vertices, known masses (e.g. the D mass) as well as energy-momentum conservation are included as constraints and the approach is the χ^2 fit with Lagrange multipliers. Figure 6.15 shows the χ^2 distributions of the decay tree fit results for both detector settings. The red arrows indicated the cut on χ^2, which is set to 40 for both cases. Both distributions are typical χ^2 distributions. The distribution of the HV-MAPS is slightly narrower than this for the hybrid pixel sensors.

In addition, the best candidate selection is performed to get for each event exactly one D^+ and one D^- which is closest to the perfect event topology by choosing the candidate with the smallest χ^2 value. Figure 6.16 shows the distributions of the D mass after truth match and all filters applied which are mentioned above. Since in the decay tree fit a mass constraint is set on the D mass, the reconstructed D mass before the fit is shown. As can be clearly seen, all background or wrongly reconstructed events vanish. There are about 200% more D candidates reconstructed with HV-MAPS, so the efficiency for the reconstruction of the whole events improves about 200% at least. Table 6.4 summarized the quantities and efficiencies of D candidates separated by steps described above.

Table 6.4: Quantities and efficiencies of D candidates.

	Sensor Type	Simulated	After Truth Match	After Filters	Total Efficiency ϵ
D^+	Hybrid Pixel Sensors	10000	1397 ± 37	333 ± 15	3.33 ± 0.15 %
	HV-MAPS	10000	2882 ± 63	926 ± 24	9.26 ± 0.24 %
D^-	Hybrid Pixel Sensors	10000	1387 ± 36	333 ± 15	3.33 ± 0.15 %
	HV-MAPS	10000	2928 ± 76	926 ± 24	9.26 ± 0.24 %

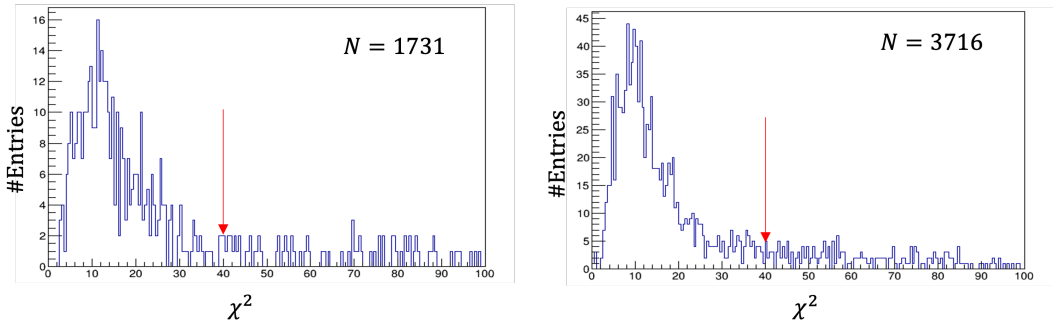


Figure 6.15: χ^2 distributions of the results of the decay tree fit. (*left*) Hybrid pixel sensors, (*right*) HV-MAPS.

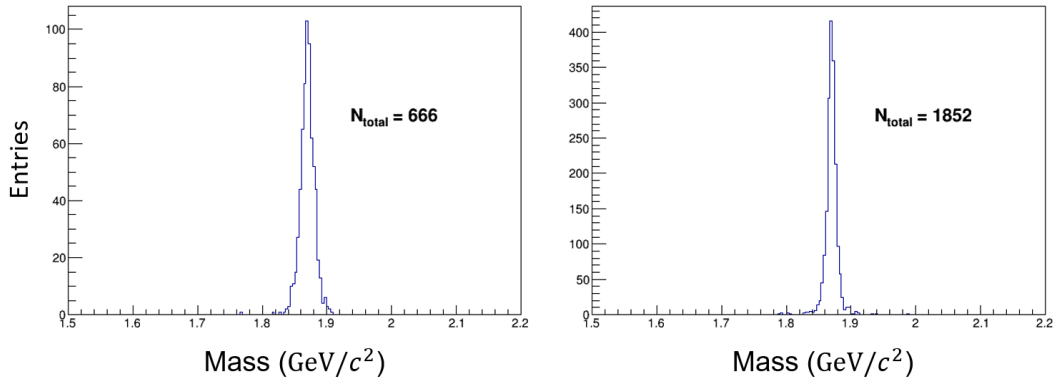


Figure 6.16: Distributions of the invariant $\pi\pi K$ mass (D mass) after MC truth match, decay tree fit with $\chi^2 < 40$ and best candidate selection. The D mass before the fit is plotted for the subset of events, which survive all selection criteria. (*left*) Hybrid pixel sensors, (*right*) HV-MAPS.

Reconstruction of the D Candidates

After combining pion and kaon tracks to D candidates, the momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of D^+ separated by hybrid pixel sensors and HV-MAPS in center-of-mass system are shown after truth match (Figure 6.17) and in Figure 6.18 in addition after the filters (decay tree fit and best candidate selection) for a more detailed comparison. The momentum distributions are quite similar in all cases. Mainly the reduction of entries around the peak vanish after applying all filters. The spatial ($\cos\theta$ and ϕ) distributions are similar after truth match, while after the filters there are more D^+ in forward direction and more D^- in backward direction with HV-MAPS. The standard deviation of the difference between the generated and the reconstructed information is used as resolution. It is calculated in the value range indicated with red arrows in Figure 6.19 and Figure 6.20. Distributions and resolutions of D^- look similar to D^+. Table 6.5 summarized the spatial and momentum resolution of D candidates for hybrid pixel sensors and HV-MAPS with/without the filters. The momentum and spatial resolutions improve about 10% for all situations with HV-MAPS.

Table 6.5: Spatial and momentum resolution of D candidates for hybrid pixel sensors and HV-MAPS with/without the filters.

	Sensor Type	w/o Filters	Resolution		
			p (MeV/c)	$\cos\theta$ ($\times 10^{-3}$)	ϕ ($\times 10^{-2}$)
D^+	Hybrid Pixel Sensors	o	52.8 ± 0.6	9.77 ± 0.14	3.07 ± 0.04
		w	11.0 ± 0.3	4.53 ± 0.18	1.08 ± 0.03
	HV-MAPS	o	47.4 ± 0.4	8.82 ± 0.10	2.80 ± 0.03
		w	10.1 ± 0.3	3.92 ± 0.15	0.88 ± 0.03
D^-	Hybrid Pixel Sensors	o	41.8 ± 0.7	8.88 ± 0.15	2.58 ± 0.04
		w	10.9 ± 0.3	5.80 ± 0.17	1.09 ± 0.03
	HV-MAPS	o	37.0 ± 0.6	6.56 ± 0.09	2.06 ± 0.02
		w	9.7 ± 0.4	4.00 ± 0.15	0.82 ± 0.03

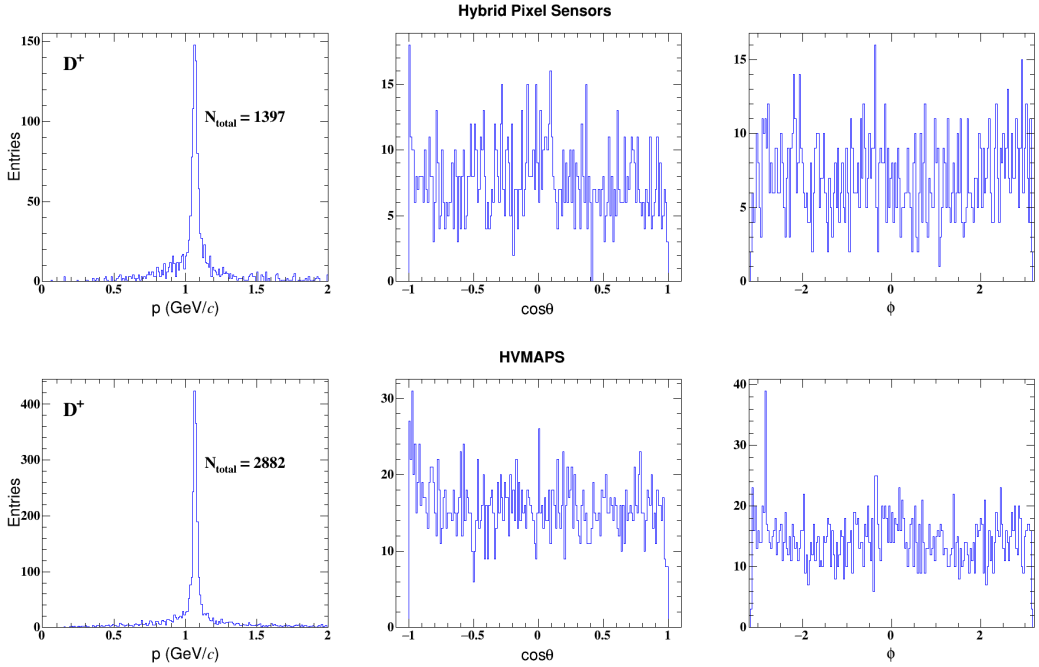


Figure 6.17: Momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of D^+ after truth match in the center mass system. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

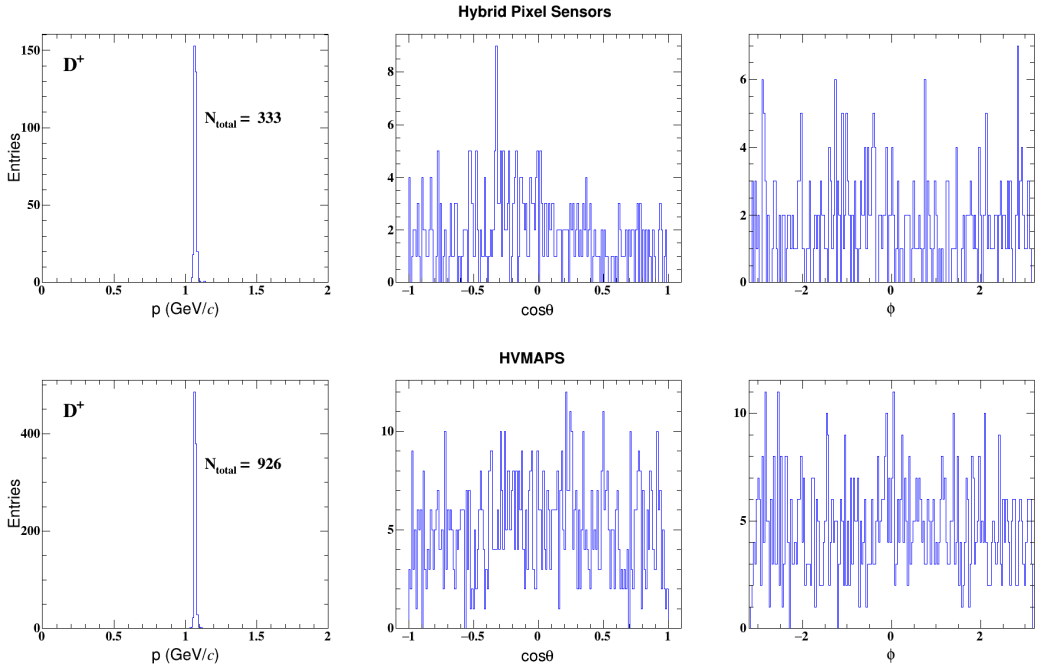


Figure 6.18: Momentum (p) and spatial ($\cos\theta$ and ϕ) distributions of D^+ after all filters in the center mass system. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

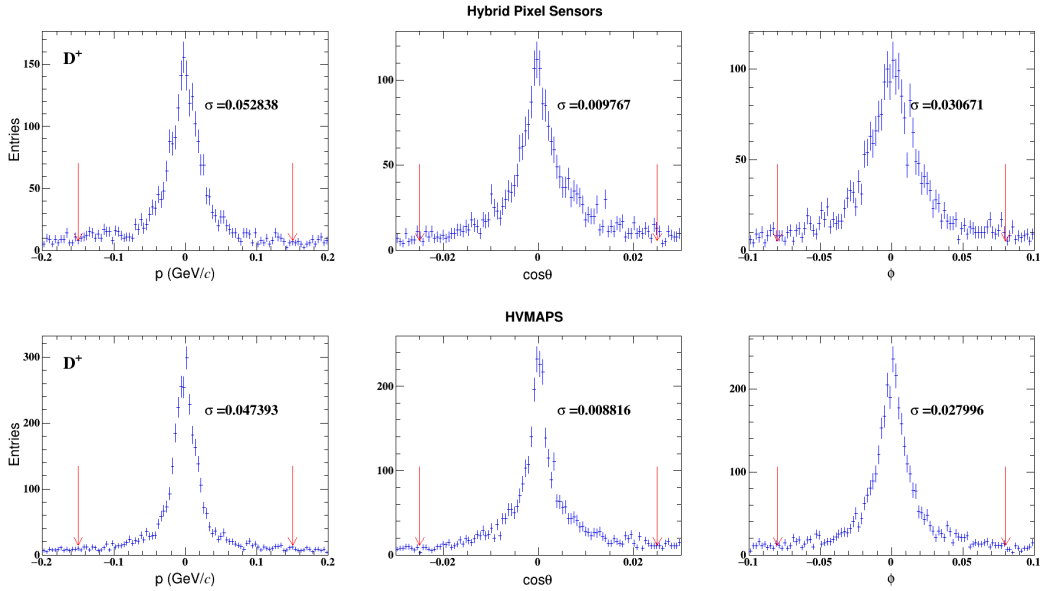


Figure 6.19: Distributions of momentum (p) and spatial ($\cos\theta$ and ϕ) resolutions of D^+ after truth match. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

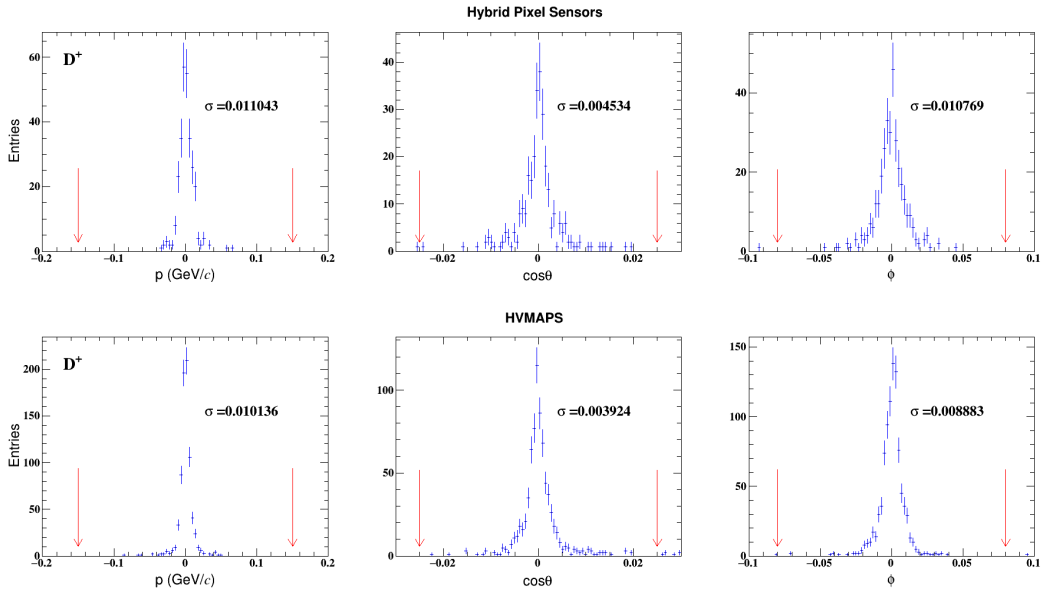


Figure 6.20: Distributions of momentum (p) and spatial ($\cos\theta$ and ϕ) resolutions of D^+ after all filters. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

6.5 Vertex Reconstruction

As the MVD is designed for precise determination of vertices, detailed knowledge of its performance is necessary. The same $\bar{p}p \rightarrow D^+D^- \rightarrow K^-2\pi^+K^+2\pi^-$ benchmark channel is used. Since the charged D meson are short lived and decay to charged kaons and pions, the properties of the primary interaction vertex and the secondary vertices are studied.

In Figure 6.21 the residual distributions of the secondary D^+ and D^- vertices after truth match in x, y and z direction are shown for hybrid pixel sensors and HV-MAPS, respectively. Also the information after all filters (truth match and decay tree fit and best candidate selection) are shown in Figure 6.22 for comparison. Again the standard deviation is used as the measure for the resolution, which is calculated from the values within the red arrows. As can be seen, the reconstruction of secondary vertices in x and y direction is better than that in z direction and the vertex resolutions improve a lot after all filters applied. After truth match, the resolutions in all directions just improve a little by using HV-MAPS instead of hybrid pixel sensors. After all filters were applied, all resolutions improve significantly by using HV-MAPS.

The corresponding distributions for the primary interaction vertex after all filters applied is shown in Figure 6.23. Since the interaction point in the MC simulation is $(0, 0, 0)$, the resolution values are the same as the reconstructed values. The standard deviation is calculated within the red arrows. Also in this case the reconstruction of the primary vertex in x and y direction is better than that in z direction. The resolutions in all directions improve also significantly by using HV-MAPS.

Table 6.6 summarizes the resolutions after reconstruction of the D^+ and D^- vertices and the primary interaction vertex. For secondary D vertices, the vertex resolution in x, y and z direction improves about 4%, 10% and 12% after truth match by using HV-MAPS instead of hybrid pixel sensors, while all values improve about 40% after all filters applied. For the primary interaction vertex, the resolution with HV-MAPS improves about 50% in comparison to the hybrid pixel sensors.

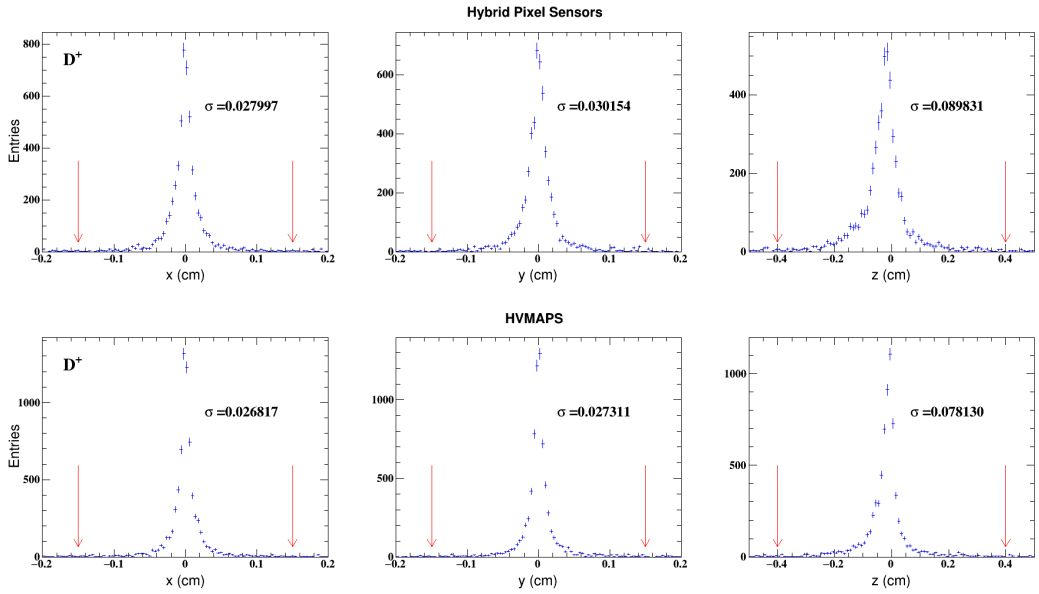


Figure 6.21: Residual distributions of the D^+ vertices in x, y and z direction after truth match. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

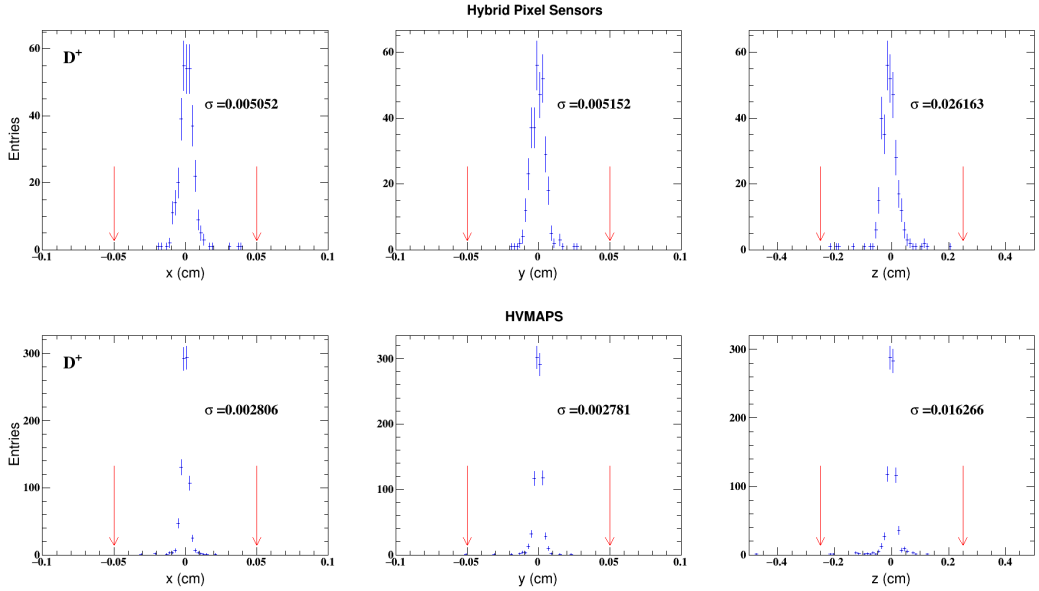


Figure 6.22: Residual distributions of the D^+ vertices in x, y and z direction after all filters. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

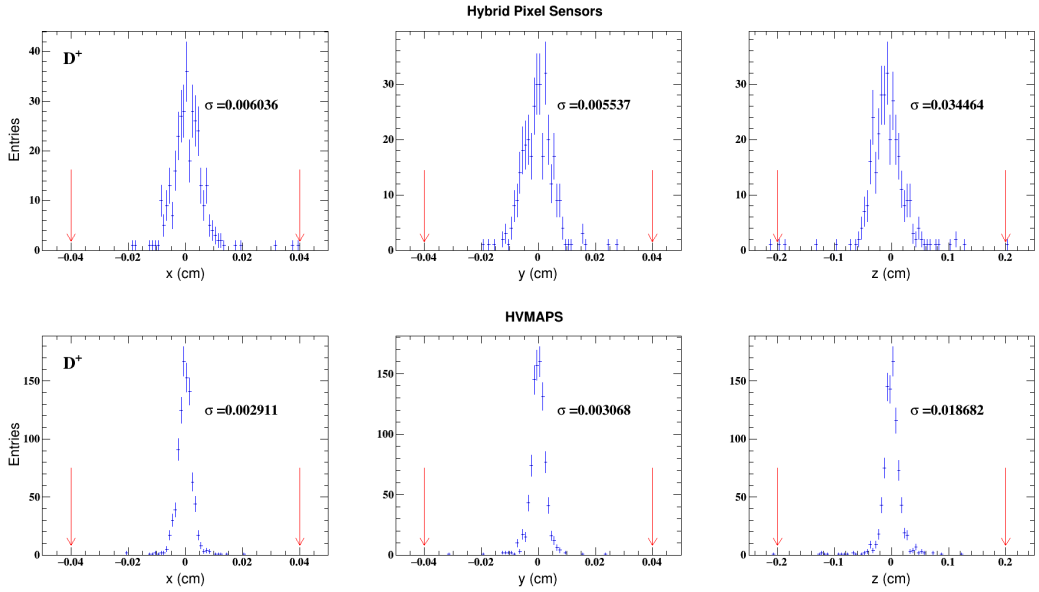


Figure 6.23: Vertex distributions of the interaction point after all filters projected in x, y and z direction. (*top*) Hybrid pixel sensors, (*bottom*) HV-MAPS.

Table 6.6: Spatial resolution of the D^+ and D^- vertices for hybrid pixel sensors and HV-MAPS with/without the filters and the interaction point with all filters.

	Sensor Type	w/o Filters	Vertex Resolution		
			$x(\mu\text{m})$	$y(\mu\text{m})$	$z(\mu\text{m})$
D^+	Hybrid Pixel Sensors	o	280 ± 3	302 ± 3	898 ± 9
		w	51 ± 2	52 ± 2	262 ± 14
	HV-MAPS	o	268 ± 3	273 ± 4	781 ± 11
		w	28 ± 1	28 ± 1	163 ± 5
D^-	Hybrid Pixel Sensors	o	255 ± 5	255 ± 5	912 ± 19
		w	60 ± 2	55 ± 2	364 ± 14
	HV-MAPS	o	248 ± 4	238 ± 4	751 ± 11
		w	30 ± 1	31 ± 1	199 ± 5
Interaction Point with Hybrid Pixel Sensors			60 ± 2	55 ± 2	345 ± 13
Interaction Point with HV-MAPS			29 ± 1	31 ± 1	187 ± 4

Chapter 7

Precise Measurement of the Restgas Distribution

The studies in Section 5.4 have clearly demonstrated the necessity of a precise measurement of the target distribution along the beam axis. Furthermore, the simulation studies in Section 6.3 have established that the upgraded MVD with HV-MAPS technology provides the necessary hardware performance, specifically the excellent spatial resolution and low material budget, to resolve the longitudinal distribution of Restgas background. However, the hardware upgrade alone is insufficient to extract the Restgas profile with the precision required for luminosity determination. To achieve this, the precise measurement of the longitudinal target profile will be performed specifically utilizing antiproton-proton ($\bar{p}p$) elastic scattering events, which serve as a high-statistics and kinematically well-understood reference channel.

A fundamental challenge lies in the event reconstruction algorithms. The standard reconstruction chain in the PandaRoot framework is primarily optimized for physics channels originating from the nominal IP at $(0, 0, 0)$. Beam-Restgas interactions, however, are distributed along the entire beam axis, often extending centimeters or decimeters away from the IP. Applying standard tracking and particle identification (PID) routines to these displaced vertices introduces systematic biases, as track parameters and error matrices are typically propagated to the origin rather than the true interaction vertex. To overcome this limitation, a dedicated reconstruction algorithm tailored for shifted IP scenarios is essential. This algorithm must accurately account for the spatial distribution of the beam-Restgas interactions, ensuring that track fitting and vertex reconstruction are performed with respect to the actual interaction points rather than the nominal IP.

7.1 Reconstruction Strategy

The accurate reconstruction of the interaction vertex distribution of the elastically scattered $\bar{p}p$ events which also originate by interactions of beam particles with Restgas profiles necessitates specialized algorithms capable of handling displaced vertices.

This section details the modifications implemented in the PandaRoot framework to achieve precise track reconstruction and particle identification for events originating away from the nominal IP.

The analysis presented in this work is based on the PandaRoot framework, version *oct19*. This specific release was selected to ensure the stability of the tracking algorithms. Newer iterations of the framework were found to exhibit regressions in the primary track reconstruction quality even for standard events at the origin, necessitating the use of the robust *oct19* release as the baseline for this study.

Tracking for Displaced Vertices: The Apollonius Algorithm

The first stage of track reconstruction is the track finding. The standard tracking algorithms in $\overline{\text{P}}$ANDA (e.g., the Barrel Track Finder) generally optimize performance by imposing a constraint that tracks must originate from the primary vertex region ($x \approx y \approx 0$). While effective for the nominal IP, this assumption leads to inefficiencies and systematic biases when reconstructing tracks from displaced vertices, such as those arising from interactions with Restgas particles or shifted beam-target collisions.

In the case of interactions with Restgas particles, the reaction vertices are distributed along the beam axis (z) and can exhibit transverse offsets. To recover these displaced tracks, which topologically resemble secondary decay products, a track finder developed for secondary vertices called Apollonius Triplet Track Finder was integrated into the reconstruction chain [147, 148].

The reconstruction of displaced tracks is implemented as a two-stage process within the PandaRoot framework. The first stage performs pattern recognition in the xy-plane using the axial STT layers. The second stage recovers the longitudinal (z) information using the skewed STT layers.

The algorithm is mathematically grounded in the classical *Problem of Apollonius* (circa 200 BC), which seeks to construct a circle that is tangent to three given circles. Unlike global Hough transforms, this algorithm constructs tracks locally from triplets of hits. The implementation of the algorithm within the PandaRoot framework addresses two main computational challenges: the combinatorial selection of hits and the ambiguity of the geometric solution.

Triplet Selection Strategy The first step of the algorithm, selecting three STT hits (h_1, h_2, h_3) to form a triplet, is critical. A brute-force combination of all available hits would result in an unmanageable combinatorial explosion. Conversely, if the selection criteria are too tight, valid tracks might be discarded, leading to a drop in efficiency.

To balance speed and performance, a multi-layered selection strategy is adopted. Rather than treating the STT as a uniform cloud of hits, the algorithm analyzes the hit distribution row by row. To maximize the geometric lever arm for curvature determination, the hits are categorized into inner, mid, and outer rows based

on their radial position. The algorithm prioritizes triplets formed by selecting one hit from each of these distinct regions (Inner → Mid → Outer). This long-baseline selection ensures that the triplet spans the maximum available track length, resulting in a stable and precise circle fit. Within this radial framework, a further reduction of candidates is achieved using a strategy inspired by Cellular Automata. The detector is segmented into angular sectors in ϕ, and spatially adjacent STT hits that form a continuous chain are grouped into segments. The search for triplet partners is then restricted to hits that are both topologically connected and ϕ-compatible. Specifically, the second and third hits must be reachable via the neighbor map, ensuring that the combination follows the particle's path through the straw tubes. In addition, the hits are required to lie within a compatible angular range in ϕ, thereby rejecting combinations that would imply unphysical "zig-zag" trajectories.

This two-tiered approach—first defining the radial strategy (Inner/Mid/Outer) to ensure quality, then using neighbor connectivity to ensure speed—allows the Apollonius tracker finder to reconstruct displaced tracks with high efficiency while keeping the computing time manageable.

Solving the Apollonius Problem Once a valid triplet is formed by selecting one hit from each of the distinct radial groups (Inner, Mid, and Outer rows), it serves as a geometric seed. The algorithm then proceeds to determine the full track parameters through a "Hypothesis and Test" approach.

Step 1: Calculation of Candidate Trajectories. The three selected drift circles define a classical Problem of Apollonius. From a mathematical standpoint, up to eight distinct circles can be constructed that are simultaneously tangent to three given circles, corresponding to the different combinations of internal and external tangency. The algorithm analytically computes the parameters (x_c, y_c, R) for all eight solutions. Each of these solutions represents a track hypothesis, reflecting the possible particle trajectories arising from the left-right ambiguities of the drift tube measurements.

Step 2: Hit Assignment (Track Growing). To identify which of these hypotheses corresponds to a physical particle, the algorithm validates each candidate against the full set of available STT hits. For every Apollonius circle, a search is performed over the axial hits, and a hit is assigned to the candidate if its measured drift distance is consistent with the circle trajectory within a predefined tolerance. Through this procedure, the purely mathematical solutions are transformed into populated track candidates.

Step 3: Selection of the Proper Circle. The final step is the selection of the proper physical trajectory among the populated candidates. This choice is based on a composite quality metric that combines topological, statistical, and geometric considerations. First, the algorithm checks the continuity of the assigned hits, requiring that they form a topologically connected and nearly gap-free chain along the straw tubes; candidates that produce fragmented or discontinuous segments are rejected. Among the remaining solutions, preference is given to the candidate that associates the largest number of hits, as a trajectory supported by more measurements is statistically more robust than one constrained only by the initial triplet.

For candidates with comparable hit counts, the geometric fit quality is evaluated by computing the quadratic distance of the assigned hits to the candidate circle,

$$Q = \sum_{i=1}^{N} (d_i - R)^2, \tag{7.1}$$

where d_i denotes the distance from the center of the i-th hit to the track center and R is the track radius. The solution that minimizes this quantity, corresponding to the best overall geometric consistency, is selected as the final track seed. This robust selection procedure ensures that the algorithm does not merely identify a mathematical circle passing through three points, but reconstructs a physically meaningful trajectory that is supported by the maximum amount of detector information.

Longitudinal Reconstruction (Skewed Straw Pz Finder)

The two-dimensional circles obtained from the Apollonius step fully define the track curvature in the xy-plane, but they do not provide any information on the longitudinal coordinate z. The reconstruction of the third dimension is therefore performed in a subsequent step by the Pz Finder, which exploits the specific geometry of the skewed STT layers that are tilted by $\pm 2.9°$ with respect to the beam axis.

Starting from the two-dimensional track candidates, the algorithm searches for hits in the skewed straw layers that are geometrically compatible with the extrapolated 2D circle. Once such hits are identified, the intersection between the cylindrical surface defined by the 2D track and the corresponding skewed straw wire which is modeled as a straight line in three-dimensional space is calculated. Owing to the stereo angle α of the skewed layers, the position of this intersection along the wire is directly correlated with the longitudinal position of the track. The z-coordinate of each intersection point is obtained according to

$$z = \frac{R \cdot \phi_{int}}{\tan(\alpha)} + z_0, \tag{7.2}$$

where ϕ_{int} denotes the azimuthal angle of the intersection and R is the track radius, α is the aforementioned stereo angle defining the wire inclination, and z_0 represents the longitudinal offset of the track (i.e., its z-coordinate at the reference azimuth $\phi = 0$).

By accumulating multiple measurements from different skewed layers, the algorithm performs a linear regression in the s-z plane, with $s = R\phi$ representing the arc length along the track. The slope of this fit determines the dip angle λ and, consequently, the longitudinal momentum component p_z, while the intercept yields the z-position of the track origin. By decoupling the pattern recognition in the transverse plane from the longitudinal reconstruction, this method enables the efficient reconstruction of tracks with arbitrary production points, making it particularly well suited for recovering Restgas events distributed along the beam axis.

Integration with MVD and GEM (Global Tracking)

The output of the combined Apollonius and Pz Finder stages is a track seed based solely on STT information. While this seed is highly effective for identifying displaced tracks, it does not yet provide the precision required for accurate vertex reconstruction. The full integration of information from all tracking sub-detectors is therefore performed in the final Kalman filtering stage.

The Apollonius algorithm supplies an initial track candidate derived from STT isochrones, which defines a preliminary trajectory but does not include the high-precision measurements from the silicon-based detectors. These track candidates are subsequently processed by the KalmanFit using the GenFit framework, which applies a statistically rigorous Kalman filter to combine measurements from different detector systems. During this procedure, the STT-based helix is extrapolated inward toward the MVD volume, where compatible pixel or strip hits are searched for and, if found, incorporated into the track. This inward propagation significantly improves the resolution of the impact parameters d_0 and z_0 and is crucial for precisely locating the interaction vertex. In parallel, the track is extrapolated outward to the GEM stations, where additional measurements refine the momentum determination, particularly for forward-going particles, leading to an improved relative momentum resolution dp/p.

As a result, the final reconstructed tracks coherently combine information from all tracking subsystems-MVD, STT, and GEM—leveraging the large acceptance and pattern recognition capability of the STT together with the superior spatial resolution of the MVD and GEM to achieve precise vertex and momentum reconstruction.

PID Step

The accurate determination of particle kinematics relies not only on precise track fitting but also on the correct association of reconstructed tracks with their physical interaction vertex during the PID step. Track candidates produced by the track finder are represented as infinite helices extending throughout the detector volume. Consequently, a crucial task of the PID step is the determination of the true physical origin of these trajectories, in order to effectively anchor and terminate them at their source. Without a well-defined starting point, essential kinematic quantities—most notably the momentum vector and the path length required for energy-loss corrections—cannot be uniquely or reliably determined.

This issue becomes particularly severe in the presence of Restgas background, where interactions occur at arbitrary locations along and within the beam pipe rather than being confined to the nominal target region. Under these conditions, the IP determination approach in the PID step must be reconsidered.

Limitations of the Standard Approach: In the standard PandaRoot workflow, a strong geometric constraint is imposed by assuming that particle trajectories originate from the beam axis near the nominal interaction point. As a result, tracks

identified by the finder are processed by a Kalman Filter and typically propagated forward by $\approx 2\,\text{cm}$, then Kalman-fitted tracks are propagated to the point of closest approach to the z-axis (POCA-to-line), which is then used to define the reference momentum vector and path length for the PID step.

While this approximation is well justified for interactions localized within the target region, it leads to significant systematic biases when applied to interactions with Restgas particles. Owing to the beam complication and the extended longitudinal density profile of the Restgas background, such interactions are distributed stochastically along the vacuum pipe and may exhibit substantial displacements in both transverse (x, y) and longitudinal (z) directions. Forcing a helical trajectory to reference the beam axis when its true origin is displaced by several centimeters can result in severe reconstruction distortions. In particular, transverse offsets may cause the geometric solver to fail or produce ambiguous solutions, while longitudinal displacements lead to an incorrect estimation of the particle's flight path through the detector material. This path-length miscalculation directly affects the energy-loss (dE/dx) correction and introduces a systematic bias in the reconstructed polar angle θ, as the momentum vector is effectively evaluated at an unphysical reference plane.

Modified POCA-First Approach: To address these topological limitations, a modified reconstruction scheme, referred to as the **POCA-First Approach**, was developed. Starting from the global track candidates seeded by the Apollonius finder and produced by the Kalman filter, the algorithm analytically computes the three-dimensional point of closest approach (POCA) of all the tracks of the elastically scattered $\bar{p}p$ events. The resulting coordinate, $\vec{v}_{\text{POCA}}$, provides an estimator of the true interaction vertex and serves as the physical anchor point for subsequent kinematic reconstruction.

Once the geometric vertex $\vec{v}_{\text{POCA}}$ is established, the Kalman-fitted tracks perform a targeted back-propagation. The track parameters are propagated from the outer detectors back specifically to the point of closest approach, to the 3D coordinate $\vec{v}_{\text{POCA}}$ (POCA-to-point), rather than to the z-axis. This POCA-to-point back-propagation effectively reconstructs the IP for displaced vertices, ensuring that the track parameters are evaluated at the actual interaction location. It is at this location that the momentum vector is evaluated, and the path length is determined for the energy loss calculation. This ensures that the PID algorithms receive kinematic inputs that correctly reflect the particle's history through the detector material and magnetic field up to the actual point of interaction.

By anchoring the track parameters to the reconstructed interaction vertex rather than an arbitrary axis, this approach effectively eliminates the systematic biases introduced by displaced vertices, thereby enabling accurate IP and momentum reconstruction and particle identification for events from beam-Restgas interactions. The validity of using this calculated geometric estimator was rigorously verified through simulation studies.

7.2 Performance Verification and Restgas Determination

With the implementation of the Apollonius triplet track finder and the POCA-First PID approach, the full reconstruction chain is established. This section validates the effectiveness of this workflow using Monte Carlo simulations and presents the final results for the Restgas profile determination.

Validation of the POCA-First Approach

The core premise of the proposed approach is that the geometric IP derived from tracking ($\vec{v}_{\mathrm{POCA}}$) is a sufficient substitute for the true vertex ($\vec{v}_{\mathrm{MC}}$) for the back-propagation, even for events significantly displaced from the nominal IP. To rigorously verify this, a simulation study was performed using 10,000 $\bar{p}p$ elastic scattering events generated by the DPM generator at a beam momentum of $p_{\mathrm{beam}} = 4.06\,\mathrm{GeV}/c$. Crucially, the simulation setup introduced displacements in all three dimensions to mimic a realistic scenario, with the true Monte Carlo vertex fixed at:

$$\vec{v}_{\mathrm{MC}} = (-0.2, 0.4, -5.0)\,\mathrm{cm}$$

The reconstruction performance was evaluated in two steps: the geometric accuracy of the vertex finder and the fidelity of the final IP reconstruction.

Geometric Accuracy: The distribution of the reconstructed POCA vertex for each event was analyzed to assess the intrinsic resolution of the tracking system without a back-propagation constraint. As shown in Figure 7.1, the POCA vertex yields a distribution centered near the true value with a finite spread due to tracking uncertainties. The reconstructed vertex coordinates were determined to be:

$$\vec{v}_{\mathrm{POCA}} = (-0.31 \pm 0.16,\ 0.42 \pm 0.04,\ -4.53 \pm 0.07)\,\mathrm{cm}.$$

Both the transverse (x, y) and longitudinal (z) reconstructions exhibit deviations and widening from the true vertex. This deviation represents the challenge of identifying the vertex purely from track topology in the absence of back-propagation to a fixed vertex.

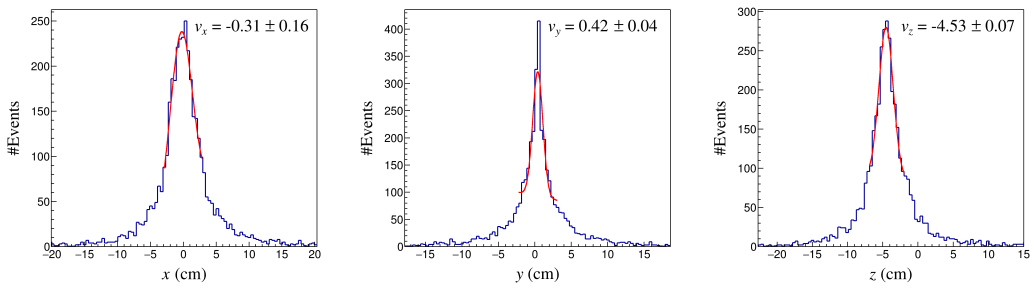


Figure 7.1: The distribution of reconstructed POCA vertices ($\vec{v}_{\mathrm{POCA}}$) for simulated events with a true vertex at $x = -0.2\,\mathrm{cm}$, $y = 0.4\,\mathrm{cm}$, $z = -5.0\,\mathrm{cm}$.

IP Consistency: However, the critical metric of the reconstruction is not the POCA vertex resolution itself, but its impact on the final physics observables. To quantify this, the full reconstruction chain was executed under two conditions: back-propagating to the MC vertex $\vec{v}_{\mathrm{MC}}$ or the POCA vertex $\vec{v}_{\mathrm{POCA}}$ evaluated for each event separately.

The results are summarized in Table 7.1. Overall, the POCA-First approach achieves performance consistent with the MC method. Remarkably, despite the initial geometric deviations of the raw POCA vertex, the final reconstructed mean values for the event after the vertex fit are virtually indistinguishable between the two methods. For instance, the final reconstructed z-position of the interaction (after full fitting) converges to $-4.92\,\mathrm{cm}$ using the POCA approach, compared to $-4.89\,\mathrm{cm}$ using the MC truth value, demonstrating an excellent recovery of the event geometry. Most importantly, the global reconstruction efficiency remains identical at about **74%** for both cases.

Table 7.1: Comparison of reconstruction results between the back-propagation to the MC vertex $\vec{v}_{\mathrm{MC}}$ and the POCA vertex $\vec{v}_{\mathrm{POCA}}$.

Parameter	**MC Truth**	**Prop to $\vec{v}_{\mathrm{MC}}$**	**Prop to $\vec{v}_{\mathrm{POCA}}$**
Vertex x [cm]	-0.200	-0.197	-0.198
Vertex y [cm]	0.400	0.401	0.399
Vertex z [cm]	-5.000	-5.001	-5.001
Global Efficiency	-	**74.2%**	**74.4%**

The visual comparison in Figure 7.2 further confirms this agreement. The IP distributions after the whole fitting procedure derived from $\vec{v}_{\mathrm{POCA}}$ overlay perfectly with those from $\vec{v}_{\mathrm{MC}}$. This quantitative evidence proves that the POCA-First approach is highly robust: the calculated POCA vertex is sufficiently accurate to anchor the track for the back-propagation, allowing for a precise spatial reconstruction of off-IP events.

To ensure that this approach does not degrade the reconstruction of standard physics channels originating from the nominal interaction point, a baseline comparison was performed using events generated exactly at the origin $(0, 0, 0)$. The results are summarized in Table 7.2. Both the standard POCA-to-axis method and the proposed POCA-First approach successfully reconstruct the primary vertex with excellent sub-millimeter precision. While the POCA-First method exhibits a slight reduction in global efficiency (82.10% compared to 84.38% for the standard baseline), this minor trade-off is highly acceptable. The negligible loss in efficiency at the nominal IP is vastly outweighed by the algorithm's capability to successfully reconstruct broadly distributed off-IP events.

Workflow for Restgas Distribution Determination

Based on the validated track finding algorithm with Apollonius triplet track finder and POCA-First approach, the complete reconstruction workflow for the determination of the Restgas profile is introduced and formalized. This specific sequence,

Table 7.2: Comparison of the reconstructed vertex coordinates and global tracking efficiencies for events generated at the nominal interaction point $(0, 0, 0)$.

Parameters	**MC Truth**	**POCA-to-axis**	**POCA-First**
Vertex x [cm]	0.0	-0.0007	-0.0006
Vertex y [cm]	0.0	-0.0001	0.0002
Vertex z [cm]	0.0	-0.0038	-0.0006
Global Efficiency	–	84.38%	82.10%

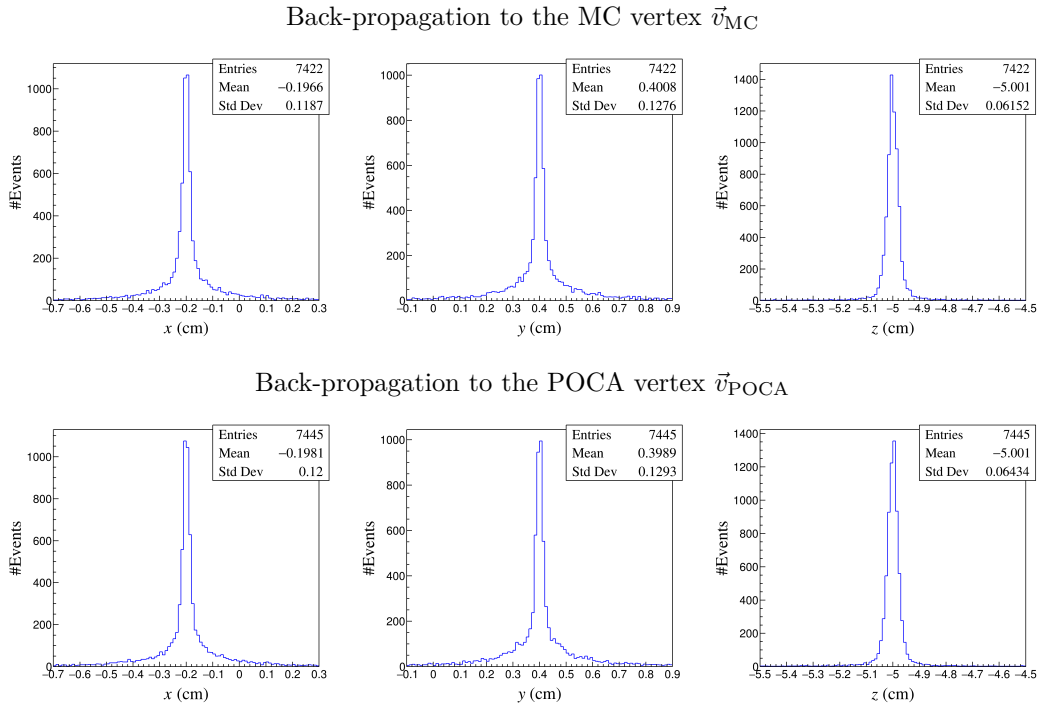


Figure 7.2: Comparison of the IP reconstruction performance by using the MC truth vertex *(top)* or the calculated $\vec{v}_{\mathrm{POCA}}$ *(bottom)*.

illustrated in Figure 7.3, outlines the data processing chain from raw signals to the final IP determination.

The workflow initiates with the processing of experimental data. The raw hits from the STT are first utilized by the Apollonius algorithm to generate initial track seeds, which are subsequently refined by the Kalman Filter with hits from MVD and GEM to produce fitted trajectories. Following the tracking stage, a first PID step is performed. At this intermediate stage, the PID step is executed without specific back-propagation, utilizing standard assumptions to provide an initial particle hypotheses. These preliminary tracks are then paired to analytically calculate the geometric POCA, providing the first reliable 3D estimate of the event vertex.

Crucially, the workflow then implements the POCA-First approach. Using the spatial coordinates of the determined POCA vertex as the definitive anchor, the original Kalman-fitted tracks are re-accessed and subjected to a second, targeted PID step.

Here, the tracks are back-propagated specifically to the POCA vertex to extract the momentum vectors and energy loss corrections. Finally, these physically corrected kinematic objects serve as the input for the global vertex fit, leading to the accurate determination of the IP and the subsequently to the Restgas density profile.

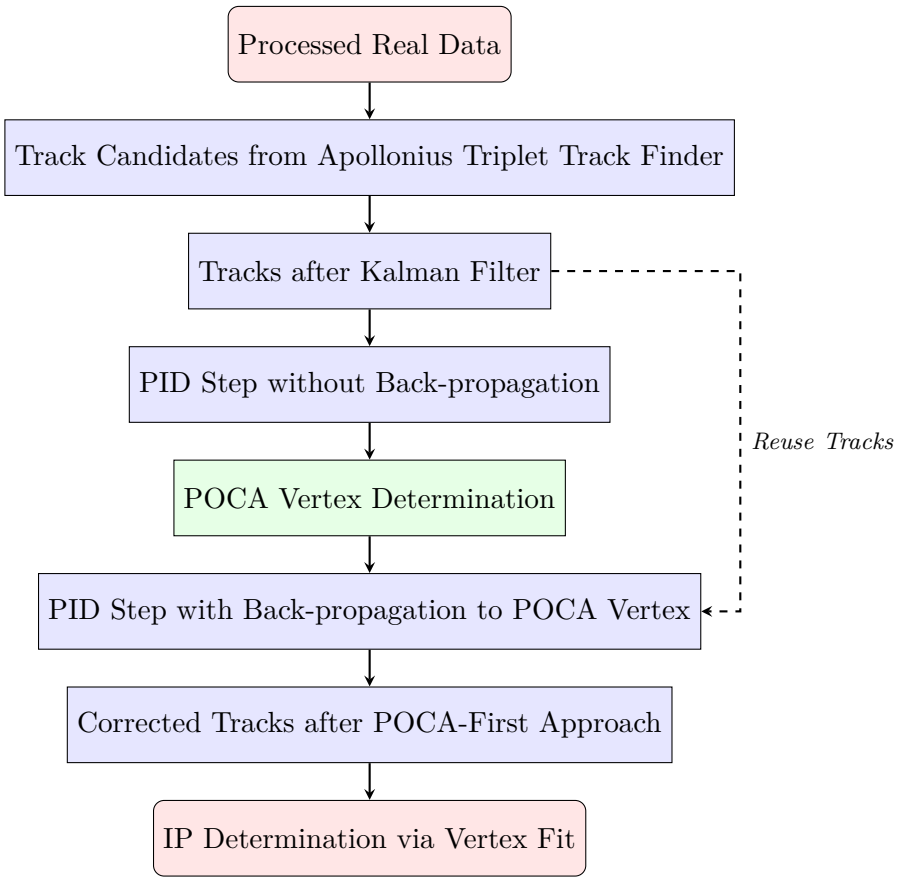


Figure 7.3: The reconstruction workflow for Restgas determination. The solid lines indicate the sequential data flow, while the dashed line represents the re-utilization of Kalman tracks for the PID step anchored to the POCA vertex.

Final Restgas Profile

By applying the formal workflow described above to the full simulation sample, the final reconstructed Restgas profile was determined. Figure 7.4 illustrates the reconstructed IP distributions in the x, y, and z directions. The reconstructed profiles accurately reflect the true geometry of the beam-Restgas interactions across all three spatial dimensions. In the transverse plane (x and y), the interaction vertices exhibit a narrow Gaussian distribution with $\sigma = 1\,\text{mm}$, characterizing the primary beam profile. Conversely, the longitudinal distribution (z) extends continuously along the whole beam pipe. This effectively demonstrates the comprehensive spatial coverage of the Restgas events successfully recovered by the POCA-First approach.

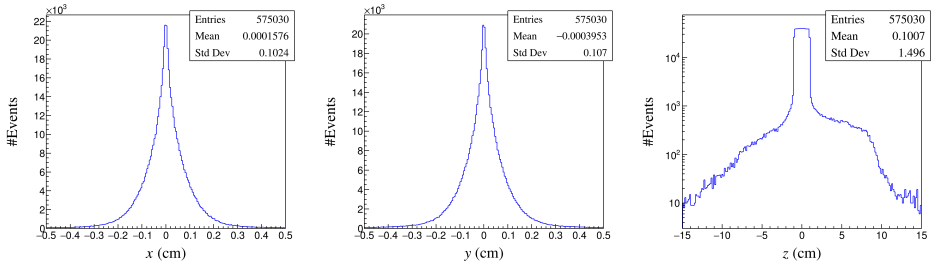


Figure 7.4: The reconstructed IP distributions in x, y, and z directions with Restgas present.

To rigorously quantify the performance of the strategy, the spatial resolution was benchmarked against an "Ideal Reference" scenario, where tracks are back-propagated directly to the true Monte Carlo vertex ($\vec{v}_{\text{MC}}$). Figure 7.5 presents the vertex resolution separately for the measured POCA-vertex *(top)* and the ideal MC-vertex *(Bottom)*. The highly similar distribution widths (σ) confirm that the POCA-First approach preserves geometric precision.

Comparing the two sets of distributions reveals a remarkable consistency. Quantitatively, the Gaussian widths of the reconstructed IP residuals using the determined $\vec{v}_{\text{POCA}}$ were measured to be $\sigma_{x,y,z} = (0.0159, 0.0163, 0.0183)\,\text{cm}$. These values are highly comparable to the residuals of $\sigma_{x,y,z} = (0.0164, 0.0167, 0.0203)\,\text{cm}$ obtained using the $\vec{v}_{\text{MC}}$. This confirms that using the measured $\vec{v}_{\text{POCA}}$ for the back-propagation is sufficiently precise.

Regarding the reconstruction efficiency, a minor reduction was observed: the global efficiency decreased from **61.4%** for using the ideal MC vertex to **59.1%** for using the measured POCA vertex. This efficiency loss of approximately 2.3 percentage points is primarily attributed to the subset of events where the tracking ambiguity prevents a stable POCA convergence. However, considering the substantial gain in geometric accuracy for off-IP events and the resolution preserved in the successful reconstructions, this marginal efficiency cost is well-justified. In conclusion, the method succeeds in delivering a high-precision determination of the Restgas profile.

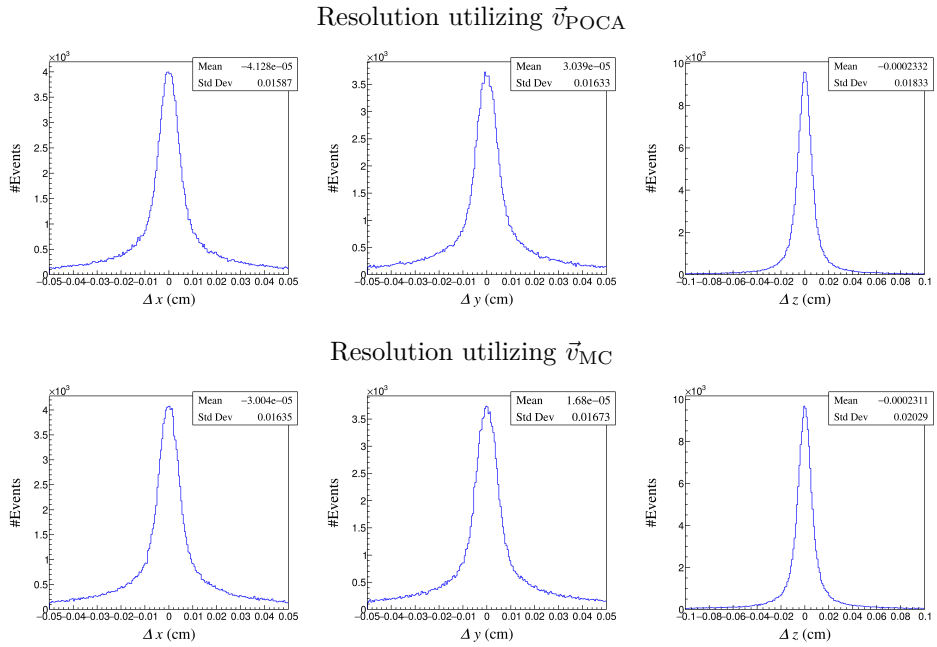


Figure 7.5: Comparison of spatial resolution $(\Delta x, \Delta y, \Delta z)$. *(top)* The resolution obtained using the measured $\vec{v}_{\mathrm{POCA}}$ for the back-propagation. *(bottom)* The ideal resolution obtained using the true $\vec{v}_{\mathrm{MC}}$ for the back-propagation.

7.3 Efficiency Analysis and Correction

The strategy described in the previous sections successfully reconstructed the IP of the single event, but the measured spectrum does not tell about the true physical density of the Restgas background. Due to the limited geometrical acceptance of the tracking system and the kinematic cuts applied during event selection, the detection probability varies significantly depending on the IP where the event originates. To extract the intrinsic Restgas profile $\rho_{gas}(z)$ from the raw reconstructed data, an efficiency correction must be applied.

Before establishing the correction function, it is essential to determine the necessary dimensionality of the efficiency map. Specifically, we must ascertain whether the detection efficiency ε exhibits a significant dependence on the transverse and longitudinal coordinates (x, y, z), which would necessitate a complex 3D correction map $\varepsilon(x, y, z)$, or if a simplified 1D curve $\varepsilon(z)$ or even a constant ε_0 suffices for the required precision.

To answer this question, a systematic simulation study was performed using the DPM generator to scan the interaction volume. The study covered a longitudinal range from $z = -10\,\mathrm{cm}$ to $z = +10\,\mathrm{cm}$ with a step size of $2.5\,\mathrm{cm}$. At each fixed longitudinal node, the efficiency was evaluated at various transverse offsets, covering $x, y \in \{\pm 0.1, \pm 0.2, \pm 0.3, \pm 0.5, \pm 0.8\}\,\mathrm{cm}$. This grid was designed to fully encompass the region of interest for beam-target and beam-Restgas interactions.

The results of this dependency study are summarized in Figure 7.6. The analysis reveals a dramatic variation in efficiency along the beam axis (z). As the interaction point shifts away from the detector center, the efficiency drops precipitously from a maximum of $\sim 81\%$ at $z = 0\,\text{cm}$ to approximately 6% at the downstream edge $z = 10\,\text{cm}$, primarily due to the acceptance coverage of the $\overline{\text{P}}$ANDA Target Spectrometer. In strong contrast, the efficiency remains remarkably stable with respect to transverse displacements. Within the central core region ($|x|, |y| < 0.5\,\text{cm}$), the variation in efficiency is negligible. Considering that the transverse profile of the real beam-Restgas interaction is tightly constrained by the beam emittance (typically well within the $\pm 0.5\,\text{cm}$ window), the transverse efficiency uniformity holds true for the experimental conditions.

Consequently, this study concludes that the transverse dependence can be safely neglected. A 1D longitudinal efficiency curve $\varepsilon(z)$ provides a sufficient and robust model for the correction of the reconstructed data. This finding allows for the precise recovery of the Restgas distribution by simply weighting the reconstructed z-spectrum by the inverse of the z-dependent efficiency, significantly simplifying the analysis framework without compromising physical accuracy.

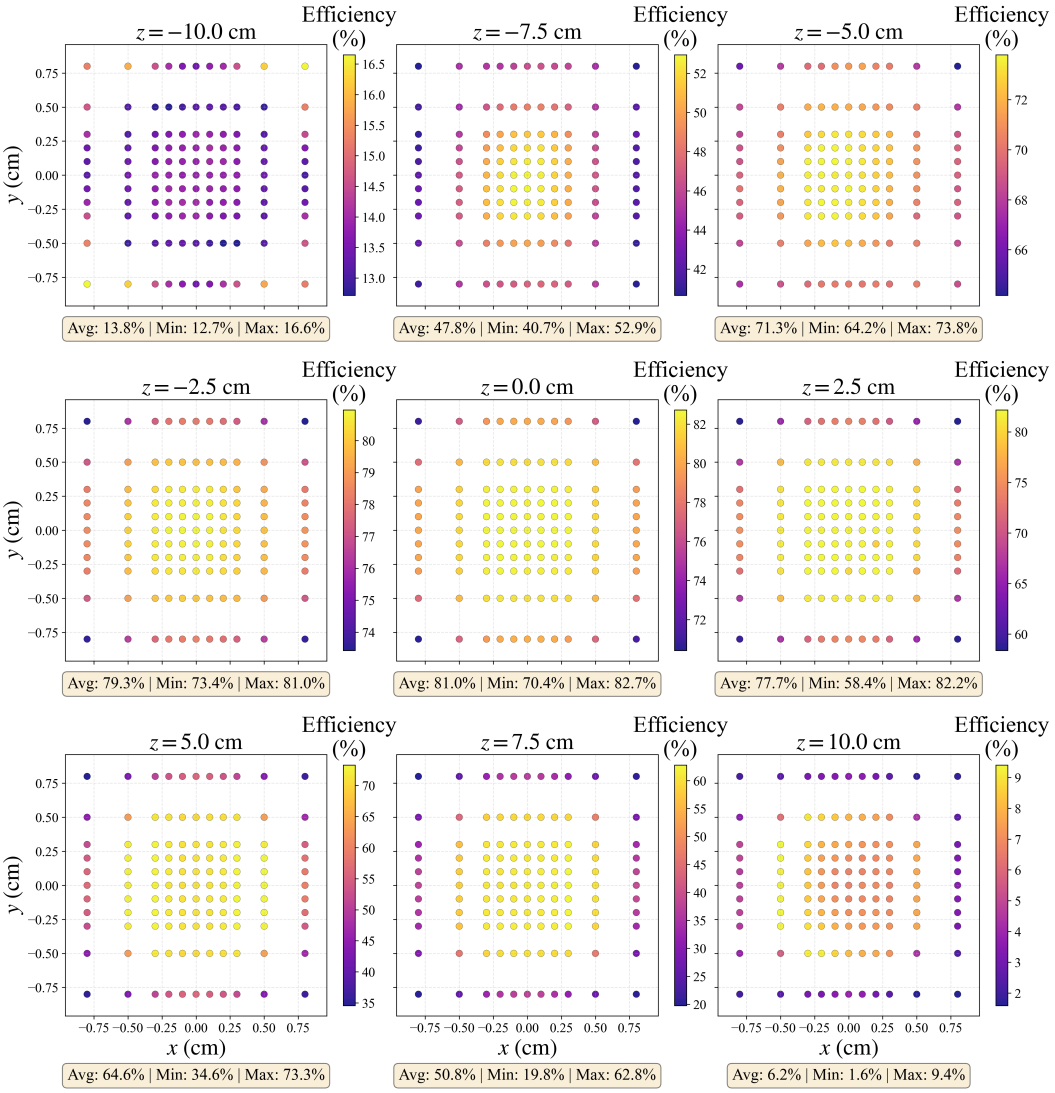


Figure 7.6: The dependence of detection efficiency on interaction position. z positions from $-10\,\text{cm}$ to $+10\,\text{cm}$ with a step size of $2.5\,\text{cm}$ are shown in x and y coordinates.

Chapter 8

Determination of the Restgas Profile

Building upon the reconstruction algorithms established in Chapter 7, specifically the Apollonius tracking and the POCA-based vertexing, this chapter evaluates the final performance of the Restgas profile measurement. The primary objective is to extract the longitudinal density distribution, $\rho_{gas}(z)$, from the reconstructed event rate. This process requires a rigorous handling of detector acceptance limits and, crucially, the correction of resolution effects arising from the extreme density gradients between the target and the Restgas background.

8.1 Definition of the Measurement Range

Unlike conventional analyses that focus solely on the immediate vicinity of the Interaction Point (IP), the Restgas determination requires a global perspective along the entire beam axis. The measurement range is explicitly defined as $z \in [-3.0, +6.0]\,\mathrm{m}$, a decision driven by the physical constraints of the $\overline{\mathrm{P}}$ANDA vacuum system and the tracking detector layout. The upstream limit at $z = -3.0\,\mathrm{m}$ is determined by the location of the powerful cryogenic pump system in the beam line. As illustrated in the vacuum profile simulation (Figure 3.14), the Restgas density drops by several orders of magnitude beyond this point due to the effective pumping, rendering the gas contribution physically negligible. Similarly, the downstream boundary at $z = +6.0\,\mathrm{m}$ is constrained by the geometrical acceptance of the Forward Tracker (FT), which features its last two layers at $z = 6075\,\mathrm{mm}$ and $z = 6545\,\mathrm{mm}$. Tracks originating significantly beyond this point cannot produce sufficient hits to form reconstructible candidates. Crucially, the vacuum distribution studies indicate that this defined region contains more than 99% of the total integrated Restgas density. Consequently, the distribution measured within this global range effectively represents the complete Restgas profile required for the luminosity determination, thereby eliminating the need for model-dependent extrapolations outside the detector acceptance.

8.2 Resolution Effects and Profile-Dependent Efficiency

A more subtle yet critical challenge in the efficiency correction arises from the finite spatial resolution of the detector, particularly at the interface between the target and the Restgas background. The particle density of the target in the central interaction region ($z \approx \pm 1\,\text{cm}$) is approximately 10^2 times higher than that of the surrounding Restgas background. This extreme density contrast, coupled with the stochastic vertex smearing, necessitates a correction strategy that goes beyond simple geometrical acceptance.

To quantify this effect, an efficiency map was generated using a realistic Monte Carlo simulation, taking the density profile provided by the Target Group as the input model. The resulting efficiency curve (shown as the orange curve in Figure 8.1) differs fundamentally from the one derived using a standard uniform density profile ($\rho_{MC} = \text{const}$, shown as the green curve). A specific feature in the realistic profile-dependent map highlights the impact of spatial resolution smearing. In the transition region immediately adjacent to the target ($z \in [-10.0, +10.0]\,\text{cm}$), the calculated efficiency $\epsilon(z) = N_{reco}/N_{gen}$ forms distinct peaks that drastically exceed unity ($\epsilon > 100\%$). This anomaly is a direct manifestation of the "spill-over" effect: due to resolution smearing, the massive migration of events out of the high-density target bins significantly outnumbers the genuine Restgas events generated in those adjacent locations. In contrast, a uniform MC distribution yields a smooth, bounded efficiency curve ($\epsilon_{uniform} \leq 1$) that severely underestimates the true efficiency in this critical transition region. In such a uniform scenario, the migration of events is inherently symmetric—the number of events smearing into a specific bin is roughly balanced by those smearing out. Consequently, applying a uniform efficiency map completely fails to capture the net migration artifact present in the real data. Doing so would incorrectly interpret the target "spill-over" events as a genuine Restgas signal, ultimately leading to a gross overestimation of the actual gas density near the interaction point.

To address this resolution smearing, mathematically deconvolving the detector resolution from the measured data would in principle be possible. However, deconvolution is notoriously sensitive to noise and often becomes mathematically ill-posed, especially in low-statistics regions. Therefore, a "forward modeling" approach is adopted. Instead of attempting to remove the resolution from the data, the resolution effects are incorporated directly into the efficiency correction. Monte Carlo samples are generated using a realistic input density profile, $\rho_{in}(z)$, which includes the high-density target peak and mimics the true physics distribution. As a result, the reconstructed events N_{reco} naturally reproduce the same resolution smearing and target-to-Restgas migration observed in the experimental data, while the generated MC events N_{gen} represent the underlying truth information. The efficiency is defined as the ratio of reconstructed to generated MC events $\varepsilon = N_{reco}/N_{gen}$, effectively acting as a convolved correction function that encapsulates both geometrical acceptance and resolution-induced migration. When this specific efficiency map is applied to the experimental data, the resolution effects present in the numerator are mathematically cancelled by those modeled in the efficiency, enabling the recovery of the true underlying Restgas profile without the numerical instabilities associated with deconvolution.

This choice, however, introduces a potential logical paradox: the generation of the correct efficiency map seems to require prior knowledge of the very Restgas distribution to be measured. This creates a circular dependency, as the input profile must be known before the measurement can be performed. In practice, this circularity is resolved by treating the measurement as an iterative process. The nominal Restgas profile provided by the Target Group, which is based on vacuum simulations and nozzle geometry, serves as the initial estimator. While this theoretical prediction may contain uncertainties regarding the absolute density or the precise shape, it provides a sufficient topological baseline to model the resolution migration. The strength of the method lies in its ability to refine this initial guess: the $\overline{\text{P}}$ANDA detector's high-precision tracking provides experimental feedback that is significantly more sensitive than the initial model.

Therefore, the critical question is not whether the initial input is perfect, but rather whether the method is robust against imperfections in the input. It must be verified if a distorted initial profile leads to a biased final measurement, or if the method converges to the true physical distribution regardless of the starting point.

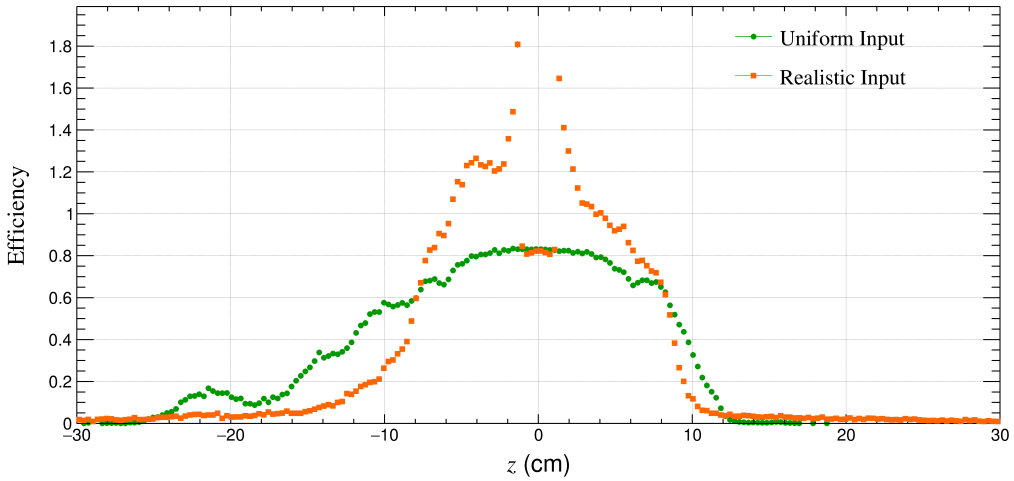


Figure 8.1: The profile-dependent efficiency map $\varepsilon(z)$ (orange) generated using the Target Group's density profile as input compared to a uniform efficiency map (green) in the central interaction region ($z \in [-30, +30]$ cm).

8.3 Check with the Ideal Case

To validate the stability of the efficiency correction and to address the circular dependency issue, systematic studies were performed. The core objective was to quantify the sensitivity of the final measurement to inaccuracies in the initial input model. This was achieved by intentionally introducing discrepancies between the "true" physical distribution (used to generate the pseudo-data) and the "input" model (used to generate the efficiency map).

The intrinsic consistency of the method was first verified under ideal conditions where the input profile used for efficiency generation matches the true physical distribution. This baseline scenario serves as a sanity check to confirm that the method can accurately recover the Restgas profile when the model is correct. Figure 8.2 presents the comparison between the reconstructed Restgas profile and the initial truth distribution over the full range of $z \in [-3.0, +6.0]$ m. This result represents the ideal control scenario where the input distribution used to generate the efficiency map is identical to the true physical distribution. To accommodate the large density gradients, the vertical axis is plotted in logarithmic scale. Additionally, the inset provides a zoomed view of the central interaction region ($z \in [-30, +30]$ cm) to visualize the reconstruction quality in the high-density zone. The reconstructed results exhibit excellent agreement with the true distribution across the entire longitudinal range. In the critical central region ($z \in [-30, +30]$ cm), the method achieves a high precision with a loss of events during reconstruction of $-0.24 \pm 0.29\%$. In the extended upstream and downstream regions, slightly larger fluctuations are observed due to the reduced statistics and lower detection efficiency. However, these deviations remain within acceptable limits. For the full range covering $z \in [-3.0, +6.0]$ m, the $1.16 \pm 1.27\%$ more events are reconstructed. This result confirms that the profile-dependent efficiency correction successfully recovers the integrated Restgas density and shape over the full experimental domain.

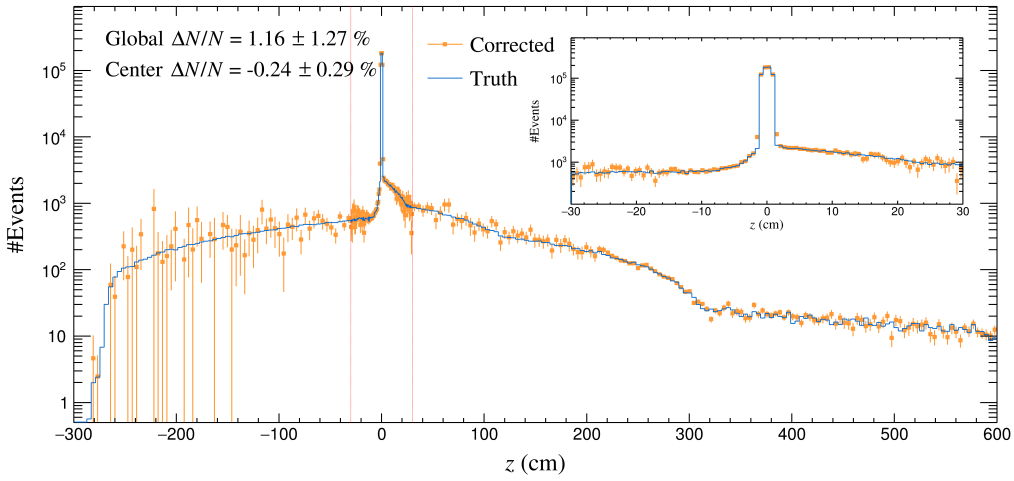


Figure 8.2: Baseline validation of the efficiency correction algorithm over the full range ($z \in [-3.0, +6.0]$ m). The main plot displays the comparison between the reconstructed Restgas density profile (orange points) and the true Monte Carlo distribution (blue solid line) on a logarithmic scale. The Red lines indicate the zoom-in region and the inset shows a zoomed-in view of the central interaction region ($z \in [-30, +30]$ cm).

8.4 Robustness Verification and Results

Moving beyond the ideal case to realistic experimental scenarios, the initial Restgas profile provided by the Target Group serves as the input for efficiency generation, but

it may deviate from the actual physical distribution. To systematically evaluate the robustness of the method against such model uncertainties, a comprehensive stress test was performed. The density of the Restgas component in the simulation was varied by 0%, ±5%, ±10%, and ±20%, while keeping the density of the central target region ($|z| < 1$ cm) constant. By combining different "True" (physics generator) and "Input" (efficiency generator) configurations, a matrix of scenarios was created, representing relative deviations between the assumed and actual Restgas content ranging from 0% to ±40%.

Figure 8.3 illustrates one specific scenario where the input model underestimates the true Restgas contribution by approximately 10% (Input: −5% vs. True: +5%). Despite this significant discrepancy between the true and input distributions, the reconstructed result remains qualitatively consistent with the baseline performance. The reconstruction aligns well with the true distribution across the entire global range. Quantitatively, the relative deviation in the high-statistics central region ($|z| < 30$ cm) is determined to be a loss in events of $-0.46 \pm 0.29\%$, while for the full range $0.71 \pm 1.28\%$ events more are reconstructed. This confirms that the algorithm effectively suppresses the bias introduced by the incorrect model for the efficiency determination (Input model). The results for the full set of scenarios are summarized in Table 8.1. The scenarios are categorized by the relative difference between the Input and True distributions.

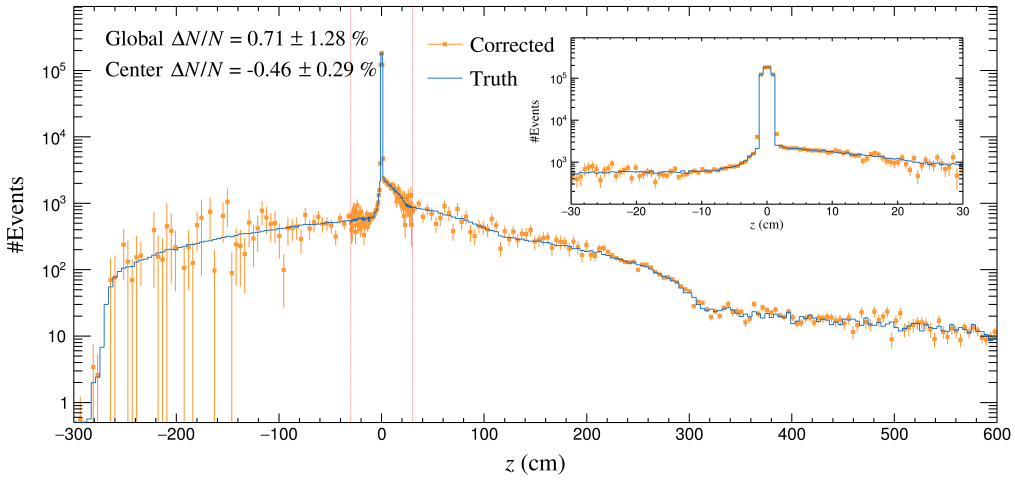


Figure 8.3: Results for the scenario with a **10% underestimation** of the Restgas distribution in the input model. The main plot displays the comparison between the reconstructed Restgas density profile (orange points) and the true Monte Carlo distribution (blue solid line) in logarithmic scale. The Red lines indicate the zoom-in region and the inset shows a zoomed-in view of the central interaction region ($z \in [-30, +30]$ cm).

Table 8.1: Summary of systematic reconstruction deviations under various mismatch scenarios between the Input model (efficiency generation) and the True physical distribution. The "Relative Difference" in the number of events $\Delta N/N$ is defined as (Input − True)/True. The deviations are evaluated for both the central window ($|z| < 30$ cm) and the whole range ($z \in [-3.0, +6.0]$ m).

Relative Diff.	True Dist. (Restgas Scale)	Input Dist. (Restgas Scale)	Global Dev. [%] $z \in [-30, +30]$ cm	Central Dev. [%] $z \in [-3.0, +6.0]$ m
0%	−20%	−20%	0.40 ± 1.01	-0.59 ± 0.25
	−10%	−10%	-1.41 ± 1.34	-0.50 ± 0.27
	−5%	−5%	-1.23 ± 1.05	-0.85 ± 0.29
	Nominal	Nominal	1.16 ± 1.27	-0.24 ± 0.29
	+5%	+5%	-2.23 ± 1.09	-0.86 ± 0.29
	+10%	+10%	-0.03 ± 1.18	-0.74 ± 0.31
	+20%	+20%	-0.41 ± 1.14	-0.65 ± 0.31
−5%	+5%	Nominal	1.55 ± 1.46	-0.70 ± 0.28
	Nominal	−5%	-2.73 ± 1.01	-0.50 ± 0.30
	−5%	−10%	-1.13 ± 1.16	-0.74 ± 0.27
	+10%	+5%	0.14 ± 1.39	-0.86 ± 0.30
+5%	−5%	Nominal	0.00 ± 1.17	-1.01 ± 0.28
	Nominal	+5%	-0.65 ± 1.31	-0.26 ± 0.30
	+5%	+10%	3.39 ± 1.37	0.15 ± 0.31
	−10%	−5%	1.16 ± 1.32	-0.41 ± 0.29
−10%	+5%	−5%	0.71 ± 1.28	-0.46 ± 0.29
	+10%	Nominal	-0.78 ± 1.32	-0.40 ± 0.29
	Nominal	−10%	-2.97 ± 1.18	-0.52 ± 0.28
	+20%	+10%	-0.46 ± 1.14	-0.51 ± 0.31
	−10%	−20%	-1.54 ± 1.38	-0.97 ± 0.26
+10%	−5%	+5%	1.44 ± 1.34	-0.51 ± 0.29
	−10%	Nominal	1.38 ± 1.21	-0.69 ± 0.28
	Nominal	+10%	-0.22 ± 1.13	0.07 ± 0.31
	+10%	+20%	3.27 ± 1.99	-0.64 ± 0.32
	−20%	−10%	2.69 ± 1.09	0.02 ± 0.26
−15%	−5%	−20%	-2.72 ± 1.18	-2.38 ± 0.26
	+5%	−10%	0.07 ± 1.49	-1.40 ± 0.28
	+10%	−5%	-2.30 ± 1.11	-1.20 ± 0.30
	+20%	+5%	-1.14 ± 1.31	-0.87 ± 0.30
+15%	−20%	−5%	4.12 ± 1.19	0.96 ± 0.28
	−10%	+5%	1.85 ± 1.21	-0.22 ± 0.29
	−5%	+10%	1.62 ± 1.18	-0.28 ± 0.30
	+5%	+20%	5.00 ± 1.96	0.04 ± 0.31

Relative Diff.	True Dist. (Restgas Scale)	Input Dist.	Global Dev. [%] $z \in [-30, +30]$ cm	Central Dev. [%] $z \in [-3.0, +6.0]$ m
−20%	+10%	−10%	-2.68 ± 1.16	-2.15 ± 0.29
	+20%	Nominal	-2.04 ± 1.14	-1.33 ± 0.29
	Nominal	−20%	-4.52 ± 1.23	-2.17 ± 0.27
+20%	−10%	+10%	3.37 ± 1.32	0.04 ± 0.29
	−20%	Nominal	5.34 ± 1.42	0.71 ± 0.27
	Nominal	+20%	3.02 ± 3.21	-0.01 ± 0.31
−25%	+5%	−20%	-1.59 ± 1.42	-2.03 ± 0.27
	+20%	−5%	-2.29 ± 1.38	-1.09 ± 0.30
+25%	−5%	+20%	3.47 ± 1.51	-0.33 ± 0.30
	−20%	+5%	5.79 ± 1.31	1.17 ± 0.28
−30%	+10%	−20%	-4.22 ± 1.16	-2.77 ± 0.28
	+20%	−10%	-1.48 ± 2.43	-2.07 ± 0.28
+30%	−10%	+20%	4.78 ± 1.84	0.00 ± 0.30
	−20%	+10%	7.29 ± 1.41	1.46 ± 0.29
−40%	+20%	−20%	-3.85 ± 2.27	-2.68 ± 0.28
+40%	−20%	+20%	12.09 ± 5.69	1.36 ± 0.29

An analysis of Table 8.1 reveals a clear trend: when the absolute deviation between the efficiency input and the true Restgas distribution is within 10%, the reconstruction quality in the central region is excellent, with the systematic deviation consistently maintained below 1%. Outside the central region, the fluctuations are slightly larger due to lower statistics but remain within acceptable limits. Even when the deviation exceeds 10%, the magnitude of the deviation is significantly smaller than the initial input bias, demonstrating that the measurement process acts as a contracting map. Based on these observations, a robust iterative workflow is established for the application to real experimental data:

1. **Initialization:** The initial efficiency map is derived from a simulation of approximately 5×10^6 events using the DPM generator, taking the density profile provided by the Target Group as the input distribution.

2. **Reconstruction:** The experimental data is reconstructed using this efficiency map.

3. **Convergence Check:** The resulting reconstructed profile ρ_{reco} is compared with the input profile ρ_{input} used for the efficiency correction. If the deviation in the central region is lower than 1% and the global deviation is below its uncertainty, the result is considered stable, and ρ_{reco} is accepted as the final target profile with Restgas distribution.

4. **Iteration:** If the deviation exceeds the threshold, the reconstructed profile ρ_{reco} is adopted as the new input model to generate an updated efficiency map. The process (Steps 2-3) is repeated until the convergence criterion is met.

Since the deviation after reconstruction is shown to be much smaller than the input deviation (contracting behavior), this iterative procedure is guaranteed to converge rapidly to the true physical distribution, ensuring a model-independent measurement of the Restgas profile.

Chapter 9

Summary of the Luminosity Determination

The capability of the $\overline{\text{P}}$ANDA detector, specifically the upgraded Micro Vertex Detector (MVD) utilizing HVMAPS technology has been established, to perform a precise, data-driven measurement of the Restgas density profile over the full range of $z \in [-3.0, +6.0]$ m. The superior vertexing resolution and reduced material budget of the sensors are critical factors that enable the effective determination of the Restgas distribution even for interactions significantly displaced from the nominal IP.

Through the implementation of a profile-dependent efficiency correction and a robust iterative workflow, the systematic distortions caused by resolution smearing and initial model inaccuracies are rigorously suppressed. This method acts as a contracting map, converging to the true physical distribution and ensuring that the final result is independent of the initial simulation input provided by the Target Group. To quantify the stability of this approach, an iterative procedure was initialized with an extreme starting condition: a 40% overestimation of the Restgas distribution. As summarized in Table 9.1, the iterative procedure converges rapidly. Within merely three steps, the global deviation drops from 12.27% to 1.30%, and the deviation in the central region vanishes to -0.22%. Consequently, the precise density profile can be derived directly from the experimental data across the entire relevant domain, providing the reliable input essential for the absolute luminosity determination.

Finally, the determined Restgas profile was integrated back into the *LuminosityFit* framework to validate the complete analysis chain. The luminosity determination was performed at a beam momentum of $p_{beam} = 4.06$ GeV/c, applying a cut within the analysis window ($|x|, |y| < 1$ cm and $|z| < 5$ cm). The relative deviation of the determined luminosity ($\Delta\mathcal{L}/\mathcal{L}$) is $(0.05 \pm 0.07)\%$ with the reconstructed target profile with Restgas background. As presented in Table 9.2, this result is consistent with the ideal case where the target is treated as point-like and the efficiency correction is based on a point-like model, which yields a deviation of $-0.036 \pm 0.07\%$ and also with the case where the target profile with Restgas background is well-known and

Table 9.1: Evolution of the discrepancy between the reconstructed and true Restgas profiles over successive iteration steps, starting from an initial 40% systematic overestimation. The central deviation is evaluated in the region $z \in [-30, +30]$ cm, while the global deviation is evaluated over the entire relevant domain for the full range $z \in [-3.0, +6.0]$ m.

Iteration Step	**Global Deviation [%]**	**Central Deviation [%]**
1	12.27 ± 5.69	1.58 ± 0.30
2	4.28 ± 1.19	1.18 ± 0.28
3	1.30 ± 1.32	-0.22 ± 0.29

totally the same as real data, which yields a deviation of $0.02 \pm 0.08\%$. This results demonstrate that the synergy of high-precision tracking and advanced fitting algorithms transforms the Restgas background from a limiting systematic background into a precisely monitored observable, enabling a robust and accurate luminosity determination for the $\overline{\text{P}}$ANDA experiment.

Table 9.2: Relative luminosity deviation ($\Delta\mathcal{L}/\mathcal{L}$) evaluated at $p_{beam} = 4.06$ GeV/c within the IP cut window ($|x|, |y| < 1$ cm, $|z| < 5$ cm), comparing different target configurations and reconstruction models.

Target Profiles	**Fit Model**	$\Delta\mathcal{L}/\mathcal{L}$ [%]
Point-like Target	Point-like Target	-0.04 ± 0.07
Target with Restgas	MC Target Distribution	0.02 ± 0.08
Target with Restgas	Determined Target Distribution	0.05 ± 0.07

Summary and Outlook

The $\overline{\text{P}}$ANDA experiment at the Facility for Antiproton and Ion Research (FAIR) represents the future precision frontier of hadron physics, addressing fundamental questions regarding the strong interaction, including the search for gluonic excitations, the spectroscopy of charmonium states, and the investigation of nucleon structure. A prerequisite for fully exploiting this discovery potential—particularly for the measurement of absolute cross-sections and the scanning of narrow resonance lineshapes—is the precise determination of the time-integrated luminosity. While the primary luminosity measurement relies on the detection of antiproton-proton elastic scattering in the Coulomb-Nuclear Interference region, the specific conditions of the $\overline{\text{P}}$ANDA fixed-target setup introduce significant challenges that render conventional point-source approximations insufficient. Specifically, the interaction of the antiproton beam with the Restgas background distributed along the beam pipe creates a non-localized background source which, if not correctly modeled, fundamentally limits the attainable experimental precision.

To overcome these limitations, this thesis established a comprehensive methodology that enables the effects of the extended interaction region to be quantitatively incorporated into the luminosity determination. Central to this approach was the refinement of the object-oriented software framework *LuminosityFit*, which implements a convolution-based model combining the theoretical differential cross section with the detector resolution and geometric acceptance. By incorporating the Dual Parton Model (DPM) to accurately describe the hadronic amplitude and introducing the concept of an "Effective Acceptance Map", the framework successfully integrates the complex, smeared geometry of the beam-Restgas overlap into the fitting procedure. Extensive validation studies demonstrated that this algorithmic approach can effectively account for distortions induced by beam-related systematic effects, including beam tilts, transverse offsets, and beam divergence, while recovering the input luminosity with a relative deviation of less than 0.5%. However, sensitivity analysis revealed a critical dependency: the accuracy of the extraction is strictly bound to the prior knowledge of the longitudinal Restgas density profile. A deviation of 10% in the assumed Restgas model was shown to induce a systematic luminosity bias of approximately 3.4%, a magnitude that exceeds the stringent precision requirements necessary for the $\overline{\text{P}}$ANDA physics program.

To constrain this dominant systematic uncertainty, the investigation extended beyond software algorithms to evaluate the feasibility of a hardware-driven solution utilizing the tracking system. The study proposed the upgrade of the Micro Vertex Detector (MVD) with High-Voltage Monolithic Active Pixel Sensors (HV-MAPS), leveraging technology originally developed for the Luminosity Detector itself. It

was demonstrated that the significantly reduced material budget of HV-MAPS, thinned to just 50 μm, minimizes the multiple Coulomb scattering that typically degrades tracking performance at low momenta. Coupled with a fine pixel pitch of $80 \times 80\,\mu\text{m}^2$, this hardware configuration achieves a longitudinal vertex resolution of approximately 20 μm, providing the necessary sensitivity to resolve the Restgas distribution which is inaccessible to the Luminosity Detector alone. To fully exploit this hardware potential, a specialized reconstruction chain was implemented, utilizing the Apollonius triplet finder for off-IP tracking and a "POCA-to-Point" back-propagation algorithm. This strategy optimized the reconstruction efficiency for events originating away from the nominal interaction point, allowing for the recovery of the physical density profile via a robust one-dimensional efficiency correction.

The culmination of this work is the integration of the measured Restgas profile back into the luminosity determination chain, establishing a self-consistent analysis cycle. Through the implementation of a profile-dependent efficiency correction and a robust iterative workflow, the target profile with Restgas background is successfully reconstructed without the reliance on the initial model. The reconstructed profile is then used to generate an "Effective Acceptance Map" for the luminosity fit, effectively suppressing the systematic uncertainty associated with the target profile with Restgas background to better than 0.5%. By proving that the Restgas distribution can be precisely reconstructed and accounted for, this research ensures that the deviation in the luminosity determination will not limit the $\overline{\text{P}}$ANDA experiment's ability to map the exotic hadron spectrum, thereby fulfilling the rigorous demands of next-generation hadron physics.

Bibliography

[1] PANDA Collaboration. *Physics Performance Report for PANDA: Strong Interaction Studies with Antiprotons*. 2009. arXiv: 0903.3905.

[2] W. Herr and B. Muratori. "Concept of Luminosity". In: *CERN Accelerator School and DESY Zeuthen: Accelerator Physics*. Sept. 2003, pp. 361–377.

[3] G. Barucca et al. "Precision resonance energy scans with the PANDA experiment at FAIR". In: *Eur. Phys. J. A* 55.3 (2019), p. 42. DOI: 10.1140/epja/i2019-12718-1.

[4] S. D. Campos and V. A. Okorokov. "Analysis of pp and $\bar{p}p$ Elastic Scattering Based on High Energy Bounds". In: *International Journal of Modern Physics A* 25.29 (2010), pp. 5333–5348. DOI: 10.1142/S0217751X10050755.

[5] A. Akram. "Towards Realistic Hyperon Reconstruction in PANDA: From Tracking with Machine Learning to Interactions with Residual Gas". PhD thesis. Uppsala U., 2023.

[6] A. Karavdina et al. "Offline Software for the PANDA Luminosity Detector". In: *Journal of Physics: Conference Series* 513.2 (June 2014), p. 022016. DOI: 10.1088/1742-6596/513/2/022016.

[7] S. Pflüger. "Precise Determination of the Luminosity with the PANDA Luminosity Detector". PhD thesis. Ruhr-Universität Bochum, 2017.

[8] Wikipedia contributors. *Standard Model — Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Standard_Model&oldid=1083767944. [Online; accessed 21-April-2022]. 2022.

[9] M. E. Peskin. *An introduction to quantum field theory*. CRC press, 2018.

[10] E. Noether. "Invariant variational problems". German. In: *Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl.* 1918 (1918), pp. 235–257.

[11] W. Pauli. "Über den Zusammenhang des Abschlusses der Elektronengruppen im Atom mit der Komplexstruktur der Spektren". In: *Zeitschrift für Physik* 31.1 (1925), pp. 765–783.

[12] Bose. "Plancks gesetz und lichtquantenhypothese". In: *Zeitschrift für Physik* 26.1 (1924), pp. 178–181.

[13] E. Wigner. "On the Consequences of the Symmetry of the Nuclear Hamiltonian on the Spectroscopy of Nuclei". In: *Phys. Rev.* 51 (2 Jan. 1937), pp. 106–119. DOI: 10.1103/PhysRev.51.106.

[14] M. Gell-Mann. "Isotopic Spin and New Unstable Particles". In: *Phys. Rev.* 92 (3 Nov. 1953), pp. 833–834. DOI: 10.1103/PhysRev.92.833.

[15] M. Gell-Mann and A. Pais. "Behavior of Neutral Particles under Charge Conjugation". In: *Phys. Rev.* 97 (5 Mar. 1955), pp. 1387–1389. DOI: 10.1103/PhysRev.97.1387.

[16] T. D. Lee and C. N. Yang. "Charge conjugation, a new quantum number G, and selection rules concerning a nucleon-antinucleon system". In: *Il Nuovo Cimento (1955-1965)* 3.4 (1956), pp. 749–753.

[17] S. L. Glashow. "The renormalizability of vector meson interactions". In: *Nuclear Physics* 10 (1959), pp. 107–117. DOI: 10.1016/0029-5582(59)90196-8.

[18] S. Weinberg. "A Model of Leptons". In: *Phys. Rev. Lett.* 19 (21 Nov. 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264.

[19] A. Salam and J. C. Ward. "Electromagnetic and weak interactions". In: *Physics Letters* 13.2 (1964), pp. 168–171. DOI: 10.1016/0031-9163(64)90711-5.

[20] G. P. Zeller et al. "Precise Determination of Electroweak Parameters in Neutrino-Nucleon Scattering". In: *Phys. Rev. Lett.* 88 (9 Feb. 2002), p. 091802. DOI: 10.1103/PhysRevLett.88.091802.

[21] ALEPH, DELPHI, L3, OPAL, SLD Collaborations et al. "Precision electroweak measurements on the Z resonance". In: *Physics Reports* 427.5 (2006), pp. 257–454. ISSN: 0370-1573. DOI: 10.1016/j.physrep.2005.12.006.

[22] K. Abe et al. "High-Precision Measurement of the Left-Right Z Boson Cross-Section Asymmetry". In: *Phys. Rev. Lett.* 84 (26 June 2000), pp. 5945–5949. DOI: 10.1103/PhysRevLett.84.5945.

[23] T. D. Lee. *Particle Physics and Introduction to Field Theory: Revised and Updated First Edition*. Contemporary concepts in physics. Taylor & Francis, 1981. ISBN: 9783718600328.

[24] M. Gell-Mann. "Symmetries of Baryons and Mesons". In: *Phys. Rev.* 125 (3 Feb. 1962), pp. 1067–1084. DOI: 10.1103/PhysRev.125.1067.

[25] D. Griffiths. *Introduction to Elementary Particles*. Physics textbook. Wiley, 2008. ISBN: 9783527618477.

[26] P. W. Higgs. "Broken Symmetries and the Masses of Gauge Bosons". In: *Phys. Rev. Lett.* 13 (16 Oct. 1964), pp. 508–509. DOI: 10.1103/PhysRevLett.13.508.

[27] F. Englert and R. Brout. "Broken Symmetry and the Mass of Gauge Vector Mesons". In: *Phys. Rev. Lett.* 13 (9 Aug. 1964), pp. 321–323. DOI: 10.1103/PhysRevLett.13.321.

[28] G. S. Guralnik, C. R. Hagen, and T. W. B. Kibble. "Global Conservation Laws and Massless Particles". In: *Phys. Rev. Lett.* 13 (20 Nov. 1964), pp. 585–587. DOI: 10.1103/PhysRevLett.13.585.

[29] ATLAS Collaboration. "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 1–29. DOI: 10.1016/j.physletb.2012.08.020.

[30] CMS Collaboration. "Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC". In: *Phys. Lett. B* 716 (2012), pp. 30–61. DOI: 10.1016/j.physletb.2012.08.021.

[31] Z. Ma. *Group Theory in Physics - the Lie Algebra (3rd ed.)* Science Press, 2015.

[32] Z. Xiao and C. Lv. *Introduction to particle physics*. Science Press, 2016.

[33] S. Navas et al. "Review of particle physics". In: *Phys. Rev. D* 110.3 (2024), p. 030001. DOI: 10.1103/PhysRevD.110.030001.

[34] D. J. Gross and F. Wilczek. "Ultraviolet Behavior of Nonabelian Gauge Theories". In: *Phys. Rev. Lett.* 30 (1973). Ed. by J. C. Taylor, pp. 1343–1346. DOI: 10.1103/PhysRevLett.30.1343.

[35] H. D. Politzer. "Reliable Perturbative Results for Strong Interactions?" In: *Phys. Rev. Lett.* 30 (26 June 1973), pp. 1346–1349. DOI: 10.1103/PhysRevLett.30.1346.

[36] F. Wilczek. "Origins of mass". In: *Open Physics* 10.5 (2012), pp. 1021–1037. DOI: doi:10.2478/s11534-012-0121-0.

[37] D. Dominguez. "A graphic impression of quarks and gluons inside the proton". In: (2019). General Photo. URL: https://cds.cern.ch/record/2678286.

[38] A. W. Thomas and W. Weise. *The Structure of the Nucleon*. Germany: Wiley, 2001. ISBN: 978-3-527-40297-7, 978-3-527-60314-5. DOI: 10.1002/352760314X.

[39] M. Gell-Mann. "A Schematic Model of Baryons and Mesons". In: *Phys. Lett.* 8 (1964), pp. 214–215. DOI: 10.1016/S0031-9163(64)92001-3.

[40] C. D. Roberts and A. G. Williams. "Dyson-Schwinger equations and their application to hadronic physics". In: *Progress in Particle and Nuclear Physics* 33 (1994), pp. 477–575. ISSN: 0146-6410. DOI: 10.1016/0146-6410(94)90049-3.

[41] M. Gell-Mann. "The Eightfold Way: A Theory of strong interaction symmetry". In: (Mar. 1961). DOI: 10.2172/4008239.

[42] Y. Ne'eman. "Derivation of strong interactions from a gauge invariance". In: *Nuclear Physics* 26.2 (1961), pp. 222–229. ISSN: 0029-5582. DOI: 10.1016/0029-5582(61)90134-1.

[43] S. L. Olsen, T. Skwarnicki, and D. Zieminska. "Nonstandard heavy mesons and baryons: Experimental evidence". In: *Reviews of Modern Physics* 90.1 (Feb. 2018). ISSN: 1539-0756. DOI: 10.1103/revmodphys.90.015003.

[44] L. R. Wollenberg. "Measurement of the Branching Fraction of the Decay Channel $J/\psi \to \bar{p}\Sigma^+ K_S^0 + c.c.$". PhD thesis. Ruhr-Universität Bochum, 2024.

[45] V. Crede and C. A. Meyer. "The experimental status of glueballs". In: *Progress in Particle and Nuclear Physics* 63.1 (2009), pp. 74–116. ISSN: 0146-6410. DOI: 10.1016/j.ppnp.2009.03.001.

[46] C. J. Morningstar and M. Peardon. "Glueball spectrum from an anisotropic lattice study". In: *Phys. Rev. D* 60 (3 July 1999), p. 034509. DOI: 10.1103/PhysRevD.60.034509.

[47] D. Horn and J. Mandula. "Model of mesons with constituent gluons". In: *Phys. Rev. D* 17 (3 Feb. 1978), pp. 898–908. DOI: 10.1103/PhysRevD.17.898.

[48] L. Liu et al. "Excited and exotic charmonium spectroscopy from lattice QCD". In: *Journal of High Energy Physics* 2012.7 (July 2012). ISSN: 1029-8479. DOI: 10.1007/jhep07(2012)126.

[49] S. L. Olsen. *XYZ Meson Spectroscopy*. 2015. arXiv: 1511.01589 [hep-ex].

[50] R. J. Jaffe. "Multiquark hadrons. I. Phenomenology of $Q^2\overline{Q}^2$ mesons". In: *Phys. Rev. D* 15 (1 Jan. 1977), pp. 267–280. DOI: 10.1103/PhysRevD.15.267.

[51] L. Maiani et al. "Diquark-antidiquark states with hidden or open charm and the nature of $X(3872)$". In: *Phys. Rev. D* 71 (1 Jan. 2005), p. 014028. DOI: 10.1103/PhysRevD.71.014028.

[52] N. A. Tornqvist. "Isospin breaking of the narrow charmonium state of Belle at 3872 MeV as a deuson". In: *Phys. Lett. B* 590 (2004), pp. 209–215. DOI: 10.1016/j.physletb.2004.03.077.

[53] F. Guo et al. "Hadronic molecules". In: *Rev. Mod. Phys.* 90 (1 Feb. 2018), p. 015004. DOI: 10.1103/RevModPhys.90.015004.

[54] S. Dubynskiy and M. B. Voloshin. "Hadro-charmonium". In: *Physics Letters B* 666.4 (2008), pp. 344–346. ISSN: 0370-2693. DOI: 10.1016/j.physletb.2008.07.086.

[55] S. Dubynskiy, A. Gorsky, and M. B. Voloshin. "Holographic hadro-quarkonium". In: *Physics Letters B* 671.1 (2009), pp. 82–86. ISSN: 0370-2693. DOI: 10.1016/j.physletb.2008.11.040.

[56] E. Braaten. "How the $Z_c(3900)$ Reveals the Spectra of Charmonium Hybrids and Tetraquarks". In: *Phys. Rev. Lett.* 111 (16 Oct. 2013), p. 162003. DOI: 10.1103/PhysRevLett.111.162003.

[57] E. Braaten, C. Langmack, and D. H. Smith. "Born-Oppenheimer approximation for the XYZ mesons". In: *Phys. Rev. D* 90 (1 July 2014), p. 014044. DOI: 10.1103/PhysRevD.90.014044.

[58] J. E. Augustin et al. "Discovery of a Narrow Resonance in e^+e^- Annihilation". In: *Phys. Rev. Lett.* 33 (1974), pp. 1406–1408. DOI: 10.1103/PhysRevLett.33.1406.

[59] J. J. Aubert et al. "Experimental Observation of a Heavy Particle J". In: *Phys. Rev. Lett.* 33 (1974), pp. 1404–1406. DOI: 10.1103/PhysRevLett.33.1404.

[60] E. Eichten et al. "The Spectrum of Charmonium". In: *Phys. Rev. Lett.* 34 (1975). [Erratum: Phys.Rev.Lett. 36, 1276 (1976)], pp. 369–372. DOI: 10.1103/PhysRevLett.34.369.

[61] S. Okubo. "Phi meson and unitary symmetry model". In: *Phys. Lett.* 5 (1963), pp. 165–168. DOI: 10.1016/S0375-9601(63)92548-9.

[62] N. Brambilla et al. "Heavy quarkonium: progress, puzzles, and opportunities". In: *Eur. Phys. J. C* 71 (2011), p. 1534. DOI: 10.1140/epjc/s10052-010-1534-9. arXiv: 1010.5827.

[63] D. M. Asner et al. "Physics at BES-III". In: *Int. J. Mod. Phys. A* 24 (2009), S1–794. arXiv: 0809.1869 [hep-ex].

[64] T. L. Collaboration et al. "The LHCb Detector at the LHC". In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08005. DOI: 10.1088/1748-0221/3/08/S08005.

[65] M. Andreotti et al. "Measurement of the resonance parameters of the $\chi_1(1^3P_1)$ and $\chi_2(1^3P_2)$ states of charmonium formed in antiproton-proton annihilations". In: *Nuclear Physics B* 717.1 (2005), pp. 34–47. ISSN: 0550-3213. DOI: 10.1016/j.nuclphysb.2005.03.042.

[66] H. H. Gutbrod et al. *FAIR Baseline Technical Report*. Jan. 2006.

[67] A. Lehrach et al. "Design Work for the High Energy Storage Ring of the Future GSI Project". In: *Proceedings of the 2005 Particle Accelerator Conference*. 2005, pp. 776–778. DOI: 10.1109/PAC.2005.1590560.

[68] M. Neubert. "Heavy quark symmetry". In: *Physics Reports* 245.5 (1994), pp. 259–395. ISSN: 0370-1573. DOI: 10.1016/0370-1573(94)90091-4.

[69] B. Aubert et al. "Observation of a Narrow Meson State Decaying to $D_s^+\pi^0$ at a Mass of 2.32 GeV/c^2". In: *Phys. Rev. Lett.* 90 (24 June 2003), p. 242001. DOI: 10.1103/PhysRevLett.90.242001.

[70] C. J. Morningstar and M. Peardon. "Glueball spectrum from an anisotropic lattice study". In: *Phys. Rev. D* 60 (3 July 1999), p. 034509. DOI: 10.1103/PhysRevD.60.034509.

[71] C. Amsler and N. A. Törnqvist. "Mesons beyond the naive quark model". In: *Physics Reports* 389 (2004), pp. 61–117. DOI: 10.1016/j.physrep.2003.09.003.

[72] M. Durante et al. "All the fun of the FAIR: fundamental physics at the facility for antiproton and ion research". In: *Physica Scripta* 94.3 (Jan. 2019), p. 033001. DOI: 10.1088/1402-4896/aaf93f.

[73] N. A. Tahir et al. "Design of a Target for Antiproton Production at the Future FAIR Facility". In: *GSI Annual Report*. 2007.

[74] GSI Helmholtzzentrum für Schwerionenforschung GmbH. *The Accelerator Facility*. 2021. URL: https://www.gsi.de/en/researchaccelerators/fair/the_machine.

[75] Q. Hu et al. "A recoil detector for the measurement of antiproton-proton elastic scattering at angles close to 90°". In: *The European Physical Journal A* 50.10 (Oct. 2014). ISSN: 1434-601X. DOI: 10.1140/epja/i2014-14156-0.

[76] T. Stöhlker et al. "SPARC Collaboration: New Strategy for Storage Ring Physics at FAIR". In: *Hyperfine Interact.* 227 (2014). Ed. by Krassimira Marinova and Magdalena Kowalska, pp. 45–53. DOI: 10.1007/s10751-014-1047-2. arXiv: 1401.7595 [physics.acc-ph].

[77] U. Kurilla. *General PANDA material for slides, logos, detector drawings, etc.* 2017.

[78] H. Stockhorst et al. "Cooling Scenario for the HESR Complex". In: *AIP Conference Proceedings* 821.1 (Mar. 2006), pp. 190–195. ISSN: 0094-243X. DOI: 10.1063/1.2190110. eprint: https://pubs.aip.org/aip/acp/article-pdf/821/1/190/11474884/190_1_online.pdf.

[79] K. Götzen. *Average Luminosities and Event Rates at PANDA*. Tech. rep. IN-IDE-2015-002. FAIR/PANDA, 2015.

[80] D. M. Welsch. "Investigation and Optimization of Transverse Non-Linear Beam Dynamics in the High-Energy Storage Ring HESR". PhD thesis. Rheinische Friedrich-Wilhelms-Universität Bonn, Mar. 2010.

[81] D. Reistad et al. "Calculations on High-energy Electron Cooling in the HESR". In: *Conf. Proc. C* 07091010 (2007). Ed. by Rainer W. Hasse and Volker R. W. Schaa, moa2c05.

[82] A. Lehrach et al. "Beam performance and luminosity limitations in the high-energy storage ring (HESR)". In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 561.2 (2006). Proceedings of the Workshop on High Intensity Beam Dynamics, pp. 289–296. ISSN: 0168-9002. DOI: 10.1016/j.nima.2006.01.017.

[83] J. Biri et al. "Beam position monitor electronics at the cooler synchrotron COSY Jülich". In: *IEEE Transactions on Nuclear Science* 41.1 (1994), pp. 221–224. DOI: 10.1109/23.281493.

[84] A. Khoukaz. "Internal targets for the PANDA experiment". In: *PoS* STORI11 (2011), p. 036. DOI: 10.22323/1.150.0036.

[85] PANDA Collaboration. *Technical Design Report for the PANDA Internal Targets (The Cluster-Jet Target and Developments for the Pellet Target)*. Tech. rep. RE-TDR-2012-002. FAIR/PANDA, 2012.

[86] A. Täschner et al. "High density cluster jet target for storage ring experiments". In: *Nucl. Instrum. Meth. A* 660.1 (2011), pp. 22–30. ISSN: 0168-9002. DOI: 10.1016/j.nima.2011.09.024.

[87] S. Vestrick et al. "The Münster cluster-jet target for the future PANDA experiment". In: *EPJ Web Conf.* 285 (2023), p. 02002. DOI: 10.1051/epjconf/202328502002.

[88] M. Büscher et al. "Production of Hydrogen, Nitrogen and Argon Pellets with the Moscow-Jülich Pellet Target". In: *International Journal of Modern Physics E* 18.02 (2009), pp. 505–510. DOI: 10.1142/S0218301309012562.

[89] B. Trostell. "Vacuum injection of hydrogen microsphere beams". In: *Nucl. Instrum. Meth. A* 362 (1995). Ed. by W. R. Lozowski and H. J. Maier, pp. 41–52. DOI: 10.1016/0168-9002(95)00302-9.

[90] C. Ekström. "Internal targets — a review". In: *Nucl. Instrum. Meth. A* 362.1 (1995), pp. 1–15. DOI: 10.1016/0168-9002(95)00240-5.

[91] PANDA Collaboration. *Technical Design Report for the PANDA Solenoid and Dipole Spectrometer Magnets*. 2009. arXiv: 0907.0169 [physics.ins-det].

[92] A. Karavdina. "Preparation for the accurate luminosity measurement by antiproton-proton elastic scattering and feasibility study for observation of h_c hadronic decay modes at the PANDA experiment". PhD thesis. Mainz U., 2015.

[93] W. Erni et al. "Technical Design Report for the $\overline{\text{P}}$ANDA (AntiProton Annihilations at Darmstadt) Straw Tube Tracker". In: *The European Physical Journal A* 49.2 (Feb. 2013). ISSN: 1434-601X. DOI: 10.1140/epja/i2013-13025-8.

[94] M. Ball et al. *Technical Design Study for the PANDA Time Projection Chamber*. 2012. arXiv: 1207.0013 [physics.ins-det].

[95] PANDA Collaboration. *Technical Design Report for the PANDA Forward Tracker*. Tech. rep. RE-TDR-2017-001. FAIR/PANDA, 2018.

[96] PANDA Collaboration. *Technical Design Report for PANDA Electromagnetic Calorimeter (EMC)*. 2008. arXiv: 0810.1216 [physics.ins-det].

[97] PANDA Collaboration. *Technical Design Report for the Panda Forward Spectrometer Calorimeter*. 2017. arXiv: 1704.02713 [physics.ins-det].

[98] PANDA Collaboration. *Technical Design Report for the PANDA Barrel DIRC Detector*. 2017. arXiv: 1710.00684 [physics.ins-det].

[99] PANDA Collaboration. *Technical Design Report for the PANDA Endcap Disc DIRC*. 2019. arXiv: 1912.12638 [physics.ins-det].

[100] PANDA Collaboration. *Forward RICH at PANDA*. Tech. rep. TA-WKS-2019-048. FAIR/PANDA, 2019.

[101] PANDA Collaboration. *Technical Design Report for the PANDA Barrel TOF*. Tech. rep. RE-TDR-2016-003. FAIR/PANDA, 2016.

[102] PANDA Collaboration. *Technical Design Report for: PANDA Forward Time of Flight detector (FToF wall)*. Tech. rep. RE-TDR-2016-004. FAIR/PANDA, 2018.

[103] PANDA Collaboration. *Technical Design Report for the PANDA Muon System*. Tech. rep. RE-TDR-2012-003. FAIR/PANDA, 2012.

[104] PANDA Collaboration. *Technical Design Report for the: PANDA Micro Vertex Detector*. 2012. arXiv: 1207.6581 [physics.ins-det].

[105] G. Lutz et al. *Semiconductor radiation detectors*. Springer, 2007.

[106] G. F. Knoll. *Radiation detection and measurement*. John Wiley & Sons, 2010.

[107] R. Kliemt. "Simulations with the PANDA Micro-Vertex-Detector". PhD thesis. Universität Bonn, 2013.

[108] PANDA Collaboration. *Technical Design Report for the PANDA Luminosity Detector*. Tech. rep. RE-TDR-2015-001. FAIR/PANDA, 2015.

[109] H. H. Leithoff. "Ein Kühlsystem für den $\bar{P}$ANDA Luminositätsdetektor und Suche nach dem Zerfall $\epsilon^+\epsilon^- \to \gamma\eta\chi_{c0}$ bei der Schwerpunktsenergie von 4.26 GeV an BESIII". doctoralthesis. Ruhr-Universität Bochum, 2020.

[110] I. Perić. "A novel monolithic pixelated particle detector implemented in high-voltage CMOS technology". In: *Nucl. Instrum. Meth. A* 582.3 (2007). VERTEX 2006, pp. 876–885. ISSN: 0168-9002. DOI: 10.1016/j.nima.2007.07.115.

[111] H. Augustin et al. "The MuPix system-on-chip for the Mu3e experiment". In: *Nucl. Instrum. Meth. A* 845 (2017). Proceedings of the Vienna Conference on Instrumentation 2016, pp. 194–198. ISSN: 0168-9002. DOI: 10.1016/j.nima.2016.06.095.

[112] S. Spataro *et al.* [PANDA collaboration]. "The PandaRoot framework for simulation, reconstruction and analysis". In: *Journal of Physics: Conference Series*. Vol. 331. 3. IOP Publishing. 2011.

[113] M. Al-Turany et al. "The FairRoot framework". In: *Journal of Physics: Conference Series* 396.2 (Dec. 2012), p. 022001. DOI: 10.1088/1742-6596/396/2/022001.

[114] R. Brun and F. Rademakers. "ROOT—An object oriented data analysis framework". In: *Nuclear instruments and methods in physics research section A* 389.1-2 (1997).

[115] I. Hrivnacova et al. "The Virtual Monte Carlo". In: *eConf* C0303241 (2003), THJT006. arXiv: cs/0306005.

[116] R. Brun et al. *GEANT: Detector Description and Simulation Tool*. CERN Program Library. Long Writeup W5013. Geneva: CERN, 1993. DOI: 10.17181/CERN.MUHF.DMJ1.

[117] S. Agostinelli et al. "Geant4—a simulation toolkit". In: *Nuclear Instruments and Methods in Physics Research Section A* 506.3 (2003).

[118] G. Hugo et al. "Latest FLUKA developments". In: *EPJ Nuclear Sciences & Technologies* 10 (Dec. 2024). DOI: 10.1051/epjn/2024023.

[119] V. Radeka and R. A. Boie. "Centroid finding method for position-sensitive detectors". In: *Nuclear Instruments and Methods* 178.2 (1980), pp. 543–554. ISSN: 0029-554X. DOI: 10.1016/0029-554X(80)90836-8.

[120] H. Xu et al. "Measurement of proton-proton elastic scattering into the Coulomb region at Pbeam = 2.5, 2.8 and 3.2 GeV/c". In: *Physics Letters B* 812 (2021), p. 136022. ISSN: 0370-2693. DOI: 10.1016/j.physletb.2020.136022.

[121] Y. Zhang et al. "Contribution of Coulomb interaction to elastic pp and $p\bar{p}$ scattering in holographic QCD". In: *Phys. Rev. D* 108 (6 Sept. 2023), p. 066001. DOI: 10.1103/PhysRevD.108.066001.

[122] T. T. Wu. "High-Energy Large-Momentum-Transfer Processes and Analyticity in Coordinate Space". In: *Phys. Rev.* 143 (4 Mar. 1966), pp. 1117–1134. DOI: 10.1103/PhysRev.143.1117.

[123] N. Amos et al. "Measurement of small-angle antiproton-proton and proton-proton elastic scattering at the CERN intersecting storage rings". In: *Nuclear Physics B* 262.4 (1985), pp. 689–714. ISSN: 0550-3213. DOI: 10.1016/0550-3213(85)90511-5.

[124] A. Capella et al. "Dual parton model". In: *Physics Reports* 236.4 (1994), pp. 225–329. ISSN: 0370-1573. DOI: 10.1016/0370-1573(94)90064-7.

[125] A. Capella. "Introduction to the Dual Parton Model". In: *Perspectives in the Structure of Hadronic Systems*. Ed. by M. N. Harakeh, J. H. Koch, and O. Scholten. Boston, MA: Springer US, 1994, pp. 287–310. ISBN: 978-1-4615-2558-5. DOI: 10.1007/978-1-4615-2558-5_12.

[126] L. Susskind. "Applications of the Dual-Parton Model to High-Energy Multiparticle Processes". In: *Phys. Rev. D* 6 (3 Aug. 1972), pp. 894–900. DOI: 10.1103/PhysRevD.6.894.

[127] P. K. Damuka, R. Aggarwal, and M. Kaur. "Dual parton model for the charged multiplicity in $p-p$ collisions at 13, 13.6 TeV and for a future LHC energy of 27 TeV". In: *Phys. Rev. D* 106 (7 Oct. 2022), p. 076017. DOI: 10.1103/PhysRevD.106.076017.

[128] A. Galoyan et al. *Parametrization of the $\bar{P}P$ Elastic Scattering Differential Cross Section Between 2 GeV/c $\leq P_{lab} \leq$ 16 GeV/c*. 2008. arXiv: 0809.3804 [hep-ex].

[129] I. Abt et al. "CATS: a cellular automaton for tracking in silicon for the HERA-B vertex detector". In: *Nucl. Instrum. Meth. A* 489.1 (2002), pp. 389–405. ISSN: 0168-9002. DOI: 10.1016/S0168-9002(02)00790-8.

[130] C. Höppner et al. "A novel generic framework for track fitting in complex detector systems". In: *Nucl. Instrum. Meth. A* 620.2 (2010), pp. 518–525. ISSN: 0168-9002. DOI: 10.1016/j.nima.2010.03.136.

[131] G. Lutz. "Optimum Track Fitting in the Presence of Multiple Scattering". In: *Nucl. Instrum. Meth. A* 273 (1988). Ed. by O. Botner, p. 349. DOI: 10.1016/0168-9002(88)90836-4.

[132] V. Blobel. "A new fast track-fit algorithm based on broken lines". In: *Nucl. Instrum. Meth. A* 566.1 (2006). TIME 2005, pp. 14–17. ISSN: 0168-9002. DOI: 10.1016/j.nima.2006.05.156.

[133] P. Billoir et al. "A parametrized Kalman filter for fast track fitting at LHCb". In: *Computer Physics Communications* 265 (2021), p. 108026. ISSN: 0010-4655. DOI: 10.1016/j.cpc.2021.108026.

[134] P. Astier et al. "Kalman filter track fits and track breakpoint analysis". In: *Nucl. Instrum. Meth. A* 450.1 (2000), pp. 138–154. ISSN: 0168-9002. DOI: 10.1016/S0168-9002(00)00154-6.

[135] A. Fontana et al. "Use of GEANE for tracking in virtual Monte Carlo". In: *Journal of Physics: Conference Series* 119.3 (July 2008), p. 032018. DOI: 10.1088/1742-6596/119/3/032018.

[136] V. Innocente, M. Maire, and E. Nagy. "GEANE: Average tracking and error propagation package". In: *Workshop on Detector and Event Simulation in High-energy Physics (MC '91)*. 1991, pp. 58–78.

[137] A. Khoukaz. *Residual Gas from PANDA Targets*. Talk at PANDA Collaboration Meeting. 2021.

[138] M. Weide. *Influence of the beam target interaction on the vacuum at the IP*. Talk at PANDA Collaboration Meeting. 2024.

[139] C. Mannweiler. *PANDA Cluster Target activities in Münster*. Talk at PANDA Collaboration Meeting. 2021.

[140] N. Berger *et al.* [Mu3e Collaboration]. "The mu3e experiment". In: *Nuclear Physics B* 248 (2014).

[141] H. Augustin et al. *MuPix10: First Results from the Final Design*. 2020. arXiv: 2012.05868 [physics.ins-det].

[142] N. Boelger. "Bau der Sensor-Module für den Prototyp des PANDA Luminositätsdetektors". MA thesis. Universität Bochum, 2021.

[143] H. Augustin et al. “MuPix: An HV–MAPS for the Mu3e Experiment”. In: *JPS Conf. Proc.* 42 (2024), p. 011020. DOI: 10.7566/JPSCP.42.011020.

[144] H. C. Augustin. *HVMAPS for Mu3e and beyond*. Talk at the Vienna Conference on Instrumentation (VCI) 2025. 2025.

[145] S. Bianco. “Tracking and vertex reconstruction with the PANDA Micro-Vertex-Detector”. PhD thesis. Universität Bonn, 2013.

[146] S. Bianco. “Characterization of the PANDA Micro-Vertex-Detector and analysis of the first data measured with a tracking station”. In: *IEEE Nuclear Science Symposuim & Medical Imaging Conference*. 2010, pp. 1149–1152. DOI: 10.1109/NSSMIC.2010.5873947.

[147] A. Alicke. “Development of fast track finding algorithms for densely packed straw tube trackers and its application to\(\Xi\)(1820) hyperon reconstruction for the PANDA experiment”. PhD thesis. Dissertation, Bochum, Ruhr-Universität Bochum, 2023.

[148] A. Alicke, T. Stockmanns, and J. Ritman. “Secondary Track Finding for PANDA”. In: FAIR next generation scientists - 7th Edition Workshop, Paralia (Greece). May 23, 2022.

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- **Date of Birth** | December 09, 1996
- **Place of Birth** | Hubei, China
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Education

- **PhD Candidate in Physics** | Sept. 2019 - Present
  Ruhr-University Bochum (Bochum, Germany)
- **B.Sc. in Physics** | Sept. 2015 - Jun. 2019
  Nankai University (Tianjin, China)
- **High School** | Sept. 2012 - Jun. 2015
  Xiangyang No.5 Middle School (Hubei, China)
- **Middle School** | Sept. 2009 - Jun. 2012
  Xiangyang Zhugeliang middle school (Hubei, China)
- **Primary School** | Sept. 2003 - Jun. 2009
  Xiangyang Renmin Road Primary School (Hubei, China)

Bachelor Thesis

- **Title** | Measurements of the absolute branching factions of the decays $D^{+(0)} \to K_S^0 \pi^{+(0)} \pi^0 \pi^0$
- **Supervisor** | Prof. Dr. Ming-gang Zhao & Prof. Dr. Hai-long Ma

Master Thesis

- **Title** | High Voltage Monolithic Active Pixel Sensors for the P$\bar{}$ANDA Micro Vertex Detector
- **Supervisor** | Prof. Dr. Miriam Fritsch

PhD Thesis

- **Title** | Luminosity Determination with Restgas Background from the Target for the PANDA Experiment
- **Supervisor** | Prof. Dr. Miriam Fritsch

Skills & Competencies

- **Programming & Software:** C++, Python (incl. NumPy, Pandas, Scikit-learn), ROOT, Linux/Shell Scripting, Git, HPC environments.
- **Modeling & Data Science:** Machine Learning (Cross-validation, Model selection), Bayesian Inference, Statistical Data Analysis, Monte-Carlo Simulations
- **Domain Expertise:** Software Development for Detectors in High Energy Physics, Optimization of the Detector Design Based on Simulation, Data analysis in HEP
- **Collaboration & Leadership:** Cross-functional collaboration with software, hardware, and data analysis teams, effective communication within large-scale international collaborations.
- **Languages:** English (Proficient), German (CEFR-B1), Chinese (Native)

Professional Experiences.

- **PhD Candidate** | Sep. 2019 – Present

Institute of Experimental Physics I, Ruhr-University Bochum, Research Group of Prof. Dr. Miriam Fritsch (Physics of Hadrons and Nuclei / PANDA Collaboration)

Leading software development for luminosity detectors and sensor optimization.

- **Research Assistant** | Jul. 2018 – Aug. 2019

  Institute of High Energy Physics, Chinese Academy of Sciences, Research Group of Prof. Dr. Hai-long Ma (BESIII Collaboration)

  Conducted branching fraction measurements using BESIII data.

- **Research Assistant** | Jun. 2017 – Sept. 2017

  European Organization for Nuclear Research (CERN) / Massachusetts Institute of Technology (MIT), Department of Prof. Dr. Samuel C. C. Ting (AMS Collaboration)

  Worked on simulation and tracker resolution for the Alpha Magnetic Spectrometer (AMS-02).

- **Research Assistant** | Apr. 2017 – May. 2018

  School of Physics, Nankai University, Research Group of Prof. Dr. Ming-gang Zhao (BESIII Collaboration)

  Conducted branching fraction measurements using BESIII data.

Selected Research Projects

Luminosity Determination with PANDA Restgas Target | Mar. 2022 – Jul. 2024

- Developed a comprehensive luminosity fitting framework for the PANDA luminosity detector, incorporating parameters for interaction point (IP) shift, beam tilt, and beam divergence.
- Integrated the PANDA restgas target module into the PandaRoot and LuminosityFit packages.
- Optimized data selection by applying spatial cuts around the IP.

PANDA Restgas Profile Determination | Jul. 2024 – Present

- Designed and implemented algorithms for track reconstruction and particle identification (PID), specifically accounting for shifted IP.
- Evaluated reconstruction efficiency and resolution along the longitudinal (z) axis and reconstructed the restgas density profile by applying efficiency and resolution corrections.

HVMAPS for the PANDA Micro Vertex Detector | Sept. 2019 – Mar. 2022

- Redesigned the PANDA MVD pixel sensor architecture, transitioning from previous Hybrid Pixel Sensors to High Voltage Monolithic Active Pixel Sensors (HVMAPS).
- Conducted comprehensive performance studies, benchmarking resolution and efficiency of the HVMAPS design against previous hybrid sensors.

Measurements of $\mathcal{B}(D^{+(0)} \to K_S^0 \pi^{+(0)} \pi^0 \pi^0)$ | Jul. 2018 – Aug. 2019

- Performed precision measurements of absolute branching fractions for the decay $D^{+(0)} \to K_S^0 \pi^{+(0)} \pi^0 \pi^0$ utilizing the double-tag method.

Measurements of $\mathcal{B}(\chi_{cJ} \to K_S^0 K_S^0 \pi^0 \pi^0)$ | Apr. 2017 – May. 2018

- Analyzed decay channels to accurately measure the absolute branching fractions of $\chi_{cJ} \to K_S^0 K_S^0 \pi^0 \pi^0$.

Improvement of tracker resolution for Proton and Helium for AMS | Jun. 2017 – Sept. 2017

- Optimized the tracker resolution modeling for Proton and Helium isotopes for the Alpha Magnetic Spectrometer (AMS-02) on the International Space Station using Monte-Carlo simulations.